Meighan I. Dillon

Geometry Through History

Euclidean, Hyperbolic, and Projective Geometries

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To my students, my teachers,
and my family

Preface

Geometry occupies a peculiar niche in both the high school and the undergraduate curricula. Undergraduate mathematics students routinely complete four-year programs without a geometry course, yet the lion's share of the standard curriculum can be motivated almost entirely by geometry. Calculus, still preponderant in the high school and undergraduate curricula, can be viewed as Euclidean geometry under the microscope. Indeed, one may be forgiven for thinking that the heavy representation of calculus in undergraduate programs is an artifact of the recognition that the peculiar nature of Euclidean geometry rests squarely on the nature of the real line.

Euclidean geometry is categorical: effectively there is one Euclidean plane, and it is $\mathbb{R}^2$. From this point of view, almost everything we teach undergraduates and high school students is geometry or at least, geometric. By neglecting to deliver dedicated geometry courses, though, we and our students may be missing out on the foundations of modern mathematics, an understanding of the role of projective geometry in contemporary mathematics and applications, and the sheer joy of interesting and challenging problems with elementary solutions.

Several years ago I developed a geometry course for future teachers, the one cohort for whom geometry is required in most US colleges and universities. This was for a non-standard education program being taught in a department that had no education faculty per se. There was no prescribed syllabus for the course and no committee to appease. I could do whatever I wanted. This book grew out of that project.

Geometry can seem wildly disparate and for my own edification, I wanted to understand where it came from and how it got here. Piecing that story together informed my choices for the course. I wanted to start with a close reading of Euclid, the better to tell the story of the two thousand year search for an explanation of the parallel postulate. With the resolution of the problem of unique parallels, an obstacle to the development of geometry was removed. An understanding of the obstacle and its removal would form the basis for learning more modern practices and subjects in geometry.

Projective geometry is at once so shocking and so accessible it seems a crime to let mathematics students graduate without having met it. I would teach it in my course, emphasizing the notion of projective geometry as a completion of Euclidean geometry. It was a goal to proceed far enough into the subject for students to appreciate the idea that some projective planes are non-Desarguesian. This would require modern algebra. I would supply the necessary tools.

The meshing of geometry and modern algebra heralded by Felix Klein's Erlangen Program would be an influence on the course. Like the shift in thinking that was essential for the resolution of the problem of the parallels, the Erlangen Program is not itself mathematics. It is a point of view, a philosophy, a framework that informed the modernization of the subject. I think everyone with an interest in mathematics should know about it.

Physics majors and computer science students appreciated the course and registered in numbers that exceeded those of the future teachers. In an early iteration of the course, one student asked if we could cover the quaternions. With the freedom I was afforded, I was able to accommodate the request. The fruit of those efforts evolved into Chapter 10.

Regardless of other changes I made from year to year, Book I from the Heath edition of *The Elements* [7] always started the course and established its trajectory. A typical approach to Euclidean geometry employs a modern point of view, starting with axioms based more or less on those of Hilbert. That approach has its merits, but anyone who takes the time to steep in Euclid's text will be richly rewarded for the effort. The book's historical significance is considerable and it can be understood by students at almost any mathematical level. Euclid's one demand is patience: the patience to read carefully, a skill every mathematics student should be required to cultivate. The attentive reader hears Euclid's voice with utter clarity from across the cold void of 2300 years. Not only do we know exactly what Euclid was writing about, it is still germane to our understanding of the world.

My assumption is that anyone using this book will read it. Readers should peruse every exercise. The only prerequisites for the book are a year of calculus, an introductory course in linear algebra, and an interest in mathematics. For the linear algebra, I have provided a refresher. This is not intended to substitute for a course in the subject.

There is some modern algebra in later chapters. The first modern algebra one encounters doing geometry arises typically in the form of symmetry groups. If one is to study planar algebraic curves, the next step is a bit steep. I have endeavored to provide that background, as well. I hope it is just enough, but not too much. If the geometry whets the reader's appetite for more modern algebra, so much the better.

Chapters on Euclid, Arabic and medieval mathematics, and the role of hyperbolic geometry in the resolution of the problem of unique parallels start the course. We turn our attention to Hilbert's *Foundations of Geometry* [3] in Chapter 4.

The point of view for the first few chapters is naïve. We pretend we know little but what Euclid tells us, and maybe a few other things from our experiences on earth as humans. Once we have paid our respects to Hilbert, we proceed as though we know what we know. We know coordinate geometry in the xy-plane. We know

polar coordinates, basic trigonometry, and how to factor a small positive integer into powers of primes.

More intricate problems in Euclidean geometry motivate Chapter 5, where we meet the Euler line and the nine point circle. One of the tools we take up there is circular inversion, which we use in Chapter 6 to study the Poincaré disk model for the hyperbolic plane. In Chapter 7, we start using linear algebra and modern algebra, which remain in the story for the rest of the book. Affine geometry is the geometry of vector spaces, and vector spaces over finite fields are interesting and accessible, so we define fields in the chapter on affine geometry. The chapter on curves includes material on UFDs.

A book like this is largely an assembly job. While writing the manuscript, I was surrounded by, and drew heavily from, stacks of wonderful books. I have mentioned Heath's edition of Euclid. There is now an edition of *The Elements* on the web, authored with care and style by David Joyce [16]. Its internal links make navigating *The Elements* easier than ever. The influence of Hartshorne's *Geometry: Euclid and Beyond* [12], which also supplies a reader for Euclid, will be evident to anyone familiar with that work. As a student, as a teacher, and while preparing this manuscript, I have turned to Fishback's classic *Projective and Euclidean Geometry* [9] again and again. Books written in the 1960s and earlier by Moise [23], and Kay [18] likewise proved sturdy companions during the preparation of this book. The chapter on curves relies heavily on Walker's text [29]. For the duration of this project, Coxeter's books [3], [4], and [5] have been at my elbow as have many more books that served as invaluable resources on curves, hyperbolic geometry, quaternions, and the history of mathematics.

The best thing about teaching is how much you learn. The same can be said for writing a book like this. It has been a tremendously rewarding experience and I am grateful to all the people who supported me in various ways as I saw the project through. My thanks go first and foremost to the students who signed up for my geometry courses. For giving me the opportunity and freedom to create and teach a course to my own taste and to prioritize completion of this project, my department chairs at Southern Polytechnic and Kennesaw State deserve mention: Andrew McMorran, Sarah Holliday, Joe DeMaio, and Sean Ellermeyer. For their support in so many different ways, I thank my husband, Steve Edwards, and children, Miles Dillon Edwards and Annabel Edwards. For encouraging me to make this book the best it could be, I am grateful to Loretta Bartolini and her associates at Springer. Finally, I am most indebted to all my teachers, but particularly my geometry teachers: the late Geoffrey Berry, at Mamaroneck High School; John Loustau, at Hunter College; John Faulkner, my geometry teacher and doctoral advisor at the University of Virginia; and Joseph M. Landsberg, my geometry teacher, colleague, and mentor during a sabbatical year at Georgia Tech. I am lucky to have had these teachers, and so many others, at various points of my life. Without knowledge of it, they contributed in myriad ways to this book.

Roswell, GA, USA Meighan I. Dillon
October 2017

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Chapter 1

The Elements of Euclid

1.1 Introduction

The word **geometry** literally means *earth measurement*. When geometry was developed by the ancients of various cultures, it was probably for earth measurement, that is, surveying. In mathematics today, geometry generally refers to the study of curves and surfaces. Different branches of geometry—differential geometry, algebraic geometry, geometric analysis—are distinguished in part by the objects of study and in part by the different sets of tools brought to bear upon the objects of study. In differential geometry, for example, the objects are curves and surfaces in complex space and the tools arise largely from differential calculus. Algebraic geometry is the study of objects that can be described using rational functions and the tools arise typically, but not always, from modern algebra.

Many mathematicians who specialize in geometry do not talk about similar triangles or alternate interior angles or angles in a circle or most of the other things you probably think of when you think about school geometry, that is, Euclidean geometry. A part of Euclidean geometry that does come up in many branches of geometry studied today is *projective geometry*. Although this is not a course in projective geometry, we will study certain projective planes. In this and in other ways, this course forms a bridge from high school geometry to more advanced studies.

The school geometry you studied was probably a course in **synthetic Euclidean geometry**. Synthetic geometry starts from first principles: definitions, and **axioms** or **postulates**, which are fundamental truths held not only to be self-evident, but actually unprovable. The heart of the subject is logical synthesis applied to the axioms in a rigorous fashion to uncover the facts—**theorems**—which relate the objects of the geometry. This remains an attractive subject for students of any age with any background because all assumptions are, ideally, clearly stated at the beginning of the program. The objects seem familiar and the relations among them rich. No previous knowledge is necessary.

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Planar Euclidean geometry may be viewed as the geometry of straight edge and compass. The straight edge is unmarked, so cannot measure length, and the compass is floppy, that is, it does not stay open once you lift it from the page. The objects and all relations among them must be produced using only a straight edge and compass applied to a flat surface. One approach to studying Euclidean geometry is constructive, and usually a portion of the high school course deals with actual constructions. It can be profitable to think of the synthetic approach, in part, as supplying detailed instructions for allowed constructions.

The work that brought synthetic geometry to the world was *The Elements of Euclid*, a detailed analysis of synthetic geometry and number theory in thirteen books dating from approximately 300 B.C. Not only has *The Elements* been studied more or less continuously since it was written, it has remained the most important source for the subject for 2300 years. From the time it was written, *The Elements* has inspired volumes of commentary and annotated texts. The earliest useful commentary, by Proclus of Alexandria, survives from around the year 450.[1] The Heath English language edition of *The Thirteen Books of the Elements* by Euclid dates from 1908. Sir Thomas L. Heath, a British classicist, whose edition contains copious historical notes and commentary, worked from a contemporary Greek edition that had been done by Heiberg.

High school students, if they have a geometry course at all, usually study textbooks based on interpretations, corrections, and adjustments to *The Elements* that have been incorporated into the subject since the time of Euclid. Many of these interpretations have noteworthy histories and pedigrees. Others appear to attempt to take the "royal road to geometry,"[2] endeavoring for a presentation fit for young learners. ([7], p. vi) Though *The Elements* starts from the most basic principles and is self-contained, it was not intended for children. It was probably a collection of results that students would have learned at Plato's Academy.

We start this course with Book I of *The Elements*, adapting from the Heath edition. Book I, and the present chapter, are entirely concerned with plane geometry. We present those results that support Euclid's proof of the Pythagorean Theorem, the penultimate proposition in Book I, and an astonishing expression of nearly the whole of what precedes it.

Exercises 1.1. The goal of these exercises is to jog your memory of high school geometry. Do them without consulting outside sources.

1. Define the following terms.

[1] Pause to ponder the timeframe: Euclid and Proclus were approximately 750 years apart, while Shakespeare and we are approximately 400 years apart.

[2] There is a legend that Ptolemy, a king of Egypt, asked Euclid, his tutor, if there was not a shorter way to learn geometry. Must one go through *The Elements*? Euclid's response was that there was "no royal road to geometry." Similar stories are attributed to other mathematicians in response to complaints from other kings trying to learn mathematics. See [7], p. 1.

(a) point
(b) line
(c) plane
(d) space
(e) triangle
(f) congruent triangles
(g) similar triangles
(h) isosceles triangle
(i) median of a triangle
(j) parallel lines
(k) perpendicular lines
(l) quadrilateral
(m) parallelogram
(n) circle

2. A **transversal** for a given curve is a line that intersects the curve non-tangentially. Suppose a transversal intersects two parallel lines in points A and B. Suppose a second transversal intersects the same two parallels in points C and D respectively. If the transversals intersect at a point

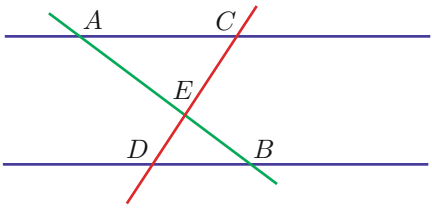


Figure 1.1: Parallel lines with transversals

E, what is the relationship between ΔAEC and ΔBED? Does it matter whether E is between the parallels, as in Fig. 1.1?

3. What is the relationship between $\angle BCD$ and the angles in ΔABC shown in Fig. 1.2?

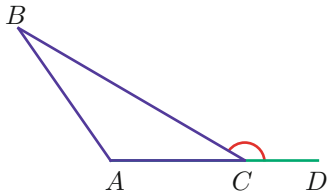


Figure 1.2: $\angle BCD$ is exterior to ΔABC

4. The points A, B, C, D in Fig. 1.3 lie on a circle. Segments AC and BD intersect at the point E. What is the relationship between ΔABE and ΔDCE?

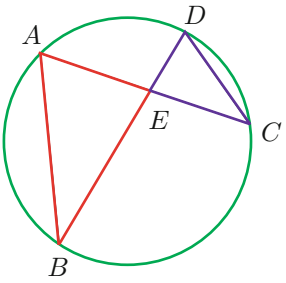


Figure 1.3: Triangles as described in Exercise 4

5. Show that triangles formed by a diagonal of a parallelogram are congruent.

6. How do you bisect an angle using a straight edge and compass?

1.2 Background on Euclid and *The Elements*

There are no contemporary accounts of Euclid's life but there are some details about Euclid that scholars have been able to cobble together indirectly and about which there is little controversy.

It is generally held that Euclid founded a school of mathematics in Alexandria, Egypt.[3] In particular, Archimedes, 287–212 B.C., sometimes described as the greatest mathematician who ever lived, studied in Alexandria at Euclid's school and cited Euclid's work in his own writing. On the other hand, *The Elements* has detailed references to the works of Eudoxus and Theaetetus. To have learned their work, Euclid would have to have gone to Plato's Academy. ([7], p. 2) Plato established the Academy outside Athens around 387 B.C.[4] There is general agreement that Euclid was too young to have studied with Plato himself, and too old to have taught Archimedes in Alexandria. This helps narrow down the dates for Euclid which are currently accepted as about 325–265 B.C.

As a student at the Academy, Euclid would have been the product of a rich tradition of careful thought, schooled in the writings and teachings of Plato, Aristotle, and the Greek geometers. Aristotle, himself a student of Plato, suggested in his own writings that his students had sources that codified the principles of mathematics and science, including geometry, that were accepted at that time. These sources presumably would have been available to Euclid as well. In other words, Euclid's was not the first geometry text, even if we restrict attention to the West. This does not diminish its greatness but it is important

[3] Alexandria is named for Alexander the Great, who, as a child, studied with Aristotle.

[4] This is the origin of the word *academic*. *Academy* is actually the name of the place where Plato set up his institution. The Academy remained in use until 526, another astonishingly long-lived force in the intellectual world.

to maintain perspective. One commonly accepted view of *The Elements* is that it pulled together what was known in geometry at the time, with the goal of proving that the five platonic solids are the only solutions to the problem of constructing a regular solid. There is room for doubt there, though, as several books of *The Elements* have nothing whatever to do with the construction of the platonic solids. ([7], p. 2) It is accepted by scholars that some of the work in *The Elements* was original. Euclid follows Aristotle's rules of logic, for instance, but his use of postulates appears to be new.

Book I: Preliminaries

The Elements starts with a set of 23 definitions. We quote from [7], p. 153–154.

Definitions

D1. A **point** is that which has no part.

D2. A **line** is a breadthless length.

D3. The **extremities of a line** are points.

D4. A **straight line** is a line which lies evenly with the points on itself.

D5. A **surface** is that which has length and breadth only.

D6. The **extremities** of a surface are lines.

D7. A **plane** surface is a surface which lies evenly with the straight lines on itself.

D8. A plane **angle** is the inclination to one another of two lines in a plane which meet one another and do not lie in a straight line.

D9. When the lines containing the angle are straight, the angle is called **rectilineal.**

D10. When a straight line set up on a straight line makes the adjacent angles equal to one another, each of the equal angles is **right**, and the straight line standing on the other is called a **perpendicular** to that on which it stands.

D11. An **obtuse angle** is an angle greater than a right angle.

D12. An **acute angle** is an angle less than a right angle.

D13. A **boundary** is that which is an extremity of anything.

D14. A **figure** is that which is contained by any boundary or boundaries.

D15. A **circle** is a plane figure contained by one line such that all the straight lines falling upon it from one point among those lying within the figure are equal to one another.

D16. And the point is called the **center** of the circle.

D17. A **diameter** of the circle is any straight line drawn through the center and terminated in both directions by the circumference of the circle, and such a straight line also bisects the circle.

D18. A **semi-circle** is the figure contained by the diameter and the circumference cut off by it. And the center of the semi-circle is that of the circle

D19. **Rectilineal figures** are those which are contained by straight lines, **trilateral figures** being those contained by three, **quadrilateral** those contained by four, and **multi-lateral** those contained by more than four straight lines.

D20. Of trilateral figures, an **equilateral triangle** is that which has its three sides equal, an **isosceles triangle**, that which has two sides equal, and a **scalene triangle** that which has its three sides unequal.

D21. Further, of trilateral figures, a **right-angled triangle** is that which has a right angle, an **obtuse-angled triangle** that which has an obtuse angle, and an **acute-angled triangle**, that which has three angles acute.

D22. Of quadrilateral figures, a **square** is that which is both equilateral and right-angled; an **oblong** that which is not equilateral but right-angled; a **rhombus** that which is equilateral but not right-angled; and a **rhomboid** that which has its opposite sides and angles equal to one another but is neither equilateral nor right-angled. All other quadrilaterals are called **trapezia**.

D23. **Parallel straight lines** are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

These are just the definitions from Book I. Some—oblong, rhombus, and rhomboid—are never used in *The Elements* at all. ([7], p. 189) Others not listed here are added in subsequent books. Others—radius of a circle, vertex of an angle, parallelogram—are missing entirely.

Mathematicians and philosophers succeeding Euclid left a wealth of commentary on the definitions, especially those of point, line, and plane. The definition of a line as the shortest distance between two points, for instance, appears to be due to Legendre, and dates to 1794.

Notice that D2 and D4 indicate a distinction between the notions of line and straight line. Euclid uses the word "line" for what we usually call a curve. Euclid's dictionary thus suggests a classification of curves into three types: lines, circles, and everything else. It is worthwhile to note that more detailed classifications of curves were of great interest to geometers through the centuries, Euclid's predecessors and contemporaries included.

The definition of angle, D8, refers to sides that are not necessarily straight lines. We are more accustomed to thinking of an angle as a rectilineal angle,

in Euclid's terminology. An angle formed by what we would call curves may be best left to the calculus. Indeed, we define the angle between curves to be the angle formed by the tangents at the point where the curves intersect. Since intersecting curves may or may not have defined tangents at a given point, this would seem an awkward start to a study of angles. But Euclid's attention is on lines and circles, so there are no such ambiguities for him.

Euclid's definition of rectilineal figure, D19, at first seems to refer to what we would call a polygon. Closer examination of D14, D15, D18, and D19, however, reveals a material distinction between the way Euclid refers to plane figures and the way we refer to plane figures. In standard usage today, for example, a triangle is a three-sided figure that forms the boundary of a region. According to Euclid's definition, a triangle is the region itself. Today we refer to Euclid's triangle as a 2-simplex. Likewise, Euclid's circle is our disk.

When we take up Euclid's propositions, we conform to certain modern conventions and terminology. By **line** we always mean a straight line. **Collinear points** lie on a single line. **Noncollinear points** are not collinear. Unlike Euclid, we distinguish lines, **line segments**, and **rays**. Formal definitions of segments and rays typically employ a notion of betweenness or ordering, a concept that Euclid does not address but does use implicitly and frequently. We study betweenness with greater attention in Chapter 4 when we consider Hilbert's approach to Euclidean geometry. For the purpose of studying the propositions in Book I of *The Elements* in the spirit of Euclid, we simply note that a segment is the portion of a line including, and lying between, two points on the line. A point A on a line ℓ determines two rays with **vertex** A: Each ray is the portion of ℓ to one side of A.

Two rays with a common vertex form an **angle** if the rays do not lie on a single line. Both the rays themselves and segments of the rays that contain the vertex are called the **legs** of the angle. While studying Euclid, we respect his convention that an angle is always less than 180° or π radians but note that Euclid does not employ numerical measures for angles or anything else. We use π as shorthand where Euclid talks of the sum of two right angles.

The word **adjacent** comes up frequently in geometry and it means slightly different things in different contexts. Typically—but not always—it refers to things that share points. Adjacent angles are non-overlapping angles with a common vertex and a common leg. Two sides of a polygon are adjacent if they intersect; the point of intersection is a **vertex** of the polygon. **Adjacent vertices** in a polygon share a side. A side and an angle in a polygon are adjacent if the side is a leg of the angle. The word **opposite** often means non-adjacent. For instance, given a side of a quadrilateral, the opposite side is the non-adjacent side. Given a vertex in a quadrilateral, the opposite vertex is the non-adjacent vertex. Likewise, given a side in a triangle, the opposite angle is the non-adjacent angle.

We use the word **polygon** to refer to the boundary of a region defined by Euclid in D19. A **triangle**, then, is a set of three noncollinear points and the line segments they determine. A **circle** is the boundary of a region Euclid describes in D15. Euclid did not have a dedicated word for the radius of a circle but we

do: A **radius** of a circle is any segment from a point on the circle to its center. Euclid studies parallelograms, but he does not define them: A **parallelogram** is a quadrilateral in which opposite sides are parallel. As we work through the propositions in Book I of *The Elements*, we introduce other modern locutions, conventions, and conveniences.

We return now to Euclid's assumptions as stated in Heath's translation. The five postulates follow the definitions. ([7], p. 154–155)

Postulates

Let the following be postulated;

I. To draw a straight line from any point to any point.

II. To produce a finite straight line continuously in a straight line.

III. To describe a circle with any center and distance.

IV. That all right angles are equal to one another.

V. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which the angles are less than two right angles.

The first three postulates establish that Euclid's concern is the geometry of a straight edge and floppy compass. Postulate IV can be read as an assumption about the homogeneity of space. For some authors, there is a question as to whether it belongs in the category of common notion, which we meet below.

Euclid's actual use of Postulate I reveals the unarticulated or **tacit assumption** that there is a *unique* line determined by a pair of points. ([7] p. 195) Postulate II says that a line segment can be extended into a line. There is another hidden assumption there, viz., that the line determined by a segment is unique. ([7], p. 196) As it turns out, the question of uniqueness—a question we address frequently in modern mathematics—is never addressed by Euclid.

Postulate V was controversial from Euclid's time through twenty succeeding centuries. The primary question was whether the fifth postulate was actually a consequence of the other four. Were Postulates I–IV enough of a foundation for Euclidean geometry? Was Postulate V actually a theorem, and not a postulate at all? Many mathematicians through the centuries thought it was and set about trying to prove it. (A brief history of attempts to prove Postulate V is given in [7], p. 202–219.) Some of the greatest minds in mathematics laid siege to this, the so-called **problem of the parallels**. Until the early nineteenth century, all attackers failed, and the controversy remained unresolved. The knot finally started coming apart when Gauss, Bolyai, and Lobachevsky, all working independently, discovered that by assuming Postulates I–IV, and assuming that Postulate V is false, one obtains a geometric system that does not have the properties of infinite, flat Euclidean space. This meant that Postulate V could be

assumed true, or it could be assumed false. This is more than a point of academic interest. Difficulty with the parallel postulate resolved with the discovery of non-Euclidean geometry and the discovery of non-Euclidean geometry upset a philosophical foundation that had been in place roughly since the time of Aristotle. Parts of that story occupy us in the next several chapters.

Euclid cites five so-called common notions in addition to the definitions and postulates. The common notions are sometimes referred to as axioms in the literature. (See p. 221 in [7].) They are distinguished from the postulates in that they are not specifically geometric, but refer to principles that apply both within and without geometry. We quote from [7], p. 155.

Common Notions

CN1. Things which are equal to the same thing are also equal to one another.

CN2. If equals be added to equals, the wholes are equal.

CN3. If equals be subtracted from equals, the remainders are equal.

CN4. Things which coincide with one another are equal to one another.

CN5. The whole is greater than the part.

Neither the postulates nor the common notions are added to in subsequent books of *The Elements*.

The common notions look strange partly because of where we stand in the history of the subject and partly because Euclid does not use symbols. We could restate CN1 and CN2 using symbols as follows:

(1) If $a = b$ and $c = b$, then $a = c$.

(2) If $a = b$, then $a + c = b + c$.

To discuss these further, we pause to consider equivalence relations, an idea that is quite modern.

Definition 1.1. Let S be a set. A relation $\sim$ on S is called an **equivalence relation** provided

(1) $\sim$ is *reflexive*, i.e., $a \sim a$ for all a in S;

(2) $\sim$ is *symmetric*, i.e., $a \sim b$ implies $b \sim a$;

(3) $\sim$ is *transitive*, i.e., if $a \sim b$ and $b \sim c$, then $a \sim c$.

Using the language of equivalence relations, we could argue that for Euclid, symmetry and reflexivity of equality are tacit assumptions. This makes CN1 the assumption that equality is transitive.

The modern notion of equality is the archetype of equivalence relations. Isomorphism is algebraic equivalence. Homeomorphism is topological equivalence.

Congruence is equivalence of geometric figures. Indeed, it is tempting to interpret Euclid's use of the word "equal" to mean "congruent." In Euclid's text, though, the word "equal" applies to comparable objects with the same measure. For example, line segments are comparable to line segments, angles are comparable to angles, and objects with defined areas are comparable to one another. Euclid's *equal lines*, then, are our *congruent line segments*. Euclid's *equal angles* are our *congruent angles*. Euclid's *equal triangles*, though, enclose equal areas. They need not be congruent. (This is evident, for example, in the wording of the fourth proposition in Book I, [7], p. 247.) Indeed, for Euclid, a triangle and a quadrilateral with the same area are *equal*. (See Proposition 44 in Book I.)

There is another sense in which our use of the word *equal* differs from Euclid's. Modern convention dictates that angles α and β are equal, for example, only if α and β are different names for the same angle. Different angles with the same measure are not equal, but **congruent**. The distinction may seem like splitting hairs but attention to the meaning of words is critical to the careful thought that modern mathematics demands.

The evaluation of Euclid's system that unfolded over the course of two thousand years was directed towards determining whether the defined terms, postulates, and common notions form a set of *complete*, *consistent*, and *independent* basic notions. The problem of the parallels was the most difficult and profound sticking point. Once that was settled, the stage was set for a final disposition of Euclid's axiomatic system, most notably with the works of Pasch, in 1882; Veronese, in 1891; and David Hilbert (1862–1943), in 1903 [13].

By the time later editions of Hilbert's *Grundlagen der Geometrie* (*Foundations of Geometry*) were published, axiomatization and abstraction in mathematics were assuming more important roles in the development of new ideas. This movement was manifested, for instance, in *The Principia Mathematica* of Russell and Whitehead, who sought to reduce all of mathematics to formal logic and set theory, and in the work of Kurt Gödel on foundations of mathematics. Bourbaki, the name for a group of mathematicians (with changing membership) based in Paris from the 1930s through the present, initiated an ambitious project of putting all of mathematics on a firm axiomatic foundation. Whether or not one prefers this approach to mathematics, it was and remains a profound influence.

Exercises 1.2. 1. What do D3 and D6 mean, in particular, in what sense might a line or a surface have "extremities"?

2. Is Euclid's definition of a circle equivalent to the more familiar definition of a circle as the set of points at a fixed distance from a fixed point?

3. Euclid's Postulate V is also called the parallel postulate although the word "parallel" does not appear in its statement.

   (a) Make a sketch to illustrate the statement of Euclid's Postulate V. Explain the connection between Euclid's postulate and parallel lines. (See [7], p. 220 for other formulations of the parallel postulate and the associated history.)

(b) What is the usual formulation for the existence of parallel lines in a plane?

4. Is similarity of triangles an equivalence relation?

5. Is parallelism of lines an equivalence relation, in other words, if we say two lines in a plane are related provided they are parallel, have we defined an equivalence relation on the lines in that plane?

1.3 Basic Constructions

We consider some of the propositions that Euclid proves in Book I of *The Elements* and some possible problems with the proofs. Our source throughout is the Heath edition [7], but as noted above, we adapt terminology and notation to reflect modern usage. Euclid's *equal lines* become our **congruent line segments**, for example, and if triangles ABC and DEF are congruent we write $\Delta ABC \cong \Delta DEF$. To distinguish a segment, a line, and a ray, we use AB for the segment with endpoints A and B, $\overline{AB}$ for the line determined by A and B, and $\overrightarrow{AB}$ for the ray emanating from vertex A through the point B. If α is an angle, triangle, or a segment that is *part* of an angle, triangle, or segment β, we invoke CN5 by writing $\alpha < \beta$ or $\beta > \alpha$. We extend this notation so that if α is congruent to an object that is part of β, we write $\alpha < \beta$.

The extent to which terminology and notation drive a flow of ideas is not always obvious so these sorts of adaptations are eschewed when the goal of studying a text is historical analysis. Our goal, though, is to appreciate Euclid's proof and approach to the Pythagorean Theorem as mathematics, while honoring their place in history. Our renovation of Euclid's text includes some trimming and some embellishing. Figures in [7] are marked only with capital letters. We edit those figures to match adjustments that we make in the text but in the interest of making some of the proofs easier to follow, we also add markings in some cases. Overall, we approach these adaptations gingerly to avoid obscuring the ideas presented in *The Elements*.

Our numbering of propositions indicates the book number and the proposition number in *The Elements*. The first proposition in Book I is then Proposition I.1, which we also denote Prop. I.1.

Most of the propositions we study in this chapter go towards Prop. I.47, the Pythagorean Theorem.

Propositions I.1–3

Proposition I.1. We may construct an equilateral triangle from a given line segment.

Proof. Given a line segment AB, form two circles: one with center A and radius AB, the other with center B and radius AB. Let C be a point where the circles

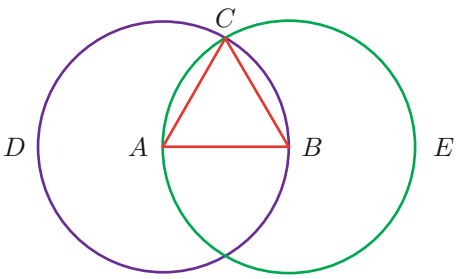


Figure 1.4: Proposition I.1: Construct an equilateral triangle from a segment

intersect. As radii of the first circle, $AB \cong AC$. As radii of the second circle, $BC \cong BA = AB$. By CN1, $AC \cong BC$. The three segments AC, AB, BC are thus congruent to one another, and the triangle ΔABC is equilateral. □

Early commentators noticed that this proof relies on the tacit assumption that the circles intersect. While the circles must overlap, the implicit assumption is that there is a point shared by the two circles where they overlap. Extensive comments elaborating on this are in [7], p. 242–3. The correction involves an assumption about the continuity—that is, the "gaplessness"— of a line or curve.

The next result gives the construction of a line segment of a given length at an arbitrary point in the plane. The proof presents an elegant solution to the problem of working without a ruler or rigid compass.

Proposition I.2. Given a line segment and any point in the plane, we may construct a congruent line segment emanating from the point.

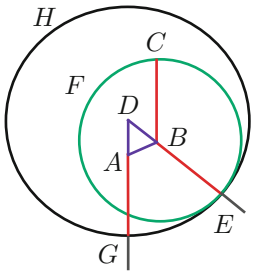


Figure 1.5: Proposition I.2: Construct a segment at a point

Proof. Let A be a point and BC be a line segment. Form the segment AB and use Prop. I.1 to construct the equilateral triangle ΔABD. Using B as

the center and BC as the radius, form a circle. Extend the segment DB to intersect the circle at E, and take another point F on this circle. As radii of CEF, $BC \cong BE$.

Form another circle with center D and radius DE. Extend the segment DA to intersect this circle at G, and take another point H on this circle. As radii of EGH, $DE \cong DG$. Since $DA \cong DB$, we can appeal to CN3 to conclude that after removing segments DA from DG and DB from DE, we get congruent remainders $AG \cong BE$. Since $BC \cong BE$, $AG \cong BC$ as desired. □

The next result finishes off the problem of copying a given segment in the plane. Where Prop. I.2 allows us to copy the segment from any point, the proof of Prop. I.3 tells us how we can control the direction of the copied segment.

Proposition I.3. Given two unequal segments, we may cut off from the longer a segment congruent to the smaller.

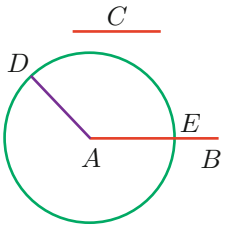


Figure 1.6: Proposition I.3: Cut a segment of a given length

Proof. Let AB and C be two unequal line segments and let AB be the longer. Use Prop. I.2 to form the segment AD congruent to C. Form the circle with center A and radius AD. Let E be the point where this circle intersects AB. Now $AE \cong C$. □

Notice that where Euclid starts with a floppy compass, by Prop. I.3 he establishes that we may assume, going forward, that our compass stays open after all.

Exercises 1.3.

1. Euclid's proofs for Props. I.2 and I.3 involve choices of specific configurations of points, lines, and circles. The proofs, in other words, speak to particular cases, ignoring others. For example, the proof of Prop. I.2 proceeds from the assumption that AB is smaller than BC. What if AB is actually longer than BC? Does that require a modification to the proof?

2. If we accept the assumption that a circle is a continuous curve, then we might notice that there is a second point of intersection of the circles in Prop. I.1, either of which could be used in the proof of that proposition or of Prop. I.2. Rework the proof of Prop. I.2 using that second point

instead of the point D in the figure. Does this choice demand a material change in the proof?

3. Does a correct proof of Prop. I.2 depend on the relative positions of the point A and the segment BC? For instance, if A is on $\overline{BC}$, does the proof still apply? Does it change depending on whether A is between B and C? What if A is an endpoint of BC, that is, what if $A = B$ or $A = C$? In this case, we ought to be able to construct a segment extending off the given segment with the same length. Does Euclid's proof apply in that case?

4. Consider the arrangement of the two line segments in Euclid's proof of Prop. I.3. What other arrangements are possible? Which other arrangements actually require a change to the proof?

5. We noted that in the proof of Prop. I.1, Euclid employs a tacit assumption, that is, that the two circles intersect. Can you identify tacit assumptions in the proofs of Props. I.2 and I.3?

1.4 Triangles: Propositions I.4–26

The first three propositions in Book I establish that we can copy a given line segment anywhere in the plane. The next two propositions start a long thread that deals with congruence questions associated to triangles. Proposition I.4 is the familiar side-angle-side (SAS) criterion for triangle congruence. (While we use SAS and other modern names for certain propositions, we note that in *The Elements*, no proposition has a name.)

The word **base** is used in the Heath edition to indicate the third side of a triangle when the other two sides are under consideration.

The statement of Prop. I.4 is the first place the word **subtend** appears in Heath's edition of Euclid. "Subtend" means to *stretch under* and dates from the late sixteenth century. Curiously, the word **hypotenuse** has the same meaning and dates from just a bit later in the sixteenth century. As *hypotenuse* does not appear at all in the Heath edition, we must conclude that Euclid did not have a special term for the side of a triangle that subtends a right angle.

Rather than say that a side subtends a particular angle, we would more likely say that a side is **opposite** the angle.

We omit the figure for Prop. I.4. You should supply one as an aid to follow the argument.

Proposition I.4 (SAS)**.** Suppose two triangles have pairs of corresponding sides congruent, and the angles between them also congruent. Then the bases of the triangles are congruent, the regions enclosed by the triangles are the same size, and the angles subtended by congruent sides are themselves congruent.

Proof. Let ΔABC and ΔDEF be triangles with $AB \cong DE$, $AC \cong DF$ and $\angle BAC \cong \angle EDF$. If ΔABC is positioned atop ΔDEF so that A is placed on the point D and AB placed on DE, then the point B must coincide with

E because $AB \cong DE$. AB coinciding with DE means the segment AC must coincide with DF because $\angle BAC \cong \angle EDF$. The point C must coincide with the point F because $AC \cong DF$.

Since B coincides with E, the base BC must coincide with EF. [If not, then the segments BC and EF enclose a space, which is impossible.] We conclude that $BC \cong EF$, that the triangles enclose regions that coincide and are thus of equal size, and that the remaining angles of ΔABC coincide with the remaining angles of ΔDEF so that $\angle ABC \cong \angle DEF$ and $\angle ACB \cong \angle DFE$. □

The proof of Prop. I.4 relies on the so-called **method of superposition**, that is, picking up one triangle and dropping it down onto another for comparison. This has generated controversy for as long as there have been commentators on Euclid. Heath et. al. aver that Euclid himself was uncomfortable about employing this argument. ([7], p. 249) In fact, Heath argues that CN4 and the bracketed material in the proof were added by later commentators to justify this step in Euclid's proof. ([7], p. 225)

Why is superposition problematic? One argument, which goes back at least to the sixteenth century, is that if superposition is indeed a legitimate method of proof, we could use it to prove many of the propositions in *The Elements* with little trouble. Consider the pains Euclid took to prove Prop. I.2, the argument goes. If superposition were allowed, could we not use it there and dispatch the proof in one line? Using superposition to prove Prop. I.4 and not some of the other propositions, perhaps belies a conflict in the underlying ideas that may have bedeviled Euclid himself.

Hilbert resolves the conflict with a bang or a whimper, depending on your point of view: in effect, he assumes SAS. (See [7], pp. 249–250 for more details regarding this and other difficulties with the proof of Prop. I.4.)

Next we consider Prop. I.5, known famously as *Pons Asinorum*, the asses' bridge. (See [3], p. 6.) The most common stories behind the name are (1) that it refers to those who cannot follow the proof, or (2) that it refers to the picture Euclid supplied for his proof. The actual origin and meaning of the name is lost, but people still enjoy puzzling over it. ([7], p. 415)

Proposition I.5 *(Pons Asinorum).* The base angles in an isosceles triangle are congruent. If the sides of an isosceles triangle are extended beyond the base, the angles formed under the base are congruent.

Proof. Let ΔABC be isosceles with $AB \cong AC$. Extending $\overrightarrow{AB}$ and $\overrightarrow{AC}$ past the base BC, we can invoke Prop. I.3 to choose D on $\overrightarrow{AB}$ and E on $\overrightarrow{AC}$ so that $AD \cong AE$. Form segments DC and EB. We have SAS for $\Delta ADC \cong \Delta AEB$, since $\angle CAD$ is common. It follows that $DC \cong EB$, $\angle ACD \cong \angle ABE$, and $\angle ADC \cong \angle AEB$. Since the whole $AD \cong AE$, and the part $AB \cong AC$, CN3 implies $BD \cong CE$.

Since $\angle BDC \cong \angle BEC$, and BC is common, we have SAS for $\Delta BDC \cong \Delta BEC$. From there, $\angle CBD \cong \angle BCE$, which proves the second assertion of

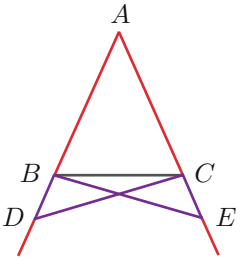


Figure 1.7: Euclid's proof of *Pons Asinorum*

the proposition. We also have $\angle BCD \cong \angle CBE$. Since $\angle ABE \cong \angle ACD$, CN3 gives us $\angle ABC \cong \angle ACB$, which proves the result. □

Pons Asinorum is significant for several reasons. First, it can be argued that this theorem is a springboard, not only for the rest of Book I of *The Elements*, but also for basic results worked up in later books and by other authors down through the centuries. This is Coxeter's approach in [3] (cf. Section 1.3), a wonderful resource in its own right. Second, references in Aristotle indicate that geometers who preceded Euclid knew *Pons Asinorum* but had a different proof, and indeed, quite a different approach to the subject altogether. This helps us understand the role of *The Elements* in history. Euclid was, to a large extent, pulling together known results but from early on in the program, he forges his own way through the thicket. (See [7], pp. 252–254.) Finally, Proclus, the commentator who came around 700 years after Euclid, relates the proof that Pappus described, an argument often advanced in high school geometry courses. (Compare also [3], Section 1.3.) Here is Pappus's proof, as quoted from [7], p. 254. Note that this does not cover the part of the theorem describing the angles under the base of the triangle.

> Let ABC be an isosceles triangle, and AB equal to AC. Let us conceive this one triangle as two triangles, and let us argue in this way. Since AB is equal to AC, and AC to AB, the two sides AB, AC are equal to the two sides AC, AB. And the angle BAC is equal to the angle CAB for it is the same. Therefore all the corresponding parts (in the triangles) are equal, namely BC to BC, the triangle ABC to the triangle ABC (i.e., ACB), the angle ABC to the angle ACB, and the angle ACB to the angle ABC, for these are the angles subtended by the equal sides AB, AC. Therefore in isosceles triangles the angles at the base are equal.

Pappus's proof is often presented without attribution and with an important modification, namely, that the triangle is picked up from the page and reflected across a median or angle or base bisector (none of which has been defined) and

dropped down again upon itself...or maybe upon a trace it has left on the page after it has been picked up. This approach is unsettling for several reasons: how can one imagine applying a physical force to pick up a triangle in order to place it back down in a new location? From this perspective, Pappus's original approach, to "conceive this one triangle as two triangles," is elegant.

Before proceeding from *Pons Asinorum*, we cite a rebuke to commentators who proposed to improve on Euclid's proof by means such as these. This is a quote of C. L. Dodgson (a.k.a. Lewis Carroll) from *Euclid and His Modern Rivals*, as cited in [3], p. 6.

> MINOS: It is proposed to prove [*Pons Asinorum*] by taking up the isosceles Triangle, turning it over, and then laying it down again upon itself.
>
> EUCLID: Surely that has too much of the Irish Bull about it, and reminds one a little too vividly of the man who walked down his own throat, to deserve a place in a strictly philosophical treatise?

Proposition I.6 is a partial converse of *Pons Asinorum*. The result is not used much in the sequel but as the first instance in *The Elements* of an indirect proof or **proof by contradiction**, the proof itself is of interest. In a proof by contradiction, we assume that the conclusion of the theorem is false and we proceed to uncover a contradiction. The contradiction may violate a common notion, a postulate, an earlier result, or a hypothesis of the theorem itself. This method of argument goes back at least to Plato and Aristotle.

Proposition I.6. If the base angles of a triangle are congruent, the sides opposite the base angles are also congruent.

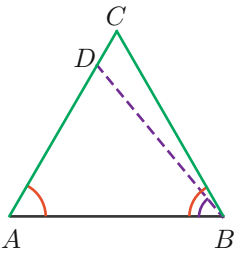


Figure 1.8: Proposition I.6: Congruent base angles imply congruent sides

Proof. Consider ΔABC where $\angle ABC \cong \angle BAC$ and suppose the conclusion is false so that one of AC or BC is longer. Assume it is AC. Choose a point D on AC so that $AD \cong BC$. This gives us SAS for $\Delta DAB \cong \Delta CAB$, which must then enclose equal areas. But as part of ΔCAB, ΔDAB must have smaller area than ΔCAB by CN5. The contradiction forces us to conclude that $AC \cong BC$. □

Proposition I.7 is not used in Book I except to prove Prop. I.8, the side-side-side (SSS) criterion for triangle congruence. Euclid's proof of SSS employs superposition again, so requires no further explanation. We assume SSS and move on to Prop. I.9, which starts a sequence of constructions familiar from high school geometry.

Proposition I.9. An angle can be bisected.

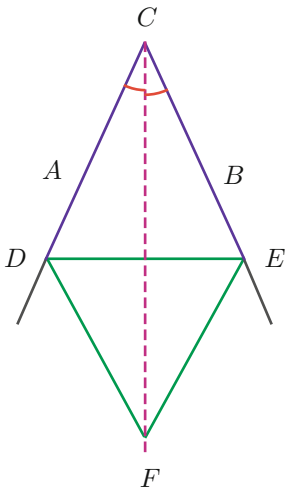


Figure 1.9: Proposition I.9: Bisect an angle

Proof. Let $\angle ACB$ be given. Choose D on $\overrightarrow{CA}$ and E on $\overrightarrow{CB}$ so that $CD \cong CE$. Construct ΔDEF, an equilateral triangle on DE. We claim that CF bisects $\angle ACB$.

By the construction, $CD \cong CE$, $CF = CF$, and $FD \cong FE$ so by SSS, $\Delta CDF \cong \Delta CEF$. As corresponding angles, $\angle FCD \cong \angle FCE$. □

Both the segment CF and the ray $\overrightarrow{CF}$ constructed in the proof of Prop. I.9 are called the **angle bisector** for $\angle ACB$. We leave it as an exercise to show that the angle bisector is essentially unique.

Proposition I.10. A line segment can be bisected.

Proof. Let AB be given. Construct ΔACB, an equilateral triangle on AB. Bisect $\angle ACB$ and let the bisector intersect AB at the point D. We have $AC \cong BC$, $\angle ACD \cong \angle BCD$, and $CD = CD$ so by SAS, $\Delta ACD \cong \Delta BCD$. As corresponding sides, $AD \cong BD$. □

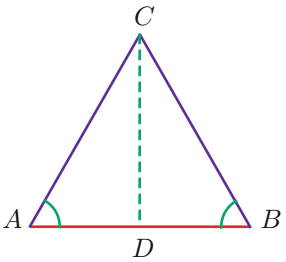


Figure 1.10: Proposition I.10: Bisect a segment

The point D, constructed in the proof of Prop. I.10, is the **midpoint** of AB. The easy proof of its uniqueness is an exercise.

The next few results deal with perpendiculars. We are accustomed to thinking about perpendiculars and parallels as two sides of the same coin, but one of the lessons of reading Euclid closely is that these ideas are not inextricably bound, even in the Euclidean plane. There is neither explicit nor implicit reliance on anything to do with parallel lines in Euclid's proofs of the next several propositions.

Proposition I.11. We can construct a perpendicular to any given line at any point on that line.

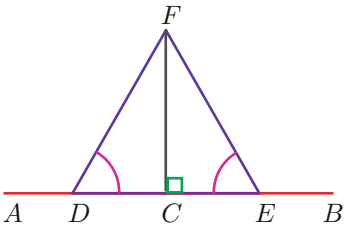


Figure 1.11: Proposition I.11: Erect a perpendicular

Proof. Suppose we want a perpendicular at C, a point on a line segment AB. Take D different from C but otherwise arbitrary on AC. Choose E on CB so that $CE \cong CD$. Construct an equilateral triangle, ΔDEF, on DE. By SAS, $\Delta DFC \cong \Delta EFC$ so $\angle DCF \cong \angle ECF$. As congruent adjacent angles, $\angle DCF$ and $\angle ECF$ must be right, so CF is perpendicular to AB. □

Notice that the proof of Prop. I.11 addresses the perpendicular—that is, the right angles—on just one side of the line $\overline{AB}$. In other words, Euclid argues that the segment CF is perpendicular to AB. The point F is on one side of $\overline{AB}$. If G on $\overline{CF}$ is on the other side of $\overline{AB}$, do we know that GC is also

perpendicular to AB? We can formulate an equivalent question: If we construct the perpendicular CF on one side of $\overline{AB}$ and the perpendicular CG on the other side of $\overline{AB}$, are we guaranteed that C, F, G are collinear? Under Euclid's assumptions, we can show that the answer is "yes" but it takes a few steps. This is part of the question of the uniqueness of the perpendicular to AB at C, which we take up in the exercises.

Proposition I.12 does not appear at all again in Book I of *The Elements* but it is particularly useful when we study properties of the plane that do not depend on Postulate V. The proof is a lovely construction that may be familiar from high school geometry.

Our presentation of the proof of Prop. I.12 includes an emphasis on Euclid's tacit assumption that a line divides the plane into two regions. This is one of several tacit assumptions that concern the relative positions of points in the plane. Assumptions of this type underlie many of the proofs in *The Elements*.

Proposition I.12. We can construct a perpendicular to any line from any point not on the line.

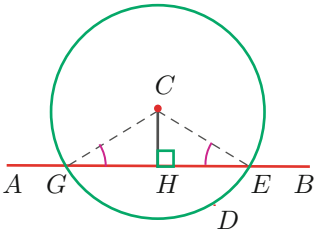


Figure 1.12: Proposition I.12: Drop a perpendicular

Proof. Suppose we wish to form a perpendicular to the line $\overline{AB}$ from a point C not on $\overline{AB}$. C must lie on one side of $\overline{AB}$; choose a point D on the other side of $\overline{AB}$. Construct the circle with center C and radius CD. Let E and G be points of intersection of the circle with $\overline{AB}$. Let H be the midpoint of EG. Since two sides of ΔGEC are radii, ΔGEC is isosceles so $\angle HGC \cong \angle HEC$. As $HG \cong HE$, we have SAS for $\Delta HGC \cong \Delta HEC$ so $\angle GHC \cong \angle EHC$. As congruent adjacent angles, $\angle GHC$ and $\angle EHC$ must be right, so CH is perpendicular to $\overline{AB}$. □

When invoking Prop. I.12, we often speak of "dropping" a perpendicular to a line from a point. The point on the line where that perpendicular intersects is called the **foot** of the perpendicular.

The proof of Prop. I.13 is the first place we run across the notion of angle addition in Euclid. Remember that Euclid does not use numerical measurements to compare angles and segments. Likewise, our use of the + notation should be taken as shorthand, not as a reference to the addition of numbers. In other words, any sums that arise in the following proofs are formal sums.

Proposition I.13. A line segment standing on a second line segment makes two right angles or angles that sum to two right angles.

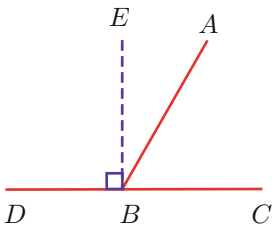


Figure 1.13: Proposition I.13: Angles that sum to two right angles

Proof. Suppose AB stands on CD as in Fig. 1.13. If $\angle ABC \cong \angle ABD$, the angles formed by the two segments are both right. If $\angle ABC \ncong \angle ABD$, stand the line segment BE perpendicular to CD so that $\angle CBE$ and $\angle DBE$ are both right angles.

We have $\angle CBE = \angle CBA + \angle ABE$. Adding $\angle DBE$ to each side, we have

$$\angle DBE + \angle CBE = \angle DBE + \angle CBA + \angle ABE \tag{1.1}$$

by CN2. Similarly, $\angle ABD = \angle ABE + \angle EBD$. Adding $\angle ABC$ to each side, we have, again by CN2,

$$\angle ABC + \angle ABD = \angle ABC + \angle ABE + \angle EBD. \tag{1.2}$$

Since the right-hand sides of (1.1) and (1.2) are the same, CN1 implies

$$\angle ABC + \angle ABD \cong \angle DBE + \angle CBE.$$

As noted above, the sum of $\angle DBE$ and $\angle CBE$ is two right angles so this completes the proof. □

Euclid did not use a special word for the relationship between angles that sum to two right angles but modern practice is to say that such angles are **supplementary**.

Proposition I.14 is the converse of Prop. I.13, although it may be hard to tell on a first reading.

Proposition I.14. Suppose two line segments, one on either side of a given line, meet the given line in a point so that the adjacent angles are supplementary. Then the two segments lie on a single line.

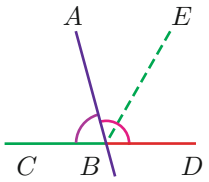


Figure 1.14: Proposition I.14: Supplementary adjacent angles

Proof. Suppose BD and BC are segments extending on either side of a given line, $\overline{AB}$. Suppose further that $\angle ABD$ and $\angle ABC$ sum to two right angles. The proposition says that B, C, D are collinear. Suppose they are not. Let E be a point on $\overline{CB}$ that lies on the same side of $\overline{AB}$ as D. Proposition I.13 guarantees that $\angle ABC$ and $\angle ABE$ sum to two right angles. Thus we have

$$\angle ABC + \angle ABE \cong \angle ABC + \angle ABD,$$

implying by CN3 that $\angle ABE \cong \angle ABD$. Since one of $\angle ABE$, $\angle ABD$ is part of the other, this violates CN5. The result then follows by contradiction. □

Vertical angles are non-adjacent angles formed by two intersecting lines. The terminology is not defined explicitly but is used in Heath's translation of *The Elements*.

Proposition I.15 (Vertical Angle Theorem)**.** Vertical angles formed by the intersection of two lines are congruent.

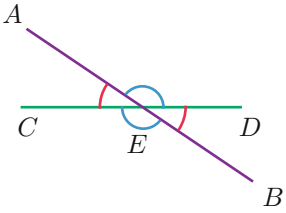


Figure 1.15: Proposition I.15: Vertical angles are congruent

Proof. Let lines $\overline{AB}$, $\overline{CD}$ intersect at E. By Prop. I.13, $\angle AEC$ and $\angle AED$ are supplementary. Similarly, $\angle AEC$ and $\angle BEC$ are supplementary. It follows by CN3 that $\angle AED \cong \angle BEC$. The same argument applies to show that $\angle AEC \cong \angle BED$. □

The Proclus edition of *The Elements* includes a corollary to the Vertical Angle Theorem stating that the angles formed by intersecting lines add up to

four right angles. Heath includes it in [7] but notes that it was likely a later addition to Euclid's text. We leave it as an easy exercise for the reader.

Proposition I.16, the Exterior Angle Theorem, starts another thread that deals with triangles. In the course of it, we see Euclid's method for constructing a copy of a given angle (Prop. I.23). That the sum of the angles in a triangle is two right angles does not come up until Prop. I.32, after a discussion of parallels that starts with Prop. I.27.

If we extend one side of a triangle at a vertex, then the extension and the second side of the triangle at that vertex form an **exterior angle** of the triangle. Note that there are two exterior angles at each vertex of a triangle.

An exterior angle is adjacent, and supplementary, to one angle in the triangle. Having fixed an exterior angle, we refer to the two non-adjacent angles in the triangle as **remote interior angles**.

Proposition I.16 (Exterior Angle Theorem)**.** If a side of a triangle is extended, then the exterior angle thus produced is greater than either of the remote interior angles.

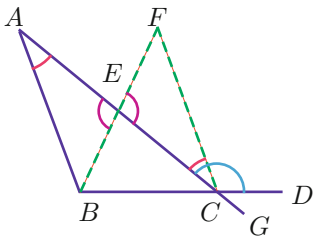


Figure 1.16: Proposition I.16: An exterior angle exceeds either remote interior angle

Proof. Consider a triangle ΔABC as in Fig. 1.16. Fix a point D on $\overrightarrow{BC}$ not on BC, and a point G on $\overrightarrow{AC}$ not on AC. The exterior angles at C are $\angle ACD$ and $\angle BCG$. By the Vertical Angle Theorem, the two are congruent so it is enough to prove that $\angle ACD$ is greater than $\angle BAC$ and that $\angle BCG$ is greater than $\angle ABC$.

Let E be the midpoint of AC. Take F on $\overrightarrow{BE}$ so that $EF \cong BE$. Form the segment FC. By the Vertical Angle Theorem, $\angle AEB \cong \angle CEF$. Since $AE \cong EC$ and $BE \cong EF$, we get SAS for $\Delta AEB \cong \Delta CEF$. As corresponding angles, $\angle BAE \cong \angle ECF$. Since $\angle ECF = \angle ACF$ is part of $\angle ACD$, CN5 guarantees that $\angle ACD > \angle ACF \cong \angle BAC$.

Repeat this argument, starting with the midpoint of BC, to show that $\angle BCG > \angle ABC$. □

The next proposition is not used in Book I of *The Elements* but it is another result that helps reveal properties of the Euclidean plane that do not depend on Postulate V.

Here we start using the symbol π to indicate the formal sum of two right angles.

Proposition I.17. The sum of two angles in a given triangle is less than π.

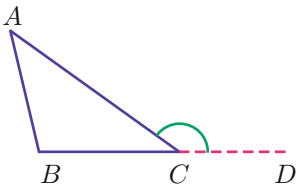


Figure 1.17: Proposition I.17: Two angles in a triangle sum to less than π

Proof. Given a triangle ΔABC, form an exterior angle at C by extending $\overrightarrow{BC}$ through a point D not on BC. We have $\angle ACD + \angle ACB = \pi$, and by the Exterior Angle Theorem, $\angle ACD > \angle ABC$, so

$$\pi = \angle ACD + \angle ACB > \angle ABC + \angle ACB.$$

Similarly, since $\angle ACD > \angle BAC$, $\pi > \angle BAC + \angle ACB$. To see that $\angle ABC$ and $\angle BAC$ sum to less than two right angles, form an exterior angle at vertex A. □

An easy corollary to Prop. I.17—not stated by Euclid—is that the sum of angles in a triangle is no greater than π. We leave its proof as an exercise.

Proposition I.18. In any triangle, the longer side subtends the larger angle.

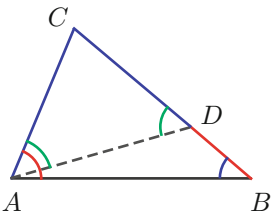


Figure 1.18: Proposition I.18: The longer side is opposite the larger angle

Proof. Let ΔABC be a triangle with $BC > AC$. We must show that $\angle CAB > \angle ABC$.

Choose D on BC so that $CD \cong AC$. Note that $\angle CAD$ is part of $\angle CAB$ so that by CN5, $\angle CAB > \angle CAD$.

By *Pons Asinorum*, $\angle CAD \cong \angle CDA$ so now we have $\angle CAB > \angle CDA$. By the Exterior Angle Theorem, $\angle CDA > \angle ABC$, giving us $\angle CAB > \angle ABC$, as was to be shown. □

Proposition I.19 is the converse of Prop. I.18. Euclid uses contradiction to prove it. Note the similarity to the proof of Prop. I.6, the converse of *Pons Asinorum*.

Proposition I.19. In any triangle, the greater angle is subtended by the greater side.

Proof. Let ΔABC be a triangle with $\angle BAC > \angle ABC$ and suppose the result is false. By *Pons Asinorum*, $CB \cong CA$ implies $\angle BAC \cong \angle ABC$ so it must be that $CB < CA$. In that case, though, Prop. I.18 ensures that $\angle BAC < \angle ABC$, a contradiction. □

The Triangle Inequality is one of the most important devices for studying the real line. For Euclid, it is strictly about the sides of a triangle. Here we use formal sums of segments for the first time.

Proposition I.20 (Triangle Inequality)**.** The length of one side of a triangle is less than the sum of the two remaining sides.

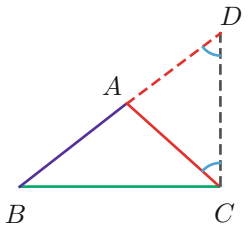


Figure 1.19: Euclid's proof of the Triangle Inequality

Proof. Consider a triangle ΔABC. Extend $\overrightarrow{BA}$ through the point D, D not on AB but chosen so that $AD \cong AC$, as in Fig. 1.19. Join points D and C. By *Pons Asinorum*, $\angle ADC \cong \angle ACD$. By CN5, $\angle BCD > \angle ACD$, thus $\angle BCD > \angle ADC$. Considering Prop. I.19 applied to ΔBCD, we have $BD > BC$. Then $BD = BA + AD \cong BA + AC > BC$. Selection of sides was arbitrary so it must be that the sum of any two sides exceeds the remaining side. □

We omit Prop. I.21, a technical result used for a single proposition in Book III that does not arise in our study.

Proposition I.22 is the converse of the Triangle Inequality. Though we have used rays for convenience in several proofs to this point, the proof of Prop. I.22 marks the first instance in which Euclid uses the concept of a ray, although he does not have a dedicated word for it.

Proposition I.22. Given three line segments for which the length of any one is exceeded by the sum of the other two, we can form a triangle with sides congruent to the three segments.

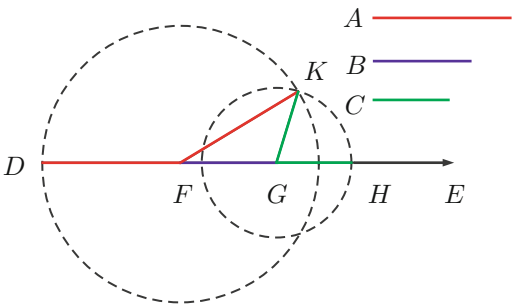


Figure 1.20: Proposition I.22: Segments that obey the Triangle Inequality

Proof. Let A, B, C be segments that satisfy the hypothesis of the proposition so that $A < B + C$, etc. Let $\overrightarrow{DE}$ be a ray emanating from some arbitrary point D. Take points F, G, H on $\overrightarrow{DE}$ so that $DF \cong A$, $FG \cong B$, and $GH \cong C$. Form two circles, the first with center F and radius DF, the second with center G and radius GH. Let K be a point of intersection of the two circles. Since $FK \cong DF \cong A$, $FG \cong B$, and $GK \cong GH \cong C$, ΔFKG is a triangle with sides of the requisite lengths. □

Euclid's argument for Prop. I.23 does not address whether the three circles constructed in the proof must overlap. Note, though, that the circles in Fig. 1.20 overlap provided $A + B + C < 2A + 2C$ which follows $B < A + C$.

Proposition I.22 implies the next proposition which finally allows us to construct an angle of a given magnitude.

Proposition I.23. Given an angle, a line, and a point on the line, we can construct a congruent angle at the point, using the line as one leg.

Proof. Let $\angle DCE$ be given and suppose we wish to construct a congruent angle with leg $\overrightarrow{AB}$ and vertex A. Join points D and E to form the triangle ΔDCE.

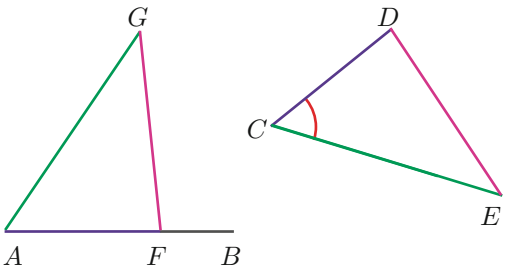


Figure 1.21: Proposition I.23: Construct an angle

Choose F on $\overrightarrow{AB}$ so that $AF \cong CD$. Now finish the construction of a triangle $\Delta AFG \cong \Delta DCE$, as in Prop. I.22. We can say $AG \cong CE$ and $FG \cong DE$. It follows that $\angle GAF \cong \angle DCE$, as desired. □

We skip next to Prop. I.26, the last of Euclid's congruence theorems for triangles. Euclid proves both the familiar angle-side-angle (ASA) criterion, as well as angle-angle-side (AAS) for triangle congruence, with no reliance, tacit or otherwise, on the angle sum in a triangle.

Proposition I.26 (ASA, AAS)**.** If triangles have two angles congruent to two angles respectively, and either the sides between the two angles congruent, or the sides opposite corresponding angles congruent, then the triangles themselves are congruent.

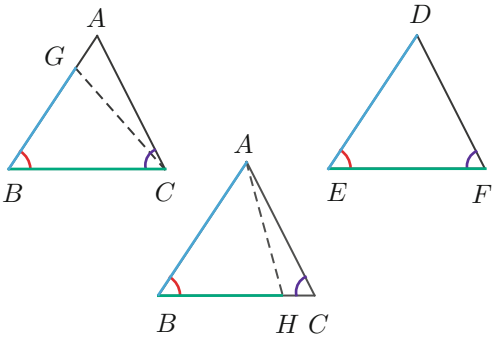


Figure 1.22: Euclid's proof of ASA and AAS

Proof. Consider triangles ΔABC and ΔDEF where $\angle ABC \cong \angle DEF$, $BC \cong EF$, and $\angle BCA \cong \angle DFE$. If $AB \cong DE$, we have SAS for $\Delta ABC \cong \Delta DEF$. Assume the result is false, then, so that $AB \ncong DE$. Without loss of generality, we may assume that $AB > DE$.

Pick G on AB so that $BG \cong ED$. We then have SAS for $\Delta GBC \cong \Delta DEF$ implying that $\angle BCG \cong \angle DFE$. But $\angle BCG$ is part of $\angle BCA \cong \angle DFE$ so $\angle BCG$ must be less than $\angle DFE$. The contradiction leads us to conclude that, in this case, $AB \cong DE$ so $\Delta ABC \cong \Delta DEF$.

Next, assume the two triangles satisfy AAS with $\angle ABC \cong \angle DEF$, $\angle ACB \cong \angle DFE$, and $AB \cong DE$. If $BC \cong EF$, we have SAS for $\Delta ABC \cong \Delta DEF$. Assume the result is false, then, with $BC > EF$. Pick H on BC so that $BH \cong EF$. This gives us SAS for $\Delta ABH \cong \Delta DEF$ implying that $\angle AHB \cong \angle DFE$. Since $\angle AHB$ is exterior to ΔACH, $\angle AHB > \angle ACH = \angle ACB \cong \angle DFE$. This contradiction forces us to conclude that $BC \cong EF$, thus, that $\Delta ABC \cong \Delta DEF$. □

We have thus far encountered tacit assumptions about the continuity of lines and curves—for example, when two circles overlap, the assumption is that they have points in common—and about the relative positions of objects in the plane—for example, if we fix a line in the plane, the remaining points in the plane form two sets, one on either side of the line. As we proceed through Book I of *The Elements*, we will run across other tacit assumptions.

Going forward, we refer to Postulates I–V, the common notions, and Euclid's tacit assumptions as **Euclid's axioms**. When we remove Postulate V from Euclid's axioms, we are left with **Euclid's neutral axioms**.

Exercises 1.4.

1. Consider the claim that Euclid's choice not to use superposition to prove Prop. I.2 is evidence that he viewed the method of superposition as suspect. Can you determine any other reason why superposition would be a poor choice for the proof of Prop. I.2?

2. Use Euclid's neutral axioms to argue that the angle bisector for a given angle is unique.

3. Use Euclid's neutral axioms to argue that the midpoint of a line segment is unique.

4. Proposition I.11 establishes that there is a perpendicular to any line at a point on that line. In this problem, you will show that the perpendicular is unique, using Euclid's neutral axioms and results from among Props. I.1–10. Here we are given a point C on a line $\overline{AB}$. As in the proof of Prop. I.11, we may assume C is the midpoint of AB.

   (a) Form both equilateral triangles on AB, one above and one below $\overline{AB}$, using Prop. I.1, as in Fig. 1.23. Say the triangles are ΔABD and ΔABE. If X is the point where DE intersects AB, argue that ΔXBD, ΔXAD, ΔXBE, and ΔXAE are all congruent. From there argue that $X = C$, and that all the angles at X are congruent, therefore, all are right.

   (b) Using Euclid's neutral axioms, argue that $CD = XD$ is the unique perpendicular to AB at C on that side of AB. From there, deduce that $CE = XE$ is the unique perpendicular on the E side of AB.

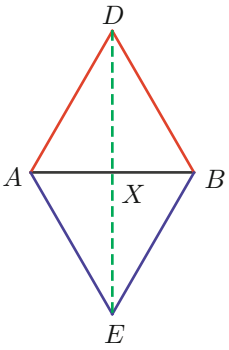


Figure 1.23: Exercise 4: The perpendicular to a segment is unique

Conclude that if we are given a line, and a point on that line, there is exactly one perpendicular line to the given line through the given point.

5. In Prop. I.12, we drop a perpendicular to a line from a point not on the line. Use Euclid's neutral axioms, and any of the propositions through I.26 (none of which depend on Prop. I.12) to formulate an argument for the uniqueness of the perpendicular.

6. Write a much shorter proof of Prop. I.13 invoking associativity of addition.

7. Show that the four angles formed by two intersecting lines add up to four right angles.

8. Show that the sum of the angles in a triangle is no greater than π.

9. Prove Prop. I.19 without using Prop. I.18.

Circles

The remainder of the exercises here concern circles and associated angles. Approach these using Euclid's neutral axioms and Props. I.1–26.

10. Euclid defines the center of a circle to be the point in the region enclosed by the circle with the property that segments from the point to the circle are all congruent. Show that the center of a circle is unique.

11. A **chord** is a line segment joining two points on a circle. Show that the perpendicular bisector of any chord passes through the center of the circle.

12. The previous exercise suggests a simple construction to locate the center of a given circle precisely. Describe that construction.

13. Suppose a circle has diameter AB intersecting a chord DE where DE does not pass through the center of the circle. Show that AB bisects DE if and only if AB and DE are perpendicular.

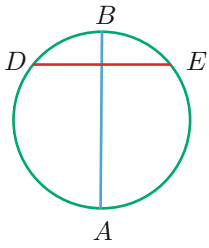


Figure 1.24: Exercise 13: Diameter and intersecting chord

1.5 Parallels: Propositions I.27–32

The first two propositions in the next thread describe sufficient conditions for lines to be parallel. Neither uses Postulate V.

Proposition I.27. If two lines are intersected by a transversal that forms congruent alternate angles, then the lines are parallel.

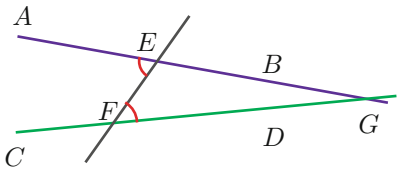


Figure 1.25: Proposition I.27: Congruent alternate angles

Proof. Let lines $\overline{AB}$, $\overline{CD}$ be given and let $\overline{EF}$ be a transversal. Assume alternate angles $\angle AEF$, $\angle EFD$ are congruent. If $\overline{AB}$ and $\overline{CD}$ are not parallel, they must meet in the direction of A and C, or the direction of B and D. Suppose they meet in the direction of B and D at a point G. Since $\angle AEF$ is exterior to the triangle ΔGEF, it must exceed $\angle EFG = \angle EFD$, a contradiction. The argument that the lines cannot meet in the direction of A and C is similar. We conclude that $\overline{AB}$ and $\overline{CD}$ are parallel. □

The statement of Prop. I.27 is the first place we see reference to **alternate angles**. Euclid does not define alternate angles but leaves it up to the reader to

infer what they are. (We normally refer to such angles as **alternate interior angles**.)

Proposition I.28. Consider two lines with a transversal. If either of the following conditions holds, then the two lines are parallel.

(1) An exterior angle and an opposite interior angle on the same side of the transversal are congruent.

(2) Two interior angles on the same side of the transversal are supplementary.

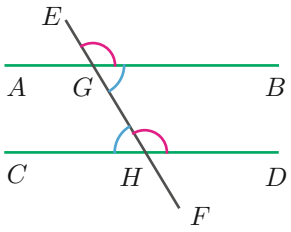


Figure 1.26: Propositions I.28 and 29: Conditions that imply lines are parallel

Proof. We assume (1) first. Referring to Fig. 1.26, suppose the exterior angle $\angle EGB$ is congruent to the opposite interior angle $\angle GHD$. By the Vertical Angle Theorem, $\angle EGB \cong \angle AGH$. This gives us alternate angles $\angle AGH \cong \angle GHD$ so by Prop. I.27, $\overline{AB}$ and $\overline{CD}$ are parallel.

Next assume (2). Say that $\angle BGH + \angle GHD = \pi$. By Prop. I.13, we also have $\angle AGH + \angle BGH = \pi$. This gets us $\angle AGH \cong \angle GHD$ by CN3. Since $\angle AGH$ and $\angle GHD$ are alternate, the result follows by Prop. I.27. □

Angles described as in Prop I.28 (1) are **corresponding angles** in modern parlance.

Notice that Props. I.27 and I.28 have the same form, viz., if a certain condition about angles formed by two lines and a transversal holds, then the lines are parallel. In Prop. I.29, where he invokes Postulate V for the first time, Euclid addresses the converse of both Prop. I.27 and I.28: If lines are parallel, then certain conditions about the angles formed by a transversal hold.

Proposition I.29. Consider two lines with a transversal. If the two lines are parallel then all of the following conditions hold.

(1) Alternate interior angles formed by the transversal are congruent.

(2) Corresponding angles formed by the transversal are congruent.

(3) The two interior angles on the same side of the transversal are supplementary.

Proof. Suppose $\overline{AB}$ and $\overline{CD}$ are parallel. Assume (1) is false so, for example, $\angle AGH > \angle GHD$ in Fig. 1.26. Since $\angle AGH + \angle BGH = \pi$, then $\angle BGH + \angle GHD < \pi$. By Postulate V, $\overline{AB}$ and $\overline{CD}$ meet on the B, D side of $\overline{EF}$, a contradiction. We conclude that $\angle AGH \cong \angle GHD$. Since this pair was an arbitrary choice of alternate interior angles, (1) must be true.

Next we argue that $\angle EGB \cong \angle GHD$. By the Vertical Angle Theorem, $\angle AGH \cong \angle EGB$. By (1), $\angle AGH \cong \angle GHD$. We conclude by CN1 that $\angle EGB \cong \angle GHD$, which proves (2).

Finally, notice that $\angle EGB + \angle BGH = \pi$ so $\angle GHD + \angle BGH = \pi$. This is enough to prove (3). □

Observation 1.2. Assume Euclid's neutral axioms. If Postulate V holds, then each of the results in Prop. I.29 is true. Conversely, if any of the three results in Prop. I.29 is true, then Postulate V holds, the proof of which we leave as an exercise. This is what we mean when we say that under Euclid's neutral axioms, Postulate V is logically equivalent to any of the three results in Prop. I.29.

Several of the succeeding propositions give conditions that are logically equivalent to Postulate V under Euclid's neutral axioms. The first is that parallelism is transitive.

Proposition I.30. Two lines parallel to a third line are parallel to one another.

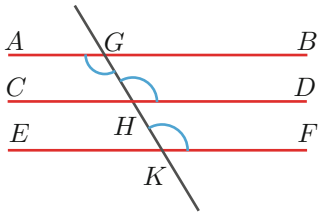


Figure 1.27: Proposition I.30: Parallelism is transitive

Proof. Suppose $\overline{AB}$ and $\overline{EF}$ are both parallel to $\overline{CD}$. Let $\overline{GK}$ be a transversal to the other three lines. Since $\overline{AB}$ is parallel to $\overline{CD}$, Prop. I.29 implies $\angle AGK \cong \angle DHG$. Since $\overline{CD}$ and $\overline{EF}$ are parallel, we also have $\angle DHG \cong \angle FKG$. By CN1, we conclude that $\angle AGK \cong \angle FKG$. By Prop. I.27, $\overline{AB}$ and $\overline{EF}$ are parallel. □

Though it comes after Euclid's first use of Postulate V, Prop. I.31 represents a pivot point between results in Book I of *The Elements* that rely on Postulate V and those that do not. The critical matter is not in what Euclid says, but in what he does not say.

Proposition I.31. Given a line and a point not on the line, we can construct a parallel to the line through the point.

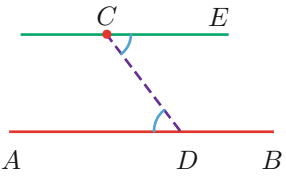


Figure 1.28: Proposition I.31: Construct a parallel to a line

Proof. Let $\overline{AB}$ be a line and C a point not on $\overline{AB}$. Take a point D on $\overline{AB}$ and form CD. Choose E opposite A relative to CD so that $\angle DCE \cong \angle ADC$ per Prop. I.23. Now we have alternate angles congruent so that by Prop. I.27, $\overline{CE}$ is parallel to $\overline{AB}$. □

Euclid's proof of Prop. I.31 suggests reliance neither on Postulate V nor on Props. I.29 or I.30, the only results in Book I to this point that themselves depend on Postulate V. The key idea that distinguishes Prop. I.31 from results that depend on Proposition V is uniqueness. As it happens, the uniqueness of the line constructed in the proof of Prop. I.31 is easy to prove using Euclid's axioms and their consequences. In other words, without Postulate V, we get a parallel to a line through a point not on the line. With Postulate V, we can establish that the parallel is unique.

Received wisdom is that Euclid considered the line constructed in Prop. I.31 to be unique. In other words, as stated, Prop. I.31 is *not* logically equivalent to Postulate V under Euclid's neutral axioms. As intended, it is logically equivalent to Postulate V.

By inserting the word "unique" before the word "parallel" in the statement of Prop. I.31, we get the **Axiom of Playfair**. Named for the Scottish mathematician and scientist John Playfair (1748–1819), Playfair's Axiom was well-known at least since Proclus. It is so important in the sequel that we state it here for reference.

Axiom of Playfair. Given a line, and a point not on that line, there is a *unique* parallel to the line through the point.

Note that we present Playfair as an assumption rather than a theorem. Since Playfair's Axiom is logically equivalent to Postulate V under Euclid's neutral axioms, it may be used as a substitute for Postulate V in Euclid's axiomatic development. For this reason, people often recite Playfair's Axiom when they claim to be reciting Postulate V.

That Playfair and Postulate V are equivalent assumptions is by no means obvious. We address their equivalence in the exercises.

Proposition I.32 contains the one fact everyone remembers about triangles, viz., that the sum of the angles is two right angles.

Proposition I.32. An exterior angle of a triangle is congruent to the sum of the remote interior angles. The sum of the interior angles of a triangle is π.

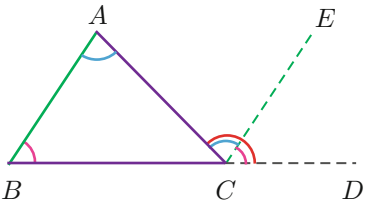


Figure 1.29: Proposition I.32: The angle sum in a triangle is π

Proof. Let the triangle ΔABC be given. Extend $\overrightarrow{BC}$ to go through a point D not on BC. We consider the exterior angle $\angle ACD$.

Construct the line $\overline{CE}$ parallel to $\overline{AB}$ through C per Prop. I.31. As alternate angles, $\angle BAC \cong \angle ACE$. By Prop. I.29, we also have $\angle ABC \cong \angle ECD$. Adding angles and invoking CN2, we get

$$\angle ABC + \angle BAC \cong \angle ACE + \angle ECD \cong \angle ACD,$$

which gives us the first part of the result. Notice now that we have only to add $\angle ACB$ to $\angle ACD$ to get π, i.e.,

$$\angle ABC + \angle BAC + \angle ACB \cong \angle ACD + \angle ACB = \pi,$$

as was to be shown. □

Note how the construction of a unique parallel is critical to the proof of Prop. I.32. Indeed, Prop. I.32 is another equivalent to Postulate V. We detail an approach to the proof of that fact in an exercise.

Exercises 1.5. 1. Formulate a purely verbal definition of *alternate angles*.

2. Figure 1.30 illustrates a construction that allows us to trisect a line segment using Euclid's axioms and their consequences. Describe the construction. Which steps depend on Postulate V or its equivalents?

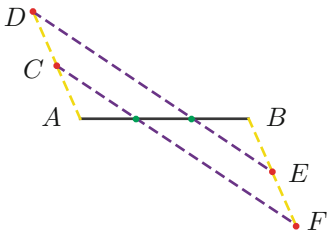


Figure 1.30: Trisecting a segment AB

Circles Revisited

The next three exercises are a continuation of our study of circles that began in Exercises 1.4. For these, use Euclid's axioms and any of Props. I.1–32.

3. Let A, B, C be points on a circle with center O, arranged as in Fig. 1.31. Show that $\angle COB \cong 2\angle CAB$. Relocate the point A—on the circle, but not to coincide with either B or C—so the result is no longer true.

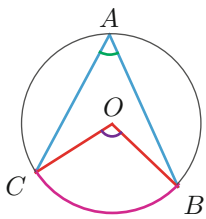


Figure 1.31: The angles of Exercise 3

4. Let A, B, C, D be points on a circle arranged as in Fig. 1.32. Show that $\angle BAC \cong \angle BDC$. Does this mean that an arc of a circle determines an angle uniquely?

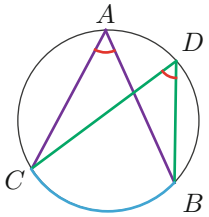


Figure 1.32: $\angle BAC$ and $\angle BDC$ of Exercise 4

5. Let A, B, C be points on a circle. Show that BC is a diameter if and only if $\angle BAC$ is right.

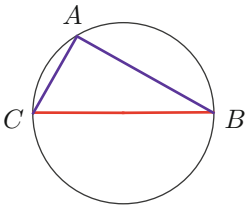


Figure 1.33: Exercise 5: BC is a diameter if and only if $\angle BAC$ is right

Equivalents to Postulate V

For the remaining exercises, assume Euclid's neutral axioms and use any of their consequences: Props. I.1–28 and results from any exercises in the previous two sections.

6. This problem, together with the proof of Prop. I.29, establishes that each of the three conclusions of Prop. I.29 is logically equivalent to Postulate V under Euclid's neutral axioms.

   (a) Assume that when a transversal intersects two parallel lines, the alternate interior angles are congruent. Show that Postulate V is true.

   (b) Assume that when a transversal intersects two parallel lines, corresponding angles are congruent. Show that Postulate V is true.

   (c) Assume that when a transversal intersects two parallel lines, the interior angles on the same side of the transversal are supplementary. Show that Postulate V is true.

7. Assume Postulate V. Show that the parallel line constructed in the proof of Prop. I.31 is unique. This shows that under Euclid's neutral axioms, Postulate V implies Playfair's Axiom.

8. Assume the Axiom of Playfair. Show that Postulate V must be true.

9. Show that Prop. I.30 (transitivity of parallelism) implies the Axiom of Playfair, thus by the previous problem, that it implies Postulate V. Conclude that under Euclid's neutral axioms, transitivity of parallelism and Postulate V are logically equivalent.

1.6 Areas: Propositions I.33–46

The propositions in this section deal with constructions and properties of parallelograms and associated triangles. We confine our attention mostly to the results that are germane to Euclid's proof of the Pythagorean Theorem.

All of the results here depend on Postulate V or one of its equivalent formulations.

Proposition I.33. Line segments which join the ends of congruent and parallel line segments in the same direction are themselves congruent and parallel.

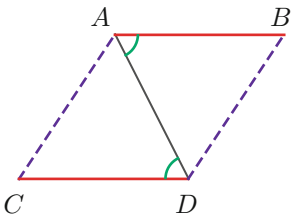


Figure 1.34: Proposition I.33: Segments joining endpoints of congruent parallel segments

Proof. Let $AB \cong CD$ be parallel line segments labeled as in Fig. 1.34 so that points A and C are both to the left of B and D. Consider also the segment AD. Since $\overline{AD}$ falls upon parallels $\overline{AB}$, $\overline{CD}$, Prop. I.29 gives us alternate interior angles $\angle BAD \cong \angle ADC$. Since AD is shared, we have SAS for $\Delta ABD \cong \Delta ACD$. As corresponding sides, $BD \cong AC$, which proves the first part of the result. As corresponding angles, $\angle CAD \cong \angle ADB$. These are alternate angles if we view $\overline{AD}$ as a transversal for $\overline{AC}$ and $\overline{BD}$. Prop. I.27 thus implies that $\overline{AC}$ and $\overline{BD}$ are parallel. □

Despite the fact that Euclid's polygon is the region enclosed by a plane figure, the statement of Prop. I.34 in [7] refers not to parallelograms, but to *parallelogrammic areas.* Note also the use of the word "diameter" for what we would call a diagonal, that is, a segment joining opposite vertices of a parallelogram. Apart from retaining that usage, we rephrase the statement of the proposition in keeping with current practice.

We leave the proof as an easy exercise but note that this is an important proposition going forward.

Proposition I.34. Opposite sides, respectively, opposite angles, are congruent in any parallelogram and a diameter of the parallelogram bisects the region enclosed.

Euclid next works through a sequence of propositions that address areas enclosed by parallelograms and triangles for figures that are "in the same parallels," that is, figures with the same height. Context makes it clear that Euclid's figures are arranged with their bases forming a single line so that opposite vertices lie on a line parallel to the bases.

Proposition I.35. Parallelograms on the same base and in the same parallels enclose equal areas.

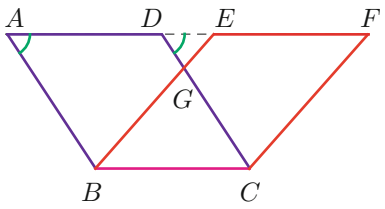


Figure 1.35: Proposition I.35: Parallelograms on the same base, in the same parallels

Proof. Suppose $ABCD$ and $BCEF$ are parallelograms that share the base BC, with $\overline{AD} = \overline{EF}$.

Proposition I.34 gives us $AD \cong BC$ and $EF \cong BC$. It follows that $AD + DE \cong EF + DE$ so $AE \cong DF$. We also have $AB \cong DC$, by Prop. I.34. Since $\overline{AF}$ falls on parallels $\overline{AB}$ and $\overline{DC}$, corresponding angles $\angle CDF$ and $\angle BAD$ are congruent. This gives us SAS for $\Delta ABE \cong \Delta DCF$. Subtract the region enclosed by ΔEDG from both to get trapezoids enclosing equal areas $ABGD$, $EFCG$. Now add the region enclosed by triangle ΔBGC to both to get the result. □

The next result is a slight generalization of the previous result.

Proposition I.36. Parallelograms on congruent bases in the same parallels enclose equal areas.

Proof. Let $ABCD$ and $EFGH$ be parallelograms with $BC \cong FG$, $\overline{BC} = \overline{FG}$, and $\overline{AD} = \overline{EH}$. Join BE and CH. Since $BC \cong FG$ and $FG \cong EH$, $BC \cong EH$. Since BC and EH are parallel and congruent, Prop. I.33 implies that BE and CH must be parallel and congruent as well. By Prop. I.35, parallelograms

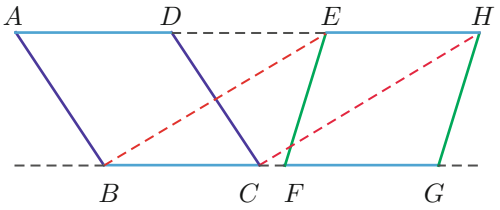


Figure 1.36: Proposition I.36: Parallelograms on congruent bases, in the same parallels

$BCEH$ and $EFGH$ enclose equal areas as do $BCEH$ and $ABCD$. It follows that the regions enclosed by $ABCD$ and $EFGH$ are equal, as desired. □

The next two results reprise Props. I.35 and I.36, this time for triangles.

Proposition I.37. Triangles on the same base and in the same parallels enclose equal areas.

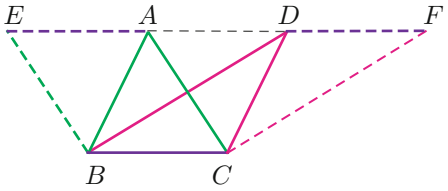


Figure 1.37: Proposition I.37: Triangles on the same base, in the same parallels

Proof. Consider triangles ΔABC and ΔBCD in which $\overline{AD}$ is parallel to $\overline{BC}$. Form a parallel to $\overline{AC}$ through B and say it goes through $\overline{AD}$ at E. Similarly, form a parallel to $\overline{BD}$ at C, going through $\overline{AD}$ at F. We have parallelograms $EBCA$ and $DBCF$ on the same base BC, so they enclose equal areas. By Prop. I.34, each area ΔABC and ΔBCD is half the area enclosed respectively by $EBCA$ and $DBCF$, so the areas enclosed by the triangles must be equal as well. □

The proof of Prop. I.37 is the first place we see Euclid employ an argument to the effect that half equal magnitudes must themselves be equal.

Proposition I.38. Triangles on congruent bases and in the same parallels enclose equal areas.

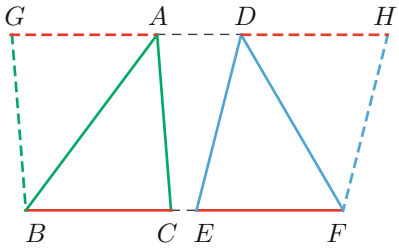


Figure 1.38: Proposition I.38: Triangles on congruent bases, in the same parallels

Proof. Consider triangles ΔABC and ΔDEF where $BC \cong EF$, $\overline{BC} = \overline{EF}$, and $\overline{BF}$ is parallel to $\overline{AD}$. Form a parallel to $\overline{AC}$ at B and say it passes through $\overline{AD}$ at G. Form a parallel to $\overline{DE}$ at F, and say it passes through $\overline{AD}$ at H. We then have parallelograms $GBCA$ and $DEFH$ on congruent bases and in the same parallels, so they enclose equal areas. By Prop. I.34, the area enclosed by ΔABC, respectively ΔDEF, is half the area enclosed respectively by $GBCA$, $DEFH$, so the areas enclosed by the triangles must be equal. □

Next is a partial converse to Prop. I.37.

Proposition I.39. Triangles enclosing equal areas on the same base and on the same side of that base are in the same parallels.

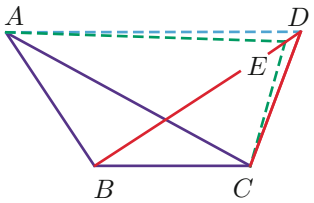


Figure 1.39: Proposition I.39: Triangles enclosing equal areas on the same base

Proof. Assume triangles ΔABC and ΔBCD enclose equal areas. We must show that $\overline{AD}$ is parallel to $\overline{BC}$. Suppose the result is false. Form the parallel to $\overline{BC}$ through A and suppose it intersects $\overline{BD}$ at the point E. Form the segment EC. Now we have ΔABC and ΔBCE on the same base in the same parallels so they must enclose equal areas. But that means ΔBCE and ΔBCD enclose equal areas. This contradicts CN5 as one of ΔBCE or ΔBCD must be part of the other. We conclude that $\overline{AD}$ is parallel to $\overline{BC}$. □

Heath notes that Heiberg established that Prop. I.40 was not in *The Elements* of Euclid but was added later on by someone who thought there should be a converse of Prop. I.38, as Prop. I.39 is a converse of Prop. I.37. We do not need it in the sequel but state it here, leaving a proof to the exercises.

Proposition I.40. Triangles enclosing equal areas on congruent bases and on the same side of those bases are in the same parallels.

Proposition I.41. If a parallelogram has the same base as a given triangle and they are in the same parallels, then the area enclosed by the parallelogram is double the area enclosed by the triangle.

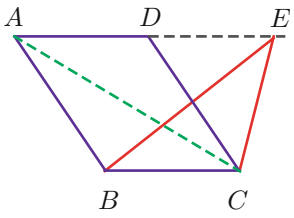


Figure 1.40: Proposition I.41: Parallelogram and triangle on the same base, in the same parallels

Proof. Consider a parallelogram $ABCD$ and a triangle ΔBCE, which are in the parallels $\overline{BC}$ and $\overline{AE}$. Form AC. By Prop. I.37, ΔABC and ΔBCE enclose equal areas. Parallelogram $ABCD$ has diagonal AC, so by Prop. I.34 it encloses a region with double the area as that enclosed by ΔABC. Since the areas enclosed by ΔABC and ΔBCE are equal, it follows that $ABCD$ encloses twice the area enclosed by ΔBCE. □

We skip next to Prop. I.46, an easy construction used in the proof of the Pythagorean Theorem.

Proposition I.46. We can construct a square on a given line segment.

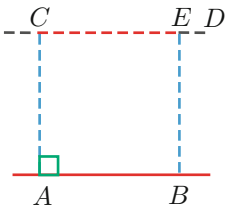


Figure 1.41: Proposition I.46: Construct a square from a segment

Proof. Suppose AB is given. Construct a perpendicular to $\overline{AB}$ at A. Choose C on this perpendicular so that $AC \cong AB$. Construct a parallel to $\overline{AB}$ through C and let D be on that parallel. Form the parallel to $\overline{AC}$ at B. Say this meets $\overline{CD}$ at E.

Note that by construction, $ABCE$ is a parallelogram. By Prop. I.34, $AB \cong CE$ and $AC \cong BE$. Since $AC \cong AB$, the four sides of $ABCE$ are all congruent. Since $\angle BAC$ is right, Prop. I.34 gives us that $\angle BEC$ is right. Since $\overline{AC}$ is a transversal for parallels $\overline{AB}$, $\overline{CE}$, Prop. I.29 implies that the angles formed by $\overline{AC}$ and $\overline{CE}$ are all right. Again invoking Prop. I.34, we see that $\angle ABE$ is right, as well, so the parallelogram is a square. □

Exercises 1.6. 1. Prove Prop. I.34.

2. Prove Prop. I.40.

3. Prove that the diagonals of a parallelogram bisect one another.

4. A polygon is **inscribed** in a circle if all of its vertices lie on the circle. Detail instructions for inscribing a square in a given circle.

5. Let $ABCD$ be a quadrilateral inscribed in a circle. Show that the sum of

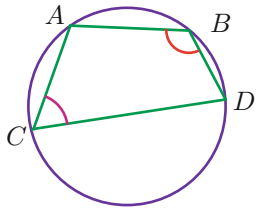


Figure 1.42: Exercise 5: Quadrilateral inscribed in a circle

opposite angles is two right angles.

6. Let ΔABC be a triangle. Let D be the midpoint of AB. Form the parallel

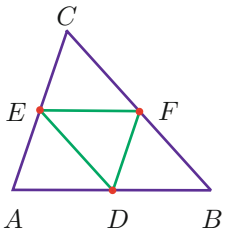


Figure 1.43: Exercise 6: ED is parallel to BC

to BC through D and say it intersects AC at E. Form the parallel to AC at D and say it intersects BC at F. Show that $\Delta DBF \cong \Delta ADE$. Show that E and F are the midpoints respectively of AC and BC. Conclude by showing that in any triangle, the line joining midpoints of two sides is parallel to the base.

7. Let ΔABC be a right triangle and let D be the midpoint of the hypotenuse, AB. Let CE be the angle bisector for $\angle ACB$. Let F be the foot of the perpendicular from C to $\overline{AB}$. Show that the triangle ΔBCD is isosceles

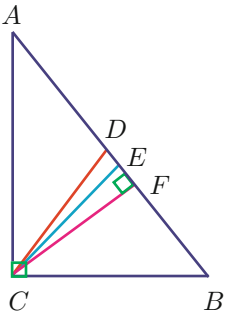


Figure 1.44: Exercise 7: D is the midpoint of AB; CE bisects $\angle ACB$

and that CE bisects $\angle DCF$. In Fig. 1.44, CD is above CF. When would their positions be reversed? When would they coincide?

8. Let ΔABC be a right triangle with hypotenuse AB. Let ℓ be perpendicu-

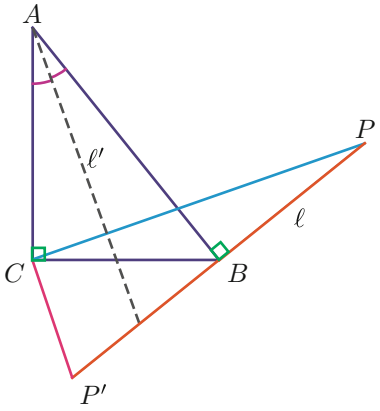


Figure 1.45: Exercise 8: ℓ is perpendicular to $\overline{AB}$

lar to $\overline{AB}$ at B. Take P and P' on ℓ so $BP \cong BC \cong BP'$, as in Fig. 1.45. Let ℓ' be the line that bisects $\angle BAC$. Show that $\overline{CP}$ is perpendicular to ℓ' and that $\overline{CP'}$ is parallel to ℓ'.

1.7 The Pythagorean Theorem

While it is debatable whether the goal of the thirteen books of *The Elements* of Euclid is the construction of the five platonic solids, there is general agreement that Book I of *The Elements* is a journey that leads to the Pythagorean Theorem. Indeed, Euclid's proof of Prop. I.47 is a crystalline application of almost every result preceding it.

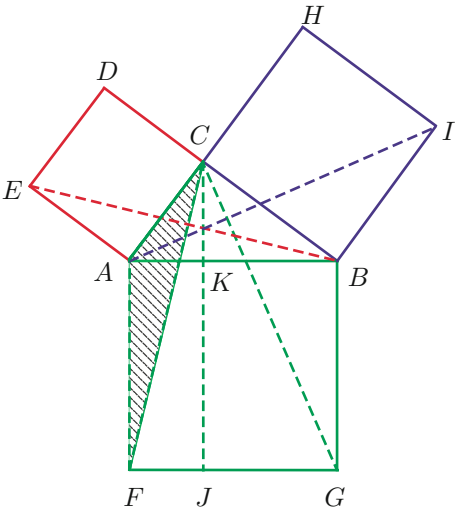


Figure 1.46: Euclid's proof of the Pythagorean Theorem

Proposition I.47 (Pythagorean Theorem)**.** The area enclosed by the square on the hypotenuse of a right triangle is the sum of the areas enclosed by the squares on the other two sides.

Proof. Referring to Fig. 1.46, let $\angle ACB$ be right. Employing Prop. I.46, we construct a square on each side of the triangle: $ABFG$ on AB, $ACDE$ on AC, and $BCHI$ on BC. By Prop. I.14, points A, C, H lie on a line, as do points B, C, D. Form the parallel to AF through C. Let J be the point where it intersects FG. We then have the parallelogram $AKJF$ and ΔCAF in the same parallels and on the same base. By Prop. I.41, $AKJF$ encloses a region of twice the area as that enclosed by ΔCAF. The same can be said for $ACDE$ and ΔAEB. Next note that $\angle BAE \cong \angle CAF$ because each is a right angle plus $\angle BAC$. This gives us SAS for $\Delta AEB \cong \Delta CAF$. It follows that the regions enclosed by $AKJF$ and $ACDE$ have the same area. The reader should

complete the proof by supplying a similar argument to show that $KBGJ$ and $BCHI$ enclose regions with the same area. The necessary triangles are shown in the figure. □

Book I of *The Elements* ends with Prop. I.48, the converse of the Pythagorean Theorem.

Proposition I.48. If the area enclosed by the square on one side of a triangle equals the sum of the areas enclosed by the squares on the remaining two sides, then the angle formed by the remaining two sides is right.

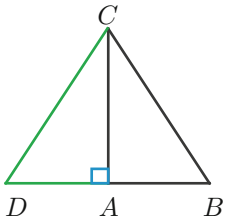


Figure 1.47: Proposition I.48: The converse of the Pythagorean Theorem

Proof. Consider ΔABC where the area enclosed by the square on BC is the sum of the areas enclosed by the squares on AB and AC. We will show that $\angle CAB$ is right.

Form $\overrightarrow{AD}$ at right angles to AC, taking D different from B so that $AD \cong AB$. Notice that the squares on AD and AB enclose equal areas. Thus the sum of the squares on AB and AC equals the sum of the squares on AD and AC. Since ΔADC is right, the sum of the squares on AD and AC is the square on CD, which must then equal the square on BC. It follows that $BC \cong CD$. Since $AB \cong AD$ and AC is common, we have SSS for $\Delta ABC \cong \Delta ADC$. As corresponding angles, $\angle CAB \cong \angle CAD$. Since $\angle CAD$ is right, we have shown what was to be proved. □

Note the difference in the way the Pythagorean Theorem is presented in *The Elements* and the way we usually discuss it. In *The Elements*, the theorem is about the areas of plane figures, whereas we typically think of the Pythagorean Theorem as about the numbers we associate to the sides of a right triangle. In the West, numbers did not become integral to geometry in a consistent way until after the sixteenth century, with the work of René Descartes. Part of what makes Euclid's work so compelling, and challenging, is the fact that he never employs numbers or any of the conveniences that numbers bring to the table.

Our goal for the next part of the course is to understand some of the mathematics developed in Europe during the middle ages and Renaissance, much of

it in the wake of work that found its way into Europe from the Arabic-speaking world. The ninth and tenth centuries were a golden age of mathematics in the early Islamic caliphates, fueled by the study and dissemination of Greek texts, including *The Elements*. The work we consider is mostly the fruit of efforts brought to bear on the problem of the parallels, the resolution of which helped precipitate the twentieth century revolutions in mathematics and physics.

Exercises 1.7. 1. Let AB be a line segment and let C be a point on the segment.

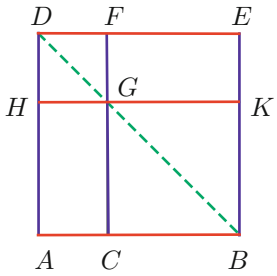


Figure 1.48: The rectangles of Exercise 1

(a) In Fig. 1.48, $ADEB$ is the square on AB, CF is parallel to AD, G is the point where CF and the segment BD intersect, and HK is the parallel to AB, through G. Using the figure as a guide, prove that the area enclosed by the square on AB is the sum of the areas enclosed by the squares on AC and CB plus twice the area of the region bounded by the rectangle with sides AC, CB.

(b) What does the theorem say algebraically?

2. Let ΔABC be an arbitrary triangle. Let D be the foot of the perpendicular

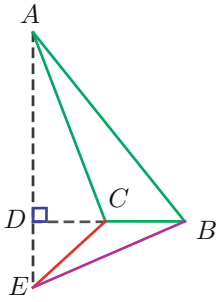


Figure 1.49: Triangles illuminate Exercise 2

from A to $\overline{BC}$. Let E be on $\overrightarrow{AD}$ but not on AD. Show that the sum of the areas bounded by squares on the segments AC and EB is equal to the sum of the areas bounded by squares on the segments AB and CE. Is it still true if $E = D$? What if E is on $\overrightarrow{DA}$ but not on DA? What if E is between A and D?

3. Let ΔABC be a triangle so that $\angle CAB$ is obtuse. Prove that the area

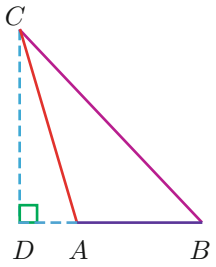


Figure 1.50: Exercise 3 generalizes the Pythagorean Theorem

enclosed by the square on BC is the sum of the areas enclosed by the squares on the other two sides of the triangle plus twice the area enclosed by the rectangle with sides AD and AB, where D is the foot of the perpendicular from C to $\overline{AB}$.

4. Now let ΔABC be a triangle for which all angles are acute. State and prove the theorem for this triangle that corresponds to the theorem for the obtuse triangle in Exercise 3. (See Fig. 1.51.)

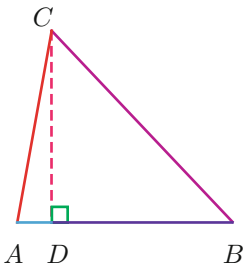


Figure 1.51: Exercise 4: A triangle with all angles acute

5. The theorems in the last two problems together constitute a statement of what famous theorem from trigonometry?

6. (For this problem, you should consult reliable, scholarly sources, for instance *Convergence*, a website of the Mathematical Association of America, or *A History of Mathematics*, by Victor Katz [17].) The theorem we know as the Pythagorean Theorem is named for the Greek thinker Pythagoras (572–497 B.C.) but was known to advanced civilizations predating the ancient Greeks. Find out what the Babylonians knew about the so-called Pythagorean Theorem. Find out what the ancient Chinese knew and when they knew it. There are several arguments for the Pythagorean Theorem found in ancient Chinese texts. Find one and compare the proof to Euclid's proof. Are there implicit assumptions in the Chinese proof?

1.8 Euclid's Assumptions

We saw in our study of Book I of *The Elements* that Euclid relies on a number of unarticulated assumptions. This may call into question the solidity of the logical foundation for Euclidean geometry. It needn't.

Starting with the proof of Prop. I.1 we can identify gaps in Euclid's work but the approach to filling those gaps does not involve correcting the results. It involves identifying a consistent set of assumptions that support Euclid's results. Except in very few cases, Euclid's work in Book I is as solid today as it was in 300 B.C. That the assumptions have changed is due largely because of changing standards of rigor. The geometry itself has not changed.

Here is a list of some tacit assumptions that underlie Book I of *The Elements*. In each case, we note how the assumption was disposed of by the time Hilbert consolidated corrections to Euclid's work in the early years of the twentieth century.

Tacit Assumptions

(1) *Uniqueness* is never addressed by Euclid, probably because it was not part of the style of writing or thinking about mathematics at the time. We have seen that in some cases in which Euclid ignores the matter of uniqueness—most dramatically in Prop. I.31—it is relatively easy to prove using Euclid's tools. In other cases, uniqueness is really a tacit assumption. We distinguish some of those cases here.

   (a) *A line is uniquely determined by two points.* This is a tacit assumption of Euclid that is expressly articulated in modern Euclidean mathematics.

   (b) *A line is uniquely determined by any line segment it contains.* This follows from the assumption above, along with an assumption about a minimal number of points that a line must have. In modern Euclidean geometry, the assumption is that a line is a copy of $\mathbb{R}$, the set of real numbers.

   (c) *The point of intersection of two lines is unique.* This is a consequence of (a).

(d) *The perpendicular to a line at a point or from a point is unique.* We saw in Exercises 1.4 that these can be proved within the framework provided by Euclid's neutral axioms.

(e) *A point and a radius determine a unique circle.* This is a tacit assumption. In Hilbert's approach to Euclidean geometry, circles are defined to imply uniqueness.

(f) *A circle has a unique center.* This can be proved within Euclid's framework.

(2) *Lines and curves are continuous extensions.* This is among the most profound of the tacit assumptions. As noted above, the modern view of a Euclidean line is that it is a copy of $\mathbb{R}$, that is, a gapless extension.

(3) *The continuity of space.* This is a tacit assumption that underlies the bisection of an angle, for example. If we identify points in the Euclidean plane with points in the Cartesian plane—a natural extension of the assumption that a line is a copy of $\mathbb{R}$—then the continuity of space follows.

(4) *Superposition is a legitimate method of proof.* Euclid uses superposition to prove SAS and SSS, key results that support the remainder of Book I. Hilbert takes something very close to SAS as an assumption, and is then able to prove both SAS and SSS.

(5) *A point partitions a line into three nonintersecting sets: the point itself, the part of the line on one side of the point, and the part of the line on the other side of the point. A line partitions the plane into three convex subsets: the line itself, the points on one side of the line, and the points on the other side of the line.* A **convex set** is one that contains all the points on a line segment joining any two points in the set. Assumptions of this ilk underlie any argument in which we choose points B, C on one side of a line, and expect all the points of BC to lie on the same side of that line. Hilbert disposes of these sorts of tacit assumptions using his axioms of order.

(6) *Triangles, circles, and parallelograms partition the plane into two subsets.* The implicit assumption that a polygon or a circle has an inside and an outside is used over and over again in *The Elements*. Hilbert deals with this using his order and incidence axioms.

(7) *Regions enclosed by triangles, circles, and parallelograms are convex.* We assume that except for the vertices, a diagonal of a parallelogram lies inside the figure. In Hilbert's treatment of Euclidean geometry, convexity of these sorts of regions is a consequence of order and incidence axioms.

(8) *If a line enters the region bounded by a triangle, it must exit the region by passing through a side of the triangle.* This assumption is on view, for example, in the proof of the Exterior Angle Theorem. The final disposition of this assumption, in Hilbert's treatment, is a consequence of an axiom credited to Pasch.

When we study Hilbert's work in Chapter 4, we adopt a systematic approach to all the recognized assumptions underlying Euclid's work.

Chapter 2

Neutral Geometry

2.1 The Problem of the Parallels

Difficulties with Euclid's fifth postulate were hinted at in *The Elements* and cited explicitly by the earliest commentators. Though it took roughly two thousand years to resolve the problem of the parallels, the result of that process was a sea change in both mathematics and physics. While it was the subject of a struggle, the problem of the parallels seemed an intractable monster, devouring egos, relationships, and careers, threatening to destabilize mathematics itself. Yet in retrospect, the struggle was so fruitful and its resolution so startling that the problem itself seems a gift.

The most difficult problems are often the most fertile. They attract the best minds, and can inspire deeply productive investigations. Many fail in their attempts on these problems and for an individual, the consequences can be devastating. In the aggregate, though, masses of failed attempts on one problem can advance the subject dramatically, revealing new perspectives and hidden worlds. The problem of the parallels is a spectacular example of this phenomenon.

Euclid's Postulate V says that if a transversal intersects a pair of lines so that on one side of the transversal, the angles formed sum to less than two right angles, then the pair of lines intersect on that side of the transversal. We have

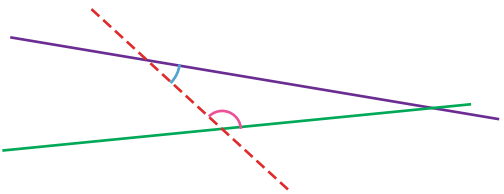


Figure 2.1: Intersecting lines and angles formed by a transversal

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M. I. Dillon, *Geometry Through History*,
https://doi.org/10.1007/978-3-319-74135-2_2

seen that Euclid's neutral axioms guarantee the existence of a parallel to a fixed line through a point in a plane. The addition of Postulate V to the axioms secures the guarantee that the constructed parallel is unique.

Prevailing wisdom in Euclid's day, and through succeeding centuries, was that in the space we experience, it was not possible to have more than one parallel to a given line through a given point in a plane. Uniqueness of parallels was viewed as an intrinsic feature of the physical world. As such, it had to be demonstrable from more fundamental principles, for instance, Euclid's neutral axioms. During the two thousand year struggle with the parallel postulate, most efforts were directed toward proving the uniqueness of parallels, either using Euclid's neutral axioms or some variation thereof. The fact that no proof could be found was not merely unsettling, it was viewed as evidence that mathematics itself was a failure.

Proclus reported that Ptolemy (100–170), author of *The Almagest*, tried to prove the contrapositive of Postulate V. Proclus debunked Ptolemy's proof and detailed one of his own. Subsequently, it too—like every other purported proof—was debunked.

Some commentators launched their attack on the problem by focusing on the definition of parallel lines. By Euclid's definition, parallels are straight lines in the same plane that do not meet, no matter how far they are extended. The most popular substitute was to define parallels as lines that were everywhere equidistant. According to this definition, lines are parallel if at any two different points on one of them, the perpendicular segments from the points to the second line are congruent. The thinking was that once we had the definition right, the necessity of the uniqueness of parallels would become evident. This approach failed because under Euclid's neutral axioms, the alternative definitions were all logically equivalent to Postulate V.

Forward several centuries to the noted English mathematician, John Wallis (1616–1703), who contributed significantly to pre-Newtonian mathematics. Wallis tried to prove Postulate V with an argument based on the assumption that to every figure there exists a noncongruent figure similar to the first. Wallis thought that Postulate III, which posits the existence of circles, implied the existence of similar figures, partly because it implies the existence of similar circles. (All circles are similar!) As it turns out, it would have been sufficient for Wallis to assume the existence of noncongruent similar triangles. But that assumption, too, is logically equivalent to Postulate V.

While discussing the mathematics that arose through assaults on the problem of the parallels, we study planes in which Euclid's neutral axioms hold. The parallel postulate—that is, Postulate V and all of its equivalents—is put aside entirely in this setting.

Definition 2.1. A **neutral plane** is a collection of points and lines that satisfy Euclid's neutral axioms. In a *neutral plane*, the parallel postulate is assumed to be neither true nor false. **Neutral geometry** is the geometry of a *neutral plane*.

A **Euclidean plane** is a collection of points and lines that satisfy Euclid's

axioms. **Euclidean geometry** is the geometry of a *Euclidean plane*.

A **hyperbolic plane** is a neutral plane in which the parallel postulate is assumed to be false. **Hyperbolic geometry** is the geometry of a *hyperbolic plane*.

Toward the end of this chapter and into the next, we uncover properties that neutral planes must have if the parallel postulate is assumed false. In Chapter 6, we discuss models for such a geometry. The existence of a model for a geometry is the key to establishing that the geometry is consistent, in other words, that its axiom system does not harbor hidden contradictions. This is what legitimizes our definition of a hyperbolic plane.

We know from Chapter 1 that Props. I.1–28 are consequences of Euclid's neutral axioms. They thus form a foundation for the study of neutral geometry. In this chapter, we press that study further to discover more about the common ground shared by Euclidean and hyperbolic geometries.

Because of its origins, hyperbolic geometry is often referred to as *non-Euclidean geometry*. This usage, though common, is unfortunate. Hyperbolic geometry is just one of many different geometries that are not Euclidean in the sense that they do not conform to all of Euclid's axioms. As we progress through this chapter, it will become evident that the body of properties that hyperbolic geometry has in common with Euclidean geometry is vast. Indeed, if we must contrast hyperbolic geometry and Euclidean geometry, it might make more sense to call hyperbolic geometry near-Euclidean.

Exercises 2.1. 1. In Chapter 1, we established the equivalence of Postulate V and several other statements under Euclid's neutral axioms. One of those statements was the Axiom of Playfair. Assume Euclid's neutral axioms.

   (a) Euclid proves Prop. I.29 using Postulate V. Prove Prop. I.29 using the Axiom of Playfair instead.

   (b) Euclid proves Prop. I.30 using Prop. I.29. Prove Prop. I.30 using the Axiom of Playfair instead.

2. Show that under Euclid's axioms, parallel lines must be equidistant.

3. Recall that triangles are **similar** provided all their angles are congruent. Show that in a plane that satisfies Euclid's axioms, there are noncongruent similar triangles.

4. Both Prop. I.32 and the Pythagorean Theorem are logically equivalent to Postulate V under Euclid's neutral axioms but the proofs are not easily accessible to the tools we have in our hands. Assume the Pythagorean Theorem is true, and use it along with Euclid's neutral axioms and their consequences at least to find a connection to Postulate V or to the Axiom of Playfair. Repeat the exercise, this time assuming Prop. I.32 instead of the Pythagorean Theorem. Finally, try to discover a connection between the Pythagorean Theorem and Prop. I.32, assuming Euclid's neutral axioms.

2.2 All For One and One For All

Gerolamo Saccheri (1667–1733) was a Jesuit priest who tried to prove Postulate V by contradiction, a common approach to the problem. The idea is that if we assume Postulate V is false, we should find that a known characteristic of space is violated. For example, if Postulate V is false, then the Pythagorean Theorem is false. This appears promising: the Pythagorean Theorem seems like a "known" characteristic of space. But under Euclid's neutral assumptions, the Pythagorean Theorem is logically equivalent to Postulate V so it is just another way of capturing the same underlying characteristic of space, namely, its "flatness." If we have to assume something, is it preferable to assume the Pythagorean Theorem, which allows us to prove Postulate V, or to assume Postulate V, as Euclid did, which allows us to prove the Pythagorean Theorem? In either case, we find ourselves assuming something we feel we ought to be able to prove.

Saccheri's attempts at the problem led him to rediscover (or rework) results that had been discovered by the Persian polymath, Omar Khayyam (1048–1131), who effected his own assault on the problem of the parallels.

Omar Khayyam was a poet, philosopher, astronomer, and mathematician who contributed materially to the early development of algebra, as well as to geometry. Khayyam's work was part of a great wave of learning that rolled through the Islamic world between the eighth and twelfth centuries, when much of Europe was experiencing what used to be called the "dark ages." During that time—the **Arabic era**, or the **Islamic era**—scholars from all over were drawn to certain cities in the Arabic-speaking world because of their translation centers. The translation centers formed around libraries where the works of Euclid and other Greek thinkers were collected, studied, advanced, and translated into Arabic. As a Persian, Omar Khayyam probably did not speak Arabic as a first language. Though Khayyam was Muslim, not all scholars of the Arabic era were. Indeed, the roles of Arabic and Islam at that time and in that place were analogous to the roles of Latin and Roman Catholicism in Europe of the middle ages. Arabic, like Latin, was the language of scholarship. Islam, like Roman Catholicism, was the religion of the rulers who supported that scholarship. The work of the so-called Arabic or Islamic thinkers eventually found its way into Roman Catholic Europe. As a member of the Society of Jesus, traditionally viewed as the most intellectual order of Catholic priests, Saccheri would have had access to some of that work and may have had access to Khayyam's study on quadrilaterals.

Some of the results of Khayyam and Saccheri are now recognized as part of hyperbolic geometry, but neither man would have seen his work in that light. Both were studying Euclid and grappling with the problem of the parallels: The discovery of hyperbolic geometry was far in the future. Indeed, Khayyam's and Saccheri's prescience in studying these results so deeply is partly what makes them intriguing.

As we proceed, assume that we are in a neutral plane, so that Euclid's neutral axioms, thus Props. I.1–28, all hold.

Definition 2.2. A **bi-right quadrilateral** is a quadrilateral with adjacent right angles. The side shared by the right angles is the **base**. The side opposite the base is the **summit**. The angles formed by the summit and adjacent sides are the **summit angles**.

A **Khayyam-Saccheri quadrilateral**, or **KS-quadrilateral**, is an isosceles bi-right quadrilateral, that is, a quadrilateral in which the opposite sides adjacent to the right angles are congruent. The **midline** of a KS-quadrilateral is the segment joining the midpoint of the base with the midpoint of the summit.

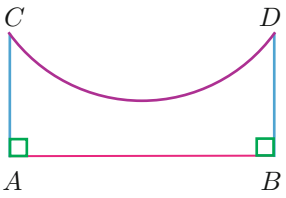


Figure 2.2: KS-quadrilateral

Since its base angles are right, *a bi-right quadrilateral has parallel sides* by Prop. I.28.

When we depict general quadrilaterals in a neutral plane, we remain open to the possibility that lines may not look the way we think straight lines ought to look. A neutral plane is not necessarily flat. If there is a slight curvature to the plane, lines that appear flat locally may have curvature detectable from a distance. We exaggerate this possibility in some of our figures. In all the pictures of quadrilaterals and triangles in this chapter, though, the sides are straight line segments *in a neutral plane*, even when they look like curves.

We leave the proof of our first lemma as an exercise. It captures two essential features of any KS-quadrilateral.

Lemma 2.3. (1) The midline of a KS-quadrilateral is perpendicular both to the base and to the summit.

(2) The summit angles of a KS-quadrilateral are congruent.

The next lemma says that a base and a midline determine a KS-quadrilateral. Recall that angles are *complements* if their sum is a right angle.

Lemma 2.4. Two KS-quadrilaterals with congruent bases and congruent midlines are themselves congruent in a neutral plane.

Proof. Let $ABCD$ and $A'B'C'D'$ be KS-quadrilaterals with bases $AB \cong A'B'$. Suppose $EF \cong E'F'$ are the respective midlines of $ABCD$ and $A'B'C'D'$, where E and E' are respective midpoints of the bases, so that, by Lemma 2.3 $\angle AEF \cong \angle A'E'F' \cong \angle CFE \cong \angle C'F'E'$ are all right. Because KS-quadrilaterals are

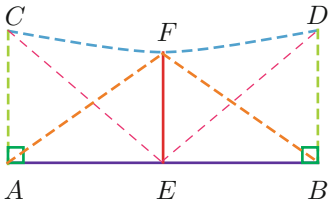


Figure 2.3: A base and a midline determine a KS-quadrilateral

isosceles and because of Lemma 2.3, we only need show that $AC \cong A'C'$, $CD \cong C'D'$, and $\angle ACF \cong \angle A'C'F'$ to prove the lemma. Further, to prove $CD \cong C'D'$, it is enough to show that $CF \cong C'F'$.

Join AF and $A'F'$. By SAS, $\Delta AEF \cong \Delta A'E'F'$. This gives us $AF \cong A'F'$ and $\angle AFE \cong \angle A'F'E'$. Using complements of congruent angles we have $\angle AFC \cong \angle A'F'C'$ and $\angle FAC \cong \angle F'A'C'$. Then by ASA, $\Delta ACF \cong \Delta A'C'F'$. This gives us $AC \cong A'C'$, $CF \cong C'F'$, and $\angle ACF \cong \angle A'C'F'$. As noted above, this is enough to establish that $ABCD \cong A'B'C'D'$. □

Compare the next theorem to Props. I.18 and 19.

Theorem 2.5. In a bi-right quadrilateral, the longer side is opposite the greater summit angle.

Proof. Let $ABCD$ be bi-right with $BD > AC$, as in Fig. 2.4. Choose E on BD

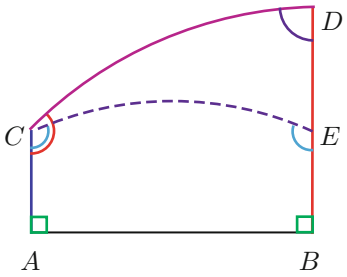


Figure 2.4: Bi-right quadrilaterals

so $BE \cong AC$. $ABCE$ is a KS-quadrilateral so $\angle ACE \cong \angle BEC$. Since $\angle ACE$ is part of $\angle ACD$, $\angle ACD > \angle ACE$. As C, D, E are noncollinear, they form a triangle, and $\angle BEC$ is exterior to it. We then have

$$\angle ACD > \angle ACE \cong \angle BEC > \angle BDC.$$

This shows that when one side is longer, the opposite summit angle is larger.

Next we assume that when one summit angle is greater, the opposite side is not longer, seeking a contradiction.

Suppose $\angle ACD > \angle BDC$. We assume $BD \leq AC$.

If $BD \cong AC$ then $ABCD$ is a KS-quadrilateral and $\angle ACD \cong \angle BDC$, a contradiction. If $BD < AC$, then by the argument above $\angle ACD < \angle BDC$, another contradiction. We conclude that $BD > AC$. □

Johann Lambert (1728–1777) studied Saccheri's work and contributed especially to an understanding of how the total angle measure in a triangle captures how far off from flat a plane is. The quadrilaterals Lambert studied had at least three right angles. These quadrilaterals were studied much earlier by ibn al-Haytham (965–1039), another one of the great polymaths of the Arabic era. (Ibn al-Haytham is also known in Europe as Alhazen.)

Definition 2.6. A quadrilateral with at least three right angles is an **ibn al-Haytham-Lambert quadrilateral**, or **HL-quadrilateral**.

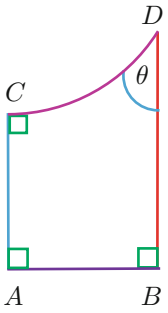


Figure 2.5: HL-quadrilateral

Corollary 2.7. A side adjacent to the fourth angle, θ, of an HL-quadrilateral is respectively greater than, congruent to, or less than its opposite side if and only if θ is respectively acute, right, or obtuse.

Proof. The result is immediate by Theorem 2.5. □

Presented with an HL-quadrilateral, we have a choice as to which side to designate as the base (AB or AC in Fig. 2.5.) Whichever it is, *the base is parallel to the summit in an HL-quadrilateral*, by Prop. I.28.

A consequence of Lemma 2.3 is that the midline of a KS-quadrilateral cuts it into two HL-quadrilaterals. The proof of the next theorem details how to go in the other direction, that is, how to "double" an HL-quadrilateral to make a KS-quadrilateral.

Theorem 2.8 (Doubling Construction)**.** An HL-quadrilateral, with summit angle θ and side AC opposite θ, determines a unique KS-quadrilateral with midline AC and summit angles congruent to θ.

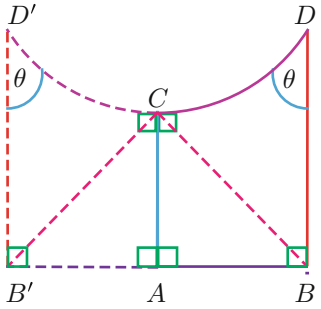


Figure 2.6: Doubling an HL-quadrilateral

Proof. Let $ABCD$ be an HL-quadrilateral with base AB, summit CD, and summit angle $\theta = \angle BDC$. Take B' different from B on $\overrightarrow{BA}$ so that $B'A \cong AB$, as in Fig. 2.6. Similarly, take D' different from D on $\overrightarrow{DC}$ so that $D'C \cong CD$. Join $B'D'$.

To verify that $B'BD'D$ is a KS-quadrilateral, we have only to show that $\angle AB'D'$ is right and that $B'D' \cong BD$. Both of these follow from arguments involving triangle congruences, as in the proof of Lemma 2.4. Indeed, since doubling AB yields a single base with midline AC, Lemma 2.4 also implies that the resulting KS-quadrilateral is unique. □

Putting the doubling construction together with Lemma 2.3, we see that we can view any KS-quadrilateral as arising via doubling an HL-quadrilateral.

Notice that an arbitrary HL-quadrilateral can be doubled to yield two generally noncongruent KS-quadrilaterals, depending on which side one chooses as base, or equivalently, which side one chooses to form the midline of the KS-quadrilateral.

The next result is immediate by the doubling construction and Theorem 2.5.

Corollary 2.9. The summit of a KS-quadrilateral is respectively greater than, congruent to, or less than the base if and only if its summit angle is respectively acute, right, or obtuse.

Definition 2.10. The **type of an angle** is acute, right, or obtuse. The **type of a KS-quadrilateral** is the type of its summit angles.

Note that a *rectangle* is a right KS-quadrilateral.

Saccheri proved an important result about the homogeneity of neutral planes, namely, that the type of one KS-quadrilateral determines the type of every KS-quadrilateral in a given neutral plane. According to [11], the historian Jeremy Gray called this the "Three Musketeers Theorem."

Following Hartshorne's treatment in [12], we approach the Three Musketeers Theorem through a sequence of lemmas.

Recall that under Euclid's neutral axioms, both the perpendicular to a line at a point, and the perpendicular to a line from a point are unique.

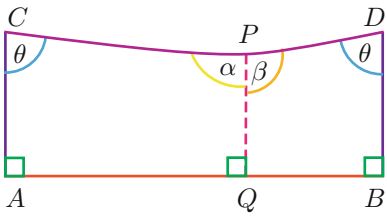


Figure 2.7: The KS-quadrilateral of Lemma 2.11

Lemma 2.11. Let $ABCD$ be a KS-quadrilateral with summit angles θ, base AB, and summit CD. Let P be a point on CD, and let Q be the foot of the perpendicular from P to AB.

(1) If $PQ < BD$, θ is acute.

(2) If $PQ \cong BD$, θ is right.

(3) If $PQ > BD$, θ is obtuse.

Proof. Let $\alpha = \angle QPC$ and $\beta = \angle QPD$ as in Fig. 2.7 so that $\alpha + \beta = \pi$. Since $AQCP$ and $QBPD$ are both bi-right, Theorem 2.5 applies to give us that if $PQ < BD$, then $\beta > \theta$, and $\alpha > \theta$. In this case, then $\pi > 2\theta$, so θ is acute. The rest of the proof is similar and we leave it as an exercise. □

Lemma 2.12. Let $ABCD$ be a KS-quadrilateral with summit angles θ and summit CD. Let P be a point on $\overline{CD}$, not on CD. Drop a perpendicular from P to $\overline{AB}$ and let Q be the foot.

(1) If $PQ > BD$, θ is acute.

(2) If $PQ \cong BD$, θ is right.

(3) If $PQ < BD$, θ is obtuse.

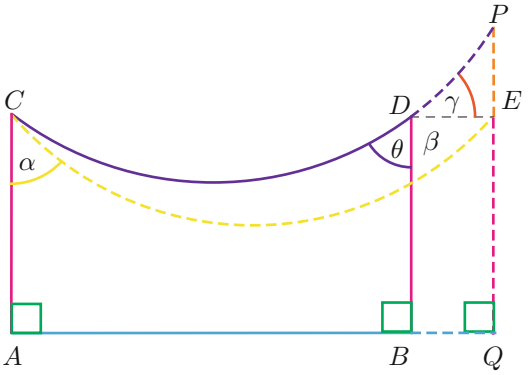


Figure 2.8: The KS-quadrilateral of Lemma 2.12, part (1)

Proof. Without loss of generality, we can take P on $\overrightarrow{CD}$, as in Figs. 2.8 and 2.9. Let E be the point on $\overrightarrow{QP}$ that satisfies $QE \cong BD$. Join C and E. In addition to the original $ABCD$, we have two new KS-quadrilaterals: $BQDE$, with summit angles congruent to $\beta = \angle BDE$, and $AQCE$, with summit angles congruent to $\alpha = \angle ACE$.

When $E \neq P$, let $\gamma = \angle EDP$. Then C, D, E are noncollinear and γ is exterior to ΔCDE.

Note that in case (1), where $PQ > BD$, E is between P and Q, in case (2), where $PQ \cong BD$, $E = P$, and in case (3) where $PQ < BD$, P is between E and Q.

Suppose we are in case (1) so that E is between P and Q as in Fig. 2.8. Here $\alpha \cong \angle QEC$ is part of $\beta \cong \angle QED$ so that $\alpha < \beta$. Notice that $\theta + \beta + \gamma = \pi$. By the Exterior Angle Theorem, we have $\gamma > \theta - \alpha = \angle DCE$, giving us

$$\pi = \theta + \beta + \gamma > \theta + \alpha + (\theta - \alpha) = 2\theta$$

which proves (1), that θ is acute.

Next consider case (2), where $PQ \cong BD$, $E = P$, and by Theorem 2.5, $\alpha \cong \beta \cong \theta$. Since $\theta = \angle BDC$ and $\beta = \angle BDP$, where C, D, P lie on a line, $\beta + \theta = 2\theta = \pi$. It follows that θ is right, which proves (2).

Finally, consider case (3), where $PQ < BD$ so that P is between E and Q as in Fig. 2.9. In this case, $\alpha \cong \angle QEC > \angle QED = \beta$. Adding the adjacent angles at D with legs that lie on $\overline{CD}$, we have $\theta + (\beta - \gamma) = \pi$. Again by the Exterior Angle Theorem, we have $\gamma > \alpha - \theta = \angle DCE$, giving us

$$\pi = \theta + \beta - \gamma < \theta + \alpha - (\alpha - \theta) = 2\theta.$$

This establishes that θ is obtuse, which proves (3). □

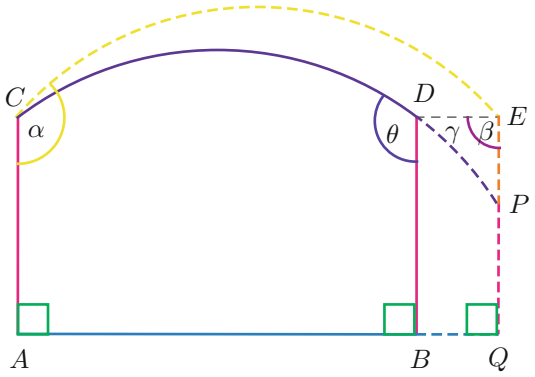


Figure 2.9: The KS-quadrilateral of Lemma 2.12, part (3)

Our proof of the Three Musketeers Theorem requires one more preparatory result.

Lemma 2.13. KS-quadrilaterals with congruent midlines in a neutral plane are of the same type.

Proof. Let $ABCD$ be a KS-quadrilateral with base AB, summit angles θ, and midline EF. Suppose $A'B'C'D'$ is another KS-quadrilateral in the same plane with base $A'B'$, summit angles α, and midline $E'F' \cong EF$. We construct a congruent copy of $A'B'C'D'$ with its base on $\overline{AB}$ and its midline coinciding with EF. In particular, choose A'' on $\overrightarrow{EA}$ and B'' on $\overrightarrow{EB}$ so $A''E \cong A'E'$ and $EB'' \cong E'B'$. (Figure 2.10 illustrates the case where $A'B' > AB$.) Set

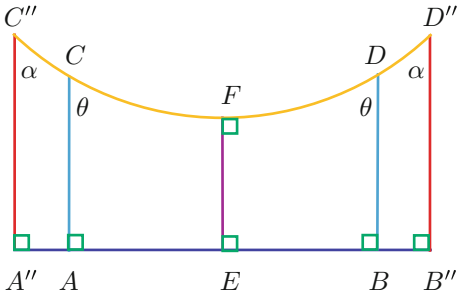


Figure 2.10: Comparing KS-quadrilaterals with the same midline

perpendiculars at A'' and B''. By Lemma 2.4, $A'B'C'D' \cong A''B''C''D''$.

Notice now that exactly one of the following must be true: $B''D'' > BD$, $B''D'' \cong BD$, or $B''D'' > BD$. Whichever it is, θ and α must be angles of the same type. Specifically, Lemmas 2.11 and 2.12 give us that $B''D'' > BD$ if and only if θ and α are both acute, $B''D'' \cong BD$ if and only if θ and α are both right, and $B''D'' < BD$ if and only if θ and α are both obtuse. It follows that $ABCD$ and $A''B''C''D''$ are of the same type. Since $A'B'C'D' \cong A''B''C''D''$, we can conclude that $ABCD$ and $A'B'C'D'$ are of the same type. □

Theorem 2.14 (Three Musketeers)**.** All KS-quadrilaterals in a given neutral plane are of the same type.

Proof. Let $ABCD$ be a KS-quadrilateral with base AB, summit angles θ, and midline EF. Let $A'B'C'D'$ be another KS-quadrilateral in the same plane with base $A'B'$ and midline $E'F'$. In light of Lemma 2.13, we assume $EF \ncong E'F'$.

We start the proof by constructing an HL-quadrilateral using a congruent copy of $E'F'$ sitting on the base of $ABCD$.

Choose G on $\overline{AB}$ so that $EG \cong E'F'$. Erect a perpendicular to $\overline{AB}$ at G and say it intersects $\overline{CD}$ at H. Notice that $EGFH$ is an HL-quadrilateral. Let β be its fourth angle. We apply the doubling construction to $EGFH$, first using EG as the base. This gives us a KS-quadrilateral $PGQH$, shown in Fig. 2.11 for the case in which $E'F' < EB$. By Lemma 2.13, $PGQH$ and $ABCD$ are

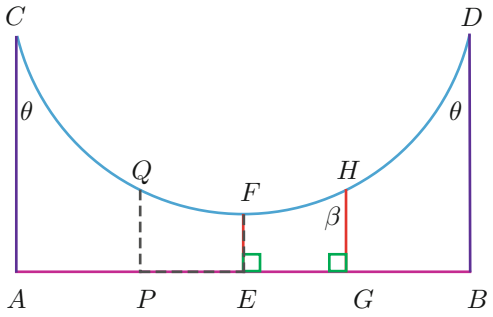


Figure 2.11: Doubling $EGFH$ using EG as the base

KS-quadrilaterals of the same type.

Next, we apply the doubling construction to $EGFH$, this time using EF as the base. This gives a KS-quadrilateral $FIHJ$, as in Fig. 2.12. Now $FIHJ$ has midline $EG \cong E'F'$ and summit angles β. We have already established that β and θ are angles of the same type, thus, $FIHJ$ and $ABCD$ are KS-quadrilaterals of the same type. By Lemma 2.13, $FIHJ$ and $A'B'C'D'$ are also KS-quadrilaterals of the same type, establishing finally that $ABCD$ and $A'B'C'D'$ are of the same type. □

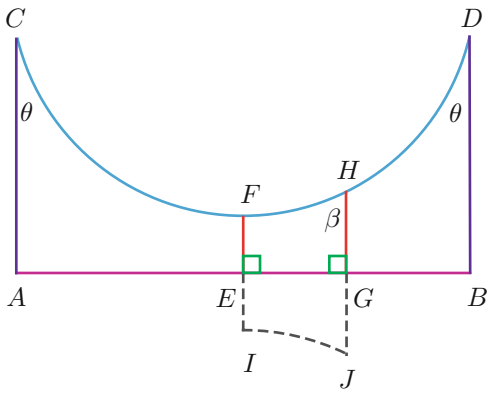


Figure 2.12: Doubling $EGFH$ using EF as the base

The Three Musketeers Theorem suggests the following definition.

Definition 2.15. The **type of a neutral plane** is acute, right, or obtuse according to whether its KS-quadrilaterals are acute, right, or obtuse.

Our work in Chapter 1 shows that in a neutral plane, the parallel postulate implies that the angle sum in a triangle is π. The converse is true as well, but we have not addressed it. The Three Musketeers Theorem is a start toward lifting the veil on the relationship between the angle sum in a triangle and the parallel postulate of Euclid.

Lemma 2.16. For any triangle, there is a KS-quadrilateral with summit angles that add up to the angle sum of the triangle.

Proof. Let ΔABC be given. Let D be the midpoint of AB and E the midpoint of AC. Form $\overline{DE}$ and drop perpendiculars to it from A, B, and C. Call the feet of these perpendiculars, respectively, F, G, H, as in Fig. 2.13. We claim that $HGCB$ is a KS-quadrilateral.

Since $AD \cong BD$, and ΔADF and ΔBDG are both right, the Vertical Angle Theorem gives us AAS for $\Delta ADF \cong \Delta BDG$. From here we can say $AF \cong BG$.

Since $AE \cong EC$, and ΔAEF and ΔCEH are both right, the Vertical Angle Theorem again gives us AAS for $\Delta AEF \cong \Delta CEH$. From here we can say $AF \cong CH$, implying $BG \cong CH$. Since $\angle BGH$ and $\angle CHG$ are both right, it follows that $HGCB$ is a KS-quadrilateral. Its summit angles are $\theta = \angle GBC \cong \angle HCB$.

The angle sum for ΔABC is

$$\angle BAC + \angle ABC + \angle ACB \cong \angle BAF + \angle FAE + \theta - \angle DBG + \theta - \angle HCE.$$

Because $\Delta DAF \cong \Delta DBG$, we have $\angle BAF = \angle DAF \cong \angle DBG$. Because $\Delta AEF \cong \Delta CEH$, we have $\angle FAE \cong \angle HCE$. It follows that the sum of the

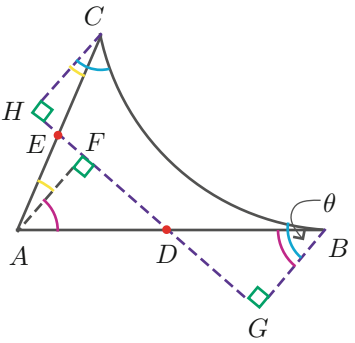


Figure 2.13: Proof of Lemma 2.16: Constructing a KS-quadrilateral from a triangle

angles in ΔABC is 2θ, the sum of the summit angles in $HGCB$, which proves the result. □

The proof of Lemma 2.16 uses AAS, which is part of Prop. I.26. It is easy to be lulled into thinking that AAS and ASA are pretty much the same: Knowing two angles in a triangle means we know the third because the angles sum to π. But we only know that this is true in a Euclidean plane. In a neutral plane, we have to understand that AAS is not a cheap consequence of ASA. Indeed, the fact that Euclid does not treat AAS as a cheap consequence of ASA and the angle sum in a triangle is one more indication that he was trying to avoid using Postulate V for as long as possible, if not altogether. Euclid's tidy housekeeping made it possible for future generations, including those of the Arabic era and those of medieval and Renaissance Europe, to draw a bright line between properties of neutral planes and properties of the Euclidean plane.

A definition will aid in our discussion as we go forward.

Definition 2.17. A triangle is of **hyperbolic type** if the sum of its angles is less than π. A triangle is of **Euclidean type** if the sum of its angles is π. A triangle is of **elliptic type** if the sum of its angles is greater than π.

Theorem 2.18. (1) All triangles in a neutral plane are of the same type.

(2) The existence of a triangle of Euclidean type in a neutral plane is equivalent to the existence of a rectangle in that plane.

Proof. (1) The type of a triangle Δ in a neutral plane determines the type of the KS-quadrilateral we construct from Δ in the proof of Lemma 2.16. By the Three Musketeers Theorem, all KS-quadrilaterals in the plane are of one type. It follows that the triangles in the plane must also be of one type.

(2) Lemma 2.16 implies that if there is a triangle of Euclidean type in a neutral plane, there is a rectangle in the plane. Conversely, the Three Musketeers Theorem implies that if there is a rectangle in the plane, every

KS-quadrilateral in the plane is a rectangle. By Lemma 2.16, this means that every triangle must be Euclidean.

□

Exercises 2.2. 1. Show that a bi-right quadrilateral has a pair of parallel sides.

2. Use Props. I.1–28 to prove Lemma 2.3.

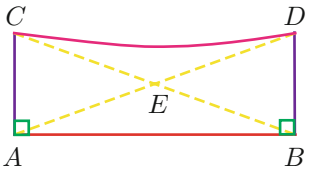


Figure 2.14: Exercise 2: Properties of KS-quadrilaterals

3. Figure 2.15 depicts two KS-quadrilaterals, $ABCD$ and $ABC'D'$ that share the midline EF. By Lemma 2.4, this is an impossible configuration in a

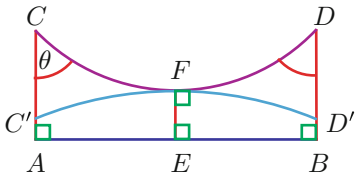


Figure 2.15: Exercise 3: KS-quadrilaterals that share a midline

neutral plane. Argue directly from Postulates I–IV that the configuration cannot exist in a neutral plane.

4. Finish the proof of Lemma 2.11.

5. Show that the converse of each statement in Lemma 2.11 is true.

6. Show that the converse of each statement in Lemma 2.12 is true.

7. Which propositions from Book I of *The Elements* are required to support the proofs of Lemma 2.11 and Lemma 2.12?

8. The proof of Lemma 2.16 details a construction of a KS-quadrilateral from a triangle. Give the reverse construction. In other words, starting with a KS-quadrilateral, describe how to construct a triangle which itself would give rise to the original KS-quadrilateral via the construction given in the proof of Lemma 2.16.

2.3 Archimedes' Axiom and The Angle Sum in a Triangle

The Three Musketeers Theorem allowed Saccheri to classify a neutral plane according to whether its KS-quadrilaterals were of acute, right, or obtuse type. Here we fine-tune the distinctions among these three classes of neutral planes by exploiting Archimedes' Axiom, considered to be among Euclid's tacit assumptions. At stake in this part of the discussion is the nature of a line, and, as we shall see, the nature of an angle.

Axiom of Archimedes. If AB and CD are line segments, then there is a positive integer n so that $nAB > CD$.

We can interpret Archimedes' Axiom as follows: By taking sufficiently many congruent copies of a segment AB, however small, and laying them end-to-end, we can construct a segment that is longer than any fixed segment CD, however large. In a sense, Archimedes' Axiom distinguishes the spatial character of a line segment from the spatial character of a point.

Archimedes' Axiom is often a starting point for courses in real analysis, the study of $\mathbb{R}$, but in itself, it does not guarantee that every line is a copy of $\mathbb{R}$.

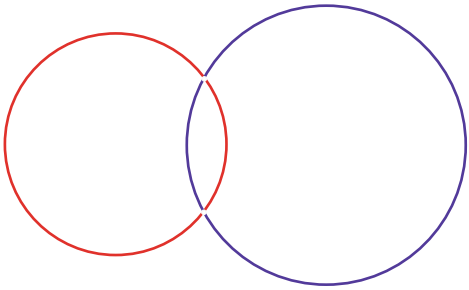


Figure 2.16: Archimedes' Axiom is not sufficient to guarantee that circles will intersect when they enclose overlapping regions.

For example, $\mathbb{Q}$, the set of rational numbers, satisfies Archimedes' Axiom. But $\mathbb{Q}$ has gaps in the sense that a sequence of rational numbers in which the terms get successively closer, for instance,

$$\{3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \ldots\}$$

does not necessarily converge in $\mathbb{Q}$. Those gaps can give rise to places where, for instance, two circles with centers at a distance from one another less than the sum of their two radii might sneak past one another without actually sharing points. According to our definition of a neutral plane, lines are without gaps because the gaplessness of lines is among Euclid's tacit assumptions. In this

section, though, we focus on Archimedes' Axiom and see what we get when we squeeze it for answers.

Saccheri recognized that in a neutral plane, it is actually impossible to have KS-quadrilaterals of obtuse type. This means that the triangles in a neutral plane are either of hyperbolic or Euclidean type. We prove this using Archimedes' Axiom, which for Saccheri, as for Euclid, would have been taken for granted. Related to this is the equivalence of Euclid's Postulate V to the existence of a triangle of Euclidean type in a given neutral plane.

We start with Saccheri's Theorem, which first requires a lemma. This is from [12].

Lemma 2.19. Archimedes' Axiom applies to angles as well as to segments in a neutral plane. In other words, if α and β are angles, there is a positive integer, n such that $n\alpha > \beta$.

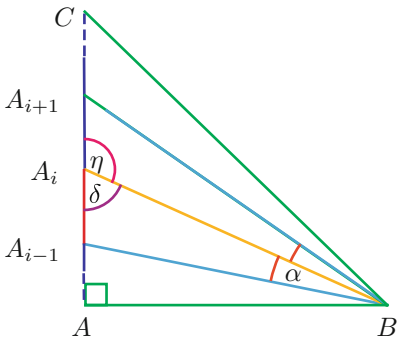


Figure 2.17: Proof of Lemma 2.19: Archimedes' Axiom for angles

Proof. Suppose $\alpha < \beta$ are given angles. Say $\beta = \angle ABC$, where A and C satisfy $AB \cong BC$. Bisect β as in the proof of Prop. I.9, by forming BD, where D is the midpoint of AC. By SAS, $\Delta ADB \cong \Delta CDB$ so $\angle CDB$ is right and $\angle CBD \cong (1/2)\beta$ is in a right triangle, ΔCDB. If we prove the lemma for $(1/2)\beta$, then there is n so that $(1/2)\beta < n\alpha$ but that implies $\beta < 2n\alpha$. In other words, it suffices to assume that β is an angle in a right triangle.

Suppose then that $\beta = \angle ABC$ where $\angle BAC$ is right. Form successive adjacent congruent copies of α at vertex B by taking $A = A_0$ and forming the first copy of α with legs BA_0, BA_1, the second with legs BA_1, BA_2, etc. After choosing $A_0 = A$, we can take successive A_is on $\overrightarrow{AC}$. This gives us a sequence of segments A_0A_1, A_1A_2, etc. as in Fig. 2.17. We claim that these segments increase in length with i, in other words $A_iA_{i+1} > A_{i-1}A_i$ for $i = 0, 1, 2, \ldots$.

Fix i and consider the three points A_{i-1}, A_i, A_{i+1}. Let $\delta = \angle A_{i-1}A_iB$ and $\eta = \angle A_{i+1}A_iB$, as in Fig. 2.17. Since η is exterior to a right triangle, ΔBAA_i,

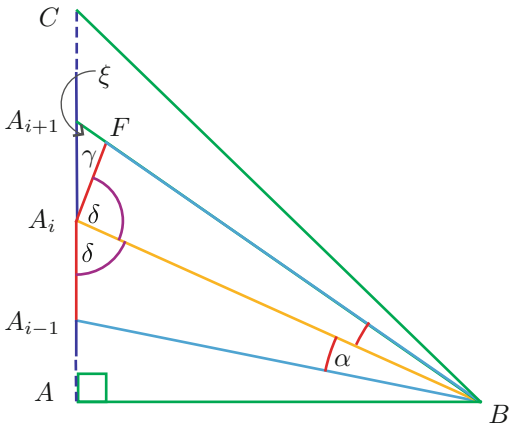


Figure 2.18: Proof of Lemma 2.19: δ is acute

it must be obtuse, so its supplement δ must be acute. Form a second copy of δ with its vertex at A_i, one leg BA_i, and the second leg on the A_{i+1} side of $\overrightarrow{BA_i}$, as in Fig. 2.18. Let F be the point on BA_{i+1} where the constructed leg intersects so that $\angle BA_iF \cong \delta$. Notice that by ASA, $\Delta A_{i-1}BA_i \cong \Delta A_iBF$.

Since $\angle A_iFA_{i+1} = \gamma$ is exterior to ΔA_iBF, $\gamma > \delta$. Further, δ is exterior to $\Delta A_iA_{i+1}B$ so $\delta > \xi = \angle A_iA_{i+1}B$. It follows that $\gamma > \xi$, so by Prop. I.19, $A_iA_{i+1} > A_iF \cong A_{i-1}A_i$, as claimed.

Now we invoke Archimedes' Axiom, taking n to satisfy $nA_0A_1 > AC$. Since $A_iA_{i+1} > A_{i-1}A_i$, we have

$$AA_n = A_0A_1 + A_1A_2 + \cdots + A_{n-1}A_n > nA_0A_1 > AC.$$

It follows that $\beta = \angle ABC$ is part of $n\alpha \cong \angle ABA_n$, so $\beta < n\alpha$, which completes the proof. □

Theorem 2.20 (Saccheri)**.** The sum of the angles in a triangle in a neutral plane is less than or equal to π.

Proof. We prove the theorem by contradiction. Suppose ΔABC has angle sum equal to $\pi + \epsilon$ for some positive ϵ.

Referring to Fig. 2.19, let the angle at A be no larger than the angles at B or C. Take D to be the midpoint of BC and proceed as in Euclid's proof of the Exterior Angle Theorem: Extend $\overrightarrow{AD}$ to the point E where $AD \cong DE$. Joining E to B, we get triangles $\Delta ADC \cong \Delta EDB$ by the Vertical Angle Theorem and SAS. Letting $\alpha = \angle BAD$, $\beta = \angle DEB$, $\gamma = \angle ACB$, $\delta = \angle ABC$, we have that the angle sum in both ΔABC and ΔABE is $\alpha + \beta + \gamma + \delta$.

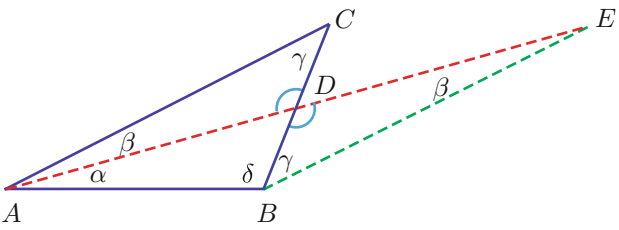


Figure 2.19: Saccheri's Theorem: $\alpha + \beta + \delta + \gamma \leq \pi$

Since α and β cannot both exceed $1/2(\alpha+\beta)$, we see that we have constructed a triangle ΔABE that has an angle that is less than or congruent to one half of $\angle BAC \cong \alpha + \beta$.

If we repeat this process starting with ΔABE, we get another triangle with angle sum $\alpha + \beta + \gamma + \delta$ that has an angle that is less than or congruent to $(1/4)(\alpha+\beta)$. Continuing in this fashion, we eventually produce a triangle with angle sum $\alpha + \beta + \gamma + \delta$ that has an angle that is less than or congruent to $(1/2^n)(\alpha + \beta) < \epsilon$. But then the sum of the remaining two angles in that triangle exceeds π, which violates Euclid's Prop. I.17. The contradiction proves the result. □

Corollary 2.21. An exterior angle of a triangle in a neutral plane equals or exceeds the sum of the remote interiors.

Proof. The sum of an exterior angle and the adjacent interior angle is π which equals or exceeds the sum of the three angles of the triangle. Thus, the exterior equals or exceeds the sum of the remote interiors. □

Postulate V and Triangles of Euclidean Type

Our next goal is to prove that in a neutral plane, Postulate V is equivalent to the condition that the angle sum is π in every triangle. We start with a preparatory result.

Lemma 2.22. Let ℓ be a line and P a point not on ℓ in a neutral plane. If there is more than one parallel to ℓ through P, then there is a triangle in the plane with angle sum strictly less than π.

Proof. Fix a line ℓ and suppose there is more than one parallel to ℓ through a point P. Drop a perpendicular from P to ℓ and let Q be the foot. Form ℓ', the line perpendicular to PQ at P. Since $\overline{PQ}$ is a transversal for ℓ and ℓ' with alternate interior angles congruent, Prop. I.27 guarantees that ℓ and ℓ' are parallel. Let ℓ'' be a second parallel to ℓ through P.

Proposition I.13 guarantees that the angles formed by ℓ'' and PQ at P add up to π. Since these angles are not right, one must be acute. Let R be a point

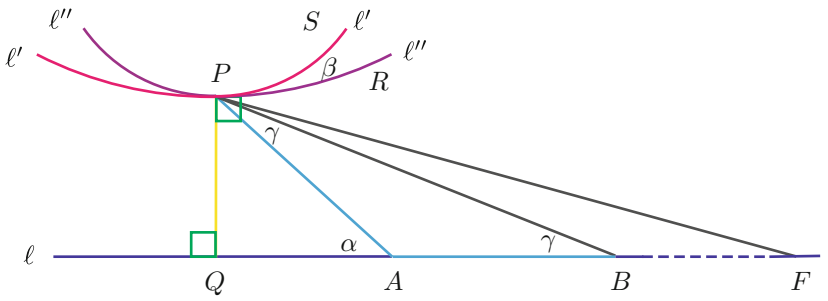


Figure 2.20: Lemma 2.22: When there is more than one parallel to a line through a point

on ℓ'' so that $\angle QPR$ is acute. Take S on ℓ' on the same side of $\overline{PQ}$ as R. Since $\angle QPR$ is part of $\angle QPS$, $\overrightarrow{PR}$ is between $\overrightarrow{PQ}$ and $\overrightarrow{PS}$. Let β be the angle with vertex P and legs $\overrightarrow{PR}$, $\overrightarrow{PS}$.

Fix a point A of ℓ on the R side of $\overline{PQ}$. Let $\alpha = \angle PAQ$. By Lemma 2.19, there is a positive integer n so that $n\beta > \alpha$. Choose a positive integer k so that $2^k > n$. We have $2^k\beta > \alpha$.

Next we form a succession of points on $\overrightarrow{QA}$.

Take B on $\overrightarrow{QA}$ but not on QA, so that $AB \cong AP$. This gives us an isosceles triangle ΔAPB. Let $\gamma = \angle ABP \cong \angle APB$. Notice that α is exterior to ΔAPB so by Corollary 2.21, $\alpha \geq 2\gamma$.

Now take B' on $\overrightarrow{AB}$ not on AB so that $BB' \cong PB$. The argument above applies to give us $\angle PBQ \geq 2\angle PB'Q$ so that

$$\alpha \geq 2\angle PBQ \geq 2^2\angle PB'Q.$$

Continuing to choose points $B'', B''', \ldots$ along ℓ, we arrive eventually at F on ℓ where $\alpha \geq 2^k\angle PFQ$. This gives us $2^k\beta > 2^k\angle PFQ$ thus $\beta > \angle PFQ$.

Since $\angle QPF$ is part of $\angle QPR$, we have $\angle QPF < \angle QPR$. Note that $\angle QPR = \pi/2 - \beta$. The sum of the angles in ΔPQF is thus less than

$$(\pi/2 - \beta) + \beta + \pi/2 = \pi,$$

which proves the lemma. □

Theorem 2.23. In a neutral plane, Euclid's Postulate V is equivalent to the condition that every triangle has angle sum equal to π.

Proof. Proposition I.32 establishes that if we assume Postulate V, then the angle sum in every triangle is π. We must prove the converse.

Theorem 2.20 says that every triangle in a neutral plane has angle sum not exceeding π. With this in mind, we see that the contrapositive of Lemma 2.22

says that if all the triangles in a neutral plane have angle sum equal to π, then a point not on a line lies on a unique parallel to the line. Such a plane satisfies the Axiom of Playfair, thus Postulate V, which proves the theorem. □

Since the Axiom of Playfair is equivalent to Euclid's Postulate V in a neutral plane, Theorem 2.23 establishes that if a neutral plane has a triangle of hyperbolic type, uniqueness of parallels must fail. But does that mean it must fail for every point-line pair, by which we mean every line and every point not on that line? The answer is "yes" but we have not quite secured a proof of that. Our next lemma gives us the connection we need.

Lemma 2.24. Let ℓ be a line and P a point not on ℓ in a neutral plane. If P lies on a unique line parallel to ℓ, then there is a triangle in the plane with angle sum π.

Proof. Suppose ℓ is a line, P a point not on ℓ, and suppose there is a unique parallel to ℓ through P. Let Q be the foot of the perpendicular from P to ℓ and let ℓ' be the unique perpendicular to PQ through P. By Prop. I.27, ℓ and ℓ' are parallel so ℓ' is our unique parallel to ℓ through P.

Take points B on ℓ' and A on ℓ on the same side of $\overline{PQ}$.

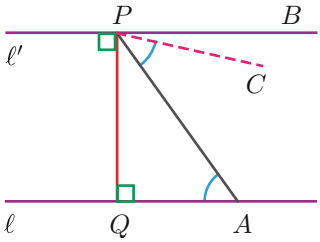


Figure 2.21: Lemma 2.24: When there is a unique parallel to a line through a point

Suppose now that the angle sum of ΔPQA is not π. We prove the lemma by arriving at a contradiction.

By Theorem 2.20, the angle sum of ΔPQA is less than π. Since $\angle PQA$ is right, $\angle QPA$ and $\angle PAQ$ must sum to less than a right angle. Choose C on the A side of $\overline{PQ}$ so that $\angle APC \cong \angle PAQ$. Notice that $\overrightarrow{PC}$ must be between PA and PB because

$$\angle QPA + \angle APB = \angle QPB$$

is right while

$$\angle QPA + \angle APC = \angle QPC \cong \angle QPA + \angle PAQ$$

is less than right. By Prop. I.27, $\angle PAQ \cong \angle APC$ implies that $\overline{PC}$ is parallel to ℓ, which violates uniqueness of ℓ' as the parallel to ℓ through P. The contradiction implies that ΔPQA must indeed have angle sum equal to π, which completes the proof. □

Theorem 2.25. Suppose that in a neutral plane there is a line, and a point not on that line, such that more than one parallel to the line passes through the point. Then for every line in that plane, and every point not on that line, there is more than one parallel to the line through the point.

Proof. Suppose we have a neutral plane in which the Axiom of Playfair fails. By Lemma 2.22, there is a triangle of hyperbolic type in this plane. By Theorem 2.18, every triangle in the plane must then be hyperbolic. Now invoke the contrapositive of Lemma 2.24: If every triangle in a neutral plane is hyperbolic, then for any line ℓ and any point P not on ℓ in the plane, P lies on more than one line parallel to ℓ. □

Angular Defect

We have accomplished three related goals at this stage. One was to establish that every neutral plane is occupied entirely by triangles of hyperbolic type, or entirely by triangles of Euclidean type. The second was to establish that a neutral plane in which the Axiom of Playfair fails is one in which uniqueness of parallels fails for every point-line pair. The third goal, one we have harbored since Chapter 1, was to establish the equivalence of Postulate V and the axiom that the angle sum of every triangle is π. At this point, the idea of a hyperbolic plane is beginning to come into focus: This is a plane in which there is more than one parallel to a given line through any point not on that line, and one in which the angle sum of every triangle is strictly less than π.

A question lingers: Do triangles of hyperbolic type, like triangles of Euclidean type, have a fixed angle sum? If the answer is no, we might refine the question to ask: In a given hyperbolic plane, is the angle sum in a triangle fixed?

Our objective in this very brief treatment is to address that question. We start with a definition.

Definition 2.26. Let Δ be a triangle in a neutral plane. The **angular defect of** Δ, $\mathcal{A}(\Delta)$, is π less the sum of the angles in Δ.

Notice that for any triangle Δ in a hyperbolic plane, $\mathcal{A}(\Delta)$ lies strictly between 0 and π.

One more definition makes it easier to state our main result about angular defect.

Definition 2.27. A line segment from a vertex to any point on the opposite side of a triangle is a **cevian**.

The next theorem reveals the sense in which angular defect is additive.

Theorem 2.28. The angular defect of a triangle is the sum of the defects of any two sub-triangles formed by a cevian.

Proof. Let ΔABC be given. Let D be a point on BC so that ΔABC subdivides into two triangles, ΔABD and ΔACD, as in Fig. 2.22. Let $\angle BAD = \alpha_1$,

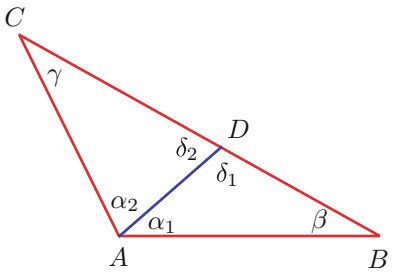


Figure 2.22: Theorem 2.28: Angular defect is additive

$\angle DAC = \alpha_2, \angle ABC = \beta, \angle BDA = \delta_1, \angle ADC = \delta_2$, and $\angle ACB = \gamma$. Since $\delta_1 + \delta_2 = \pi$, we have

$$\mathcal{A}(\Delta ABC) = \pi - (\alpha_1 + \alpha_2 + \beta + \gamma) =$$

$$\pi + (\delta_1 + \delta_2) - (\alpha_1 + \delta_1 + \beta) - (\alpha_2 + \delta_2 + \gamma) =$$

$$\pi - (\alpha_1 + \delta_1 + \beta) + \pi - (\alpha_2 + \delta_2 + \gamma) =$$

$$\mathcal{A}(\Delta ABD) + \mathcal{A}(\Delta ADC).$$

□

The theorem makes it clear that if we subdivide a triangle using successive cevians, eventually we arrive a triangle with a small defect, where "small" means smaller than a fixed positive number.

Corollary 2.29. Fix a hyperbolic plane. For any value of $\epsilon > 0$, there is a triangle Δ in the plane with $\mathcal{A}(\Delta) < \epsilon$.

Proof. Suppose we are given $\epsilon > 0$. Let Δ be an arbitrary triangle in our hyperbolic plane and suppose $\mathcal{A}(\Delta) = \delta > \epsilon$. According to Archimedes' Axiom, there is a positive integer n such that $n\epsilon > \delta$. Choose a positive integer k so that $2^k > n$.

Choose a cevian for Δ to form sub-triangles Δ_1 and Δ_1'. Assume $\mathcal{A}(\Delta_1) \leq \mathcal{A}(\Delta_1')$ so that $\mathcal{A}(\Delta_1) \leq (1/2)\delta$. Repeat this process k times, if necessary, to arrive at a sub-triangle of our original Δ, Δ_k where $\mathcal{A}(\Delta_k) \leq (1/2^k)\delta < \epsilon$. □

We study properties of hyperbolic planes in more detail in the next chapter.

Exercises 2.3. 1. Use Saccheri's Theorem to show that in a neutral plane, a KS-quadrilateral cannot have obtuse summit angles.

2. Show that in a neutral plane, an HL-quadrilateral cannot have an obtuse angle.

3. Discuss the following statement: In a hyperbolic plane, the sum of the angles in a triangle is fixed and always strictly less than two right angles.

4. Consider a neutral plane in which a pair of parallel lines is equidistant. Argue that the plane is Euclidean.

5. Consider a neutral plane in which a pair of parallel lines has two common perpendiculars. Argue that the plane is Euclidean.

6. By Euclid's definition, a *polygon* is the region bounded by a finite set of line segments that make a closed, non-self-intersecting path in a plane. The line segments are the *sides* of the polygon. The point where two sides



Figure 2.23: A closed, self-intersecting path does not define a polygon

meet is a *vertex* of the polygon. An n-gon is a polygon with n vertices. A polygon is *convex* provided the line segment determined by any pair of points in the region lies entirely in the polygon.

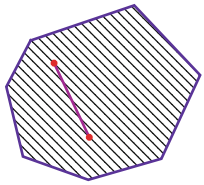


Figure 2.24: Convex n-gon

(a) Show that the smallest n for which there exist n-gons in a neutral plane is 3. (A 3-gon is usually called a triangle.) Your argument should employ Euclid's postulates.

(b) Can you sketch a nonconvex triangle? If not, what is stopping you? Can you articulate the difficulty with enough care to argue that in a neutral plane a triangle is necessarily convex?

(c) Try to formulate a definition of "convex n-gon" that is based on a statement about the vertices of the n-gon.

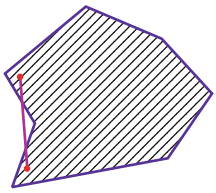


Figure 2.25: Nonconvex n-gon

(d) Show that an n-gon has n sides.

(e) A 4-gon is usually called a quadrilateral. Quadrilaterals may or may not be convex. Are the results about quadrilaterals in Book I of *The Elements* necessarily true for nonconvex quadrilaterals?

(f) Let P be a convex quadrilateral. A line segment joining two nonadjacent vertices is called a *diagonal* of P. Argue that a diagonal subdivides a convex quadrilateral into two triangles. Would the same definition of diagonal for a nonconvex quadrilateral subdivide the quadrilateral into two triangles as well?

(g) Verify that the sum of the interior angles in a convex n-gon is less than or equal to $(n-2)\pi$.

(h) If P is a convex n-gon, its *angular defect* is

$$\mathcal{A}(P) = (n-2)\pi - (\theta_1 + \ldots + \theta_n)$$

where the θ_is are the interior angles in the n-gon. Show that the angular defect of a convex quadrilateral is the sum of the defects of the triangles we obtain by decomposing the quadrilateral using a diagonal. Show that the defect is the same regardless of which diagonal we use.

(i) Let P be an arbitrary convex n-gon where $n > 3$. We approach subdividing P into triangles, that is, *triangulating* P, by identifying adjacent vertices A and B and side AB. Since P has more than three vertices, A shares an edge with a third vertex C that does not share an edge with B. Connect B to C and consider the m-gon that remains if you now disregard the triangular region bounded by ΔABC. What is m? Can you turn this into a process that we can use to triangulate any polygon? The really interesting result is that the angular defect of the polygon is the sum of the defects of the triangles, independent of which triangulation you use. See if you can find a way to approach a proof of that statement.

2.4 A Word About the Geometry on a Sphere

A neutral plane in which every triangle is of hyperbolic type is hyperbolic. A plane that contains a triangle of elliptic type is sometimes called *elliptic*. By Saccheri's Theorem, such a plane is not neutral. In other words, there must be postulates among Euclid's neutral axioms that are violated in an elliptic plane. Though they do not satisfy Postulates I–IV, elliptic planes are of interest in their own right: In some sense, they go back to the earliest scientific endeavors of humans. Navigating on earth is navigating a surface that is nearly spherical, after all, and the surface of a sphere is a model for an elliptic plane.

Spherical geometry is the geometry of points and *great circles* on the surface of a sphere. Here, we are thinking of a sphere as sitting in Euclidean 3-space, that is, $\mathbb{R}^3$. For specificity, think of the sphere as the collection of points $(x, y, z) \in \mathbb{R}^3$ such that $x^2 + y^2 + z^2 = 1$. This is the **unit sphere**, S^2, **in** $\mathbb{R}^3$. Two points on S^2 determine a *great circle*, that is, a circle on the sphere, centered at $(0, 0, 0)$. Great circles on S^2 play the role of lines in spherical geometry and this is where we notice the first departure from Euclid: Two points at the ends of a diameter of the sphere—that is, *antipodal points*—determine many great circles on S^2, hence, many spherical lines. The geometry on the surface of a sphere does not satisfy the tacit assumption of uniqueness in Euclid's first postulate.

Every pair of great circles has a point of intersection so there are no parallels in spherical geometry. As for Archimedes' Axiom, that also has to be jettisoned: Lines are not arbitrarily extensible.

Consider three noncollinear points on the surface of a sphere, that is, three points which do not all lie on the same great circle. The three great circles these points determine form the extended sides of a triangle on S^2 for which the points are vertices. Formed this way, triangles both large and small can be had on the surface of the sphere. If we think of the sphere as a globe representing Earth, and we take the equator, the great circle defined by the Prime Meridian (0° longitude), and the great circle determined by the 90° longitude line, then we have the sides of a triangle for which every angle is a right angle. Clearly, this fails the Exterior Angle Theorem of Euclid, an important engine in the theory of Euclidean geometry.

The geometry of the surface of S^2 is one example of an elliptic geometry. Projective planes are also elliptic. We turn our attention to these in Chapter 8.

Chapter 3

The Hyperbolic Plane

3.1 Introduction

The last chapter showed us what we can say if we assume the first four postulates of Euclid while maintaining a neutral attitude toward the fifth postulate. There are two possibilities: the Euclidean case, where every triangle is Euclidean, and the hyperbolic case, where every triangle has a positive angular defect. In this chapter, we consider properties of the plane in the latter case.

Hyperbolic geometry was first revealed by Bolyai and Lobachevsky. In 1829 and 1832, respectively and independently, Nicolai Lobachevsky (1792–1856) and János Bolyai (1802–1860) published work on hyperbolic geometries that got to the bottom of the controversy surrounding the parallel postulate. With their discoveries, Bolyai and Lobachevsky suggested that the failure of the mathematics community to prove the parallel postulate was not because of a lack of insight, but because it was possible to have a consistent geometry in which the parallel postulate was false. Their work showed that Euclidean geometry—the geometry associated to the Pythagorean Theorem, the geometry that seemed to describe our everyday experience of space—is based on an assumption. The existence of a unique parallel to a given line in a plane is not an essential truth about space. Rather, it is an axiom that can either be assumed true or assumed false. With the work of Lobachevsky and Bolyai, it appeared that Euclid's approach to the parallel postulate was correct all along, simply because other geometries were possible.

This may not seem like much to those of us living in the twenty-first century, but to thinkers of the early-mid-nineteenth century, it was unfathomable. In the face of irrefutable evidence to the contrary, the vast majority of people in the mathematics community clung to the notion that the parallel postulate could be proved. At the time and to the extent that it was made public, the work of Bolyai and Lobachevsky was all but ignored. It would be another thirty-five years before Beltrami determined that models of hyperbolic planes could be constructed inside Euclidean space and, most importantly, that hyperbolic

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M. I. Dillon, *Geometry Through History*,
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geometry was consistent if and only if Euclidean geometry was consistent. That was the step that finally put the controversy to rest.

Our goal in this chapter is to push the underlying hypotheses to reveal some curious features of the hyperbolic plane. In Chapter 6, after developing more Euclidean geometry, we study models of hyperbolic planes.

Axioms

The single difference in the axioms that underlie Euclidean plane geometry and the axioms that underlie hyperbolic plane geometry is, as Coxeter put it in [3], "the vital word *not*." Euclid's neutral axioms hold for both geometries. In Euclidean geometry, we assume the parallel postulate, which is equivalent to the statement that for every line there is a unique parallel through a given point not on the line. In hyperbolic geometry, we assume the negation of the parallel postulate, which is equivalent to the statement that for some line there is more than one parallel through some point not on the line.

Additional properties of the plane articulated by later mathematicians to address the problem of unstated assumptions underlying results from *The Elements* arise in this chapter with a bit more insistence than we have seen to this point. In Chapter 4, where we study Hilbert's *Foundations of Geometry*, we tackle the problem of Euclid's unstated assumptions in earnest. The nature of the arguments and the features of the plane that we scrutinize in this chapter, though, warrant a few words about assumptions regarding betweenness and assumptions regarding the role of numbers in geometry.

We start with an assumption we have used many times. It is simply about how lines separate points in a plane.

Axiom 3.1. Let ℓ be a line in the plane. The points of the plane not on ℓ form nonintersecting convex sets called the **sides of the line**. Moreover, when a second line m intersects ℓ at P, and S, S' are points on m with P between them, $\overrightarrow{PS}$ is on one side of ℓ, and $\overrightarrow{PS'}$ is on the other side of ℓ.

The axiom guarantees that when a line m intersects a line ℓ, m passes from one side of ℓ to the other.

We frequently use Axiom 3.1 when referring to sides of a segment or a ray, by which we mean sides of the line containing the segment or ray.

Next is an axiom that at first blush appears to come from out of the blue.

Axiom of Completeness. Every nonempty subset of $\mathbb{R}$ with an upper bound has a least upper bound in $\mathbb{R}$.

A challenge for modern readers studying Euclid is to engage in the material without reducing geometry to numerical and algebraic expression, something we become accustomed to doing in analytic geometry. As of the late 19th century, though, the prevailing view of Euclidean geometry has been that it is based on numbers, specifically, numbers that we associate to points on a line. If the points on every line enjoy a one-to-one correspondence with the elements in $\mathbb{R}$,

completeness guarantees that a line has no gaps, no missing points, no empty spaces, however small. Though we avoid coordinatizing lines in this chapter, there are instances in which we assign numerical values to angles. Properties of $\mathbb{R}$, in these cases, give us insight into the associated geometric configurations.

Let $\mathbb{R}_+$ designate the set of all positive real numbers, and $\mathbb{Z}_+$ be the set of positive integers.

Theorem 3.2. The Axiom of Completeness implies Archimedes' Axiom.

Proof. Suppose for some $x \in \mathbb{R}_+$, $S = \{nx \mid n \in \mathbb{Z}_+\}$ has an upper bound, y. The Completeness Axiom guarantees that S must then have a least upper bound, b. The implication is that there is $m \in \mathbb{Z}_+$ so that $mx > b - x/2$: otherwise, $b - x/2 < b$ would be an upper bound for S. Fixing such a positive integer m, we have $(m + 1/2)x > b$. Since $x > 0$,

$$(m + 1)x > (m + 1/2)x > b,$$

but that contradicts b as an upper bound for S. It follows that S can have no upper bound. We have shown, then, that given any $x, y \in \mathbb{R}_+$, there is $n \in \mathbb{Z}_+$ such that $nx > y$. This is Archimedes' Axiom. □

We saw in Chapter 2 that if Archimedes' Axiom holds in a neutral plane, there is a version of it that applies to angles. Our next axiom gives us a more direct link between angles and numbers.

Axiom 3.3. Let ℓ be a line and P a point not on ℓ in a neutral plane. Let Q be the foot of the perpendicular from P to ℓ and let $\overrightarrow{PS}$ be perpendicular to PQ. Let R be a point on ℓ, different from Q, on the S side of PQ. For every

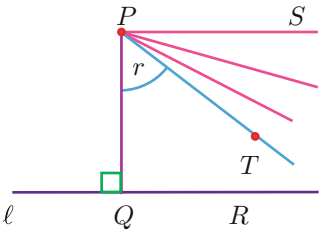


Figure 3.1: Axiom 3.3: Angles and rays in 1-1 correspondence

$r \in (0, \pi/2)$, there is exactly one ray $\overrightarrow{PT}$ lying between PQ and $\overrightarrow{PS}$, so that $\angle QPT = r$.

Axiom 3.3 is related to Prop. I.23. Recall that Prop. I.23 guarantees that we can form a congruent copy of any angle $\angle ABC$ starting with a fixed point D as the vertex, and using a line $\overline{DE}$. Forming the congruent copy of $\angle ABC$

amounts to forming a ray $\overrightarrow{DF}$ for the second leg of the copy, $\angle EDF$. Indeed, there are actually four different rays we could employ to form a copy of $\angle ABC$ using $\overline{DE}$ as one leg. There are two choices of ray on $\overline{DE}$ with vertex D. For each of those choices, there are two sides of $\overline{DE}$ where we can form the second leg of the copy. We proceed assuming that once we have chosen a ray $\overrightarrow{DE}$ and a side of $\overline{DE}$, there is a unique ray $\overrightarrow{DF}$ such that $\angle ABC \cong \angle EDF$. Moreover, if we form a congruent copy of a smaller angle using one leg and the vertex of a larger angle, we assume that we can take the formed ray to lie between the rays forming the legs of the larger angle.

The next assumption post-dates the work of Bolyai and Lobachevsky. As in matters of betweenness, we enjoy a more formal introduction to Pasch's Axiom in Chapter 4. Like Axiom 3.3, and the Axiom of Completeness, Pasch's Axiom is implicit in Euclid's work.

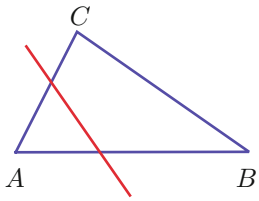


Figure 3.2: Pasch's Axiom

Axiom of Pasch. A line that does not meet a vertex but that does intersect one side of a triangle must intersect a second side of the triangle.

One more result in this direction helps to clarify some of the properties that we rely on in our analysis of the hyperbolic plane. We offer a proof of this in the spirit of Euclid, i.e., using tacit assumptions, in this case about the nature of the region enclosed by a triangle.

Theorem 3.4 (Crossbar)**.** Suppose a line ℓ passes through a vertex of a triangle and one point interior to the region bounded by the triangle. Then ℓ must pass through the side opposite the vertex it contains.

Proof. Let ℓ contain the point A and a point D interior to the region bounded by the triangle ΔABC. Archimedes' Axiom implies that $\overrightarrow{AD}$ is infinitely extensible so it must pass out of the region bounded by ΔABC. Since it passes through AB at A and AC at A, it must exit the region bounded by ΔABC at some point on BC. Since BC, and thus $\overline{BC}$, contain no points interior to ΔABC, $\overline{AD}$ must be different from $\overline{BC}$ meaning that $\overrightarrow{AD}$ must pass through a point between B and C. □

Exercises 3.1. 1. It is suggested above that when all the assumptions of Euclid are properly detailed, it becomes apparent that a Euclidean line is a copy of $\mathbb{R}$. Euclid never references numerical values in Book I of *The Elements*, but numerical ideas must be tacit in Book I. With this in mind, determine how one may construct a line segment that we can identify with $\sqrt{2}$ in Euclid's plane.

2. Show that if there is more than one parallel to a line ℓ through a point P in a hyperbolic plane, then there are infinitely many parallels to ℓ through P. This shows that by inserting the word "not" into Playfair's Axiom—as in, there is not a unique parallel to every line through every point not on the line—we do not get just two or three parallels, we actually get infinitely many parallels to some line.

3. What exactly are the assumptions employed in our proof of the Crossbar Theorem? What do we mean, exactly, by the "interior" of the region bounded by a triangle?

3.2 Polygons, Perpendiculars, and Parallels

Our setting here is a hyperbolic plane. Before proceeding with our investigations, we pause to collect conditions that we now know to be equivalent to Postulate V in a neutral plane.

Observation 3.5. Each of the following conditions is equivalent to Postulate V in a neutral plane.

1. The Axiom of Playfair
2. There is a line in the plane and a point not on the line such that there is a unique parallel to the line through the point.
3. Prop. I.29, which guarantees that alternate interior angles are congruent, that corresponding angles are congruent, and that interior angles on the same side of the transversal are supplementary, when the angles are formed by parallels and a transversal
4. Transitivity of Parallelism (Prop. I.30)
5. The angle sum in one triangle is π.
6. The angle sum in every triangle is π.
7. The angular defect of every triangle is 0.
8. The plane contains a triangle with angular defect 0.
9. Every KS-quadrilateral is a rectangle.
10. There is a rectangle in the plane.

11. Parallel lines are always equidistant.

12. There is a pair of equidistant parallel lines.

13. Every pair of parallel lines has two common perpendiculars.

14. One pair of parallel lines has two common perpendiculars.

Since each condition is equivalent to Postulate V, each condition is equivalent to every other condition.

A hyperbolic plane is a neutral plane in which none of these conditions holds. Thus for instance, every triangle in a hyperbolic plane has positive angular defect. Every KS-quadrilateral in a hyperbolic plane is acute. No two parallel lines in a hyperbolic plane have more than one common perpendicular. The distance between a pair of parallel lines in a hyperbolic plane is not necessarily the same from point to point, etc.

Saccheri's Theorem implies that KS-quadrilaterals in a hyperbolic plane are acute. The next theorem addresses the nature of more general quadrilaterals in a hyperbolic plane.

Theorem 3.6. The sum of the angles in a quadrilateral in a hyperbolic plane is strictly less than 2π.

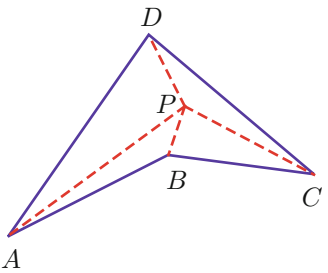


Figure 3.3: Quadrilaterals have angular defect in a hyperbolic plane.

Proof. Let $ABCD$ be a quadrilateral. Pick any point P interior to the region bounded by the quadrilateral and not on any line determined by a side. Join P to each vertex. By choosing a single side of $ABCD$ to go with P we determine a triangle. P thus subdivides $ABCD$ into four triangles. In each of those four triangles, the angles not at P are subangles of the interior angles at the vertices of $ABCD$. Note also that one may use Prop. I.13 to argue that the angles around P add up to 2π. The sum of the angles interior to $ABCD$ is thus the sum of the twelve angles associated to the four triangles, less 2π. If the angular

defect of the ith triangle is $\mathcal{A}_i$, for each of our four triangles, $\mathcal{A}_i > 0$ and, as claimed, the angle sum of $ABCD$ is

$$\pi - \mathcal{A}_1 + \pi - \mathcal{A}_2 + \pi - \mathcal{A}_3 + \pi - \mathcal{A}_4 - 2\pi = 2\pi - (\mathcal{A}_1 + \mathcal{A}_2 + \mathcal{A}_3 + \mathcal{A}_4).$$

□

The next result captures one of the most striking features of a hyperbolic plane. Note that congruence in a hyperbolic plane is no different from congruence in a Euclidean plane: Triangles are congruent if there is a correspondence between their vertices so that corresponding angles and sides are themselves congruent. On the other hand, triangles are *similar* provided there is a correspondence between their vertices so that corresponding sides are in the same proportion. By Euclid's Props. VI.4 and 5, triangles ΔABC and $\Delta A'B'C'$ are similar if and only if we can arrange the labels so that $\angle ABC \cong \angle A'B'C'$, $\angle ACB \cong \angle A'C'B'$, and $\angle BAC \cong \angle B'A'C'$. The next theorem thus says that there are no similar, noncongruent triangles in a hyperbolic plane.

Theorem 3.7. If triangles in a hyperbolic plane have the same angle measures, then the triangles are congruent.

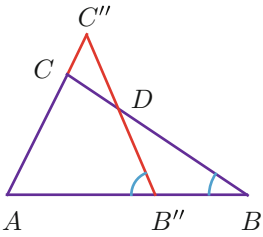


Figure 3.4: Proof of Theorem 3.7: Similar triangles are congruent

Proof. Suppose ΔABC and $\Delta A'B'C'$ satisfy $\angle ABC \cong \angle A'B'C'$, $\angle ACB \cong \angle A'C'B'$, and $\angle BAC \cong \angle B'A'C'$. We assume the triangles are not congruent and arrive at a contradiction. Notice that if we have one pair of congruent corresponding sides then the triangles are congruent by ASA. Suppose then that $AB > A'B'$. Take B'' on AB so that $AB'' \cong A'B'$. Take C'' on $\overrightarrow{AC}$ so that $AC'' \cong A'C'$. By SAS, $\Delta AB''C'' \cong \Delta A'B'C'$. This gives us $\angle AB''C'' \cong \angle A'B'C' \cong \angle ABC$.

If $AC'' > AC$, as in Fig. 3.4, then BC intersects AC'' so by Pasch's Axiom, $\overline{BC}$ must also intersect AB'' or $B''C''$. Since $AB > A'B' \cong AB''$, B, which is on $\overline{AB''}$, is not on AB'' so the line $\overline{BC}$ must intersect the triangle side $B''C''$ at a point D. We then have a triangle $\Delta B''BD$ with exterior angle $\angle AB''C'' \cong \angle B''BD$, a violation of Prop. I.16. We are forced to conclude that $AC'' < AC$, as in Fig. 3.5.

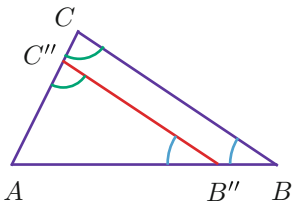


Figure 3.5: Proof of Theorem 3.7: $AC'' < AC$

In this case, we get a quadrilateral, $B''BC''C$, in which $\angle B''BC$ and $\angle BB''C''$ sum to π, and $\angle C''CB$ and $\angle CC''B''$ sum to π. The angles thus sum to 2π, a violation of Theorem 3.6.

This establishes that $AB > A'B'$ leads to a contradiction. Reversing the roles of A, A' and B, B', we see that $AB \cong A'B'$. As noted above, we then have $\Delta ABC \cong \Delta A'B'C'$ by ASA. □

Similar figures have identical proportions at different scales. For instance, a model ship constructed "to scale" is a smaller, but identically proportioned, version of the original. If the two ships were mathematical objects, we would say they were similar. Theorem 3.7 tells us that in a hyperbolic plane, any attempt to scale a triangle must result in distortion. The lengths of the sides are locked in by the angles.

The common ground between hyperbolic and Euclidean planes falls away immediately upon arrival at Prop. I.29 in *The Elements*. Proposition I.29 guarantees that a transversal falling across a pair of parallel lines forms congruent alternate interior angles, congruent corresponding angles, and supplementary interior angles on the same side of the transversal. In other words, what we do not understand in a hyperbolic plane is the behavior of its parallel lines.

Parallels

Recall that by Theorem 2.25, if Playfair's Axiom fails for one point-line pair in a neutral plane, then it fails for every point-line pair in that plane. Given any line, then, and any point not on that line in a hyperbolic plane, we are guaranteed a second parallel to the line through the point.

Lemma 3.8. Let ℓ be a line in a hyperbolic plane, P a point not on ℓ, and Q the foot of the perpendicular from P to ℓ. There is then a ray $\overrightarrow{PS}$ such that $\angle QPS$ is acute, $\overrightarrow{PS}$ is parallel to ℓ, and if U is any point on the S side of PQ with $\angle QPU < \angle QPS$, $\overrightarrow{PU}$ intersects ℓ. In other words, on one side of PQ, there is a minimum $\angle QPS < \pi/2$ such that $\overrightarrow{PS}$ is parallel to ℓ.

Proof. Let ℓ, P, and Q be as in the hypotheses of the lemma. We saw in the proof of Lemma 2.22 that there is a ray $\overrightarrow{PR}$ which does not intersect ℓ, where $\angle QPR$ is acute.

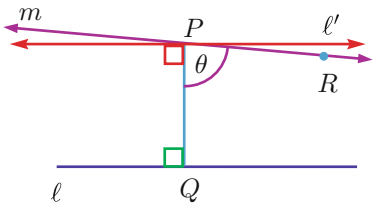


Figure 3.6: Rays parallel to a given line through a point P

Taking T to designate an arbitrary point in the plane on the R side of PQ, let

$$K = \{\varphi = \angle QPT \mid \overrightarrow{PT} \text{ intersects } \ell\}.$$

Notice that the set K is bounded above by θ. By the Completeness Axiom, K has a least upper bound, call it $\theta_P = \angle QPS$. We claim that θ_P is not in K, that is, that $\overrightarrow{PS}$ does not intersect ℓ.

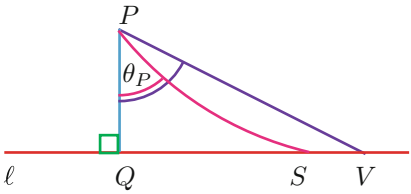


Figure 3.7: Proof of Lemma 3.8: θ_P is the least upper bound of K

Suppose it did so that we many assume that S is on ℓ. Take V on ℓ, so that $\overrightarrow{PS}$ is between $\overrightarrow{PV}$ and PQ. We then have $\angle QPV > \theta_P$. But θ_P is an upper bound for K, so $\overrightarrow{PV}$ has to be parallel to ℓ. There is the contradiction, forcing us to conclude that $\overrightarrow{PS}$ is indeed parallel to ℓ.

We have established that $\theta_P = \angle QPS$ is the minimum angle on the R side of PQ so that $\overrightarrow{PS}$ is parallel to ℓ. This completes the proof. □

Lemma 3.8 suggests the following definition.

Definition 3.9. Let ℓ be a line in a hyperbolic plane, P a point not on ℓ. Let Q be the foot of the perpendicular from P to ℓ. Let S be a point in the plane

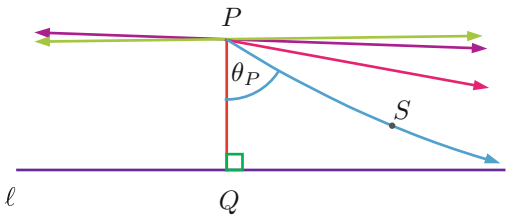


Figure 3.8: Angle of parallelism θ_P and limiting parallel $\overrightarrow{PS}$ at P

so that $\theta_P = \angle QPS$ is least such that $\overrightarrow{PS}$ is parallel to ℓ. Then θ_P is an **angle of parallelism to ℓ at P** and $\overrightarrow{PS}$ is a **limiting parallel ray to ℓ at P.**

Let $\overrightarrow{PT}$ be the unique ray on the S side of PQ that is perpendicular to PQ. $\overrightarrow{PT}$ and all rays emanating from P and strictly between $\overrightarrow{PT}$ and $\overrightarrow{PS}$ are called **hyperparallels** to ℓ at P. The collection of all hyperparallel and limiting parallel rays to ℓ at P are the **parallel rays to ℓ at P.**

It is an easy exercise to show that if $\overrightarrow{PS}$ is a parallel ray to ℓ at P, then $\overline{PS}$ is itself parallel to ℓ.

Given a line and a point not on the line, we determine easily that there is a limiting parallel ray on each side of the perpendicular from the point to the line. It is not difficult to see that the limiting parallel rays on either side of a point P cannot belong to the same line. But what about the angles of parallelism on the two sides of a point? Must they be congruent? Another easy argument establishes that they must. We state this as a lemma, leaving the proof as an exercise.

Lemma 3.10. Let ℓ be a line, and P a point not on ℓ in a hyperbolic plane. Let Q be the foot of the perpendicular from P to ℓ. Given a limiting parallel ray $\overrightarrow{PS}$ to ℓ at P, there is a second limiting parallel ray to ℓ at P, $\overrightarrow{PS'}$, where PQ is between S and S'. The points S', P, S are noncollinear and $\angle QPS' \cong \angle QPS$.

Lemma 3.8 and Lemma 3.10 handled the dirty work so the following needs no further justification.

Theorem 3.11. Given a line and a point not on the line in a hyperbolic plane, there is a unique limiting parallel ray to the line emanating from the point on each side of the perpendicular from the point to the line. The limiting parallel rays on the two sides of the perpendicular form congruent acute angles with the perpendicular.

The perpendicular to a line from a point remains convenient in an ongoing discussion of limiting parallels. The next result suggests that it is not essential. We leave its proof as an exercise.

Corollary 3.12. Let ℓ be a line, P a point not on ℓ, $\overrightarrow{PS}$ parallel to ℓ, and Q an arbitrary point on ℓ, in a hyperbolic plane. Then $\overrightarrow{PS}$ is a limiting parallel to ℓ at P if and only if every ray emanating from P and falling between PQ and $\overrightarrow{PS}$ intersects ℓ.

Given that the notion of a limiting parallel ray has a "sidedness" built into it, the following terminology will be useful as we proceed.

Definition 3.13. Let ℓ be a line with distinct points S, P, S' where P is between S and S'. $\overrightarrow{PS}$ and $\overrightarrow{PS'}$ are **opposite rays**. If T is a point on ℓ between P and S, $\overrightarrow{TS}$ is a **sub-ray** of $\overrightarrow{PS}$ and $\overrightarrow{PS}$ is a **super-ray** of $\overrightarrow{TS}$. We also say that $\overrightarrow{PS}$

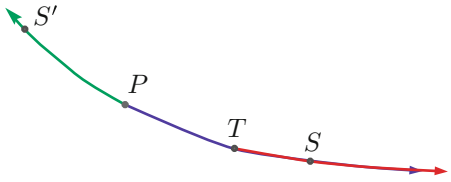


Figure 3.9: $\overrightarrow{TS}$ is a sub-ray of $\overrightarrow{PS}$. $\overrightarrow{PS}$ is a super-ray of $\overrightarrow{TS}$. $\overrightarrow{PS'}$ and $\overrightarrow{PS}$ are opposite rays.

is a **sub-ray of** ℓ. A ray $\overrightarrow{k}$ and any sub-ray of $\overrightarrow{k}$ have the **same direction**. If $\overrightarrow{k}$ and $\overrightarrow{h}$ are sub-rays of a line ℓ but neither $\overrightarrow{k}$ nor $\overrightarrow{h}$ is a sub-ray of the other, then $\overrightarrow{k}$ and $\overrightarrow{h}$ have **opposite directions**.

Some questions arise naturally at this stage. To what extent is a limiting parallel ray to a line tied to the point where we define it in a hyperbolic plane? In other words, if $\overrightarrow{PS}$ is a limiting parallel ray to ℓ at P, and $\overrightarrow{TS}$ is a sub-ray of $\overrightarrow{PS}$, is $\overrightarrow{TS}$ a limiting parallel ray to ℓ at T? There is a question of *symmetry*: If a line m contains a limiting parallel ray to a line ℓ in a hyperbolic plane, does ℓ contain a limiting parallel ray to the line m? Finally, Theorem 3.7 suggests a crucial link between the notions of angle and length in a hyperbolic plane. How should we expect angles of parallelism to change when we change the distance between a point-line pair in a hyperbolic plane?

The remainder of this chapter is dedicated to these and related questions. Our first lemma addresses the last question.

Lemma 3.14. Let ℓ and ℓ' be lines in a hyperbolic plane. Let P and P' be points not on the respective lines. Let Q and Q' be the feet of the perpendiculars from P and P' respectively to ℓ and ℓ'. Let θ_P and $\theta_{P'}$ be angles of parallelism from P and P' respectively to ℓ and ℓ'.

(1) If $PQ \cong P'Q'$, then $\theta_P \cong \theta_{P'}$.

(2) If $PQ < P'Q'$, then $\theta_P \geq \theta_{P'}$.

Proof. Take ℓ, ℓ', P, P', Q, Q', and θ_P, $\theta_{P'}$ all as in the hypotheses of the lemma.

(1) Suppose $PQ \cong P'Q'$ but that $\theta_P \ncong \theta_{P'}$. Assume $\theta_P < \theta_{P'}$. Form a ray at P' at the angle θ_P to $P'Q'$. This ray must intersect ℓ', say at the point S'. Now take S on ℓ so that $QS \cong Q'S'$. By SAS, $\Delta QPS \cong \Delta Q'P'S'$, but that means

$$\theta_P \cong \angle SPQ < \theta_P,$$

which is absurd. It follows that if $\theta_P \ncong \theta_{P'}$, $\theta_P \geq \theta_{P'}$. Reversing the roles of θ_P and $\theta_{P'}$ in the first part of the argument, we can conclude that this also leads to a contradiction so that $\theta_P \cong \theta_{P'}$.

(2) Now suppose $PQ < P'Q'$. Let P'' be on $\overrightarrow{QP}$ so that $QP'' \cong Q'P'$. Form

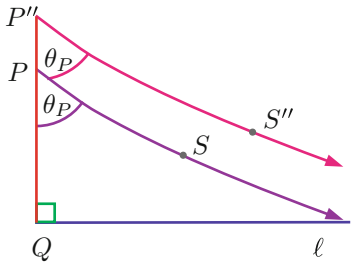


Figure 3.10: Lemma 3.14, Part 2: $PQ < P'Q'$

rays $\overrightarrow{PS}$, $\overrightarrow{P''S''}$ on the same side of PQ so that

$$\angle QPS \cong \angle QP''S'' \cong \theta_P,$$

as in Fig. 3.10. By Prop. I.28, $\overrightarrow{PS}$ and $\overrightarrow{P''S''}$ are parallel to one another. We claim that $\overrightarrow{P''S''}$ cannot intersect ℓ. Suppose by way of contradiction that it does. We can then assume that S'' is on ℓ. Now we have a triangle $\Delta QP''S''$ with $\overline{PS}$ intersecting QP''. As $\overrightarrow{PS}$ is a limiting parallel to ℓ, $\overline{PS}$ must be parallel to ℓ, so $\overline{PS}$ cannot intersect ℓ. By Pasch's Axiom, it must then intersect $P''S''$. But that contradicts $\overrightarrow{PS}$ and $\overrightarrow{P''S''}$ parallel. The contradiction establishes our claim which implies that θ_P is an upper bound for the angles at which rays from P'' are parallel to ℓ. In other words, $\theta_{P''} \cong \theta_{P'} \leq \theta_P$.

□

Next we address the stability of a limiting ray, that is, whether a limiting ray is a limiting ray from any point on the ray in a hyperbolic plane.

Lemma 3.15. Let ℓ be a line and let $\overrightarrow{PS}$ be a limiting parallel ray to ℓ at a point P in a hyperbolic plane. If $\overrightarrow{TS}$ is a sub-ray of $\overrightarrow{PS}$, then $\overrightarrow{TS}$ is a limiting parallel ray to ℓ at T.

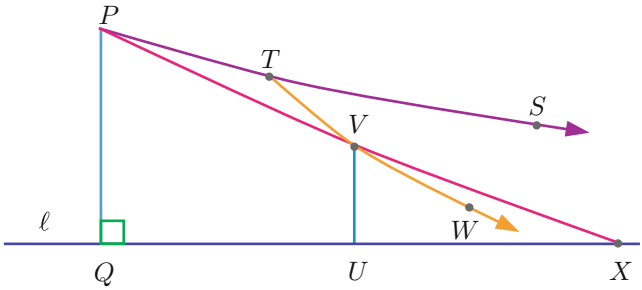


Figure 3.11: Proof that a sub-ray of a limiting parallel is a limiting parallel

Proof. Let ℓ, $\overrightarrow{PS}$, and T be as in the hypotheses of the lemma. Let Q be the foot of the perpendicular from P to ℓ. Since $\overline{TS} = \overline{PS}$, it is clear that $\overrightarrow{TS}$ cannot intersect ℓ. We must show that any ray emanating from T on the ℓ side of $\overrightarrow{PS}$ and on the S side of PQ intersects ℓ. Let $\overrightarrow{TV}$ be such a ray. Take W on $\overrightarrow{TV}$ not on TV.

Notice that $\overrightarrow{PV}$ is between $\overrightarrow{PS}$ and ℓ. It follows that $\overrightarrow{PV}$ must intersect ℓ, say at X. Note also that except for T and V, TV lies on the $\overrightarrow{PS}$ side of $\overrightarrow{PV}$, and $\overrightarrow{VW}$ on the ℓ side of $\overrightarrow{PV}$.

Let U be the foot of the perpendicular from V to ℓ. We now have a triangle, ΔVUX. Notice that $\overline{TV}$ passes through a vertex and points interior to the region bounded by the triangle, in particular, as noted above, points on the ℓ side of VX. By the Crossbar theorem, $\overline{TV}$ must intersect UX. This establishes that $\overrightarrow{TV}$ intersects ℓ, thus proves the lemma. □

The direction of a limiting parallel ray remains important but with the next result, we fully sever the connection between a particular vertex and the nature of a ray as a limiting parallel to a given line.

Theorem 3.16. Let ℓ be a line in a hyperbolic plane. A ray $\overrightarrow{k}$ is a limiting parallel to ℓ if and only if every sub-ray and every super-ray of $\overrightarrow{k}$ is a limiting parallel to ℓ.

Proof. Let ℓ be a line in a hyperbolic plane, and $\overrightarrow{TS}$ a limiting parallel to ℓ. Lemma 3.15 establishes that every sub-ray of $\overrightarrow{TS}$ is a limiting parallel ray to ℓ.

Let $\overrightarrow{PS}$ be a super-ray of $\overrightarrow{TS}$. Let Q be the foot of the perpendicular from P to ℓ. Form $\overrightarrow{PV}$ between $\overrightarrow{PS}$ and PQ. We claim that $\overrightarrow{PV}$ intersects ℓ. That will establish that $\overrightarrow{PS}$ is a limiting parallel to ℓ.

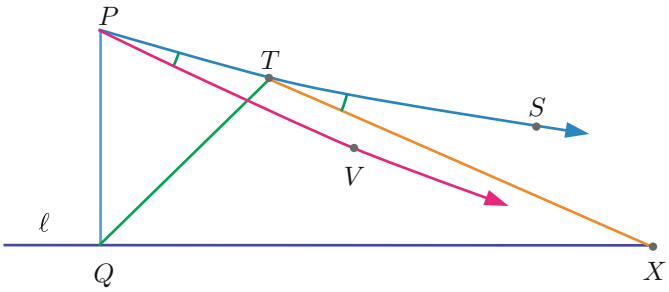


Figure 3.12: Proof that a super-ray of a limiting parallel is a limiting parallel

Consider ΔQPT and notice that since it is exterior, $\angle QTS > \angle QPT$. We also have $\angle QPT > \angle VPT$ so employing Prop. I.23 and Euclid's tacit assumptions, we form a congruent copy of $\angle VPT$ with leg TS, vertex T, and formed leg $\overrightarrow{TX}$ between $\overrightarrow{TS}$ and ℓ. Since $\overrightarrow{TS}$ is a limiting parallel, $\overrightarrow{TX}$ intersects ℓ. We can then assume X is on ℓ.

Now we have triangle ΔQTX. Since $\overrightarrow{PV}$ is on the $\ell = \overline{QX}$ side of $\overrightarrow{TS}$, it intersects QT. By Pasch's Axiom, $\overrightarrow{PV}$ must then intersect either TX or QX. Note now that since $\angle TPV \cong \angle STX$, $\overline{PV}$ and $\overline{TX}$ are parallel. It follows that $\overrightarrow{PV}$ intersects QX, thus ℓ, as desired. We conclude that if $\overrightarrow{k}$ is a limiting parallel to ℓ, then every super-ray of $\overrightarrow{k}$ is a limiting parallel to ℓ.

Now suppose every super-ray of $\overrightarrow{k}$ is a limiting parallel to ℓ. In this case, $\overrightarrow{k}$ is a sub-ray of a limiting parallel so by Lemma 3.15, $\overrightarrow{k}$ is itself a limiting parallel.

If every sub-ray of $\overrightarrow{k}$ is a limiting parallel to ℓ, then since $\overrightarrow{k}$ is a super-ray of a limiting parallel to ℓ, the argument above establishes that $\overrightarrow{k}$ is itself a limiting parallel to ℓ. □

Next we deal with symmetry. A lemma smooths the way.

Lemma 3.17. Let $\overrightarrow{PS}$ be a limiting parallel ray to the line ℓ in a hyperbolic plane. Let Q be the foot of the perpendicular from P to ℓ. There is then a sub-ray of ℓ, $\overrightarrow{WX}$, and on the same side of PQ a sub-ray of $\overrightarrow{PS}$, $\overrightarrow{UT}$, so that $\angle XWU \cong \angle TUW$.

Proof. Let $\overrightarrow{PS}$, ℓ, and Q be as in the hypotheses of the lemma. Bisect $\angle QPS$. Since the angle thus formed at P is less than the angle of parallelism to ℓ at P,

the bisector intersects ℓ, say at the point D. This gives us a triangle, ΔDQP. We are guaranteed by the Crossbar Theorem that the angle bisector of $\angle DQP$ intersects PD. Let R be that point of intersection.

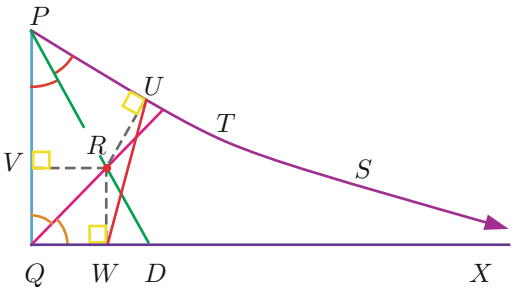


Figure 3.13: Proof that congruent interior angles can be formed by limiting parallels and a transversal

Drop perpendiculars from R to $\overline{PS}$, $\overline{PQ}$, and $\overline{QD}$, and call the respective feet U, V, W. It is easy to see that if R, U, W are collinear, we are done. Suppose then that R, U, W are not collinear.

By AAS, we have triangle congruences $\Delta RWQ \cong \Delta RVQ$ and $\Delta RVP \cong \Delta RUP$. From these we get $RW \cong RV$, and $RV \cong RU$, so that $RU \cong RW$. Now we have an isosceles triangle, ΔURW. By *Pons Asinorum*, $\angle RUW \cong \angle RWU$. Take T on $\overrightarrow{PS}$ and X on $\overrightarrow{QW}$, both on the side of UW opposite PQ. Then rays $\overrightarrow{WX}$ and $\overrightarrow{UT}$ give us $\angle TUW$ and $\angle XWU$, which are supplements of congruent angles, thus themselves congruent. That proves the lemma. □

The lemma gives us a better visual sense of what it means for rays to be limiting parallels in a hyperbolic plane: Limiting parallel rays always have a transversal that forms congruent interior angles on the same side of the transversal.

Theorem 3.18. $\overrightarrow{PS}$ is a limiting parallel to a line ℓ in a hyperbolic plane if and only if ℓ has a sub-ray $\overrightarrow{QT}$ that is a limiting parallel to $\overline{PS}$.

Proof. Suppose $\overrightarrow{PS}$ is a limiting parallel to ℓ in a hyperbolic plane. We must show that ℓ has a sub-ray that is a limiting parallel ray to $\overline{PS}$. In view of Lemma 3.17, we can take Q and T on ℓ, so that T and S are on the same side of PQ, and so that $\angle SPQ \cong \angle TQP$. We claim that $\overrightarrow{QT}$ is a limiting parallel ray to $\overline{PS}$. Let $\overrightarrow{QU}$ fall between $\overrightarrow{QT}$ and PQ. We must show that $\overrightarrow{QU}$ intersects $\overrightarrow{PS}$.

Form a congruent copy of $\angle UQT$ at vertex P, using $\overrightarrow{PS}$ as one leg, and forming the leg $\overrightarrow{PX}$ between $\overrightarrow{PS}$ and PQ. Because $\overrightarrow{PS}$ is a limiting parallel

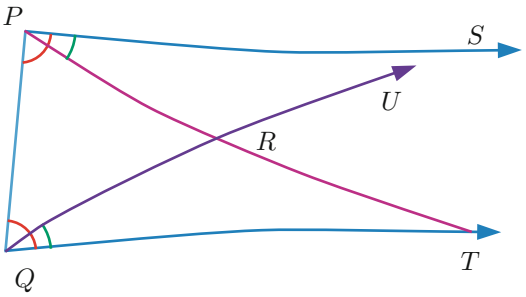


Figure 3.14: The symmetry of limiting parallelism

ray to ℓ, $\overrightarrow{PX}$ must intersect ℓ. Without loss of generality, say the point of intersection is T.

Consider triangle ΔTQP. The Crossbar Theorem guarantees that $\overrightarrow{QU}$ intersects PT. Let that point of intersection be R.

By Prop. I.6, ΔQRP is isosceles so $PR \cong QR$. By the Vertical Angle Theorem, $\angle QRT \cong \angle PRU$. We also have $\angle TQR \cong \angle SPR$. This gives us ASA for ΔQRT and ΔPRY for some point Y on both $\overrightarrow{PS}$ and $\overrightarrow{QU}$. In other words, $\overrightarrow{QU}$ intersects $\overrightarrow{PS}$ at Y. This establishes that $\overrightarrow{QT}$ is a limiting parallel to $\overline{PS}$.

Now suppose ℓ has a sub-ray $\overrightarrow{QT}$ that is a limiting parallel ray to $\overline{PS}$. Repeat the argument we just advanced, substituting $\overrightarrow{QT}$ for $\overrightarrow{PS}$ and $\overline{PS}$ for ℓ. □

The theorem establishes symmetry of limiting parallelism in a hyperbolic plane which supports the following definition.

Definition 3.19. Let ℓ and m be lines in a hyperbolic plane such that ℓ has a sub-ray that is a limiting parallel ray to m. Then ℓ and m are **limiting parallel lines**.

We have seen that even in Euclidean geometry, it is easy to assign a kind of false equivalence between the existence of perpendiculars and the existence of parallels. Next we tease these ideas apart in a hyperbolic plane. We start with a lemma that helps us place the significance of a common perpendicular to two lines in a hyperbolic plane.

Lemma 3.20. If lines ℓ and m in a hyperbolic plane have a common perpendicular, then the line segment PQ perpendicular to both lines is the shortest segment joining a point of ℓ to a point of m. In other words, if RS is any other segment with $R \in \ell$ and $S \in m$, then $PQ \leq RS$.

Proof. Suppose ℓ and m are lines with a common perpendicular, PQ, in a hyperbolic plane. Say P is on m and Q is on ℓ. Pick points A and B on m,

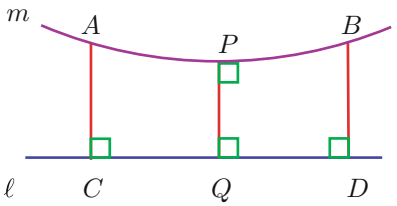


Figure 3.15: When lines have a common perpendicular

on either side of P, so that $AP \cong BP$. Let C be the foot of the perpendicular from A to ℓ and let D be the foot of the perpendicular from B to ℓ. It is an easy exercise to show that $ABCD$ is a KS-quadrilateral, which establishes that $AC > PQ$, and indeed, that PQ is shorter than any other perpendicular dropped to ℓ from a point on m. Another easy exercise is to show that the shortest segment from a point on m to ℓ is the perpendicular from that point to ℓ. These two statements, the proofs of which we leave to the reader, together show that PQ is shorter than any other segment joining a point of m to a point of ℓ. □

Lemma 3.21. If ℓ and m are limiting parallel lines in a hyperbolic plane, then they have no common perpendiculars.

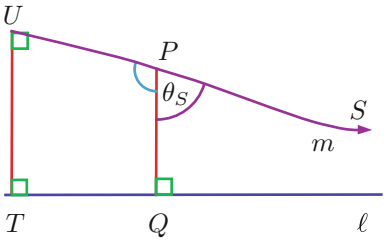


Figure 3.16: Limiting parallels cannot have a common perpendicular

Proof. Suppose ℓ and m are limiting parallel lines in a hyperbolic plane and that $\overrightarrow{PS}$, a sub-ray of m, is a limiting parallel ray to ℓ. Let Q be the foot of the perpendicular from P to ℓ. Let $\theta_S = \angle QPS$ be the angle of parallelism to ℓ at P. Suppose there is U on m, T on ℓ so that UT is perpendicular to both ℓ and m.

If U is on the side of PQ opposite θ_S, then $TQUP$ is an HL-quadrilateral so $\angle UPQ$ is acute. But this is impossible as $\angle QPS = \theta_S$ is acute and $\angle UPQ +$

$\angle QPS = \pi$. If U is on the same side of PQ as S—say U is between P and S—then by Lemma 3.15 $\overrightarrow{US}$ is a limiting parallel ray to ℓ at U, implying that the angle $\angle TUS$ is acute, again contradicting our assumption of perpendicularity. We are forced to conclude that there can be no line perpendicular to both ℓ and $\overline{PS}$, which proves the result. □

The discussion about perpendiculars helps refine our understanding of the distance between parallel lines in a hyperbolic plane. Because there are no rectangles in a hyperbolic plane, two lines cannot remain a fixed distance apart as we move from point to point along either one of the lines. Thus, if the lines are parallel, the distance between them varies in one of two ways: When the lines have a common perpendicular, there is a single, well-defined point at which they have their closest approach. If we imagine moving along one of the lines, we would see the other line get closer then get farther away as we move through the point of closest approach. When parallel lines do not have a common perpendicular, there is no point of closest approach. Though the lines never intersect, they must approach each other asymptotically.

A proof of the next theorem is easy with the results we have at hand. (The converse is also true but the proof is beyond the scope of our work here. See [26], for example.)

Theorem 3.22. (The Hyperparallel Theorem)**.** If two lines in a hyperbolic plane have a common perpendicular, they are hyperparallel.

Proof. If two lines in a hyperbolic plane have a common perpendicular, then Prop. I.27 implies they are parallel. By Lemma 3.21, the two lines cannot serve as limiting parallels for one another so they must be hyperparallel. □

The next result revisits Prop. I.28 in the context of hyperbolic planes.

Proposition 3.23. If a transversal to lines ℓ and m forms congruent alternate interior angles in a hyperbolic plane, then ℓ and m are hyperparallel.

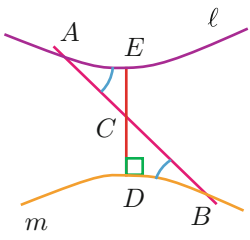


Figure 3.17: Proposition I.28 for hyperbolic planes

Proof. Suppose lines ℓ and m in a hyperbolic plane have a transversal that forms congruent alternate interior angles, say at vertices A on ℓ and B on m. Let C

be the midpoint of the segment AB and let D be the foot of the perpendicular from C to m. Extend CD to intersect ℓ at E. Our congruent angles are then $\angle CAE \cong \angle CBD$. Note that we have congruent vertical angles at C: $\angle ACE \cong \angle BCD$. By ASA, $\Delta ACE \cong \Delta BCD$. This gives us that $\angle DEA$ is right. By the Hyperparallel Theorem, ℓ and m are hyperparallel. □

Exercises 3.2. 1. Show that in the proof of Theorem 3.6, the angles formed around P add up to 2π.

2. Prove that if $\overrightarrow{PS}$ is a parallel ray to ℓ at P, then $\overline{PS}$ is itself parallel to ℓ.

3. Prove Lemma 3.10.

4. Prove Corollary 3.12.

5. Show that the figure $ABCD$ in Fig. 3.15 is a KS-quadrilateral.

6. Let P be a point not on a line ℓ in a hyperbolic plane. Show that the perpendicular from P to ℓ is the shortest segment from P to any point on ℓ.

7. Argue that there cannot be more than one common perpendicular to two lines in a hyperbolic plane.

8. Fix two points O and A in a hyperbolic plane. The collection of points B such that $OB \cong OA$ is the circle with center O and radius OA. Let AB be a diameter of a circle in a hyperbolic plane, in other words, let A, B be points on the circle so that the center of the circle, O, is on AB. Let C be some other point on the circle. Show that $\angle ACB$ is acute.

9. Let ΔABC be a triangle in a hyperbolic plane. Let I be the midpoint of AB and J the midpoint of BC. From each vertex, drop a perpendicular to IJ. Label the respective feet Q_A, Q_B, Q_C, as in Fig. 3.18. (Compare to the configuration we worked with for the proof of Lemma 2.16.) Verify

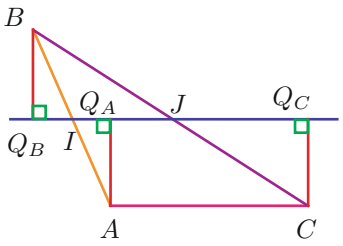


Figure 3.18: A KS-quadrilateral from a triangle, as in Exercise 9

that the area enclosed by AQ_ACQ_C is the same as the area enclosed by ΔABC.

Chapter 4

Hilbert's *Grundlagen*

Hilbert's goal in studying *The Elements* was to sort out the objects Euclid considered, the assumptions about them, and the relationships among them, all while respecting standards of rigor as recognized by the mathematics community of the late nineteenth century. Hilbert sought a description of the foundations of Euclidean geometry that would yield all the familiar theorems: The system had to be *complete*. The axioms could not contradict one another: The system had to be *consistent*. No axiom could follow from any combination of the others: The axioms had to be *independent*.

We have seen that there were tacit assumptions in Euclid's system that were recognized by the earliest commentators, some probably by Euclid himself. Whatever misgivings anyone may have had about *The Elements*, Euclid's work set a standard for rigor respected not only during the classical era, but for generations to come. Standards of rigor change, though, and during the two thousand years between the assembling of *The Elements* and the resolution of the problem of the parallels, the notion of what constituted a proper axiomatic system itself underwent an evolution.

Despite the tacit assumptions, Postulate V presented the biggest problem in Euclid's axiom system. Though Euclid was right about the parallel postulate all along, establishing that he was right was a tricky business. There were philosophical, cultural, and psychological barriers to overcome. The last nail in the coffin for the problem of the parallels was not the work of Lobachevsky and Bolyai on hyperbolic geometry, but the work of Beltrami, Klein, and Poincaré on models for hyperbolic geometry. As it would turn out, those models were tied so closely to Euclidean geometry that it became clear to anyone who bothered to look that the consistency of Euclidean geometry and the consistency of hyperbolic geometry were equivalent. Once everyone could agree on that, the stage was set for overhauling the axiomatic system that underlay Euclidean geometry.

Hilbert's *Grundlagen der Geometrie* (*Foundations of Geometry*) is a set of lecture notes for a course Hilbert gave in 1898–1899. Hilbert's work built on that of earlier mathematicians, going back to antiquity. It was long recognized,

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for example, that it was impossible to explain something like a straight line "by any regular definition which does not introduce words already containing in themselves, by implication, the notion to be defined." [7], p. 168 (cf. Aristotle's remark about defining the prior with the posterior.) Predecessors of Hilbert started with material approximations to terms such as *point* and *line*. Though he was certainly not the first one to think about using undefined terms, Hilbert advanced the necessity of starting with terms that were and would remain, frankly, undefined. It was the axioms that capture the nature of the elements of a geometry, not the definitions. This may seem like a triviality but it was an important idea in the early development of logic and set theory and, in some sense, goes back to Kant, whom Hilbert quotes before the introduction of *Foundations of Geometry*. [13], p. 2.

All human knowledge thus begins with intuitions, proceeds thence to concepts, and ends with ideas.

Kant, *Critique of Pure Reason,* "Elements of Transcendentalism," *Second Part*, II.

Hilbert singled out the following undefined terms referring to objects: *points*, denoted A, B, C, etc., *lines*, denoted a, b, c, etc., *planes*, denoted α, β, γ, etc., and *space.* The points are called the **elements of line geometry**. Points and lines are the **elements of plane geometry**. Points, lines, and planes are the **elements of space geometry.** The undefined terms referring to relations among points, lines, planes, and spaces, are *lies on*, *between*, and *congruent*. Hilbert's axioms are organized into groups according to what relationships they address. There are five groups of axioms altogether: Axioms of Incidence, Axioms of Order, Axioms of Congruence, Axiom of Parallels, and Axioms of Continuity. We reproduce them here, in some cases with slight modifications, from the translation of *Grundlagen* in [13]. While Hilbert's concerns were with both planar and spatial geometry, we focus here on the former, omitting some of the results Hilbert cites in the spatial setting.

Hilbert's convention, which we follow, is that when we refer to points and lines with language such as, "A and B are points" it is to be understood that A and B are distinct.

4.1 Axioms of Incidence

Axiom I.1. Two points A and B determine a line a that contains them.

Axiom I.2. The line determined by two points is unique.

We noted in Chapter 1 that points that lie on a single line are **collinear** and points that do not lie on a single line are **noncollinear.** Hilbert did not use these words in his axioms but did later on in the text. We use them here as we have all along.

Axiom I.3. Every line contains at least two points. There exist three noncollinear points.

Axiom I.3 is a guarantee against triviality, in other words, that points, lines, and planes are all different sorts of objects.

A set of points and lines that satisfies Axioms I.1–3 forms an example of an **incidence plane**. **Incidence geometries** are the geometries of incidence planes.

The rest of the incidence axioms specify how a plane sits in space.

Axiom I.4. Every set of three noncollinear points is contained in a plane. Every plane contains at least one point.

Axiom I.5. The plane determined by three noncollinear points is unique.

Axiom I.6. If two points of a line a lie in a plane α, then all of the points of a lie in α.

Axiom I.7. If two planes have a point in common, then they have at least one more point in common.

Objects that lie in a single plane are **coplanar**. Objects that are not in a single plane are **noncoplanar**.

Axiom I.8. There exist four noncoplanar points.

Modern set theory dates from the late nineteenth century with the work of Georg Cantor (1845–1918), who was senior to Hilbert by seventeen years. Hilbert admired Cantor's work and knew set theory but did not use its language and conventions in the *Grundlagen*. Although we will not deal with set theory in a deep way, its basic ideas and notation are irresistible in this context. While Euclid and the thinkers of classical antiquity may have thought of points and lines as having specific characters—albeit difficult to put into words—we treat points and lines as pure abstractions: points are elements in sets. Lines and planes are sets of points. If A is a point lying on the line a or the plane α, we write $A \in a$ or $A \in \alpha$. If the line a lies in the plane α, then the points of a form a subset of α so we write $a \subset \alpha$. If the point A belongs to the lines a and b, the line a and plane α, or the planes α and β, write $A \in a \cap b$, $A \in a \cap \alpha$, or $A \in \alpha \cap \beta$.

The exercises explore incidence geometries that satisfy Hilbert's axioms, but Hilbert's is just one approach to incidence geometries. Starting with a different set of axioms, we may be led to different, possibly very rich, incidence geometries. In a later chapter, we study projective geometry, another example of an incidence geometry.

A **model** for a plane geometry is a specific set with members that play the roles of points and lines, and relations that play the roles of any undefined relations in the system, so that the axioms of the geometry are satisfied. For example, in the exercises, you will show that $\mathbb{R}^2$ is a model for a plane that satisfies Hilbert's incidence axioms for a plane. Two models are **isomorphic** provided the underlying sets have the same *cardinality*—that is, there is a bijection from one to the other—the models satisfy the same axioms, and there is a

bijection from one set to the other that respects the axioms. Isomorphic models are identical except for the names given their elements and relations.

It is possible to have nonisomorphic geometries that satisfy the same set of axioms. When that cannot happen, when all models for a given geometry are isomorphic, we say the geometry is **categorical**. It turns out that the Euclidean plane is categorical. We leave it as an exercise to show that all incidence planes with three points are isomorphic. If you add another point, things get more interesting.

Exercises 4.1. 1. Use Hilbert's Axioms I.1–8 to prove the following.

(a) There are at least two lines through a given point.

(b) The intersection of two lines in a plane is a single point or empty.

(c) The intersection of two planes is empty or exactly one line and no other points.

(d) Let α be a plane and a a line that does not lie in α. The intersection of a and α is empty or a single point.

(e) Let a be a line and A a point not on a. There is exactly one plane containing a and A.

(f) Let a and b be intersecting lines. There is exactly one plane containing both a and b.

In the following exercises, an **incidence plane** is a collection of points and lines satisfying Axioms I.1–3.

2. What is the minimum number of points in an incidence plane? What is the minimum number of lines?

3. Suppose an incidence plane has four points.

(a) What is the minimum number of lines it must it have?

(b) Is there a maximum number of lines in a plane with four points?

(c) How many nonisomorphic incidence planes with exactly four points can you identify?

4. Show that $\mathbb{R}^2$ is a model for an incidence plane if we take points, lines, and incidence in the usual sense.

5. Consider a geometry in which the points are lines in $\mathbb{R}^2$, and the lines are points in $\mathbb{R}^2$. Does this satisfy Axioms I.1–3? If not, which axioms fail?

6. Consider a geometry in which points are played by lines through the origin in $\mathbb{R}^3$ and lines are played by planes through the origin in $\mathbb{R}^3$. Does this give us an incidence plane?

4.2 Axioms of Order

Line segments and angles are not undefined terms in Hilbert's system but the notion of *betweenness* is and it underlies a great many of the familiar ideas from Euclidean geometry. Indeed, some view Euclid's omission of reference to betweenness as one of the biggest shortcomings of *The Elements*.

Hilbert credits Moritz Pasch (1843–1930) for this set of axioms. [13] We met an axiom named for Pasch in Chapter 3.

When the point B lies between points A and C, we write $A * B * C$. We continue to use the notation $\overline{AB}$ for the line determined by points A and B.

Axiom II.1. If $A * B * C$ then A, B, C are collinear and $C * B * A$.

Axiom II.2. If A and C are two points there exists a point B with $A * C * B$.

The **line segment** $AB = BA$ is the set of points A and B together with all points between A and B. When $A * C * B$, C is **inside** AB and B is **outside** AC. A and B are the **endpoints** of AB.

Read the second order axiom carefully. It does not guarantee that there is a point between any two given points, but does guarantee that $\overline{AB}$ contains a point outside AB. That there must be a point between any two given points actually follows from another axiom below.

Hilbert defines a triangle as a member of the larger class of polygons. Recall the definition we have been using since Chapter 1.

Definition 4.1. A **triangle** ΔABC is a collection of three noncollinear points, A, B, and C, called the **vertices** of the triangle, and the line segments they determine. Those segments, AB, BC, AC are called the **sides** of the triangle.

Axiom II.3. Of any three points on a line, there is no more than one that lies between the other two.

We repeat Pasch's Axiom here for reference. Note that if A, B, C are noncollinear points, they determine a unique plane by Axioms I.4 and 5. Hilbert uses ABC to designate that plane.

Axiom II.4 (Pasch's Axiom). Let A, B, C be noncollinear points and let a be a line in the plane ABC that does not contain any of A, B, or C but that passes through AB. Then a must also pass through AC or BC.

The axioms to this point allow us to prove several things that seem intuitively obvious when we talk about *space*. For us, space typically means $\mathbb{R}^3$, but remember that neither Euclid nor Hilbert was discussing $\mathbb{R}^3$ explicitly. Hilbert in fact remarked famously that his axiom system should be as applicable to chairs, tables, and bottles of beer, as to points, lines, and planes. [27] We are not discussing a particular model here. We are discussing the properties that different models have in common.

We consider some consequences of the Axioms of Incidence and Order. The proofs of these are not obvious, which is typical at the beginning of the development of an axiomatic system. We work through a few of the proofs as they

appear in Hilbert's notes, just to get a sense of the nature of the material. Some details are left as exercises.

Theorem 4.2. In a plane satisfying Axioms I–II, there is a point between any two given points.

Proof. Let α be a plane satisfying Axioms I–II. Fix a line $a \subset \alpha$. Let A, B be points on a, as in Fig. 4.1. We know there is $C \in \alpha$ not on a. Take D so $A*C*D$

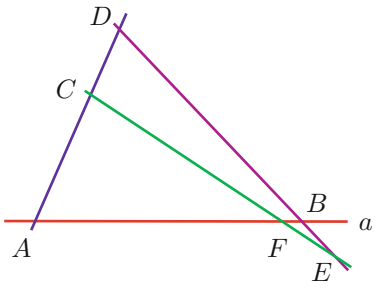


Figure 4.1: Proof that there is a point between any two points

and take E so that $D*B*E$, as in Fig. 4.1. The line $\overline{CE}$ intersects AD, but not at an endpoint so by Pasch's Axiom, it must intersect AB or BD. The latter is impossible since $\overline{CE} \cap \overline{BD} = E$. Then if $\overline{CE} \cap AB = F$, $A * F * B$. □

Theorem 4.3. Given three points on a line in a plane that satisfies Axioms I–II, one point must lie between the other two.

Proof. Let A, B, C lie on a line $a \subset \alpha$, where α is a plane satisfying Axioms I–II. Suppose A is not between B and C and that C is not between A and B. We will use successive applications of Pasch's Axiom to conclude that $A * B * C$.

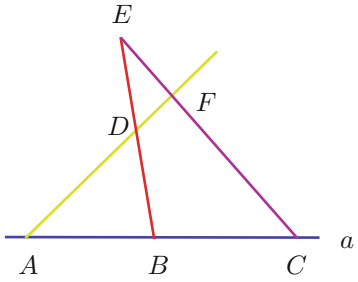


Figure 4.2: First application of Pasch's Axiom in Theorem 4.3

Let D be a point not on a. Take E on $\overline{BD}$ so that $B * D * E$. Since $A \in a$ is outside BC it is not on a side of ΔBCE. Apply Pasch's Axiom to ΔBCE and the line $\overline{AD}$: $\overline{AD}$ enters ΔBCE at $D \in BE$, so $\overline{AD}$ must pass through CE, say at the point F, as in Fig. 4.2. This gives us $C * F * E$.

Note now that C is not on any side of ΔAEB, and repeat the argument, this time for triangle ΔAEB and the line $\overline{CD}$, as in Fig. 4.3. We can conclude

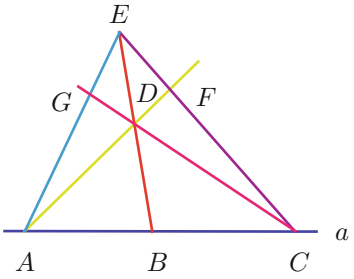


Figure 4.3: Second application of Pasch's Axiom proving Theorem 4.3

that $\overline{CD}$ intersects AE at a point G, giving us $A * G * E$. Since $\overline{CG}$ enters ΔAEF through side AE, it must exit through side AF. But $\overline{CG} \cap \overline{AF} = D$, so $A * D * F$. A final application of Pasch's Axiom to ΔAFC and $\overline{ED}$ gives us B on AC, as desired. □

The next theorem follows by repeated application of the previous theorem.

Theorem 4.4. Suppose α is a plane that satisfies Axioms I–II. Given any finite number of points on a line in α, it is possible to order the points $A, B, C, \ldots, K$ so that $A * B * C * D * \cdots * K$.

The next result follows Theorem 4.2.

Theorem 4.5. Between any two points on a line in a plane that satisfies Axioms I–II, there exist an infinite number of points.

We have seen in our earlier work that betweenness gives us the notions of a point partitioning a line, and a line partitioning the plane.

Definition 4.6. Let α be a plane satisfying Axioms I–II. Let A, B, O, C be four points on a line $a \subset \alpha$ where $A * O * C$ and $B * O * C$. The points A, B are then said to lie **on one and the same side of** O and the points A, C are said to lie on **different sides** of the point O. The totality of points of a that lie on one and same side of O is called a **ray** emanating from O.

We continue to use $\overrightarrow{OA}$ to designate the ray emanating from O through A. We also follow Hilbert and use letters k, h, j to designate rays. If k is a ray, the line it determines is $\overline{k}$.

Next we consider how a line partitions a plane in Hilbert's system. We leave the proof as an exercise.

Theorem 4.7. Let α be a plane that satisfies Axioms I–II and let a be a line in α. The points of α not on a fall into two disjoint sets characterized by the following properties: (1) every point A of one set determines with each point B of the other set a segment AB where $AB \cap a \neq \emptyset$; (2) two points A, B of one and the same set determine AB where $AB \cap a = \emptyset$.

Definition 4.8. The disjoint sets associated to a line $a \subset \alpha$ are the **sides of the line**. Points A and B not on a are on the **opposite sides** of a provided the segment $AB \cap a \neq \emptyset$. Points A and B not on a are on **the same side** of a provided $AB \cap a = \emptyset$.

The Axioms of Order allow us to sort out relative positions of points and lines so that we can define polygons, for instance, and then think of a polygon as dividing a plane into two regions: one inside the polygon, the other outside it.

Exercises 4.2.

1. Write out the proof of Theorem 4.2 supplying references to the axioms that support each statement in the theorem. For example, which axiom guarantees that A and B determine a line? Which guarantees that there is a point not on that line?

2. In the proof of Theorem 4.3, why can't $\overline{AD}$ pass through BC? Why can't $\overline{CG}$ pass through EF?

3. Let α be a plane satisfying Axioms I–II and consider a line $a \subset \alpha$. Fix a point $O \in a$. Give a formal proof the fact that O partitions the remaining points of a into two disjoint sets.

4. Prove Theorem 4.7.

5. State and prove an analog of Theorem 4.7 referring to the manner in which a plane partitions the points of a space satisfying Axioms I–II. Reframe Definition 4.8 to define the sides of a plane in space.

4.3 Axioms of Congruence

When we start thinking about segments, angles, and polygons, the notion of congruence comes up. It is one of Hilbert's undefined terms. We continue to use the symbol $\cong$ but we must be careful to understand that unless stated explicitly, assumptions we have employed to this point when using this symbol do not apply here. In particular, we cannot assume that congruence is an equivalence relation, or even that it is reflexive ($AB \cong AB$). Matters like these are precisely what Hilbert was addressing using the greatest economy in terms of assumptions.

The first axiom of congruence allows the possibility of constructing congruent segments. Recall that Euclid manages this problem in Props. I.2 and I.3.

Axiom III.1. Given a segment AB on a line a and a line a' with a point $A' \in a'$, we can find points $B' \in a'$ on either side of A' so that $AB \cong A'B'$.

The second axiom is a form of the first of Euclid's Common Notions.

Axiom III.2. If $AB \cong CD$ and $EF \cong CD$ then $AB \cong EF$.

Axiom III.2 is sufficient to imply that segment congruence is an equivalence relation. We leave the proof as an exercise.

The next axiom allows addition of segments, as did Euclid's CN 2.

Axiom III.3. On the line a let AB and BC be segments where $AB \cap BC = B$. On the same or another line a' let $A'B'$ and $B'C'$ be two segments where $A'B' \cap B'C' = B'$. If $AB \cong A'B'$ and $BC \cong B'C'$ then $AC \cong A'C'$.

Definition 4.9. Let α be a plane and h, k any two distinct rays emanating from O in α, with $\overline{h} \neq \overline{k}$. The pair of rays h, k is called an **angle** and is denoted $\angle(h,k) = \angle(k,h)$. The point O is the **vertex** of the angle and the rays k and h are its **legs**.

Note that, as with segments, we are not treating angles as having sense or direction. Thus $\angle(h,k)$ and $\angle(k,h)$ denote the same angle.

When the vertex of an angle is A and its legs are $\overrightarrow{AB}$ and $\overrightarrow{AC}$, we continue to use the notation $\angle BAC$ for the angle. Notice though that according to Hilbert's definition, an angle is simply a pair of rays. It does not include its vertex.

Straight angles and angles that exceed straight angles are excluded by this definition,[1] as they are in Euclid.

Definition 4.10. The **interior of an angle** $\angle AOB$ is the intersection of the A side of $\overline{OB}$ and the B side of $\overline{OA}$. The **exterior** of $\angle AOB$ is every other point in the plane except O, $\overrightarrow{OA}$, and $\overrightarrow{OB}$.

Next is a variation on the Crossbar Theorem.

Lemma 4.11 (Crossbar Theorem). Suppose the ray j shares its vertex with and is interior to $\angle(h,k)$. Then every segment HK, where $H \in h$ and $K \in K$ has a nonempty intersection with j.

Proof. Suppose h, j, k, H, K are as in the hypotheses of the lemma. That j is between h and k implies that H is on one side of $\overline{j}$ and K is on the other. According to Theorem 4.7, this means HK intersects $\overline{j}$. We must show that the intersection point is on j.

Notice that h, j are on the same side of $\overline{k}$ so the inside points of HK must be on the j side of $\overline{k}$. That forces $HK \cap \overline{j} \in j$. □

[1] In [13], it says that "obtuse angles" are excluded by this definition. This may be an error in the translation.

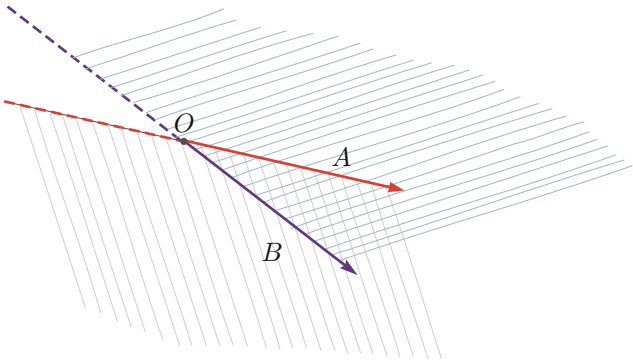


Figure 4.4: The interior of $\angle AOB$

Note that an angle partitions the points of a plane into its vertex, the rays that determine the angle, the interior of the angle, and the exterior of the angle.

Hilbert's treatment of angle congruence does not parallel his treatment of segment congruence. He starts this thread with an axiom that includes the assumption that angle congruence is reflexive.

Axiom III.4. Every angle in a given plane can be constructed on a given side of a given ray in a uniquely determined way. Moreover, every angle is congruent to itself.

The next axiom establishes a connection between segment congruence and angle congruence.

Axiom III.5. If ΔABC and $\Delta A'B'C'$ are triangles with $AB \cong A'B'$, $\angle BAC \cong \angle B'A'C'$, and $AC \cong A'C'$, then the congruence $\angle ABC \cong \angle A'B'C'$ is also satisfied.

We leave it as an exercise to show that under the hypotheses of Axiom III.5, we also have $\angle BCA \cong \angle B'C'A'$. Axiom III.5 thus implies that if we have SAS for triangles, we get congruence of corresponding angles. In other words, SAS implies AAA. Note that we do not yet have that SAS implies triangle congruence. In fact, we need a definition of triangle congruence at this point.

If it is not ambiguous, we use the notation $\angle A$ for an angle with vertex A.

Definition 4.12. Triangles ΔABC and $\Delta A'B'C'$ are **congruent** provided we can label their vertices so that $AB \cong A'B'$, $BC \cong B'C'$, $CA \cong C'A'$, and $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$.

Before proving the congruence theorems for triangles, Hilbert addresses *Pons Asinorum*, essentially using Pappus's proof. The idea is this: If we think of ΔABC also as ΔCBA, then if ΔABC is isosceles with $AB \cong CB$, ΔABC and ΔCBA enjoy SAS.

Theorem 4.13. The base angles of an isosceles triangle are congruent.

Proof. Given isosceles ΔABC with $AB \cong CB$, we have $\angle B \cong \angle B$ (Axiom III.4), $BC \cong BA$ so by Axiom III.5, $\angle C \cong \angle A$. □

Hilbert's first congruence theorem for triangles establishes that SAS not only implies AAA, but also SSS, thus triangle congruence. Recall that Euclid used superposition to prove that SAS and SSS imply triangle congruence.

Theorem 4.14 (SAS). If ΔABC and $\Delta A'B'C'$ satisfy $AB \cong A'B'$, $\angle B \cong \angle B'$, $BC \cong B'C'$, then $\Delta ABC \cong \Delta A'B'C'$.

Proof. Axiom III.5 implies $\angle A \cong A'$ and $\angle C \cong \angle C'$. It remains to show $AC \cong A'C'$.

Choose C'' on $\overrightarrow{A'C'}$ so that $A'C'' \cong AC$ as in Fig. 4.5. We leave it as an exercise to show that C'' is uniquely determined.

Notice we have SAS for triangles ΔABC and $\Delta A'B'C''$: $AC \cong A'C''$, $\angle A \cong \angle A'$, and $AB \cong A'B'$. Axiom III.5 implies that $\angle B \cong \angle A'B'C''$. By assumption, we also have $\angle B \cong \angle A'B'C'$. Unless $C' = C''$, uniqueness of angle construction as given in Axiom III.4 is violated. We conclude that $C' = C''$ so $AC \cong A'C'$. □

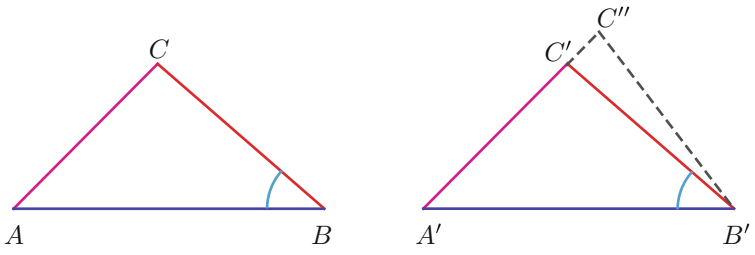


Figure 4.5: Proofs of SAS and ASA

Theorem 4.15 (ASA). If $\angle A \cong \angle A'$, $AB \cong A'B'$ and $\angle B \cong \angle B'$, then $\Delta ABC \cong \Delta A'B'C'$.

Proof. Figure 4.5 also illustrates this proof. Choose C'' exactly as in the last proof and note that this gives us SAS for ΔABC and $\Delta A'B'C''$ as $AC \cong A'C''$, $\angle A \cong \angle A'$ and $AB \cong A'B'$. SAS gives us $\Delta ABC \cong \Delta A'B'C''$ which in turn implies that $\angle B \cong \angle A'B'C''$. By assumption, $\angle B \cong \angle A'B'C'$ so, again, unless $C' = C''$, uniqueness of angle construction is violated. We conclude that $AC \cong A'C'$, thus that $\Delta ABC \cong \Delta A'B'C'$ by SAS. □

Euclid defined a right angle in terms of lines impinging on one another: If a line standing on a straight line makes adjacent angles equal to one another, each of the equal angles is right. Hilbert's approach is a bit different. For reference, we include Hilbert's definition of supplementary angles.

Definition 4.16. Two angles with a common vertex and a common leg are **supplementary** provided their other legs form a line. A **right angle** is an angle congruent to one of its supplements.

Theorem 4.17. Supplements of congruent angles are congruent.

Proof. Suppose $\angle B \cong \angle B'$. As in Fig. 4.6, we can choose points A and A', C

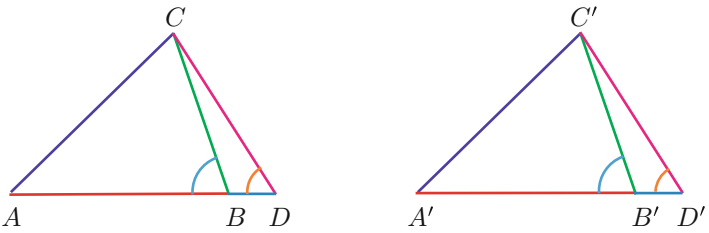


Figure 4.6: Supplements of congruent angles are congruent

and C' on the legs of the two angles so that $AB \cong A'B'$ and $BC \cong B'C'$. By SAS, $\Delta ABC \cong \Delta A'B'C'$.

Now choose points D, D' with $A * B * D$, $A' * B' * D'$, and $BD \cong B'D'$. Again by SAS, $\Delta ADC \cong \Delta A'D'C'$. This gives us $\angle D \cong \angle D'$, in turn implying that $CD \cong C'D'$. This gives us SAS for ΔCBD and $\Delta C'B'D'$.

We conclude that $\angle DBC \cong \angle D'B'C'$, as desired. □

Corollary 4.18. Vertical angles are congruent.

Proof. Consider Fig. 4.7. There are two pairs of vertical angles in the figure:

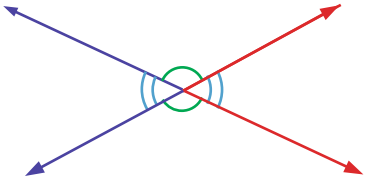


Figure 4.7: Vertical angles are congruent

one pair has double arcs, the other has single arcs. Pick one angle with double arcs. Note it is supplementary to each of the two angles with a single arc. By the previous theorem, the angles with single arcs are thus congruent. If you pick an angle with a single arc, the same logic leads to the conclusion that the double arc angles are congruent as well. □

The next lemma allows us to add and subtract congruent angles. The reader should supply a picture to help understand the hypotheses as well as the proof of the lemma. Note that a definition of the interior of an angle is an important element in the proof.

Lemma 4.19. Let h, k, j be rays emanating from a point O. Let h', k', j' be rays emanating from a point O'. Suppose that h, k are on the same side (respectively, different sides) of j and that h', k' are on the same side (respectively, different sides) of j'. If $\angle(h,k) \cong \angle(h',k')$ and $\angle(k,j) \cong \angle(k',j')$, then $\angle(h,j) \cong \angle(h',j')$.

Proof. We complete the proof for the case where h, k are on the same side of j and h', k' are on the same side of j', leaving the other case to the reader.

Changing labels if necessary, we can suppose h is interior to $\angle(k,j)$. Pick $K \in k$, $J \in j$ and choose $K' \in k'$, $J' \in j'$ so that $OK \cong O'K'$, $OJ \cong O'J'$. We then have SAS for $\Delta KOJ \cong \Delta K'O'J'$.

Since h is interior to $\angle(k,j)$, $KJ \cap h = H$, a unique point. Choose $H' \in h'$ to satisfy $OH \cong O'H'$. Now we have SAS for $\Delta KOH \cong \Delta K'O'H'$.

We claim that $H' \in \overline{K'J'}$.

Since $\Delta KOH \cong \Delta K'O'H'$, $\angle OKH = \angle OKJ \cong \angle O'K'H'$. Since $\Delta KOJ \cong \Delta K'O'J'$, $\angle OKJ = \angle OKH \cong \angle O'K'J'$. Since H' and K' are on the same side of j', Axiom III.4 implies $\angle O'K'J' = \angle O'K'H'$ implying that $\overline{K'J'} = \overline{K'H'}$, i.e., $H' \in \overline{K'J'}$ as claimed.

Since the triangle congruences give us $KJ \cong K'J'$, and $KH \cong K'H'$, Axiom III.3 allows us to conclude that $HJ \cong H'J'$. Using supplements of $\angle OHK \cong \angle O'H'K'$, we have $\angle OHJ \cong \angle O'H'J'$. Since $OH \cong O'H'$ by construction, we have ASA for $\Delta OJH \cong \Delta O'J'H'$, implying that $\angle(h,j) \cong \angle(h',j')$, as desired. □

We now have a procedure that allows us to construct a right angle on a given line.

Theorem 4.20. There is a right angle at any point on any line.

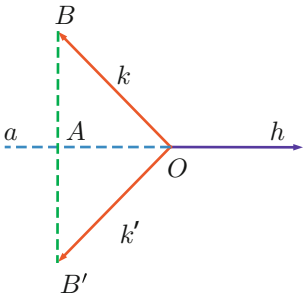


Figure 4.8: Construct a right angle

Proof. Given a point O on a line a, let h be one ray on a emanating from O. Choose a point B not on a and let k be the ray emanating from O passing through B. Construct an angle congruent to $\angle(h,k)$ with vertex O and leg on a, by taking a ray k' emanating from O on the side of a opposite B. Choose B' on k' so that $OB \cong OB'$. Let A be the point where a intersects BB', as in Fig. 4.8.

Consider two cases.

Case 1: Suppose $A = O$. Here $\angle(h,k)$ and $\angle(h,k')$ are supplements and congruent so are right angles.

Case 2: If $A \neq O$ then we have SAS for ΔAOB and $\Delta AOB'$ since $AO \cong AO$, $\angle AOB \cong \angle AOB'$ by construction, and $OB \cong OB'$, also by construction. The implication is that $\angle OAB \cong \angle OAB'$. Notice that these are supplements, thus must be right angles. □

The next result helps us approach SSS.

Lemma 4.21. Let a be a line through points A and B. Let C_1 and C_2 be on opposite sides of a and suppose $AC_1 \cong AC_2$ and $BC_1 \cong BC_2$. Then $\angle ABC_1 \cong \angle ABC_2$.

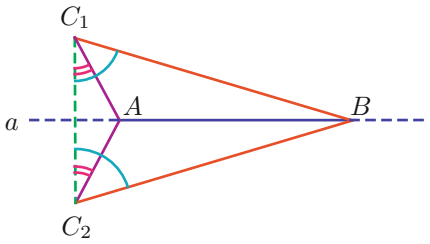


Figure 4.9: When C_1C_2 does not intersect AB

Proof. [2]Figure 4.9 shows one arrangement of points that satisfies the hypotheses of the lemma. Using *Pons Asinorum*, we get congruent angles as marked. We invoke Lemma 4.19 to subtract congruent angles so that $\angle BC_1A \cong \angle BC_2A$. Now SAS applies to give us $\Delta BC_1A \cong \Delta BC_2A$. As corresponding angles, $\angle ABC_1 \cong \angle ABC_2$.

The reader should complete the proof by addressing cases in which C_1C_2 intersects AB, at an endpoint or otherwise. □

We have not yet shown that angle congruence is symmetric or that it is transitive. In the proof of the third triangle congruence theorem, we actually work around transitivity.

[2]I am indebted to the late Wilson Stothers for pointing out errors in a previous version of this proof.

Theorem 4.22 (SSS). If corresponding sides of two triangles are congruent, then corresponding angles are congruent as well so that the triangles themselves are congruent.

Proof. Suppose ΔABC and $\Delta A'B'C'$ are triangles with corresponding sides congruent, as in Fig. 4.10. Construct two angles congruent to $\angle BAC$ with

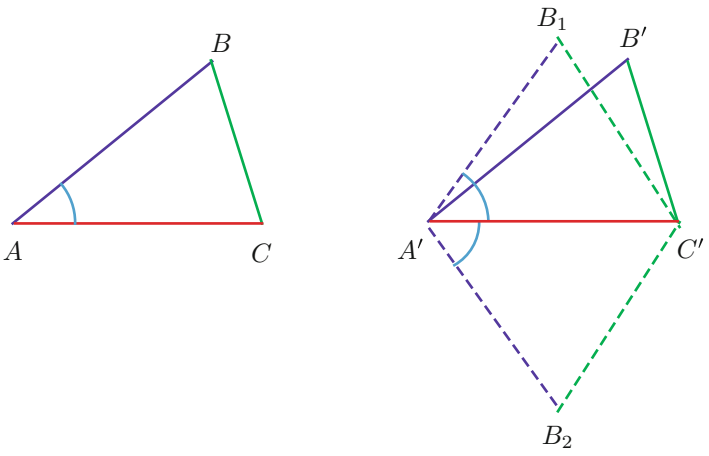


Figure 4.10: SSS for triangle congruence

vertex A' and leg $\overrightarrow{A'C'}$, one angle on each side of $\overrightarrow{A'C'}$. B' is on same side of $A'C'$ as one of the rays making these angles. On that ray, take the point B_1 so that $AB \cong A'B_1$. Take B_2 on the ray that forms the congruent copy of $\angle BAC$ on the other side of $\overrightarrow{A'C'}$ so that $AB \cong A'B_2$. We have $AC \cong A'C'$ by assumption, $\angle BAC \cong \angle B_1A'C'$ and $\angle BAC \cong \angle B_2A'C'$ by construction, $AB \cong A'B_1$ and $AB \cong A'B_2$, also by construction. This gives us SAS for $\Delta ABC \cong \Delta A'B_1C'$ and $\Delta ABC \cong \Delta A'B_2C'$.

It follows from there that $BC \cong B_1C'$ and $BC \cong B_2C'$ implying $B_1C' \cong B_2C'$, by Axiom III.2. By Lemma 4.21, $\angle B_1A'C' \cong \angle B_2A'C'$. Lemma 4.21 also applies to $\Delta A'B'C'$ and $\Delta A'B_2C'$ so that $\angle B'A'C' \cong \angle B_2A'C'$. Uniqueness of angle construction implies that $B' = B_1$. This gives us $\Delta ABC \cong \Delta A'B_1C' = \Delta A'B'C'$, as desired. □

Corollary 4.23. Triangle congruence is an equivalence relation.

Proof. The statement about triangle congruence follows SSS because segment congruence is an equivalence relation. □

It is a bit more work to show that the SSS criterion for triangle congruence also implies that angle congruence is symmetric and transitive. The next result in this direction is analogous to Axiom III.2.

Theorem 4.24. If $\angle(h,k) \cong \angle(h'',k'')$ and $\angle(h',k') \cong \angle(h'',k'')$ then $\angle(h,k) \cong \angle(h',k')$.

Proof. Label the vertices of the angles in question O, O', O'' respectively. On h, choose any point, A. Choose A' and A'' on h', h'' respectively so that $OA \cong O'A' \cong O''A''$. Choose a point B on k and B', B'' on k', k'' respectively so that $OB \cong O'B' \cong O''B''$. This gives us SAS for $\Delta AOB \cong \Delta A''O''B''$ and $\Delta A'O'B' \cong \Delta A''O''B''$, thus SSS. SSS allows us to say $\Delta AOB \cong \Delta A'O'B'$, implying $\angle(h,k) \cong \angle(h',k')$ as desired. □

Corollary 4.25. Angle congruence is an equivalence relation.

Proof. Since $\angle(h',k') \cong \angle(h',k')$, $\angle(h,k) \cong \angle(h',k')$ implies by Theorem 4.24 that $\angle(h',k') \cong \angle(h,k)$. Another application of the theorem then gives us transitivity. □

The Exterior Angle Theorem is a linchpin of Euclidean geometry. To get to it, Hilbert first sorts out the ground rules for comparing angles.

We start this thread with a result related to Lemma 4.19.

Lemma 4.26. Suppose $\angle(h,k) \cong \angle(h',k')$, where $\angle(h,k)$ has vertex O, and $\angle(h',k')$ has vertex O'. Let j be a ray emanating from O interior to $\angle(h,k)$. The unique ray j' emanating from O' on the k' side of h' that satisfies $\angle(h,j) \cong \angle(h',j')$ and $\angle(j,k) \cong \angle(j',k')$ is interior to $\angle(h,k)$.

Proof. Let h,k,j,O and h',k',j',O' be as in the hypotheses of the lemma. To show j' is interior to $\angle(h',k')$, we will prove that if H' is a point on h' and K' is a point on k', then $H'K'$ intersects j'.

Taking such an $H' \in h'$ and $K' \in k'$, choose $H \in h$ and $K \in k$ so that $OH \cong O'H'$ and $OK \cong O'K'$. This gives us SAS for $\Delta OHK \cong \Delta O'H'K'$. Since j is interior to $\angle(h,k)$, the Crossbar Theorem guarantees that it intersects HK. Let $J = HK \cap j$. Pick $J' \in j'$ so that $OJ \cong O'J'$. This gets us SAS for $\Delta OHJ \cong \Delta O'H'J'$. Then $\angle OHJ = \angle OHK \cong \angle O'H'K'$ and $\angle OHK = \angle OHJ \cong \angle O'H'J'$. By uniqueness of angle construction, $\angle O'H'K' = \angle O'H'J'$. We conclude that $J' \in H'K'$, which proves the result. □

Theorem 4.27. Let $\angle(h,k)$ and $\angle(h',j')$ be given, $\angle(h,k)$ with vertex O, $\angle(h',j')$ with vertex O'. Suppose j is the ray emanating from O on the k side of h such that $\angle(h,j) \cong \angle(h',j')$. Suppose k' emanates from O' on the j' side of h' such that $\angle(h,k) \cong \angle(h',k')$. Then j is interior to $\angle(h,k)$ if and only if k' is exterior to $\angle(h',j')$.

Proof. Suppose everything is as hypothesized, with j interior to $\angle(h,k)$, but with k' interior to $\angle(h',j')$. By Lemma 4.26, there is a unique ray k'' emanating from O, interior to $\angle(h,j)$ with $\angle(h,k'') \cong \angle(h',k')$. Since $\angle(h',k') \cong \angle(h,k)$, and both k and k'' are on the j side of h, this violates uniqueness of angle construction. We conclude that j interior to $\angle(h,k)$ implies k' is exterior to $\angle(h',j')$. Changing labels, we get the result in the other direction as well. □

Let $\angle(h,k)$ and $\angle(h',j')$ be as in the theorem. We write $\angle(h,k) < \angle(h',j')$ provided the construction of an angle congruent to $\angle(h,k)$ using h' and a ray emanating from O' on the j' side of h' yields a ray interior to $\angle(h',j')$. It is clear then that for any two angles $\angle(h,k)$ and $\angle(h',k')$, one and only one of the following statements is true: $\angle(h,k) < \angle(h',k')$, $\angle(h,k) \cong \angle(h',k')$, $\angle(h,k) > \angle(h',k')$.

Theorem 4.28. All right angles are congruent.

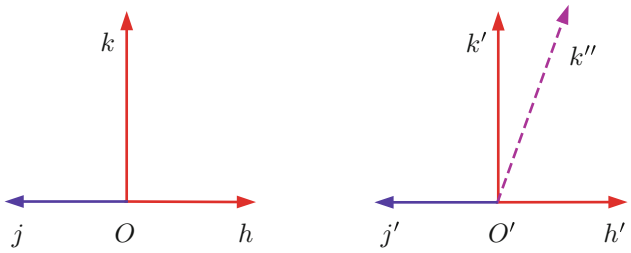


Figure 4.11: Right angles are congruent

Proof. A right angle is defined as one congruent to its supplement. Let $\angle(h,k)$, with vertex O, and $\angle(h',k')$ with vertex O', be right angles with supplements $\angle(k,j)$ and $\angle(k',j')$ respectively. Suppose $\angle(h,k) < \angle(h',k')$. Let k'' emanate from O' on the k' side of h' so that $\angle(h,k) \cong \angle(h',k'')$. Note that k'' is interior to $\angle(h',k')$ and exterior to the supplement $\angle(k',j')$. This gives us $\angle(k,j) \cong \angle(h,k) < \angle(h',k') \cong \angle(k',j')$ as well as $\angle(k,j) \cong \angle(k'',j') > \angle(k',j')$ which is a contradiction, proving the result. □

Definition 4.29. An angle greater than its supplement is **obtuse**. An angle less than its supplement is **acute**.

Definition 4.30. Let ΔABC be a triangle. The **interior angles** of the triangle are $\angle ABC$, $\angle BCA$, and $\angle BAC$. The **exterior angles** of the triangle are the supplements of the interior angles. Given $\angle CAD$ exterior to ΔABC, we say that $\angle ABC$ and $\angle ACB$ are the **remote interior angles**.

We have everything we need now for Hilbert's proof of the Exterior Angle Theorem.

Theorem 4.31. An exterior angle of a triangle is greater than either of the remote interior angles.

Proof. Given ΔABC, choose $D \in \overline{AB}$ with $B * A * D$ and $AD \cong CB$. Suppose the exterior angle $\angle CAD$ is congruent to $\angle ACB$, as in Fig. 4.12. Then $\Delta ACD \cong \Delta ACB$ by SAS, which implies that $\angle ACD \cong \angle CAB$.

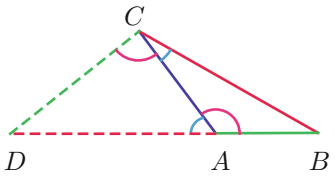


Figure 4.12: An exterior angle cannot equal a remote interior angle

By congruence of supplements, we also have that $\angle CAB$ is congruent to the supplement of $\angle ACB$ which then must actually be $\angle ACD$. But that means D belongs both to $\overline{AB}$ and to $\overline{BC}$ so $B = D$, which is absurd. We conclude that $\angle CAD \ncong \angle ACB$.

Suppose next that $\angle CAD < \angle ACB$. Construct an angle congruent to

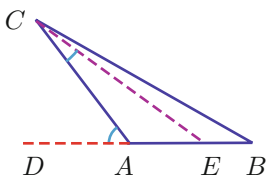


Figure 4.13: An exterior angle cannot be less than a remote interior angle

$\angle CAD$ on the B side of $\overrightarrow{CA}$, using C as the vertex, and $\overrightarrow{CA}$ as one leg, as in Fig. 4.13. Then we have a ray interior to $\angle ACB$ that intersects AB at a point E. This gives us a triangle ΔAEC with exterior angle $\angle CAD$ congruent to interior angle $\angle ACE$, which, as shown above, is impossible.

We conclude that $\angle CAD > \angle ACB$.

To see that $\angle CAD > \angle ABC$, notice that the angle vertical to $\angle CAD$ is congruent to $\angle CAD$ and it has the same relationship to $\angle ABC$ that $\angle CAD$ has to $\angle ACB$. □

The Exterior Angle Theorem has a number of important corollaries. One of these is AAS, which is part of Prop. I.26 in *The Elements*. We leave the proof as an exercise.

Corollary 4.32 (AAS). If ΔABC and $\Delta A'B'C'$ satisfy $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $BC \cong B'C'$, then $\angle C \cong \angle C'$, implying $\Delta ABC \cong \Delta A'B'C'$.

Theorem 4.33. Every segment can be bisected.

Proof. Let AB be given and let C be a point not on $\overline{AB}$. Using B as a vertex,

$\overrightarrow{BA}$ as a leg, and the side of $\overline{BA}$ opposite C, construct an angle congruent to $\angle BAC$. Pick the point D on the constructed leg that satisfies $BD \cong AC$.

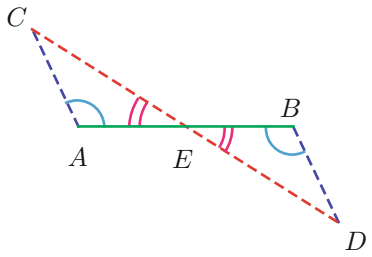


Figure 4.14: Segments can be bisected

Since C and D lie on opposite sides of $\overline{AB}$, CD has a nonempty intersection with $\overline{AB}$. Call that intersection point E. We claim that E cannot coincide with A or B. If it did coincide with A, say, then $\angle BAC$ is exterior to ΔBAD. But $\angle ABD$, which is interior to ΔBAD, was constructed to be congruent to $\angle BAC$.

Note now that $\Delta ACE \cong \Delta BDE$ by AAS: $\angle AEC \cong BED$, since they are vertical angles; $\angle EAC \cong \angle EBD$ by construction; and $AC \cong BD$ by construction. We conclude that $AE \cong EB$, thus, that CD bisects AB. □

Corollary 4.34. Every angle can be bisected.

Proof. Let $\angle(h, k)$ have vertex O. Pick $A \in h$ and $B \in k$ so that $OA \cong OB$. Then ΔAOB is isosceles. Let AB have midpoint C. *Pons Asinorum* gives us SAS for $\Delta ACO \cong \Delta BCO$. It follows that $\angle AOC \cong \angle BOC$, hence, that OC bisects $\angle(h, k)$. □

As we noted above, AAS is part of Prop. I.26 in *The Elements*, a bit more than half-way through Book I. At that point, Euclid starts a thread about parallel lines. Hilbert takes up parallelism next as well.

Exercises 4.3. 1. How is congruence different from equality?

2. Prove that every segment is congruent to itself, i.e., that segment congruence is reflexive. Prove that segment congruence is also symmetric and transitive.

3. Prove uniqueness of segment construction using uniqueness of angle construction and Axiom III.5. Start with the assumption that a segment congruent to AB can be constructed two ways from a point A' on a given line.

4. Axiom III.5 says that if triangles ΔABC and $\Delta A'B'C'$ satisfy $AB \cong A'B'$, $\angle BAC \cong \angle B'A'C'$, and $AC \cong A'C'$, then $\angle ABC \cong \angle A'B'C'$. Show that this implies $\angle BCA \cong \angle B'C'A'$, as well.

5. Argue that the point C'' constructed in the proof of SAS is uniquely determined.

6. Supply a picture for the proof of Lemma 4.19. Prove the other case, and supply a picture for that as well.

7. Finish the proof of Lemma 4.21.

8. Recall that if $\mathcal{R}$ is any region of a plane with the property that the segment joining any two points in $\mathcal{R}$ is itself entirely contained in $\mathcal{R}$, then $\mathcal{R}$ is **convex**.

   (a) Show that the intersection of convex sets is convex.

   (b) By part (a), the interior of an angle is convex. Is the exterior of an angle convex as well?

9. The **interior of a triangle** ΔABC is the intersection of the A side of $\overline{BC}$, the B side of $\overline{AC}$, and the C side of $\overline{AB}$. (Note then that the interior of a triangle is convex.) Show that the interior of ΔABC is the intersection of the interiors of two of its angles.

10. Let $\angle(h, k)$ be an angle with vertex O. Let H be a point of h and K a point of k. Show that all the points inside HK lie in the interior of $\angle(h, k)$.

11. Fix $\angle(h, k)$ with vertex O. Let j be any ray different from h and k, emanating from O. Show that all the points on j lie either entirely interior to or entirely exterior to $\angle(h, k)$.

12. Prove the following corollaries to Theorem 4.31.

    (a) In every triangle, the greater angle lies opposite the greater side. (Hint: Start by picking a point on the greater side that cuts off a segment with the length of a smaller side.)

    (b) A triangle with two equal angles is isosceles.

    (c) Corollary 4.32.

4.4 The Axiom of Parallels

The controversy that swirled around Euclid's fifth postulate swept up the definition of parallel lines so that, by Hilbert's time, there were many different definitions to choose from. In the spirit of firming up the foundation for Euclid's work, Hilbert stayed with Euclid's definition.

Definition 4.35. Two lines are **parallel** provided they lie in the same plane and do not intersect.

We start this thread by constructing parallel lines in a plane. Euclid's construction is Prop. I.31.

Theorem 4.36. Given a line in a plane and a point not on the line, there is a parallel to the line through the point.

Proof. Let α be a plane, a a line in α, C a point in α not on a. Let A and B be points on a. Form $\overline{AC}$ and take D so that $A * C * D$. Construct a copy of $\angle BAC$, using vertex C, and leg $\overrightarrow{CD}$, forming the constructed leg on the B side of $\overline{AC}$. Axiom III.4 guarantees that the constructed leg is unique. Let c be the line it determines.

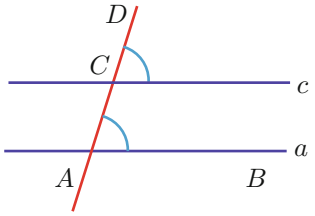


Figure 4.15: Construct a parallel to a given line

We claim that a and c are parallel. Suppose not and that $E = a \cap c$. Notice then that A, C, and E are noncollinear, thus, form the vertices of a triangle. One of the interior angles would be $\angle BAC$ and the constructed angle at C would be exterior. But the constructed angle is congruent to $\angle BAC$, a contradiction of the Exterior Angle Theorem. We conclude that a and c must be parallel, as claimed, which completes the proof. □

Can there be more than one parallel to a through B? The parallel postulate says no. It is important to realize that there is nothing in Hilbert's system to this point that forces uniqueness. All of Hilbert's results so far hold in a neutral plane, thus, in a hyperbolic plane.

Axiom IV (Euclid's Axiom)**.** Let a be any line and A a point not on it. There is at most one line in the plane, determined by a and A, that passes through A and does not intersect a.

The construction in the proof of Theorem 4.36 is then the unique parallel to a through B.

If the line a is parallel to the line b we write $a \| b$. It is easy to see that this relation is symmetric. We leave it as an exercise to show that parallelism is transitive, that is, if $a \| b$ and $b \| c$, then $a \| c$.

Hilbert's next theorem subsumes Props. I.27, 28, and 29.

Theorem 4.37. If two parallel lines are intersected by a transversal, then the corresponding and alternate angles are congruent. Conversely, congruence of the corresponding or alternate angles formed by a transversal intersecting two lines implies that the two lines are parallel.

Proof. Let a and b be parallel lines intersected at points A and B respectively by a third line, c. Take C so that $A * B * C$. Take points $A' \in a$, $B' \in b$ on the same side of c, as in Fig. 4.16. Consider the corresponding angles $\angle A'AC$ and

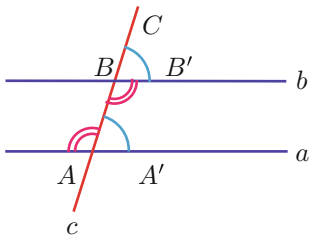


Figure 4.16: Hilbert's version of Props. I.27, 28, and 29

$\angle B'BC$.

Employing the construction in the proof of Theorem 4.36, we can form a line through B parallel to a, so that the angle at B corresponding to $\angle A'AC$ is congruent to $\angle A'AC$. By uniqueness of parallels, b must be the line we formed. Thus $\angle B'BC \cong \angle A'AC$.

To get congruence of the other sets of corresponding angles and alternating angles, use supplements and verticals.

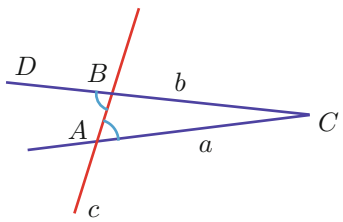


Figure 4.17: Congruent alternate interior angles

Now suppose that a and b are lines intersected by the line c so that corresponding and alternate angles are congruent. Let $a \cap c = A$ and $b \cap c = B$. Suppose a and b intersect with $a \cap b = C$, as in Fig. 4.17. Let D be on the side of c opposite C. We then have ΔABC with interior angle $\angle CAB$ congruent to the alternate angle $\angle DBA$, which is exterior to ΔABC, which violates the Exterior Angle Theorem. The contradiction implies that when a transversal forms congruent corresponding and alternate angles with a pair of lines, the lines must be parallel. □

Theorem 4.38. The angles of a triangle add up to two right angles.

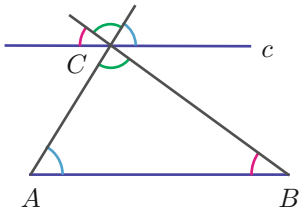


Figure 4.18: Angles in a triangle add up to a straight line

Proof. Let ΔABC be given. Let c be the parallel to $\overline{AB}$ through C, as in Fig. 4.18. Note that we get congruent angles to $\angle A$, $\angle B$, $\angle C$ as shown, the angles congruent to $\angle A$ and $\angle B$ because of parallels, the one congruent to $\angle C$ as a vertical angle. The result follows since the constructed angles add up to a straight angle, that is, two right angles. □

It is at this point that Hilbert introduces circles.

Definition 4.39. If O is any point in a plane α then the collection of all points A in α for which the segments OA are congruent to each other is called a **circle**. O is the **center of the circle**. If A is on the circle, OA is a **radius** of the circle.

Euclid uses circles but only incidentally in Book I. Book III is the more systematic study. As in *The Elements*, a few results from the beginning of Hilbert's program provide enough heft to prove theorems about circles that may be familiar from high school geometry. We have already seen some of these as exercises in Chapter 1. A couple of them are reprised below, this time to be done within Hilbert's framework.

Exercises 4.4. 1. Show that the parallel relation is transitive.

2. Theorem 4.20 shows how to construct a right angle to a given line. Give a construction for dropping a perpendicular from a given point to a given line.

3. Give a construction for forming a perpendicular to a line at a given point on that line.

4. Without appealing to Theorem 4.38, show that an exterior angle of a triangle equals the sum of the two remote interior angles.

5. Let a, b, and c be lines in a plane. Suppose $a \| b$. Show that if c intersects a, it also intersects b.

6. Show that the perpendicular bisectors of sides of a triangle intersect in a single point.

7. Show that three noncollinear points determine a circle uniquely.

8. A pair of points A, B on a circle determines the **chord** AB as well as two **arcs** of the circle: One arc is the set of points of the circle on one side of $\overline{AB}$, the other arc is the set of points of the circle on the other side of $\overline{AB}$. Three points A, B, C on a circle determine the **inscribed angles** $\angle ABC$, $\angle BCA$, and $\angle BAC$. The angle $\angle ACB$ is inscribed **over the chord** AB. By taking all the vertices on one side of $\overline{AB}$ we get one set of angles inscribed over AB. By taking vertices on the other side of $\overline{AB}$, we get a second set of angles inscribed over AB.

   Show that inscribed angles over the same chord in a circle are congruent provided the vertices of the angles are on the same arc of the circle determined by the chord. (See Fig. 4.19.)

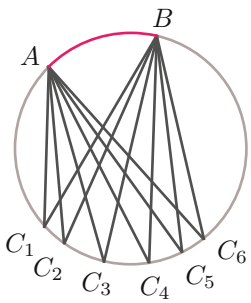


Figure 4.19: Inscribed angles $\angle AC_iB$ are all congruent

9. Fix a circle in the plane with center O and radius OA. Any point B in the plane with $OB < OA$ is **interior** to the circle. Show that if segments constructed from a point interior to a circle to the circle itself are congruent, then that point must be the center of the circle.

4.5 Axioms of Continuity

Hilbert's last two axioms address the nature of a line. We repeat Archimedes' Axiom in Hilbert's (translated) words from [13].

Axiom V.1 (Archimedes' Axiom). If AB and CD are any segments then there exists a number n such that n segments CD constructed contiguously (end-to-end) from A, along the ray from A through B, will pass beyond the point B.

Hilbert describes a model that satisfies all of Axioms I–IV and Axiom V.1. In doing so, he establishes that these axioms form a consistent system, that is, one without internal contradictions. The model Hilbert describes is based on Ω, the field consisting of numbers that arise by applying five operations, finitely many times, to the number 1. The operations are addition, subtraction, multiplication, division, and $\sqrt{1+\omega^2}$, where ω is any number already in the set. (For example, $\sqrt{2}$, $\sqrt{3}$ are in Ω but i where $i^2 = -1$ is not.) The points of the model are ordered pairs (x, y) where $x, y \in \Omega$. A line in the model is a set of points (x, y) with coordinates that satisfy $ux + vy + w = 0$, for some fixed $u, v, w \in \Omega$, u, v not both zero. If, instead of Ω, we take x, y in $\mathbb{R}$, we get the standard Cartesian plane, which is a nonisomorphic model that also satisfies Axioms I–V.1.

Indeed, Hilbert points out that there are infinitely many different (that is, nonisomorphic) geometries that satisfy Axioms I–IV and V.1 but that there is only one geometry that satisfies all of Axioms I–IV, V.1, and V.2, where Axiom V.2 is the Line Completeness Axiom. We cite the Axiom of Line Completeness exactly as it appears in [13].

Axiom V.2 (Axiom of Line Completeness)**.** An extension of a set of points on a line with its order and congruence relations that would preserve the relations existing among the original elements as well as the fundamental properties of line order and congruence that follow from Axioms I–III and V.1 is impossible.

The Axiom of Line Completeness guarantees, for instance, that when a line passes from one side to the other side of a second line, it must share a point with the second line. This is the axiom that makes Euclid's proof of Prop. I.1 valid.

Much has been written about geometries that satisfy Hilbert's Axioms I–IV, V.1 and these are often labeled "Euclidean geometries." Indeed, non-Archimedean Euclidean geometries satisfy Axioms I–IV only. For some authors, the key idea that makes Euclidean geometry Euclidean is uniqueness of parallels.

The complete set of Hilbert's axioms is sufficient for all of Euclid's results. With the two so-called continuity axioms, Euclidean geometry reduces to coordinate geometry over $\mathbb{R}$. Note, though, that Axioms V.1 and 2 mean that any line in a Euclidean plane can be viewed as a copy of $\mathbb{R}$. Some may disagree, but this suggests that the study of Euclidean geometry in some sense reduces to a study of the calculus. Perhaps this is why calculus gained ascendency in high school and university mathematics instruction through the 20th century, possibly at the expense of geometry.

Exercises 4.5.

1. Discuss precisely how to assign coordinates to points on a Euclidean line. Be as detailed as possible about your use of the axioms to do so.

2. Hilbert discusses the comparison of angles with no reference to numbers. If the plane is coordinatized by Ω, for example, how, if at all, is angle measure affected?

Chapter 5

More Euclidean Geometry

5.1 Euclid Beyond Book I

Our interest in this chapter is in circular inversion and the nine point circle, constructions in planar Euclidean geometry that were developed after the Arabic era. We assume all of Hilbert's axioms and their consequences as expressed in Book I of *The Elements*. Circular inversion, a lovely topic in its own right, proves useful in understanding models of the hyperbolic plane. The nine point circle, which is about triangles and the lines, centers, and circles naturally associated to triangles, typifies work done in Euclidean geometry in Europe during the eighteenth and nineteenth centuries. With circular inversion and the nine point circle as goals, our journey winds naturally through some favorite classical ideas in Euclidean geometry.

The results we study from *The Elements* are labeled by book and proposition number, even in cases where the statements and proofs diverge from Euclid's. It is important to understand that some of these concepts had already started germinating by Euclid's time. The notion of the *power of a point* is an example. Though not fully exploited until the nineteenth century, this is an idea that has been studied by generations of mathematics students, more or less continuously, via *The Elements*, since 300 B.C.

We pick up with *The Elements* in Book III, which Euclid devotes to circles. Some of these results should look familiar: they were exercises in previous chapters! Euclid does not deal with similarity until Book VI, after the theory of proportions, the subject of Book V. Since we use similar triangles to prove some of the results in this chapter, we dip into Books V and VI, mainly to appreciate the context within which similarity was developed by Euclid. Indeed, similarity makes a brief appearance here before we take it on properly in Section 5.2.

We start with Prop. III.1, which says that we can find the center of a circle. The proof we cite here is directly from *The Elements*. It gives insight into circles and the lines that intersect them transversally. It also has several important corollaries, not necessarily noted by Euclid explicitly.

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M. I. Dillon, *Geometry Through History*,
https://doi.org/10.1007/978-3-319-74135-2_5

Proposition III.1. Given a circle, we can determine its center.

Proof. Let A and B be points on a given circle. Let C be the midpoint of AB. Erect a perpendicular at C and say it intersects the circle at points D and E. Let F be the midpoint of DE. We claim that F is the center of the circle.

Assume F is not the center and that the point G is the center. We have $GA \cong GB$ as radii, and $AC \cong CB$, since C is the midpoint of AB. By SSS, $\Delta AGC \cong \Delta BGC$, which implies that $\angle GCA \cong \angle GCB$ is right. As $\angle FCA \cong \angle FCB$ is right, this violates CN 5 so F must be the center, as originally claimed. □

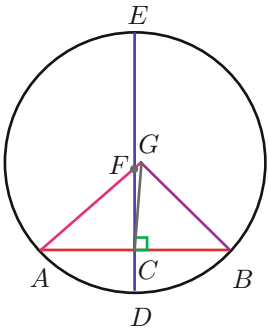


Figure 5.1: The midpoint of DE must be the center of the circle

As we remarked at the end of Chapter 1, the notion of uniqueness was not addressed explicitly in *The Elements*. The proof of Prop. III.1 suggests the following.

Corollary 5.1. The center of a circle is unique.

The proof of the next corollary is also embedded in the proof of Prop. III.1.

Corollary 5.2. The perpendicular bisector of a chord in a circle passes through the center of the circle.

One more corollary of the proof of Prop. III.1 suggests the possibilities for how a line and a circle can intersect.

Corollary 5.3. A line cannot intersect a circle in more than two points.

Proof. Suppose a line ℓ intersects a circle $\mathcal{C}$ in three distinct points A, B, and C. Invoking Hilbert, we can say that one of the three points is between the other two. If $A * B * C$, then AB and AC must have different midpoints. As perpendiculars to ℓ, the perpendicular bisectors of AB and AC must be parallel by Prop. I.28. But both perpendicular bisectors must run through the center of $\mathcal{C}$, giving us a common point on distinct parallel lines. The contradiction implies that a circle and a line intersect in at most two points. □

Our proof of Prop. III.2 is similar to Euclid's but we continue to employ Hilbert's constructs.

Recall that if $\mathcal{C}$ is a circle with center O, a point C is *interior* to $\mathcal{C}$ provided $OC < OA$ for any point $A \in \mathcal{C}$.

Proposition III.2. Let A and B be points on a circle $\mathcal{C}$. The inside points of the chord AB are all interior to $\mathcal{C}$.

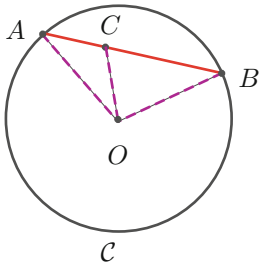


Figure 5.2: The points inside a chord are interior to a circle

Proof. Let $\mathcal{C}$ be a circle with center O and points A and B. Let C be inside AB. It is enough to show that $OC < OA$.

Suppose AB goes through O. By definition, O is interior to the circle. Say $C \neq O$ is some other point inside AB. Then either $A*C*O*B$ or $A*O*C*B$. In the first case, $AO > OC$. In the second case, $OB \cong OA > OC$.

Suppose next that AB does not go through O. We then have a triangle ΔAOB. If $OC \cong OA$, then C is on $\mathcal{C}$ but that is impossible by Corollary 5.3. Suppose then that $OC > OA \cong OB$. We seek a contradiction.

By Prop. I.18, $\angle OAC = \angle OAB > \angle ACO$ and $\angle OBC = \angle OBA > \angle BCO$. Notice though that $\angle ACO + \angle BCO = \pi$ so that $\angle OAB + \angle OBA > \pi$. This violates Prop. I.17, from which we must conclude that $OC < OA$. □

Our proof of Prop. III.2 needs only slight modification to serve as proof of the following. We leave the details as an exercise.

Corollary 5.4. The region enclosed by a circle is convex.

If $\mathcal{C}$ is a circle with center O and $A \in \mathcal{C}$, then any segment $OB \cong OA$ is a *radius* of $\mathcal{C}$. It is often convenient to let a single symbol, a, stand for any radius of a circle.

Lemma 5.5. If A and B are points interior to a circle with radius a, then $AB < 2a$.

Proof. If A and B are interior to a circle with radius a and center O, then $AO < a$ and $BO < a$. By the Triangle Inequality, $AB < AO + BO < 2a$. □

A useful consequence of Lemma 5.5 is the following analog of the Crossbar Theorem. This result is in the spirit of Hilbert's work.

Corollary 5.6. If a line contains one point on a circle and one point interior to the circle then the line passes through the circle in a second point.

Proof. Suppose $\mathcal{C}$ is a circle with center O and that ℓ contains a point $A \in \mathcal{C}$ and a point B interior to $\mathcal{C}$. By Hilbert's version of Archimedes' Axiom, $\overrightarrow{AB} \subset \ell$ exceeds $2OA$, which by Lemma 5.5 exceeds the distance between any two points interior to $\mathcal{C}$. It follows that ℓ contains points E, $A * B * E$, that are exterior to $\mathcal{C}$. By the Axiom of Line Completeness, $\overrightarrow{AB}$ must pass through a point on $\mathcal{C}$. □

We return now to Euclid.

Proposition III.3. A line that passes through the center of a circle bisects a chord not passing through the center of the circle if and only if it is at right angles to the chord.

Proof. Let $\mathcal{C}$ be a circle with center O. Let AB be a chord that does not pass through O. Let DE be a chord that does pass through O and that intersects AB at F. Notice that $AO \cong BO$.

If DE bisects AB, then $AF \cong BF$ so by SSS we have $\Delta AFO \cong \Delta BFO$. It follows that $\angle AFO \cong \angle BFO$ must be right.

Now suppose $\angle AFO \cong \angle BFO$ is right. Since ΔABO is isosceles, $\angle OAB \cong \angle OBA$. We then have AAS for $\Delta AFO \cong \Delta BFO$ so $AF \cong BF$, in other words, DE bisects AB. □

Next is a result we use several times in the sequel. The proof is a straightforward application of Props. I.27 (or 28) and I.29, thus depends on Postulate V. We leave the details to the reader.

Lemma 5.7. Perpendiculars to intersecting lines must themselves intersect.

Departing from Euclid's approach, we apply Lemma 5.7 to prove the next two propositions.

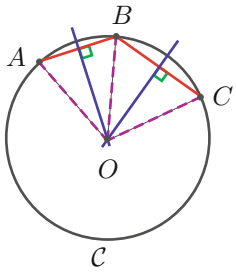


Figure 5.3: Three noncollinear points determine a circle

Proposition III.10. Three noncollinear points determine a unique circle.

Proof. Let A, B, C be noncollinear. Form AB and BC, as in Fig. 5.3. By Lemma 5.7, their perpendicular bisectors meet in a point O. Using SAS congruences, we get $OA \cong OB \cong OC$, so A, B, C determine the circle $\mathcal{C}$ with center O and radius OA. Corollary 5.2 implies that any circle through A, B, C must have its center at O. Moreover its radius must equal the distance from O to A, B, and C, so it must be $\mathcal{C}$. □

Proposition III.25. An arc of a circle determines the circle uniquely.

Proof. By Corollary 5.3, an arc of a circle cannot be a line so it must contain three noncollinear points. By Prop. III.10, there is only one circle containing those three points so it must be the circle from which the arc was drawn. □

When it comes to the relative positions of a line and a circle in the plane, Corollary 5.3 implies three possibilities: there is no point of intersection between the line and the circle, there is exactly one point of intersection, or there are exactly two points of intersection. We have encountered the idea of **transverse intersections** several times: A line that shares exactly two points with a circle intersects the circle transversely. A **tangent** to a circle at a point P is a line that intersects the circle at P and only at P.

Proposition III.16, 18, 19. A tangent to a circle at a given point is the unique line perpendicular to the radius at that point.

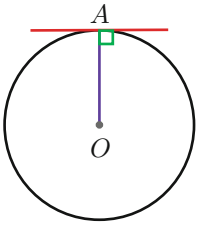


Figure 5.4: The tangent to a circle is perpendicular to the radius at that point

Proof. Let $\mathcal{C}$ be a circle with center O, and let A be a point on $\mathcal{C}$. Let ℓ be tangent to $\mathcal{C}$ at A. Suppose that OA is not perpendicular to ℓ. If we drop a perpendicular to ℓ from O, it must have foot Q different from A. Since all the points of ℓ except A lie outside $\mathcal{C}$, $OQ > OA$. Apply Prop. I.18 to ΔOAQ to get $\angle OAQ > \angle OQA$. As $\angle OQA$ is right, this violates Prop. I.17. We conclude that OA is the perpendicular to ℓ at A.

Next, suppose ℓ is perpendicular to OA at A but that it intersects $\mathcal{C}$ at a second point B. Then ΔOAB is isosceles, so $\angle OAB \cong \angle OBA$. But that again violates Prop. I.17, because $\angle OAB$ and $\angle OBA$ are both right. We conclude

that ℓ is tangent to $\mathcal{C}$. By the uniqueness of a perpendicular to a line at a point, this proves the result. □

We saw in Exercise 4.4(8) that there are two families of inscribed angles associated to any chord in a circle. We formalize the associated definitions here. Some of these are definitions that Euclid omitted.

Definition 5.8. Let $\mathcal{C}$ be a circle with center O and points $B, C \in \mathcal{C}$. BC determines two arcs of $\mathcal{C}$. If $O \in BC$, each arc is a **semi-circle** of $\mathcal{C}$. If $O \notin BC$, the arc of $\mathcal{C}$ on the O side of $\overline{BC}$ is the **major arc** determined by BC. The arc of $\mathcal{C}$ on the opposite side of $\overline{BC}$ is the **minor arc**.

Let A be a third point on $\mathcal{C}$. Then $\angle BAC$ is an **angle inscribed over the chord** BC. If A is on the major arc of $\mathcal{C}$ determined by BC, then the minor arc subtends $\angle BAC$. In this case we say the minor arc is the **base** of $\angle BAC$ or the **base on the circumference of** $\mathcal{C}$ for $\angle BAC$. If A is on the minor arc of $\mathcal{C}$ determined by BC, then the major arc subtends $\angle BAC$. In this case we say the major arc is the **base** of $\angle BAC$ or the **base on the circumference of** $\mathcal{C}$ for $\angle BAC$.

A **central angle** in $\mathcal{C}$ is any angle with vertex O. The base on the circumference of $\mathcal{C}$ for any central angle is the minor arc determined by BC.

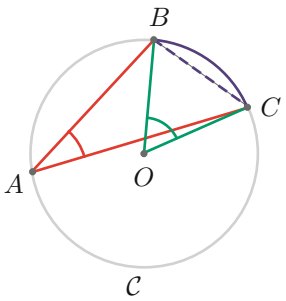


Figure 5.5: The inscribed angle $\angle BAC$ is on the same base on the circumference of $\mathcal{C}$ as the central angle $\angle BOC$

Note that a chord that does not pass through the center of a circle determines a unique central angle.

Proposition III.20. A central angle in a circle is double any inscribed angle with the same base on the circumference.

Proof. Consider the circle $\mathcal{C}$ with center O and points A, B, C, on the circumference, A on the major arc determined by BC. We must show that $\angle BOC \cong 2\angle BAC$. First we consider the case where O is interior to $\angle BAC$, as shown in Fig. 5.6.

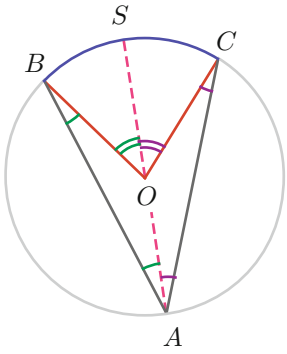


Figure 5.6: The angle at O is central, the angle at A is inscribed

Form AS, the diameter of $\mathcal{C}$ at A. Then $\angle BOC = \angle BOS + \angle COS$ and $\angle BAC = \angle BAO + \angle CAO$.

Now $\angle BOS$ is exterior to ΔAOB and $\angle COS$ is exterior to ΔAOC. Note further that ΔAOB and ΔAOC are both isosceles so that $\angle BAO \cong \angle ABO$ and $\angle CAO \cong \angle ACO$. With these observations and Prop. I.32, we have $\angle BOS \cong 2\angle BAO$ and $\angle COS \cong 2\angle CAO$. Adding we get $\angle BOC \cong 2\angle BAC$, as desired.

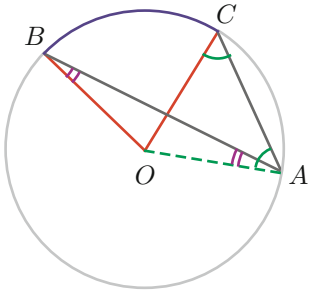


Figure 5.7: Prop. III.20, when O is exterior to $\angle BAC$

Next we consider the case where O is exterior to $\angle BAC$, as shown in Fig. 5.7. Here we have isosceles triangles ΔAOC and ΔAOB as in the previous case but now $\angle BAC = \angle CAO - \angle BAO$, and $\angle BOC = \angle AOB - \angle AOC$. We invoke Prop. I.32 in this case as well to say $\angle AOB \cong \pi - 2\angle BAO$ and $\angle AOC \cong \pi - 2\angle CAO$ so that

$$\angle BOC \cong \pi - 2\angle BAO - (\pi - 2\angle CAO) = 2(\angle CAO - \angle BAO) = 2\angle BAC,$$

as desired. □

The proof of the next proposition was Exercise 4.4.8. Note that Prop. III.20 implies the case in which the inscribed angles are subtended by a minor arc.

Proposition III.21. Inscribed angles on the same base of the circumference of a circle are congruent.

The third inscribed angle theorem is Prop. III.32. Here the comparison is between the angle determined by a chord of a circle and the tangent at one endpoint, and the inscribed angles "in the alternate segments of the circle." [7] There is an arc of the circle interior to an angle formed by a chord and a tangent at one endpoint. (See Fig. 5.8.) An inscribed angle **in the alternate segment** has its vertex on the other arc of the circle.

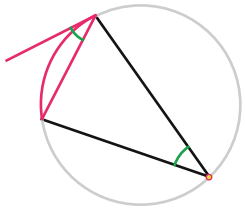


Figure 5.8: Comparing angles associated to a chord

We supply an outline of the proof for Prop. III.32 in Exercise 7 below and ask the reader to fill in details.

Proposition III.32. The angle formed by a tangent to a circle and a chord, one endpoint of which is the point of tangency, is congruent to the inscribed angles in the alternate segment.

The next theorem is not in Euclid.

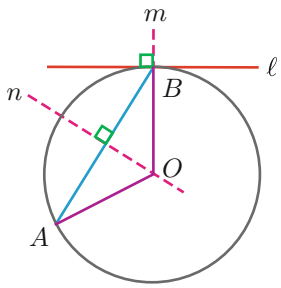


Figure 5.9: Two points and a line through one of them determine a circle

Theorem 5.9. Given points A, B and a line ℓ through B not through A, there is a unique circle through A and B with tangent ℓ.

Proof. Given the hypotheses of the theorem, take m to be perpendicular to ℓ at B. If $m = \overline{AB}$, then the circle with center at the midpoint O of AB and radius OA has tangent ℓ at B. If $m \neq \overline{AB}$, let n be the perpendicular bisector of AB. By Lemma 5.7, m and n intersect, say at O. It is easy to check that $OA \cong OB$, which is the radius of a circle with center O, passing through A and B, with tangent ℓ at B. Uniqueness follows by the fact that the radius of a circle through B with tangent ℓ must be perpendicular to ℓ, and if the circle goes through A, the center must also lie on the perpendicular bisector of AB. □

Notice the idea that the last result shares with our version of Prop. III.10: With three independent pieces of information, we can specify a unique circle. This is exactly what we do when we define a circle in the Cartesian plane using a polynomial of the form

$$(x - x_0)^2 + (y - y_0)^2 = a^2.$$

The polynomial requires three parameters: a, the radius of the circle, x_0 and y_0, the coordinates of the center of the circle. There are several different ways of arriving at the equation, depending on the information we are given, just as there are several different ways of arriving at an equation for a line in the plane, depending on whether we are given two points, or a point and a slope, or a point and a line perpendicular to the line we want, etc.

Proposition III.5, 6. Distinct intersecting circles have different centers.

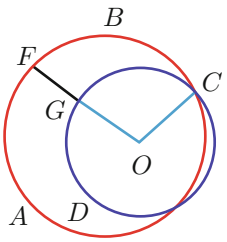


Figure 5.10: Intersecting circles have different centers

Proof. Let circles ABC, CDG intersect at C, as in Fig. 5.10. Suppose the circles have a common center, O so that $OC \cong OG$. Extend OG through F on ABC. Now $OF \cong OG$, the whole equal to the part, a violation of CN 5. The contradiction proves that the circles cannot have a common center. □

We turn our attention now to an important construction in the discussion of circular inversion. In Theorem 5.9, we are given two points and a line through one and must argue that there is an associated circle. In Prop. III.17, we are given a circle and a point not on it, and must argue that there is an associated tangent. The proof here is Euclid's.

Proposition III.17. Given any point P exterior to a circle $\mathcal{C}$, we can construct a tangent to $\mathcal{C}$ from P.

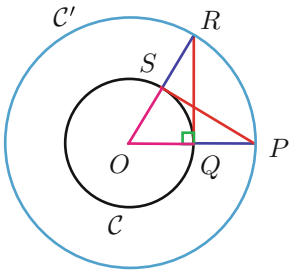


Figure 5.11: Construct a tangent to a circle from a point outside the circle

Proof. If O is the center of $\mathcal{C}$, construct OP and form $\mathcal{C}'$, the circle concentric with $\mathcal{C}$ with radius OP, as in Fig. 5.11. Let Q be the point where OP intersects $\mathcal{C}$. Erect a perpendicular to OP at Q and say it intersects $\mathcal{C}'$ at R. Join OR and say it intersects $\mathcal{C}$ at S. Join SP. The claim is that SP is tangent to $\mathcal{C}$.

We have $OR \cong OP$, as radii of $\mathcal{C}'$ and $OQ \cong OS$, as radii of $\mathcal{C}$. By SAS, we get $\Delta ROQ \cong \Delta POS$. Now ΔROQ contains $\angle OQR$, which is right and corresponds to $\angle OSP$. This gives us OS and SP perpendicular, thus, that SP is tangent to $\mathcal{C}$ as desired. □

We study similarity with proper ceremony in Section 5.2 but we make brief reference to the notion of similar triangles here. Recall for now that ΔABC and $\Delta A'B'C'$ are similar provided there is a labeling of the vertices so that

$$AB/A'B' = BC/B'C' = AC/A'C'.$$

We noted in Chapter 3 that by Props. VI.4 and 5, triangles in the Euclidean plane are similar if and only if their corresponding angles are congruent.

Consider two circles in the plane: $\mathcal{C}$ with center O and radius a, and $\mathcal{C}'$ with center O' and radius a'. Suppose $a \neq a'$, that the circles do not intersect, and that one circle is not inside the other, as in Fig. 5.12. Choose a point A on $\mathcal{C}$ but not on $\overline{OO'}$. Take A' on $\mathcal{C}'$ so that OA and $O'A'$ are parallel and on the same side of the line $\overline{OO'}$. Since $a \neq a'$, $\overline{AA'}$ intersects $\overline{OO'}$ in a point P. The point P is an **external center of similitude** of $\mathcal{C}$ and $\mathcal{C}'$. It is easy to argue that ΔAOP and $\Delta A'O'P$ are similar. From there, we have $OP/O'P = a/a'$. Note that if we construct the tangent to $\mathcal{C}$ from P, we must also get the tangent to $\mathcal{C}'$ from P. The designation *external* means that P is outside the segment OO'.

Circles positioned as $\mathcal{C}$ and $\mathcal{C}'$ also have centers of similitude between them, as in Fig. 5.13. This is an **internal center of similitude**. We leave it as an exercise to find that point.

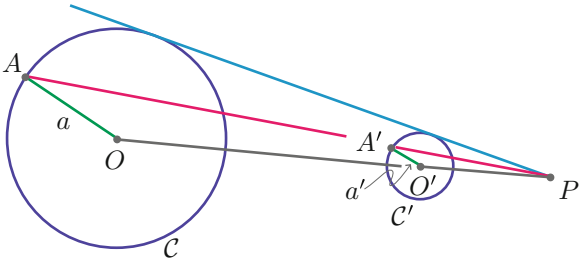


Figure 5.12: An external center of similitude for two circles

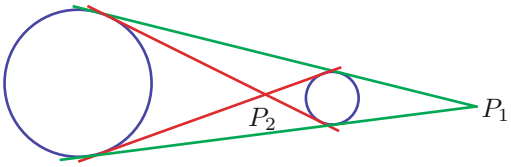


Figure 5.13: Internal and external centers of similitude

Euclid's topic in Book IV is inscribing and circumscribing circles about various figures. The next proposition appears to be a restatement of Prop. III.10, but our statement of Prop. III.10 looks quite different from Euclid's and the concern here is very different.

Proposition IV.5. The vertices of a triangle determine a circle, i.e., a triangle may be circumscribed by a circle.

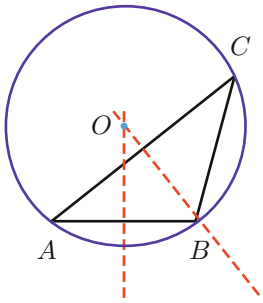


Figure 5.14: The circumcircle of a triangle

Proof. The vertices of a triangle constitute a set of three noncollinear points so this is just a restatement of our version of Prop. III.10 above. □

The circle given in Prop. IV.5 and shown in Fig. 5.14 is the **circumcircle** of the triangle. Its center is the **circumcenter** of the triangle. This is just the beginning of a study of the centers of a triangle. We take up that thread below, after working out the theory of circular inversion.

Exercises 5.1. 1. A circle partitions the plane into three nonintersecting sets: the points on the circle itself, its interior, and its **exterior**, the points at a distance greater than the radius from its center. Which of Hilbert's axioms allow us to make these designations?

2. In the proof of Prop. III.1, we erect a perpendicular to a chord at a point interior to the circle then claim that this perpendicular intersects the circle in two points. This would appear to be a tacit assumption of Euclid. We have claimed that Hilbert's axiom system accounts for Euclid's tacit assumptions. What part of Hilbert's axiom system guarantees that we actually get those points of intersection?

3. Look up the statement of Prop. III.10 in the Heath edition of *The Elements* and argue that our statement is equivalent.

4. Argue from Hilbert's axioms and/or theorems that an arc of a circle contains at least three noncollinear points.

5. Use algebra to find a Cartesian equation of the circle in $\mathbb{R}^2$ that passes through points $(1, 1)$, $(2, -1)$, and $(3, -2)$.

6. Prove that an inscribed angle subtended by a semi-circle is right.

7. This exercise provides steps that lead to a proof of Prop. III.32. Let $\mathcal{C}$ be a circle with center O and points A and B.

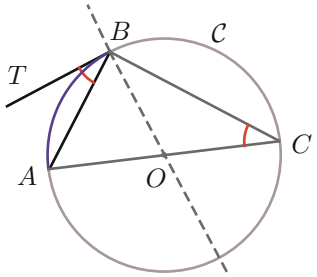


Figure 5.15: Exercise 7, part (a): when AC is a diameter

(a) Form the diameter of $\mathcal{C}$ at A, AC, and form the tangent line to $\mathcal{C}$ at B, ℓ. Take a point T on ℓ on the A side of $\overline{OB}$. Show that $\angle ABT \cong \angle ACB$. Note that the congruent angles are associated to alternate arcs of $\mathcal{C}$.

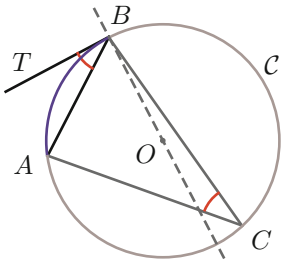


Figure 5.16: Exercise 7, part (b): AC need not be a diameter

(b) Argue next that AC need not be a diameter. In other words, for any chord AC on the arc alternate to the one determined by $\angle ABT$, $\angle ABT \cong \angle ACB$. This completes the proof of Prop. III.32.

8. Show that ΔAOP and $\Delta A'O'P$, as given in Fig. 5.12 are similar.

9. Let P be the external center of similitude of $\mathcal{C}$ and $\mathcal{C}'$ as in Fig. 5.12. Argue that the tangent from P to $\mathcal{C}$ is also the tangent from P to $\mathcal{C}'$.

10. Find P_2, the internal center of similitude for $\mathcal{C}$ and $\mathcal{C}'$ as shown in Fig. 5.13. Show that the tangent from P_2 to $\mathcal{C}$ is also tangent to $\mathcal{C}'$.

11. Consider a circle centered at O with tangents at points A and B that intersect at S.

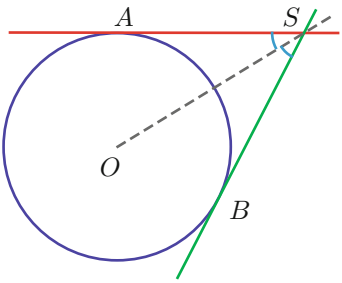


Figure 5.17: Exercise 11: Circle with two intersecting tangents

(a) Show that OS bisects $\angle ASB$. (See Fig. 5.17.)

(b) Show that $AS \cong BS$.

12. Consider a circle with chords AB and CD intersecting at right angles at the point E. Show that the perpendicular from E to AC bisects the segment BD. (See Fig. 5.18.)

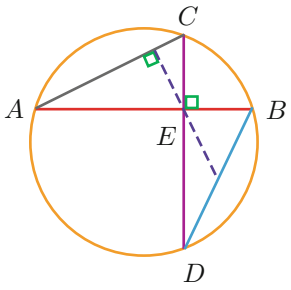


Figure 5.18: Exercise 12: Circle with perpendicular chords

13. Consider a circle with center O and parallel tangents at two points. Suppose a third tangent line intersects the two parallels at points A and B. Show that the circle with diameter AB passes through O. (See Fig. 5.19.)

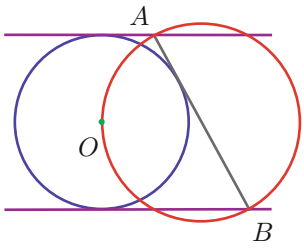


Figure 5.19: Exercise 13: Circle with three tangents, two parallel

5.2 Similar Triangles and the Power of a Point

"Power of a point" is a nineteenth century locution, probably due to Jakob Steiner (1796–1863), but it refers to an idea that comes directly from Euclid. One reason for our interest in the power of a point is its role in circular inversion.

The origin of the notion of the power of a point is Prop. III.36. Similarity makes the proof of that proposition easier and we depart from Euclid to employ it here, with some justification. By Theorem 3.7, and the fact that the parallel postulate holds in a neutral plane if and only if the plane is not hyperbolic, the parallel postulate is equivalent to the existence of noncongruent similar triangles. As the results in this section are endemic to the Euclidean plane, similarity is a natural tool to bring to bear in our proofs.

We start with a few of the definitions from Book V, experiencing the challenge of comparing quantities without attaching numbers to them.

Definition V.1. A magnitude is a **part** of a magnitude, the less of the greater, when it measures the greater.

Although we will not use Definition V.1 in the sequel, it is of interest to see how Euclid tackles the idea of magnitude. Much has been written about Euclid's use of the word *part* in this definition, for instance. (Compare CN 5.)

Definition V.3. A **ratio** is a sort of relation in respect of size between two magnitudes of the same kind.

A ratio is the comparison of two like attributes: length of line segment AB to length of line segment CD, area of ΔABC to area of rectangle $DEFG$. It does not allow comparison of length to area, for instance. There are other relations between magnitudes so "ratio is [one] sort of relation."

Definition V.6. Let magnitudes which have the same ratio be called **proportional**.

Where a ratio is the comparison of two like attributes, proportional magnitudes involve at least four attributes. In other words, length of AB to length of CD is a ratio. We say AB, CD are proportional to EF, GH if the ratios AB to CD, and EF to GH are the same. This is enough to define similar figures.

Definition VI.1. Similar rectilineal figures have their angles equal and the sides about equal angles proportional.

Note that corresponding sides are opposite congruent angles in similar triangles, though Euclid does not say so. Proposition VI.21 says that similarity is transitive. It is easy to see that it is, in fact, an equivalence relation, as long as we are willing to consider a triangle to be similar to itself.

The first result from Book VI extends Prop. I.37, which itself starts a thread in Book I about comparing areas of triangles and parallelograms in the same parallels. We omit the proof, noting that it depends on results from Book V about relationships among ratios, all of which are familiar to us from working with ratios of numbers.

Proposition VI.1. Triangles and parallelograms which are under the same height are to one another as their bases.

The second proposition in Book VI is a pleasing application of another proposition from the same thread in Book I. The demonstration of this result also relies on Book V, but this time we include a proof. To distinguish between a segment and its length, we now start using the notation $|AB|$ for the latter. Similarly, $|\Delta ABC|$ is the area of a triangle ΔABC.

Proposition VI.2. A line drawn parallel to one side of a triangle will cut the sides of the triangle proportionately. Conversely, if the sides of a triangle are cut proportionately, the line joining the points of the section will be parallel to the remaining side of the triangle.

Proof. Consider ΔABC and points D, E with $A*D*B$, $A*E*C$, and $DE \| BC$ as in Fig. 5.20. Since ΔBDE and ΔCDE have coincident bases and are in the

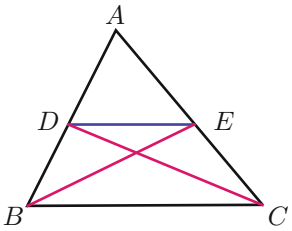


Figure 5.20: Prop. VI.2: Cutting a triangle proportionately

same parallels, they have equal areas. We can say that

$$|\Delta BDE|/|\Delta ADE| = |\Delta CDE|/|\Delta ADE|.$$

Next, view AD as the base of ΔADE and BD as the base of ΔBDE. These triangles share the vertex E so are in the same parallels. By Prop. VI.1,

$$|\Delta BDE|/|\Delta ADE| = |BD|/|AD|.$$

Similarly, $|\Delta CDE|/|\Delta ADE| = |CE|/|EA|$, implying $|BD|/|AD| = |CE|/|AE|$, which proves the first part of the result.

Now suppose $A*D*B$ and $A*E*C$ where $|BD|/|AD| = |CE|/|AE|$. We claim $DE \| BC$.

We have $|\Delta BDE|/|\Delta ADE| = |BD|/|AD|$. Similarly, $|\Delta CDE|/|\Delta ADE| = |CE|/|AE|$. It follows that $|\Delta BDE|/|\Delta ADE| = |\Delta CDE|/|\Delta ADE|$ so

$$|\Delta BDE| = |\Delta CDE|.$$

As ΔBDE and ΔCDE have a common base, they must be in the same parallels (Prop. I.39), which proves the result. □

The next proposition is often called SAS for similar triangles. We leave the proof as an exercise.

Proposition VI.6. If triangles have one angle equal, and sides about that angle proportional, then the triangles are similar.

The rules for manipulating and equating areas of different geometric configurations are worked out in Book II of *The Elements*. Euclid's goal in Book II is apparently Prop. II.14, where it is established that one can always find a square that encloses the same area as a given polygon. We saw Prop. II.4 in Exercise 1.7(1a).

Book II is often referred to as *geometric algebra* because its results can be captured quite neatly in algebraic equations. Succumbing to the siren call, we may say that Prop. II.4 says

$$(a+b)^2 = a^2 + b^2 + 2ab. \tag{5.1}$$

The problem with this is that *algebra* refers to rules that were first developed by Muhammad ibn Musa al-Khwarizmi (780–850) in the original algebra text, 1100 years after Euclid wrote *The Elements*. While algebra makes it easy for us to track what Euclid is doing in Book II, the propositions look nothing at like Equation (5.1). Euclid's statement of Prop. II.4, as quoted from [7], is

> If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

We can mediate our expression of Prop. II.4 as Equation (5.1) by saying that a and b are the lengths we cut from a line, so the square on the whole is $(a+b)^2$, etc., but we have to realize that none of the symbols used in Equation (5.1) would have been familiar to Euclid and that the manipulations we do with ease after years of practice would have been completely foreign to the ancient Greeks. More importantly, the use of algebra introduces a level of abstraction into the picture that is far removed from Euclid's work.

The benefits of using algebraic manipulation to gain geometric insight at this stage of our study can be debated. But as we are climbing out of *The Elements* and finding our way into mathematics that was worked out well after the Arabic era, we will proceed using the techniques of classical algebra that most of us learned in middle school. Accordingly, we now start to think of $|PA||PB|$ both as the area of the rectangle with sides PA and PB and as a numerical value for that area.

The symbol $\sim$ indicates similarity.

Proposition III.36 Fix a circle and let P be any point in the plane. Let A and A' be points on the circle and suppose $\overline{PA}$ intersects the circle at B and $\overline{PA'}$ intersects the circle at B'. Then $|PA||PB| = |PA'||PB'|$.

Proof. If P is a point on the circle, then both products are zero.

If P is not on the circle and both lines $\overline{PA}$, $\overline{PA'}$ intersect the circle transversely, then we have one of the configurations in Fig. 5.21. Note that $\Delta PA'B \sim \Delta PAB'$ whether P is inside or outside the circle: The angles in the triangles at P are the same, or vertical, and in each picture, $\angle B \cong \angle B'$ by Prop. III.21. The result then follows by comparing corresponding sides in the similar triangles.

If $\overline{PA'}$ is tangent to the circle, as in Fig. 5.22, think of $A' = B' = T$ as the (double) point of intersection so $|PA'||PB'| = |PT|^2$. Join P to the circle center O and say $\overline{PO}$ intersects the circle at C and D. Then ΔPTO is right, with hypotenuse PO so by the Pythagorean Theorem, $|PO|^2 = |PT|^2 + |TO|^2$. Since $|TO| = |DO|$ is the radius of the circle,

$$|PO| - |TO| = |PC|$$

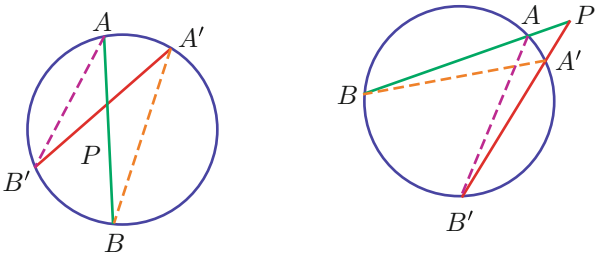


Figure 5.21: Proof of Prop. III.36 for transverse lines

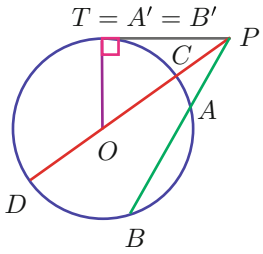


Figure 5.22: Proof of Prop. III.36 in the tangent case

and

$$|PO| + |TO| = |PD|.$$

Putting these together we get

$$|PT|^2 = |PO|^2 - |TO|^2 = (|PO| - |TO|)(|PO| + |TO|) = |PC||PD|.$$

If AB is a diameter, we are done. If not, then by the first case, $|PA||PB| = |PC||PD|$, so that $|PA||PB| = |PT|^2$, as desired. □

The significance of the result is that it implies that a circle associates a well-defined number to each point P in the plane. Let $\mathcal{C}$ be a circle with center O and radius a. If P is a point exterior to $\mathcal{C}$, and T is the point of tangency from P to $\mathcal{C}$, it is easy to verify that $|PT|^2 = |PO|^2 - a^2$, where a is the radius of the circle. If P is a point interior to the circle, and AB is a chord of $\mathcal{C}$ with $A * P * B$, then $|PA||PB| = a^2 - |PO|^2$. (Verification of these statements is left to the exercises.) Of course, if P is point on the circle, $|PO|^2 - a^2 = 0$. This suggests the following.

Definition 5.10. The **power of a point** P with respect to a circle $\mathcal{C}$ with center O and radius a is $||PO|^2 - a^2|$.

We can interpret Prop. III.36 now to say that the power of a point P with respect to a circle $\mathcal{C}$, is $|PA||PB|$ where $A, B \in \mathcal{C}$ and $P \in \overline{AB}$.

Exercises 5.2. 1. In reference to the proof of Prop. VI.2, and using the same sorts of tools, show that $|AB|/|AD| = |AC|/|AE|$. (See Fig. 5.20.)

2. Referring to Fig. 5.23, show that $|AE|/|EC| = |BE|/|ED|$ if and only if AB is parallel to DC.

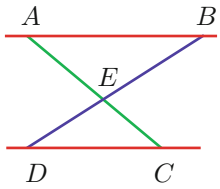


Figure 5.23: Exercise 2: Segments cut proportionately

3. Prove Prop. VI.2 using whatever you know about similar triangles, from inside or outside this course.

4. Show that in Fig. 5.24, $\Delta ADE \sim \Delta ABC$ if and only if $DE \| BC$ if and only if $|AD|/|AC| = |AE|/|AB| = |DE|/|CB|$.

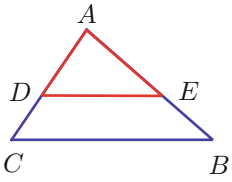


Figure 5.24: $\Delta ADE \sim \Delta ABC \Leftrightarrow DE \| CB$

5. Prove Prop. VI.6.

6. Let $\mathcal{C}$ be a circle with center O and radius a. Let P be an arbitrary point in the plane, and let A be a point on the circle, $A \neq P$. Consider the line $\ell = \overline{PA}$. If ℓ is tangent to $\mathcal{C}$, let $B = A$. If ℓ intersects $\mathcal{C}$ transversely, let $B \neq A$ be the second point of intersection. Show that $|PA||PB| = ||PO|^2 - a^2|$.

7. Let $\mathcal{C}$ be the circle in $\mathbb{R}^2$ given by $(x-1)^2 + (y+2)^2 = 4$. Use algebra and analytic geometry to calculate the power of $P = (3, -5)$ with respect to $\mathcal{C}$. Is P on $\mathcal{C}$, interior to $\mathcal{C}$, or exterior to $\mathcal{C}$?

5.3 Circular Inversion

Here we not only become freer in our use of arithmetic, we actually use polar and rectangular coordinates to understand mappings called *circular inversions*. Before proceeding, recall the following definition.

Definition 5.11. Let $\mathcal{S}$ and $\mathcal{T}$ be sets. A **mapping** $f : \mathcal{S} \to \mathcal{T}$ is a rule that associates to each element s in $\mathcal{S}$ exactly one element $f(s)$ in $\mathcal{T}$. We say f **maps** $\mathcal{S}$ **into** $\mathcal{T}$ and we write $s \mapsto f(s)$. $\mathcal{S}$ is the **domain** of the mapping f and $\mathcal{T}$ is the **codomain** of f. The set $\{t \in \mathcal{T} \,|\, t = f(s) \text{ for some } s \in \mathcal{S}\} \subseteq \mathcal{T}$ is the **range** of f. If $A \subseteq \mathcal{S}$, we write $f(A) = \{f(a) \,|\, a \in A\} \subseteq \mathcal{T}$. If $f(\mathcal{S}) = \mathcal{T}$, then f is **onto** or **surjective**. If $f(s_1) = f(s_2)$ implies $s_1 = s_2$, f is **one-to-one** or **injective**. If f is both one-to-one and onto, f is **bijective**.

Mappings are also called *functions*, although some people prefer to use the latter only for mappings into $\mathbb{R}$.

A circular inversion is a mapping defined on a subset of the Euclidean plane. When the context is clear, we refer to circular inversions simply as *inversions*.

Inversions were studied by Jakob Steiner during the 1830s, after the discovery of non-Euclidean geometries and before the discoveries of models for the hyperbolic plane based on Euclidean geometry. A rich and interesting topic in its own right, circular inversion will help us with an application involving triangles and then again when we study the Poincaré disk model for the hyperbolic plane.

Let O be a fixed point in the Euclidean plane, $\mathcal{E}$. The **punctured plane** is the set $\mathcal{E} \setminus O$, that is, the collection of all points in $\mathcal{E}$ except the point O.

Definition 5.12. Let $\mathcal{C}$ be a circle in $\mathcal{E}$ with center O and radius a. The **circular inversion** through $\mathcal{C}$ is the mapping

$$f : \mathcal{E} \setminus O \longrightarrow \mathcal{E} \setminus O$$

defined by $f(P) = P'$, where P' is the unique point on $\overrightarrow{OP}$ that satisfies $|OP||OP'| = a^2$. $\mathcal{C}$ is called **the circle of inversion**, O is the **center of inversion**, and a is the **radius of inversion**.

We will see that on inversion through $\mathcal{C}$, points exterior to the circle are mapped to points interior to the circle, and vice versa. By puncturing the plane, we are able to use circular inversion to turn the plane "inside out" around the circle.

Observation 5.13. Constructing the inverse of a point $P \in \mathcal{E} \setminus O$ is straightforward. Let f be the inversion determined by the circle $\mathcal{C}$ with center O and radius a.

(1) If P is interior to $\mathcal{C}$, erect the perpendicular to OP at P. Pick one point on $\mathcal{C}$ where the perpendicular intersects and call it T. (See Fig. 5.25.) Join radius OT. Construct the tangent to $\mathcal{C}$ at T. By Prop. III.16, 18, 19, that

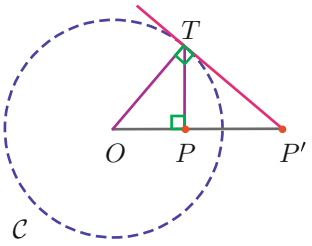


Figure 5.25: The inverse of a point with respect to a circle

tangent is perpendicular to OT. The line segment PT is interior to the right angle made by OT and the tangent so the angle between PT and the tangent is less than right. The parallel postulate then implies that the tangent intersects $\overrightarrow{OP}$. Let P' be the point of intersection.

Note that $\Delta OTP' \sim \Delta OPT$, as both are right and share $\angle P'OT$. It follows that $|OT|/|OP| = |OP'|/|OT|$. As $|OT| = a$, it follows that $|OP||OP'| = a^2$, thus, that $f(P) = P'$.

(2) If P' is exterior to $\mathcal{C}$ and we want $f(P')$, reverse the construction above: Construct a tangent to $\mathcal{C}$ from P', per Prop. III.17, and let T be the point of tangency. Drop a perpendicular from T to OP'. Let the foot of that perpendicular be P. It is an easy exercise to show that $f(P') = P$.

Our first lemma is a straightforward application of the definition. We leave the proof as an exercise.

Lemma 5.14. Let f be the inversion determined by a circle $\mathcal{C}$ with center O. Then

(1) for all points $P \in \mathcal{C}$, $f(P) = P$.

(2) If P is interior to $\mathcal{C}$, then $f(P)$ is exterior to $\mathcal{C}$.

(3) For all $P \in \mathcal{E} \setminus O$, $f(f(P)) = P$.

(4) f is bijective.

Item (3) says that circular inversion is its own inverse. Any such mapping is an **involution**. A more familiar example of an involution is Euclidean reflection. In fact, it may be helpful to think of circular inversion as reflection through a circle.

We consider certain distinguished sets of points in $\mathcal{E} \setminus O$ and what happens to them under circular inversion. Note that a line or a circle in $\mathcal{E}$ that does not go through O is respectively a line or a circle in $\mathcal{E} \setminus O$. A line or a circle in $\mathcal{E}$ that does go through O corresponds respectively to a punctured line or a punctured circle in $\mathcal{E} \setminus O$.

Lemma 5.15. Let f be the inversion determined by a circle $\mathcal{C}$ with center O. If ℓ is a line in $\mathcal{E}$ through O, then $f(\ell \setminus O) = \ell \setminus O$.

Proof. Let f and O be as in the hypotheses. Given $P \in \ell \setminus O$, we have $f(P) \in \overrightarrow{OP} \subset \ell \setminus O$ which implies $f(\ell \setminus O) \subseteq \ell \setminus O$. Since f is bijective, this is sufficient to prove the lemma. □

Observation 5.16. We consider what happens to the vertices of a triangle under circular inversion when one of them is the center of the inversion.

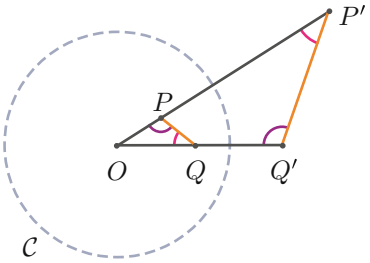


Figure 5.26: A triangle with one vertex at O under inversion

Let O, P, Q be noncollinear points in the plane, O the center of an inversion f. Let $f(P) = P'$, and $f(Q) = Q'$. Since $P' \in \overline{OP}$ and $Q' \in \overline{OQ}$, it is clear that O, P', Q' are noncollinear. What is the relationship between ΔOPQ and $\Delta OP'Q'$?

Since $|OP||OP'| = |OQ||OQ'|$, we have $|OP|/|OQ| = |OQ'|/|OP'|$. Note also that $\angle POQ \cong \angle Q'OP'$. By Prop. VI.6, $\Delta OPQ \sim \Delta OQ'P'$.

Finally, notice that our discussion here is about the vertices of a triangle, not about its sides. Since two of the sides of the original triangle include the point O, we will see below that the punctured sides are mapped to rays. In particular, $OP \setminus O$ is mapped to $\overrightarrow{P'T'}$, where $O * P' * T'$. We will also find that as a line segment, the third side is mapped to an arc of a punctured circle.

Polar and Rectangular Coordinates

As promised, we now start employing coordinates to facilitate our study of circular inversions. We define a given coordinate system with respect to a specific circular inversion.

Let O be the center of a circular inversion f. Fix a ray $\overrightarrow{OA}$ in $\mathcal{E}$ and call this the **polar axis of $\mathcal{E}$ with origin** O. We associate to every point $P \in \mathcal{E}$ an ordered pair of numbers, its **polar coordinates** $\mathrm{Pol}(r, \theta)$, where $r = |OP|$ and θ is the angle formed by $\overrightarrow{OP}$ and the polar axis. (θ is positive if measured in the counterclockwise direction, negative if measured in the clockwise direction.)

Notice that $r = 0$ corresponds to the point O in $\mathcal{E}$. For points in the punctured plane, $r > 0$.

This polar coordinate system allows us to describe f as the mapping

$$\text{Pol}(r, \theta) \mapsto \text{Pol}\left(\frac{a^2}{r}, \theta\right),$$

where a is the radius of inversion. Though r cannot equal zero, it can be close to zero, in which case a^2/r is a large positive number.

Identifying the polar axis with the positive x-axis, we get an associated *rectangular coordinate system* for $\mathcal{E}$: the positive y-axis is at an angle $\pi/2$ measured counterclockwise from the positive x-axis. The negative x-axis is at an angle $\pi/2$ measured counterclockwise from the positive y-axis, etc. The ordered pair (x, y) associated to a point $P \in \mathcal{C}$ are its **Cartesian** or **rectangular coordinates**. When we think of the points of $\mathcal{E}$ as having rectangular coordinates, we call $\mathcal{E}$ the **Cartesian plane** or the xy-plane.

Recall the conversion equations that allow us to toggle between Cartesian and polar coordinates:

$$r^2 = x^2 + y^2, \; x = r\cos\theta, \; y = r\sin\theta.$$

We test drive our coordinate systems by using them to understand the effect of an inversion f with center O and radius a on the line ℓ given in rectangular coordinates by $x = a$. Notice that ℓ is tangent to the circle of inversion $\mathcal{C}$ at the point $(a, 0)$. In polar coordinates, ℓ is given by

$$r\cos\theta = a, \quad \text{or } r = a\sec\theta, \; -\pi/2 < \theta < \pi/2.$$

Under inversion, ℓ thus maps to the set of points that satisfy polar equations

$$\frac{a^2}{r} = a\sec\theta, \quad \text{or } r = a\cos\theta, \; -\pi/2 < \theta < \pi/2.$$

Notice that $r \neq 0$.

We convert $r = a\cos\theta$ to rectangular equations, squaring both sides to get

$$r^2 = a^2\cos^2\theta = ra\cos\theta \;\text{ so that }\; x^2 + y^2 = ax,$$

an equation for a circle in the xy-plane. Completing the square we convert to

$$\left(x - \frac{a}{2}\right)^2 + y^2 = \frac{a^2}{4}. \tag{5.2}$$

This is an equation for the circle $\mathcal{C}'$, which has center $\left(\frac{a}{2}, 0\right)$ and radius $\frac{a}{2}$. Notice that $\mathcal{C}'$ is tangent to $\mathcal{C}$ at $(a, 0)$ and that the remaining points of $\mathcal{C}'$ are interior to $\mathcal{C}$. Indeed, Equation (5.2) describes a circle that goes through the origin in $\mathcal{E}$, but in the conversion to rectangular coordinates, we notice that θ strictly between $-\pi/2$ and $\pi/2$ means to $r \neq 0$ so that $(x, y) = (r\cos\theta, r\sin\theta) \neq (0, 0)$. This establishes that $f(\ell)$ is a punctured circle.

It turns out that the way f works on ℓ is a clue to the story of how circular inversion works on arbitrary lines.

Theorem 5.17. Let f be the inversion determined by a circle $\mathcal{C}$ with center O and radius a. The image of a line in $\mathcal{E} \setminus O$ under f is a punctured circle.

Proof. Let ℓ be a line in $\mathcal{E}$ not containing O and let A be the foot of the perpendicular from O to ℓ. Take $\overrightarrow{OA}$ to be the polar axis of $\mathcal{E}$ with origin at O. The equation for ℓ then has the form $x = b$ (rectangular) or $r\cos\theta = b$ (polar),

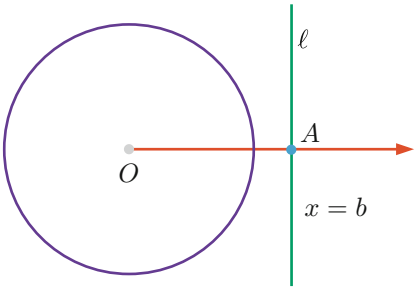


Figure 5.27: A line in $\mathcal{E} \setminus O$ determines a polar axis, $\overrightarrow{OA}$

where $b = |OA|$ or $b = -|OA|$. Notice again that $-\pi/2 < \theta < \pi/2$.

Inverting through $\mathcal{C}$ with radius a, we map ℓ to the set of points given in polar coordinates by

$$\left(\frac{a^2}{r}\right)\cos\theta = b,\ -\pi/2 < \theta < \pi/2$$

or

$$r = \left(\frac{a^2}{b}\right)\cos\theta,\ -\pi/2 < \theta < \pi/2.$$

Converting back to rectangular coordinates, we have $x^2 + y^2 = \left(\frac{a^2}{b}\right)x$, an equation for a circle. To see what its center and radius are we rearrange and complete the square to get

$$\left(x - \frac{a^2}{2b}\right)^2 + y^2 = \left(\frac{a^2}{2b}\right)^2.$$

This is an equation for the circle with center $\left(\frac{a^2}{2b}, 0\right)$ and radius $\frac{a^2}{2b}$. Since $(x, y) \neq (0, 0)$, ℓ is mapped to a punctured circle as claimed. □

Since inversion is involutive, the next result is immediate.

Theorem 5.18. Let f be circular inversion with center O. Let $\mathcal{C}'$ be a circle in $\mathcal{E}$ that passes through O. The image of $\mathcal{C}' \setminus O$ under f is a line.

Enjoying some poetic license, we think of circular inversion as swapping the "endpoints" of a line in $\mathcal{E} \setminus O$ with O.

Theorem 5.19. Let f be the inversion determined by a circle $\mathcal{C}$ with center O. If $\mathcal{C}'$ is a circle in $\mathcal{E} \setminus O$, then $f(\mathcal{C}')$ is a circle in $\mathcal{E} \setminus O$.

Proof. We continue to use a polar coordinate system on $\mathcal{E}$ based on a ray emanating from O, this time through the center of $\mathcal{C}'$. Since $\mathcal{C}'$ does not go through O, it can be described in rectangular coordinates as the set of points (x, y), that satisfy

$$x^2 + y^2 - \alpha x - \beta y + \gamma = 0, \text{ where } \gamma \neq 0, \text{ and } \frac{\alpha^2 + \beta^2}{4} - \gamma > 0, \tag{5.3}$$

for some $\alpha, \beta \geq 0$. (We leave the verification of that as an exercise.) Converting to polar coordinates we get

$$r^2 - \alpha r \cos\theta - \beta r \sin\theta + \gamma = 0,$$

which maps to the object given by

$$\frac{a^4}{r^2} - \alpha\left(\frac{a^2}{r}\right)\cos\theta - \beta\left(\frac{a^2}{r}\right)\sin\theta + \gamma = 0.$$

Multiply through by r^2/γ to get

$$\frac{a^4}{\gamma} - \left(\frac{\alpha a^2}{\gamma}\right) r\cos\theta - \left(\frac{\beta a^2}{\gamma}\right) r\sin\theta + r^2 = 0.$$

Letting $\alpha' = \alpha a^2/\gamma$, $\beta' = \beta a^2/\gamma$, $\gamma' = a^4/\gamma$, and converting back to rectangular coordinates we get

$$\gamma' - \alpha' x - \beta' y + x^2 + y^2 = 0, \text{ where } \gamma' \neq 0.$$

We leave it as an exercise to verify that $\frac{\alpha'^2+\beta'^2}{4} - \gamma' > 0$. Comparing to Equation (5.3), we see this is an equation for another circle that does not go through O, which proves the theorem. □

Now in $\mathcal{E}\setminus O$ we have four types of objects of interest: lines, circles, punctured lines, and punctured circles. Following [23], we call these ***k*-sets**. We have proved the following theorem.

Theorem 5.20. Inversion maps k-sets to k-sets.

Orthogonal Circles

Intersecting circles are **orthogonal** if their tangents at the points of intersection are orthogonal. We leave the proof of the next lemma as an exercise.

Lemma 5.21. (1) The tangent of one circle at the point of intersection of a pair of orthogonal circles passes through the center of the other circle.

(2) Circles are orthogonal if and only if a tangent to one from the center of the other passes through a point of intersection of the circles.

(3) Orthogonal circles intersect in two points.

(4) If $\mathcal{C}$ and $\mathcal{C}'$ are orthogonal circles, the center of $\mathcal{C}$ is exterior to $\mathcal{C}'$.

(5) If one circle passes through the center of a second circle, the circles cannot be orthogonal.

If a mapping f sends each point in $\mathcal{R}$ to a point in $\mathcal{R}$, that is, if $f(\mathcal{R}) \subseteq \mathcal{R}$, we say f **fixes $\mathcal{R}$ as a set**.

The role of orthogonal circles in the story of circular inversion is analogous to the role of orthogonal lines in the story of Euclidean reflection through a line. As reflection through a line ℓ fixes lines orthogonal to ℓ, we will see that inversion through a circle $\mathcal{C}$ fixes circles orthogonal to $\mathcal{C}$, as well as the interiors and exteriors of circles orthogonal to $\mathcal{C}$.

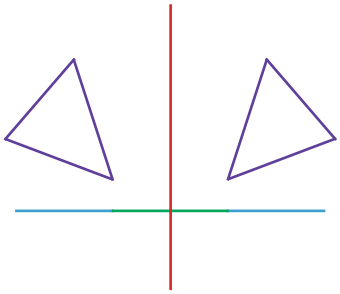


Figure 5.28: Euclidean reflection fixes orthogonal lines

Definition 5.22. The **open disk** defined by a circle $\mathcal{C}$ with center O and radius a is the collection of all points X interior to $\mathcal{C}$, that is, with $|OX| < a$.

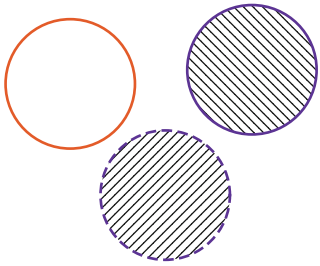


Figure 5.29: Circle, open disk, closed disk

The **closed disk** determined by the circle is the collection of points X with $|OX| \leq a$, that is, the union of the open disk with the circle itself.

Theorem 5.23. Inversion with respect to a given circle leaves any orthogonal circle fixed as a set and it leaves the open disk determined by any orthogonal circle fixed as a set.

Proof. Let $\mathcal{C}$ be a circle with center O and radius a, and let f be inversion with respect to $\mathcal{C}$. Let circle $\mathcal{C}'$ with center O' and radius a' be orthogonal to $\mathcal{C}$. Let P and Q be the points of intersection of the two circles so that $f(P) = P$ and $f(Q) = Q$. Let P' be a third point on $\mathcal{C}'$. Notice that P' cannot be on PQ by Prop. III.2. Say that $\overrightarrow{OP'}$ passes through $\mathcal{C}'$ at Q'. Applying the power of a point to O with respect to $\mathcal{C}'$, we have

$$|OP'||OQ'| = |OP||OQ| = a^2,$$

which implies $f(Q') = P'$. Since f maps circles in $\mathcal{E} \setminus O$ to circles in $\mathcal{E} \setminus O$ with $f(P) = P$, $f(Q) = Q$, and $f(Q') = P'$, and since three noncollinear points determine a circle uniquely, it follows that $f(\mathcal{C}') = \mathcal{C}'$.

Now suppose R is a point interior to $\mathcal{C}'$. Form $\overrightarrow{OR}$ and say it intersects $\mathcal{C}'$ at points T_1' and T_2'. As we have seen, $f(T_1') = T_2'$. Since $f(R)$ is on $\overrightarrow{OR}$, it is sufficient to show that $f(R)$ is between $f(T_1') = T_2'$ and $f(T_2') = T_1'$ to establish that points interior to $\mathcal{C}'$ are inverted to points interior to $\mathcal{C}'$.

Since $T_1' * R * T_2'$ on $\overrightarrow{OR}$, without loss of generality we may assume

$$|OT_1'| < |OR| < |OT_2'|.$$

Multiply the inequality by $|Of(R)|$ to get

$$|Of(R)||OT_1'| < |Of(R)||OR| < |Of(R)||OT_2'|. \tag{5.4}$$

Now

$$|Of(R)||OR| = a^2 = |OT_1'||OT_2'|,$$

so dividing the first two expressions in (5.4) by $|OT_1'|$, we get $|Of(R)| < |OT_2'|$. Dividing the second two expressions in (5.4) by $|OT_2'|$ we get $|OT_1'| < |Of(R)|$. These two taken together imply $T_1' * f(R) * T_2'$, which completes the argument that f leaves the interior of $\mathcal{C}'$ fixed as a set. □

The next result highlights the connections between orthogonal circles, inversion, and the power of a point.

Theorem 5.24. A circle passes through both a point and its inverse with respect to another circle if and only if the two circles are orthogonal.

Proof. Let $\mathcal{C}$ be a circle with center O and radius a. Let $\mathcal{C}'$ be a circle passing through P, a point different from O. Say the inverse of P with respect to $\mathcal{C}$ is P' and suppose $P' \in \mathcal{C}'$. Let T be a point of tangency from O to $\mathcal{C}'$. Applying the power of a point to O relative to $\mathcal{C}'$, we have $|OT|^2 = |OP||OP'| = a^2$, giving us $|OT| = a$. It follows that T is on $\mathcal{C}$, making it a point of intersection of $\mathcal{C}$ and $\mathcal{C}'$. By Lemma 5.21, $\mathcal{C}$ and $\mathcal{C}'$ must then be orthogonal.

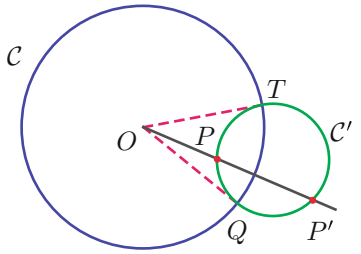


Figure 5.30: Circle passing through a point and its inverse

Next suppose $\mathcal{C}$ and $\mathcal{C}'$ are orthogonal, with points of intersection T and Q. Take a third point P on $\mathcal{C}'$. We must verify that the inverse of P with respect to $\mathcal{C}$ is on $\mathcal{C}'$. Since OP is not tangent to $\mathcal{C}'$, $\overline{OP}$ passes through $\mathcal{C}'$ twice. Let P' be the second point of intersection. Applying the power of a point to O with respect to $\mathcal{C}'$, we get $|OT|^2 = |OP||OP'|$. Since $|OT|^2 = a^2$, P' is the inverse of P with respect to $\mathcal{C}$. □

A fact we will not prove is that circular inversion is **conformal**, that is, that it maps angles to congruent angles. What makes that a tricky business is that it holds for angles between arbitrary curves. We consider a less general, nonetheless useful and informative result.

Theorem 5.25. An angle formed by two k-sets in $\mathcal{E} \setminus O$ is mapped to a congruent angle under any inversion with center O.

Proof. It is easy to check that there are nine different k-set pairs that can form an angle made by two k-sets in the punctured plane. (The case of two punctured lines is not actually a case to consider because in $\mathcal{E} \setminus O$, two punctured lines do not intersect.)

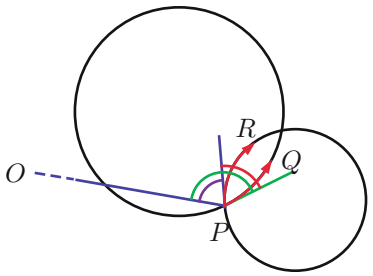


Figure 5.31: An arbitrary angle formed by k-sets

Say the type of an angle is determined by the types of k-sets that form its legs. For example, one type of angle is formed by a punctured circle and a punctured line. Under inversion, that type is mapped to an angle of a different

type, viz., one formed by a line and a punctured line. We call these two types of angle an *inversive pair*. We leave it as an exercise to work out what and how many inversive pairs there are. Notice that the involutive property of inversion implies that we need only show that the theorem is true for one type of angle. This cuts down on the number of cases we need consider to prove the theorem.

Notice next that an angle formed, for example, by two circles in $\mathcal{E} \setminus O$ can be viewed as the difference of an angle formed by a punctured line and one circle, and the same punctured line and the second circle, as shown in Fig. 5.31: $\angle QPR = \angle OPQ - \angle OPR$.

Similarly, if two punctured circles share a point $P \in \mathcal{E} \setminus O$, then the angle formed by the circles at P is a sum of angles, one between the punctured line through P and the first punctured circle, the second between the punctured line through P and the second punctured circle. Indeed, we claim that every angle formed by two k-sets in $\mathcal{E} \setminus O$ is either the inverse of, or a sum or difference of, angles of two types: those formed by a punctured circle and a punctured line, and those formed by a circle and a punctured line. We leave the final disposition of that claim as an exercise and go on to prove that angles formed in $\mathcal{E} \setminus O$ by a punctured circle and a punctured line, or by a circle and a punctured line, are preserved under inversion. In light of our claim, this is sufficient to prove the theorem.

Let O be the center of an inversion on $\mathcal{E} \setminus O$.

First we consider an angle with one leg on an arc of a punctured circle and the second leg on a punctured line in $\mathcal{E} \setminus O$.

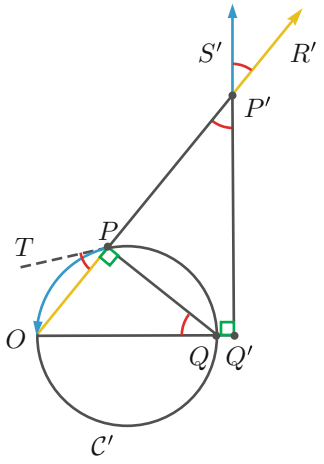


Figure 5.32: The angle between a punctured circle and a punctured line and its inverse

Let $\mathcal{C}'$ be a circle and let ℓ be a line, both passing through O in $\mathcal{E}$. Suppose ℓ also passes through $P \in \mathcal{C}'$. We consider the angle with vertex P and legs $\overset{\frown}{OP}$ (the arc), and OP (the chord). Note that this is actually the angle formed at P by the tangent to $\mathcal{C}'$ and the chord OP. Let $\overrightarrow{PT}$ be tangent to $\mathcal{C}'$ at P. The angle we care about is then $\angle OPT$, which is illustrated in Fig. 5.32.

Let OQ be the diameter of $\mathcal{C}'$ through O. Let P', Q' be the inverses of P and Q. We leave it as an exercise to show that the inverse of $OP \setminus O$ is $\overrightarrow{P'R'}$, and that the inverse of $\overset{\frown}{OP} \setminus O$ is $\overrightarrow{P'S'}$, both as indicated in Fig. 5.32. This means that $\angle OPT$ is mapped under the inversion to $\angle R'P'S'$. We must show then that $\angle OPT \cong \angle R'P'S'$.

By Prop. III.32, $\angle OPT \cong \angle OQP$. In Observation 5.16, we established that $\angle OQP \cong \angle OP'Q'$. This gives us $\angle OPT \cong \angle OP'Q'$. By vertical angles, $\angle OP'Q' \cong \angle R'P'S'$ giving us $\angle OPT \cong \angle R'P'S'$, as desired.

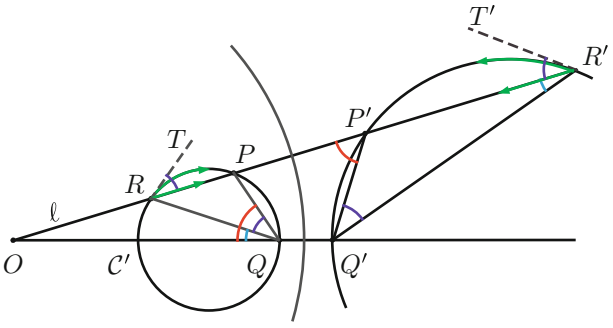


Figure 5.33: The angle between a circle and a punctured line and its inverse

Next we consider an angle formed by an arc of a circle $\mathcal{C}'$ and a punctured line $\ell \setminus O$ in $\mathcal{E} \setminus O$ as in Fig. 5.33. Let $\ell \setminus O$ pass through $\mathcal{C}'$ at R and P and suppose the angle in question is $\angle PRT$, where $\overrightarrow{RT}$ is tangent to $\mathcal{C}'$ at R. Let $\overrightarrow{OQ}$ run through the center of $\mathcal{C}'$. We can take Q on $\mathcal{C}'$. Let P', Q', R', T' be the inverses of P, Q, R, T respectively. We must show that $\angle PRT \cong \angle P'R'T'$.

By Prop. III.32, $\angle PRT \cong \angle PQR$, which itself is the difference between $\angle PQO$ and $\angle RQO$. Now we can apply Observation 5.16 to ΔPQO and ΔRQO to get $\angle PQO \cong \angle Q'P'O$ and $\angle RQO \cong \angle Q'R'O$. This gives us

$$\angle PRT \cong \angle PQR \cong \angle PQO - \angle RQO \cong \angle Q'P'O - \angle Q'R'O.$$

Notice now that $\angle Q'P'O$ is exterior to $\Delta P'Q'R'$ and that $\angle Q'R'O = \angle Q'R'P'$. Proposition I.32 then gives us $\angle Q'P'O - \angle Q'R'O \cong \angle P'Q'R'$, so that $\angle PRT \cong \angle P'Q'R'$. Another application of Prop. III.32 gives us $\angle P'Q'R' \cong \angle P'R'T'$, as was to be shown. □

Exercises 5.3. 1. Let $f : \mathcal{S} \to \mathcal{T}$ be a mapping of sets.

(a) Show that if $A \subseteq B \subseteq \mathcal{S}$ then $f(A) \subseteq f(B) \subseteq \mathcal{T}$.

(b) Show that if $A \cap B$ is nonempty, then $f(A) \cap f(B)$ is nonempty.

(c) Show that any mapping is onto its range.

2. Referring to Observation 5.13 (2), show that $f(P') = P$.

3. Prove Lemma 5.14.

4. This problem refers to the proof of Theorem 5.19. Verify that a circle in $\mathcal{E} \setminus O$ can be described by an equation of the form given in Equation (5.3). Verify the necessity of the conditions given on α, β, and γ. Verify that if α, β, and γ satisfy the requisite conditions, so do α', β', and γ'.

5. Show that a punctured circle $\mathcal{C}'$ in $\mathcal{E} \setminus O$ is mapped by inversion with center O to a line parallel to $\ell \setminus O$, where ℓ is the tangent to $\mathcal{C}'$ through O in $\mathcal{E}$.

6. Prove Lemma 5.21.

7. Let f be inversion through the circle $\mathcal{C}$ with center O. Let P be a point interior to $\mathcal{C}$. Erect the perpendicular to $\overline{OP}$ at P and let T be a point on $\mathcal{C}$ where that perpendicular intersects $\mathcal{C}$. Show that the circle centered at $P' = f(P)$ with radius $P'T$ is orthogonal to $\mathcal{C}$.

8. What is the difference between Theorem 5.23 and the statement that if $\mathcal{C}$ and $\mathcal{C}'$ are orthogonal, then inversion through $\mathcal{C}$ leaves the closed disk associated to $\mathcal{C}'$ fixed?

9. Show that two circles are orthogonal if and only if the sum of the squares of their radii is the square of the distance between their centers.

10. Consider a circle $\mathcal{C}'$ in $\mathcal{E}$ given by $x^2 + y^2 - \alpha x + \gamma = 0$, where $\alpha > 0$ and $\alpha^2/4 > \gamma$. Show that $\mathcal{C}'$ is orthogonal to $\mathcal{C}$ given by $x^2 + y^2 = a^2$ if and only if $\gamma = a^2$.

11. (a) List all the inversive pairs of angles formed by k-sets in $\mathcal{E} \setminus O$. (See the proof of Theorem 5.25.)

(b) How many different inversive pairs are there? Include the ones like punctured circle-line that are paired with themselves.

(c) Argue that each inversive pair has an angle type that is a sum or difference of angles formed by a punctured line and a punctured circle, or by a circle and a punctured line.

12. Referring to Fig. 5.32, let P, P' be inverses under f, a circular inversion with center O. Show that f maps $OP \setminus O$ to $\overrightarrow{P'R'}$ and $\overset{\frown}{OP} \setminus O$ to $\overrightarrow{P'S'}$, where R' and S' are as shown in the figure.

5.4 Triangles, Centers, and Circles

There are several points that can be considered the center of a triangle. These points are also centers for certain circles associated to the triangle, among them the nine point circle. In this section, we explore some of the ideas that lead into and flow out of the construction of the nine point circle. We also get a glimpse of how inversion can be brought to bear in a proof of Feuerbach's Theorem, a gem from the early nineteenth century that illuminates a beautiful relationship among some of the circles associated to a triangle. Much of this material is from [5].

We first ran across cevians in Chapter 2. These special line segments are named for the seventeenth century mathematician, Giovanni Ceva. Though Ceva is credited with the theorem below, like so much else that we run across in geometry since the Greek era, it seems to go back to the Arabic period of the eleventh century.

Recall (Definition 2.26) that a **cevian** is a line segment from a vertex of a triangle to the opposite side. By Prop. VI.1, the areas of the triangles created by a cevian and the lengths of the segments of the side it intersects are proportional.

The proof we have here of Ceva's Theorem uses the following fact, which is simple to prove, but not necessarily something you would know off-hand:

$$\frac{x}{y} = \frac{a}{b} \implies \frac{(x-a)}{(y-b)} = \frac{x}{y}. \tag{5.5}$$

(Proposition V.12 is a related result.)

Theorem 5.26 (Ceva's Theorem). Given a triangle ΔABC, and three cevians AX, BY, CZ, the cevians are concurrent if and only if

$$\frac{|BX|}{|CX|}\frac{|CY|}{|AY|}\frac{|AZ|}{|BZ|} = 1.$$

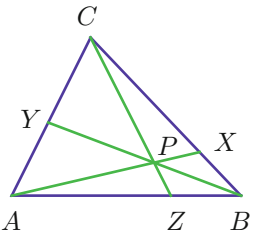


Figure 5.34: Concurrent cevians

Proof. Suppose cevians AX, BY, CZ are concurrent at P. Referring to Fig. 5.34, invoking Prop. VI.1, and employing (5.5), we have

$$\frac{|BX|}{|CX|} = \frac{|\Delta ABX|}{|\Delta ACX|} = \frac{|\Delta BPX|}{|\Delta CPX|} = \frac{|\Delta ABX| - |\Delta BPX|}{|\Delta ACX| - |\Delta CPX|} = \frac{|\Delta ABP|}{|\Delta ACP|}.$$

The same logic applies to give us

$$\frac{|CY|}{|AY|} = \frac{|\Delta BCP|}{|\Delta ABP|} \text{ and } \frac{|AZ|}{|BZ|} = \frac{|\Delta ACP|}{|\Delta BCP|}$$

so that

$$\frac{|BX|}{|CX|}\frac{|CY|}{|AY|}\frac{|AZ|}{|BZ|} = \frac{|\Delta ABP|}{|\Delta ACP|}\frac{|\Delta BCP|}{|\Delta ABP|}\frac{|\Delta ACP|}{|\Delta BCP|} = 1.$$

Conversely, suppose cevians AX, BY, CZ are given with $\frac{|BX|}{|CX|}\frac{|CY|}{|AY|}\frac{|AZ|}{|BZ|} = 1$. Let P be the point where AX and BY intersect and let Z' be the point on AB where the cevian from C through P intersects. We just showed that

$$\frac{|BX|}{|CX|}\frac{|CY|}{|AY|}\frac{|AZ'|}{|BZ'|} = 1$$

implying that $\frac{|AZ'|}{|BZ'|} = \frac{|AZ|}{|BZ|}$. In other words, Z and Z' cut the segment AB into identical ratios, thus must be the same point. □

Ceva's Theorem is still true if we define the sides of a triangle to be lines rather than line segments.

Where Ceva's Theorem is about points of concurrency of lines associated to a triangle, Menelaus' Theorem is about collinearity of points associated to a triangle.

Theorem 5.27 (Menelaus' Theorem). If ΔABC is a triangle and X, Y, Z are points, respectively, on $\overline{AC}$, $\overline{BC}$, and $\overline{AB}$, then X, Y, Z are collinear if and only if

$$\frac{|AZ|}{|BZ|}\frac{|BY|}{|CY|}\frac{|CX|}{|AX|} = 1.$$

Proof. Assume X, Y, Z are collinear and call the line they lie on ℓ. Drop perpendiculars from A, B, C to ℓ and say the feet of the perpendiculars are, respectively, L, M, N. Because of all the similar triangles, we can say, for instance that $|AZ|/|BZ| = |AL|/|BN|$. We leave it as an exercise to follow through on the proof from here. □

Definition 5.28. The **medians** of a triangle are the cevians determined by the midpoints of its sides.

Ceva's Theorem gives us the next result nearly immediately. We leave details to the reader.

Corollary 5.29. The medians of a triangle are concurrent.

Definition 5.30. The **centroid** of a triangle is the point of concurrency of its medians.

The centroid is the geometric balance point of the triangle. A classic calculus problem is to show that the centroid is the center of mass of a triangle of uniform mass density.

Looking at Fig. 5.35, we see that the medians of a triangle subdivide the figure into six triangles that enclose non-overlapping regions. We leave it as an exercise to show that these six triangles all have the same area. From there, it is easy to show the following.

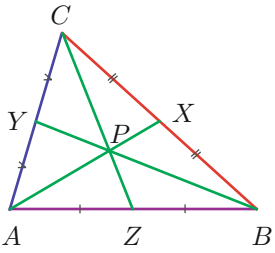


Figure 5.35: Medians are concurrent

Theorem 5.31. The centroid of a triangle is 2/3 the distance along a given median from its vertex to the opposite side.

Proof. Let P be the median of ΔABC, and let X be the midpoint of BC and Z be the midpoint of AB. By the remark preceding the theorem, we have

$$|\Delta PAZ| = |\Delta PBZ| = |\Delta PBX|,$$

where the triangles are as shown in Fig. 5.35. It follows that ΔABX has three times the area of, say, ΔPBX. By Prop. VI.1, $|\Delta ABX|/|\Delta PBX| = |AX|/|PX|$, so

$$|AX| = |AP| + |PX| = 3|PX|$$

that is, $|AP| = 2|PX|$ as was to be shown. □

Definition 5.32. The **medial triangle** for ΔABC is the triangle with vertices at the midpoints of the sides of ΔABC.

We distinguish ΔABC from its medial triangle by calling ΔABC the **parent triangle**.

If ΔXYZ is the medial triangle for parent ΔABC, Prop. VI.2 says that $XY \| AB$, since XY cuts AC and BC proportionately. Likewise, $XZ \| AC$ and $YZ \| BC$. Indeed, joining midpoints of sides of a triangle produces several different pairs of similar triangles. (See Fig. 5.36.) We leave it as an exercise to show that the medial triangle is actually congruent to three other triangles, namely ΔAYZ, ΔBXZ, and ΔCXY.

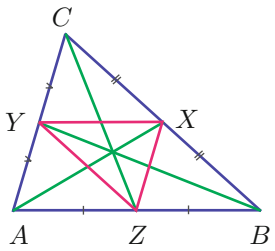


Figure 5.36: The medial triangle ΔXYZ

Figure 5.36 also shows a parallelogram at each vertex of the parent triangle, for example, $AZXY$. The median AX is a diagonal of $AZXY$, its other diagonal being YZ. Proof that the diagonals of a parallelogram bisect one another was Exercise 1.6.3. Using that result, we can say that the point where AX intersects YZ is the midpoint of YZ. It follows that AX is double the length of, and runs through, a median of the medial triangle ΔXYZ. Indeed, all the medians of ΔABC contain, and are twice the length of, the medians of ΔXYZ. We see then that the centroids of the medial triangle and its parent triangle coincide.

Definition 5.33. An **altitude** of a triangle is the perpendicular from a vertex to the line determined by the opposite side. A **perpendicular bisector** of a triangle is the perpendicular line positioned at the midpoint of a side of the triangle.

Altitudes are considered cevians. Perpendicular bisectors are not.

Using the definition of the medial triangle and properties of cevians, we can show that the altitudes of a medial triangle are the perpendicular bisectors of the parent triangle. We leave the details as an exercise.

Since the sides of a triangle are chords of its circumcircle, Corollary 5.2 implies that the perpendicular bisectors of a triangle intersect at its circumcenter. This observation establishes that the altitudes of a medial triangle meet at the circumcenter of the parent triangle. (See Fig. 5.37.) In general, the altitudes of a triangle are concurrent. (We leave that as an exercise!)

Definition 5.34. The **orthocenter** of a triangle is the point of concurrency of its altitudes.

It is easy to check that if the centroid and circumcenter of a triangle coincide, the triangle is equilateral. (The verification is yet another exercise!) If the centroid and the circumcenter are distinct, they determine the so-called **Euler line** of the triangle, named for Leonhard Euler (1707–1783), the Swiss mathematician who dominated European mathematics during the eighteenth century. Often cited as the most prolific mathematician in history, Euler made significant contributions to geometry, trigonometry, number theory, and analysis.

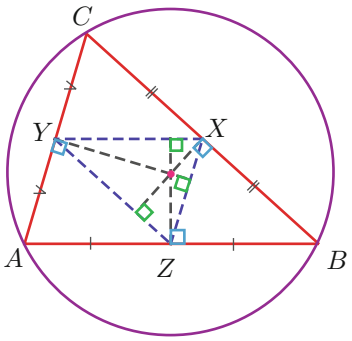


Figure 5.37: The orthocenter of the medial triangle is the circumcenter of the parent triangle

The Euler line of a triangle passes through several interesting points, aside from the centroid and circumcenter.

Theorem 5.35. The orthocenter of a triangle is on its Euler line.

Proof. If the triangle is equilateral, the theorem is vacuous so assume the triangle ΔABC is not equilateral, thus, that the centroid P and the circumcenter O determine a line. Let Z be the midpoint of AB, as in Fig. 5.38. Take H on

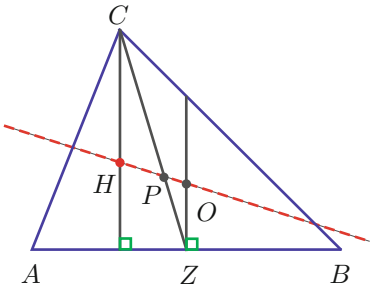


Figure 5.38: The Euler line

$\overline{OP}$, on the C side of $\overline{OZ}$, so that $|PH| = 2|OP|$ and recall that $|CP| = 2|PZ|$. By vertical angles at P, we can invoke Prop. VI.6 to get $\Delta CPH \sim \Delta ZPO$. This gives us $\angle HCP \cong \angle OZP$, so that, by Prop. I.27, $\overline{CH} \| \overline{OZ}$. This shows that $\overline{CH}$ is actually the altitude from C. Since we can turn the triangle around and do the same trick with either of the other vertices and the midpoint of the opposite side, we realize that all the altitudes must pass through H, in other words, that H is the orthocenter. □

The centroid of a triangle is always interior to the triangle, but the orthocenter and the circumcenter need not be.

Definition 5.36. The **feet of a triangle** are the points where the altitudes intersect the lines determined by the sides of the triangle. The feet of a given triangle are the vertices for the associated **orthic triangle**. The **orthic circle** associated to a given triangle is the circumcircle for its orthic triangle.

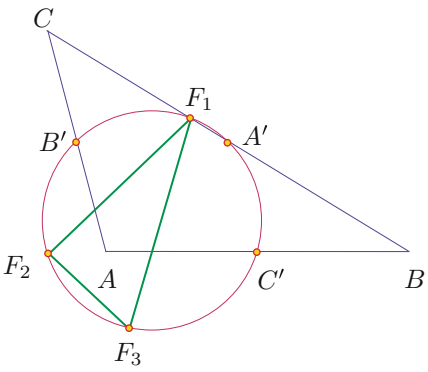


Figure 5.39: The orthic triangle, orthic circle, and midpoints of sides of ΔABC

Theorem 5.37 (Nine Point Circle Theorem)**.** The orthic circle of a triangle passes through the midpoints of the sides of the triangle as well as the midpoints of the segments joining the vertices to the orthocenter of the triangle.

Proof Let ΔABC be given and let A' be the midpoint of BC, B' be the midpoint of AC, and C' be the midpoint of AB. Let H be the orthocenter and let A'' be the midpoint of AH, B'' the midpoint of BH, and C'' the midpoint of CH. Since $B'C'$ cuts sides AB and AC in half, Prop. VI.2 implies $B'C' \| BC$. Next consider ΔBHC. Since B'' and C'' cut BH and CH in half, $B''C'' \| BC$ as well. Considering ΔAHC, we see that since B' cuts AC in half and C'' cuts CH in half, $B'C'' \| AH$. Similarly, $C'B'' \| AH$. This gives us a parallelogram, $C'B'C''B''$, as in Fig. 5.40.

Since $C'B''$ and $B'C''$ are both parallel to AH which is perpendicular to BC, which is parallel to $B''C''$, we see that $C'B'C''B''$ is actually a rectangle, thus, $B''C' \cong B'C''$.

Next, we turn our attention to the quadrilateral $A''B''A'B'$. An argument similar to the one just advanced reveals that this is a rectangle as well. It shares a diameter with $C'B'C''B''$, viz., $B'B''$. The three diameters $A'A''$, $B'B''$, $C'C''$ are concurrent so are diameters of a circle with center at this point of concurrency. Since this circle passes through the midpoints of the sides of

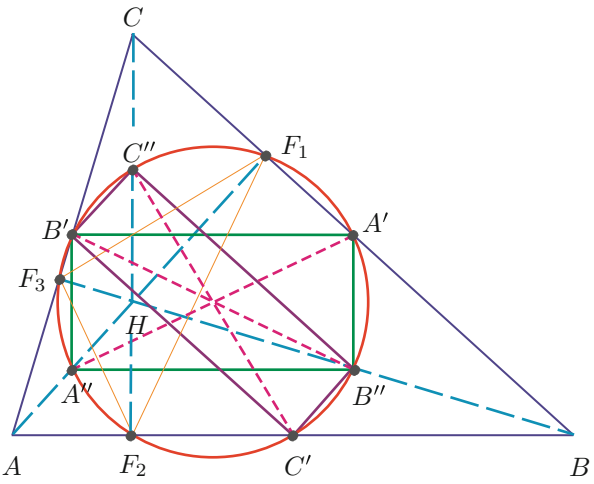


Figure 5.40: The Nine Point Circle Theorem

ΔABC, and the midpoints of the segment from each vertex to H, we just have to show that it goes through the feet of the altitudes.

Let F_1 be the foot of the altitude from A to BC. Since $\angle A''F_1A'$ is right, F_1 must lie on the circle with diameter $A''A'$. The same argument applies to the other two feet, F_2, F_3. This completes the proof. □

The Nine Point Circle Theorem guarantees that the orthic circle of a triangle passes through nine distinguished points. The orthic circle is thus more typically referred to as the **nine point circle**.

We have considered perpendicular bisectors of sides of a triangle, which intersect at the circumcenter, and altitudes of a triangle, which intersect at the orthocenter. We have not yet considered the angle bisectors in a triangle.

We start with a lemma that has more general applications. Here we use the notion of the **distance from a point to a line**, which is the length of the perpendicular line segment from the point to the line.

Lemma 5.38 (1) An angle bisector is equidistant from the legs of the angle it bisects. (2) Angle bisectors are perpendicular if and only if the underlying angles are supplements.

Proof Let h be the bisector for $\angle QOQ'$. Choose any point P on h. Form the perpendiculars from P to the legs of the angle. We may assume that the feet are Q and Q'. Since we have AAS for $\Delta QOP \cong \Delta Q'OP$, $PQ \cong PQ'$. This proves (1).

Take R on $\overline{OQ}$ so that $R*O*Q$. Note that $\angle ROQ'$ and $\angle Q'OQ$ are supplements. Let k bisect $\angle ROQ'$. We wish to show that h and k are perpendicular.

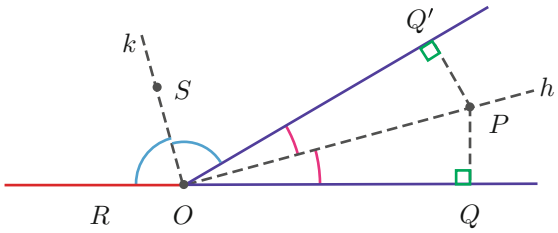


Figure 5.41: Angle bisectors for supplements

Take S on k. We have

$$2\angle SOQ' + 2\angle POQ' \cong \angle ROQ' + \angle Q'OQ = \pi,$$

as illustrated in Fig. 5.41. Then

$$\angle SOQ' + \angle POQ' = \angle(h,k) = \pi/2,$$

as desired.

Finally, suppose h is the bisector for $\angle QOQ'$, k is the bisector for $\angle Q'OR$, and that $\angle(h,k)$ is right. We wish to show that $\angle QOQ' + \angle Q'OR = \pi$. Let S be a point on k and P a point on h. We have

$$\pi/2 = \angle POS = \angle POQ' + \angle Q'OS$$

so

$$\pi \cong 2\angle POQ' + 2\angle Q'OS \cong \angle QOQ' + \angle Q'OR,$$

as desired. This completes the proof of (2). □

Theorem 5.39 The angle bisectors of a triangle are concurrent. The point of concurrency, I, is equidistant from the sides of the triangle. The perpendicular line segments from I to the sides of the triangle thus form radii of a circle. The sides of the triangle are tangents to that circle.

Proof Let ΔABC be given with angle bisectors at A and C. We leave it as an exercise to show that I, the point of intersection of the two angle bisectors, must be interior to the triangle.

Since I is on the bisector of $\angle BAC$, Lemma 5.38 implies that it is equidistant from AB and AC. Since I is on the bisector of $\angle ACB$, it is also equidistant from AC and BC. We conclude that I is equidistant from all three sides of ΔABC. Let r be the distance from I to any side of ΔABC.

Form the segment BI. We must show that BI is a bisector for $\angle ABC$. This will establish that the three angle bisectors of ΔABC are concurrent.

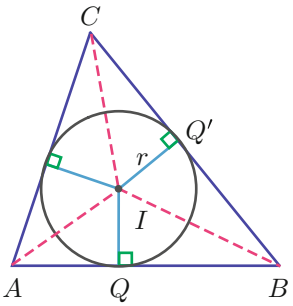


Figure 5.42: Proof that angle bisectors are concurrent

Let Q and Q' be the feet of the perpendiculars from I to AB and BC respectively, as in Fig. 5.42. Consider that ΔBIQ and $\Delta BIQ'$ are right triangles with two congruent sides: BI and $IQ \cong IQ'$. By the Pythagorean Theorem, $BQ \cong BQ'$, giving us SSS for $\Delta BIQ \cong \Delta BIQ'$. It follows that $\angle QBI \cong \angle Q'BI$, which verifies that BI bisects $\angle ABC$.

Finally, since the point of concurrency of the angle bisectors is equidistant from all three sides of the triangle, Props. III.16, 18, 19 imply that those sides are tangents for the circle with center I and radius r. □

Definition 5.40 The **incenter** I of a triangle ΔABC is the point of intersection of its angle bisectors. The **incircle** for ΔABC has center I and radius IQ, where Q is the foot of the perpendicular from I to AB.

Next we consider the **excircles** of a triangle. We need another lemma before we define or construct excircles.

Lemma 5.41 The angle bisector of any exterior angle in a triangle intersects every interior angle bisector.

Proof When the interior and exterior angles are at the same vertex, the vertex itself is the point of intersection of the bisectors. Consider then ΔABC with bisectors at $\angle BAC$ and at $\angle CBD$, where $A * B * D$. (See Fig. 5.43.) We established in Lemma 5.38 that the bisectors at $\angle ABC$ and $\angle CBD$ are perpendicular. Since $\angle BAC$ and $\angle ABC$ cannot be supplementary, Lemma 5.38 guarantees that their bisectors are not perpendicular. We conclude that the bisector of $\angle CBD$ cannot be parallel to the bisector of $\angle BAC$, which proves the result. □

Given a triangle ΔABC, let I_a be the point of intersection of the bisector of $\angle BAC$ and the bisector of $\angle CBD$, where $A * B * D$, as in Fig. 5.43. As a point on the bisector of $\angle BAC$, I_a is equidistant from $\overline{AB}$ and $\overline{AC}$. As a point on the bisector of $\angle CBD$, I_a is also equidistant from $\overline{BC}$ and $\overline{AB}$. We conclude that I_a is equidistant from all three sides of ΔABC, where we view the sides

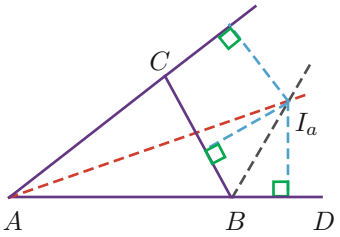


Figure 5.43: Bisectors of interior and exterior angles in a triangle

of the triangle as lines, instead of line segments. Let r be the distance from I_a to any of the sides of ΔABC. The **excircle** determined by $\angle BAC$ is the circle with center I_a and radius r.

The two other excircles are constructed analogously.

Like the incircle, an excircle is tangent to all three sides of the triangle. Though it may appear to be so for a particular triangle, the incircle does not necessarily share a point with any of the excircles. Feuerbach's Theorem, however, states a much more surprising result.

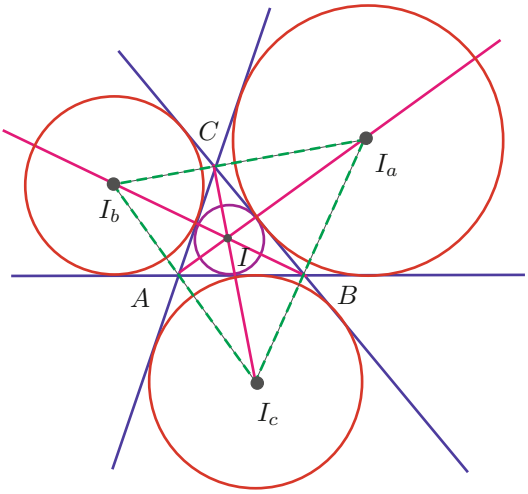


Figure 5.44: The incircle is not necessarily tangent to the excircles!

We provide a sketch of the proof of Feuerbach's Theorem, indicating how circular inversion can be applied. The diligent reader is encouraged to flesh out the details of the proof.

Theorem 5.42 (Feuerbach's Theorem)**.** The nine point circle is tangent to the incircle and the excircles.

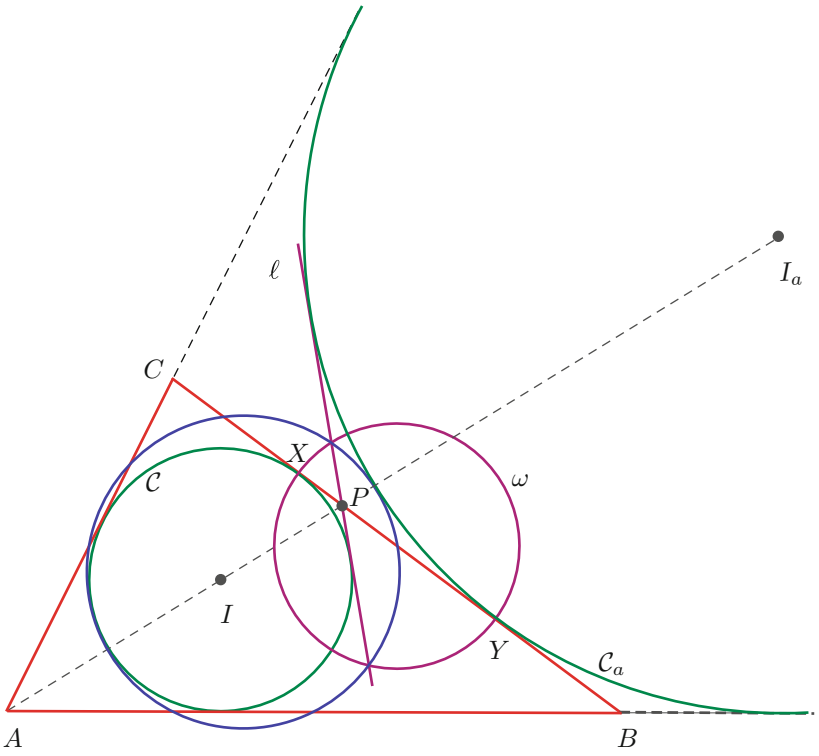


Figure 5.45: Proof of Feuerbach's Theorem

Sketch of proof Consider ΔABC with incircle $\mathcal{C}$ centered at I and one excircle, $\mathcal{C}_a$, centered at I_a, determined by $\angle BAC$, as in Fig. 5.45. The sides of the triangle form three of the four tangents common to $\mathcal{C}$ and $\mathcal{C}_a$. The fourth tangent, ℓ, passes through P, the internal center of similitude of $\mathcal{C}$ and $\mathcal{C}_a$.

Let X be the point on $\mathcal{C}$ where it is tangent to BC, and let Y be the point on $\mathcal{C}_a$ where it is tangent to BC. Take ω to be the circle with diameter XY. Since $\mathcal{C}$ and $\mathcal{C}_a$ are orthogonal to ω, inversion through ω leaves the incircle and the excircle fixed. The nine point circle, shown in blue, passes through the center of ω, so is mapped to a line under the inversion. Since it also passes through the points of intersection of ℓ and ω, the nine point circle must in fact be mapped to ℓ under inversion. Since ℓ is tangent to the circles $\mathcal{C}$, and $\mathcal{C}_a$, the nine point circle must be as well. □

Exercises 5.4 1. Show that if $x/y = a/b$ then $(x-a)/(y-b) = x/y$.

2. In the proof of Ceva's Theorem we refer to the point where two cevians in a triangle intersect. Why must two cevians intersect?

3. Does the proof of Ceva's Theorem, as given in this section, depend on the points X, Y, Z being on the sides of the triangle or will the proof, as written, hold when the points are on the lines determined by the sides of the triangle?

4. Finish the proof of Menelaus' Theorem. Include a sketch.

5. Verify that Corollary 5.29 follows Ceva's Theorem.

6. Show that the medians of a triangle subdivide it into six triangles of equal area.

7. Let ΔABC be isosceles with $AB \cong BC$. Show that the median from B is also the perpendicular bisector of AC.

8. Verify that the altitudes of a medial triangle are the perpendicular bisectors of the parent triangle.

9. This problem refers to Fig. 5.36.

   (a) Find as many similar, noncongruent, triangles as you can in the figure.

   (b) Show that $\Delta AYZ \cong \Delta BXZ \cong \Delta CXY \cong \Delta XYZ$.

10. Show that the altitudes of a triangle are concurrent. (Hint: For a given triangle ΔABC, describe how to construct a parent triangle, $\Delta A'B'C'$, so that ΔABC is the medial triangle for $\Delta A'B'C'$.)

11. Show that two angle bisectors of a triangle must intersect at a point interior to the triangle. (Hint: Look back at Exercise 4.3(9).

12. Show that if the centroid and the circumcenter of a triangle coincide, then the triangle is equilateral.

13. Show that if a triangle has two equal medians, it is isosceles.

14. Complete the details required to finish the proof of Feuerbach's Theorem using circular inversion. Be sure to justify all the statements in the sketch of the proof, including the one that says the nine point circle for ΔABC passes through the center of ω, and the one that says the nine point circle passes through the points of intersection of ℓ and ω.

5.5 Summary

What follows is a list of some of the points, lines, and circles that are naturally associated to a triangle.

1. A **cevian** is a line segment joining a vertex to the opposite side of a triangle.

2. A **median** is a cevian that joins a vertex of a triangle with the midpoint of the opposite side. The medians of a triangle are concurrent. The point of concurrency is the **centroid**. The centroid is located 2/3 the distance along a median from a vertex to the opposite side. The triangle determined by the midpoints of the sides of the parent triangle is the **medial triangle**. The centroid of the medial triangle is the same as the centroid of the parent.

3. A **perpendicular bisector** of a triangle is the perpendicular formed at the midpoint of a side of the triangle. The perpendicular bisectors are concurrent. The point of concurrency is the **circumcenter** of the triangle. It is the center of the **circumcircle**, which is the circle determined by the three vertices of the triangle.

4. An **altitude** is the perpendicular from a vertex to the line determined by the opposite side of a triangle. The altitudes of a triangle are concurrent. The point of concurrency is the **orthocenter**. The altitudes of a triangle are the perpendicular bisectors of the medial triangle.

5. The centroid, circumcenter, and orthocenter are collinear. The line they determine is the **Euler line**.

6. The feet of the altitudes of a triangle are the vertices of the **orthic triangle**. The circumcircle of the orthic triangle is the **orthic circle**, also known as the **nine point circle**. The other distinguished points we have discussed on the nine point circle are the midpoints of the sides of the triangle, and the midpoint of each segment from a vertex to the orthocenter.

7. The angle bisectors in a triangle are concurrent. The point of concurrency is the **incenter**. It is the center of the **incircle**, which is the circle that can be inscribed in the triangle so that the sides of the triangle are tangents to the circle.

8. The angle bisector for an exterior angle in a triangle, and the angle bisector for a remote interior angle, intersect at an **excenter**. An excenter is the center of an **excircle** which is tangent to all three lines determined by the sides of a triangle.

9. **Feuerbach's Theorem** says that the nine point circle is tangent to the incircle and all of the excircles.

Chapter 6

Models for the Hyperbolic Plane

6.1 Introduction

The role of models in geometry is tied to the subject's historical connection to axiomatics. While *The Elements* of Euclid is the archetype of an axiomatic system, the real plane, $\mathbb{R}^2$, is the archetype of a model for an axiomatic system, in this case, that of the Euclidean plane. The existence of a model for an axiomatic system is proof that the system is consistent.

We have seen that Euclidean and hyperbolic geometries share so much that we might think of hyperbolic geometry as nearly Euclidean. Indeed, Theorem 2.28 implies that when we look at a region of the hyperbolic plane that is sufficiently small, the angular defect of a triangle is so small as to be indetectable. In practical terms, if space were hyperbolic, our perception of space as Euclidean may be correct, if human scale is "sufficiently small."

Carl Friedrich Gauss (1777–1855) claimed to have discovered, and secreted, Bolyai's and Lobachevsky's results years before the latter two made their work public. He may well have. Gauss was the leading mathematician of his time and he worked on a broad array of problems, among them the problem of the parallels. His influence was such that he could make or break a career and he probably broke Bolyai's career—and spirit—by brushing off the younger man's work when he saw it by saying, in effect, "Yes this is great work. I know because I did it years ago." The story is that he kept his work on non-Euclidean geometries secret because he feared the outcry that would attend its public release.

As we work through our discussion, we will see that the connections between Euclidean and hyperbolic geometries go beyond formalities. Models for the hyperbolic plane and for the Euclidean plane can be constructed from one another. Establishing this was the critical step in the resolution of the problem of the parallels. The relative consistency of hyperbolic geometry meant that the axioms for hyperbolic geometry are consistent if and only if the axioms

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for Euclidean geometry are consistent. Understanding this is what enabled the mathematics community to move on, once and for all, from the problem of the parallels.

6.2 Beltrami

Eugenio Beltrami (1835–1900) in 1868 published two articles about hyperbolic geometry and surfaces of constant negative curvature. In the first, he described how to realize the geometry of Lobachevsky on a surface of negative curvature. The date is significant: 1868 is also the year in which Riemann's work on geometry was published.

Georg Bernhard Riemann (1826–1866) studied with Gauss in Göttingen and received his doctoral degree there in 1851. To teach in the university, Riemann had to complete a second project called the *habilitation*. The habilitation culminated in a lecture to the faculty, its topic chosen by the examiners from among different subjects supplied by the candidate. Riemann's topic, chosen by Gauss, was foundations of geometry. The lecture he gave included ideas about generalized surfaces called manifolds, higher-dimensional spaces, and the notion that a surface has intrinsic properties, that is, properties that are independent of any surrounding space. Riemann's work was to have an enormous impact on the mathematics of the nineteenth and twentieth centuries. Indeed, its influence is still felt today. Before the force of his blow could be fully effected, though, Riemann's ideas had to circulate through the community. By 1868, they were known well enough abroad to give Beltrami the fortitude to go ahead with his account about realizing Lobachevsky's geometry. (See [10] and [17].)

The discovery and dissemination of ideas associated to the resolution of the parallel postulate is a story rich in irony and this part of the tale is no exception. Lambert, the eighteenth century mathematician whom we met in the context of ibn al-Haytham-Lambert quadrilaterals, is associated with several ideas in mathematics, among them the problem of the parallels. Quite apart from that, he also studied and developed the theory of hyperbolic trigonometric functions. Though Lambert viewed his assault on the problem of the parallel postulate as a failure, his hyperbolic trigonometry turned out to be a key element in work leading up to Beltrami's first model of a non-Euclidean plane. While Lambert had both things in his hands—the problem of the parallels and hyperbolic trigonometry—the connection between the two escaped him entirely. It would be another 100 years before the problem of the parallels was resolved. (See [17] for details.)

Beltrami's paper starts with an analytic description of a line element on a surface of *constant negative curvature*. There is not a great deal that we will say about this; we want simply to establish some context. Curvature, by which we mean *Gaussian curvature*, is a measure of how a surface bends. As expected, a plane has zero curvature but so does the surface of a cylinder: A cylinder can be "uwrapped" and laid flat on a plane. A sphere has constant positive curvature: constant, because the curvature does not change from point

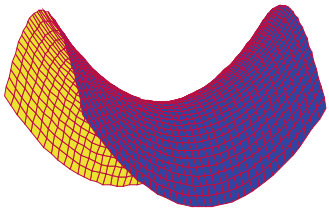


Figure 6.1: A saddle has negative curvature

to point, positive, because if you make a cut in a sphere in an attempt to lay it flat on the plane, pieces of the sphere on either side of the cut have to be pulled apart. The surface of a sphere has too little "material" to form something flat. A hyperbolic paraboloid—a saddle—has negative curvature. (See Fig. 6.1.) If you make a cut in a saddle in an attempt to lay it flat on the plane, pieces of the saddle on either side of the cut have to be pulled together: The surface has too much "material" to form something flat.

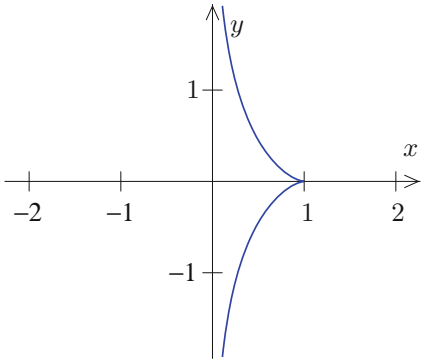


Figure 6.2: The tractrix defined by $x = 1/\cosh t, y = t - \tanh t$

The *pseudosphere*, or *tractoid*, is a surface of constant negative curvature that can be generated by rotating a curve called a *tractrix* about its vertical axis. (See Fig. 6.2.) Beltrami identified a particular region on this surface, and used differential geometry to establish that Lobachevsky's geometry is realized there. An important device in his discussion is the fact that the region of interest on the surface can be mapped onto an open disk in the plane. The circle defining the disk in the plane is the *limit circle*. (See Fig. 6.3.) Its points correspond to points "at infinity" on the surface, indicated in pink in Fig. 6.3. Points outside

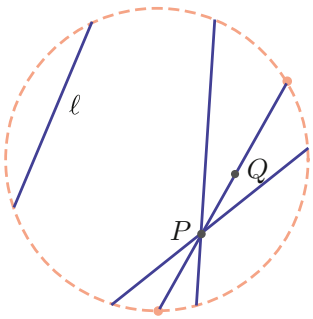


Figure 6.3: The Klein-Beltrami model for the hyperbolic plane

the disk are not part of the model. Lines on the surface are represented by chords of the limit circle. In Fig. 6.3, the line determined by points P and Q is shown, as are several parallels to the line ℓ through the point P. This is the model often referred to as the *Klein-Beltrami model* for the hyperbolic plane.

If Beltrami's work established the relative consistency of Euclidean and non-Euclidean geometry—that is, hyperbolic geometry—Klein's work established the relative consistency of projective geometry, spherical geometry, Euclidean geometry, and non-Euclidean geometry. [28] At this point in our tale, though, the most accessible model for a hyperbolic plane is the Poincaré disk.

Henri Poincaré (1854–1912) started working in hyperbolic geometry when he was thinking about quadratic forms, solutions to certain differential equations, and complex analysis, areas of mathematics that at that time, were not seen as part of geometry. Poincaré's random insight was that the transformations he had been working on were also transformations of the hyperbolic plane. Once his eyes were opened, Poincaré saw the relevance of non-Euclidean geometry all over mathematics. He quickly set about bringing hyperbolic geometry to bear in several different areas of mathematics, including number theory. Non-Euclidean geometry, which until Poincaré's time had seemed a bizarre curiosity, was suddenly everywhere. Poincaré's work might properly signal the true arrival of hyperbolic geometry into mainstream mathematics. (See [28].)

6.3 The Poincaré Disk

There are two models attributed to Poincaré: the upper half-plane model and the disk model. We study the disk model. Like the upper half-plane model, the disk model is constructed within the Euclidean plane, which we designate $\mathcal{E}$. Since $\mathcal{E}$ is modeled by $\mathbb{R}^2$, and we will be concerned with certain mappings from $\mathcal{E}$ to $\mathbb{R}$, calculus is available to us and we employ it to study the Poincaré disk.

Let $\mathcal{C}$ be the circle with center O and radius 1. The choice of 1 for the

radius of $\mathcal{C}$ is arbitrary. Let $\mathcal{D}$ be the open disk in $\mathcal{E}$ associated to $\mathcal{C}$. $\mathcal{D}$ is the foundation for our model of a hyperbolic plane. When we think of $\mathcal{D}$ as modeling the hyperbolic plane, we designate it $\mathcal{P}$.

The points of $\mathcal{P}$ are the points of $\mathcal{D}$. Lines in the Poincaré disk are of two types: **type-1 lines** are intersections of diameters of $\mathcal{C}$ with $\mathcal{D}$, and **type-2 lines** are intersections of circles orthogonal to $\mathcal{C}$ with $\mathcal{D}$.

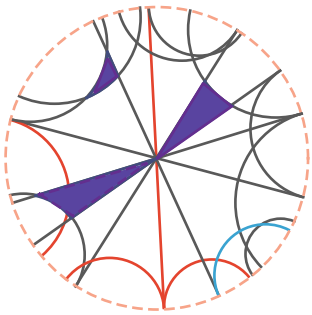


Figure 6.4: The Poincaré disk model

Proceeding through our discussion, we will be dealing with $\mathcal{D}$ sometimes as a subset of $\mathcal{E}$—which, when convenient, we can model using $\mathbb{R}^2$, with which we have some familiarity—and sometimes as modeling $\mathcal{P}$, which we are endeavoring to understand. Going forward, every subset of $\mathcal{D}$ wears two hats: one Euclidean, and one hyperbolic. We must be careful to distinguish which hat it wears when we pick up an object to discuss its nature.

We continue to refer to Euclidean lines simply as lines. Following [23], we refer to type-1 and type-2 lines together as **L-lines** ("L" for Lobachevsky).

A goal for us right now is understanding what it means for one point to be between two other points in our model, $\mathcal{P}$. Once we establish that we have a well-behaved notion of betweenness in $\mathcal{P}$, we will have license to talk about L-segments. In other words, if A and B are points on an L-line ℓ, then the associated L-segment is the collection of points $X \in \ell$ so that $X = A$, or $A * X * B$, or $X = B$.

Angles in $\mathcal{P}$ are identical to angles in $\mathcal{E}$. The Euclidean tangents at the point of intersection of two L-lines determine the angles formed by the L-lines.

Figure 6.4 shows some L-lines and regions in $\mathcal{P}$. The red L-lines are all parallel. The light blue arc, a type-2 line, is parallel to three of the red L-lines, but not to the fourth. This is significant: In a neutral plane, transitivity of parallelism is one of the conditions equivalent to the parallel postulate. That parallelism is not transitive is thus a defining feature of a hyperbolic plane. Three points in $\mathcal{P}$ that do not lie on a single L-line determine an **L-triangle**. The filled areas in the figure are regions enclosed by L-triangles.

Our first task is to show that $\mathcal{P}$ satisfies the first postulate of Euclid, that is, Axioms I.1–2 of Hilbert. These are the only incidence axioms we need. The

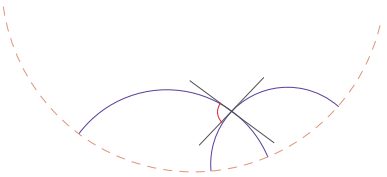


Figure 6.5: An angle between L-lines is an angle between their tangents

others are either automatic, since our model employs Euclidean points, or irrelevant, since we are discussing a plane.

Lemma 6.1. Two points in $\mathcal{P}$ determine a unique L-line.

Proof. Let P and Q be points in $\mathcal{P}$. If P belongs to a Euclidean diameter of $\mathcal{C}$ that runs through Q, then that diameter determines a unique type-1 line containing P and Q. Suppose in this case that there were also a type-2 line containing P and Q. That L-line would be determined by a circle $\mathcal{C}'$ in $\mathcal{E}$ orthogonal to $\mathcal{C}$. By Lemma 5.21, both P and Q would have to be different from O. Under inversion through $\mathcal{C}$, $\overline{PQ} \setminus O$ is mapped to itself. As an orthogonal to $\mathcal{C}$, $\mathcal{C}'$ is also mapped to itself under inversion through $\mathcal{C}$. If the $\mathcal{C}$ inverses of P and Q are P' and Q' respectively, this means P, Q, P', Q' would belong both to a line in $\mathcal{E}$ and to a circle in $\mathcal{E}$, a violation of Prop. III.10. We conclude that if P and Q lie on a diameter of $\mathcal{C}$, there is a unique L-line that contains the two points.

Next suppose P and Q do not lie on a diameter of $\mathcal{C}$. In this case neither P nor Q can be O. We must show that there is a single type-2 line containing them. Let P' be the $\mathcal{C}$ inverse of P so that the Euclidean line determined by P and P' is $\overline{OP}$. Then P, Q, and P' are noncollinear, so by Prop. III.10, they determine a unique circle, $\mathcal{C}'$. By Theorem 5.24, $\mathcal{C}$ and $\mathcal{C}'$ are orthogonal, thus, in this case, P and Q determine a type-2 line. Also by Theorem 5.24, any circle orthogonal to $\mathcal{C}$ and containing P and Q must contain their inverses, so the L-line determined by P and Q is unique, again by Prop. III.10. □

Let ℓ be an L-line in $\mathcal{P}$. In $\mathcal{E}$, the Euclidean arc or line segment that determines ℓ intersects $\mathcal{C}$ in two points. We call these the **points at infinity** of ℓ. Note that these are Euclidean points that are not in $\mathcal{P}$.

We can establish that both the axioms of order and the continuous nature of a line hold in $\mathcal{P}$ by coordinatizing L-lines using $\mathbb{R}$. Once we coordinatize lines, several of Hilbert's axioms will be verified immediately.

First, we consider a notion of non-Euclidean distance in $\mathcal{P}$.

Distance in the Poincaré Disk

A measure of distance on a set that conforms to our understanding of how distance ought to work is usually defined by a *metric*.

Definition 6.2. A **metric** on a set $\mathcal{S}$ is a mapping $d : \mathcal{S} \times \mathcal{S} \to \mathbb{R}$ that satisfies the following for all x, y, z in $\mathcal{S}$:

(1) $d(x, y) \geq 0$ with equality only if $x = y$;

(2) $d(x, y) = d(y, x)$;

(3) $d(x, z) \leq d(x, y) + d(y, z)$.

When $\mathcal{S}$ has a metric d, $(\mathcal{S}, d)$ is a **metric space**.

The last condition of Definition 6.2 is a version of Euclid's Triangle Inequality, Prop. I.20. In this case, the inequality is not strict because x, y, z are not necessarily vertices of a triangle. Going forward, we use "triangle inequality" to refer to condition (3) in Definition 6.2, or to

$$|a + b| \leq |a| + |b|,$$

the triangle inequality as it arises in $\mathbb{R}$, or to the inequality in the next lemma.

Lemma 6.3. Let x, y, z be points in a metric space $(\mathcal{S}, d)$. Then

$$d(x, y) \geq |d(x, z) - d(y, z)|.$$

Proof. Since $d(x, z) \leq d(x, y) + d(y, z)$, we have $d(x, z) - d(y, z) \leq d(x, y)$. On the other hand, $d(y, z) \leq d(x, y) + d(x, z)$ implies $d(y, z) - d(x, z) \leq d(x, y)$. Putting the two together, we get the result. □

The archetype for a metric space is $\mathbb{R}^n$ with

$$d(P, Q) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{6.1}$$

where $P = (x_1, x_2, \ldots, x_n)$ and $Q = (y_1, y_2, \ldots, y_n)$. Notice that in $\mathbb{R}$, (6.1) becomes

$$d(x, y) = |x - y|.$$

Proofs that (6.1) satisfies the triangle inequality abound in the literature. We leave it as an exercise to prove the case for $n = 1$. Equation (6.1) defines the **Euclidean metric** on $\mathbb{R}^n$.

Lemma 6.4. Let $(\mathcal{S}, d)$ be a metric space. Let $A \subseteq \mathcal{S}$. Then (A, d) is also a metric space.

Proof. Let $x, y \in A$. Since $x, y \in \mathcal{S}$, properties (1) and (2) of Definition 6.2 hold. Now let z be an element in A as well. Since x, y, z all belong to $\mathcal{S}$, property (3) of Definition 6.2 must hold as well. □

Since $\mathcal{D}$ is a subset of a metric space, $\mathbb{R}^2$, it is also a metric space under the Euclidean metric. The problem with this is that Postulate II will not hold in this metric space. We need a metric on $\mathcal{P}$ that will allow us to stretch line segments indefinitely.

Definition 6.5. Let P and Q be points in $\mathcal{P}$, and let ℓ be the L-line they determine. Let S and T be the points at infinity of ℓ. The **L-distance between P and Q** is

$$L(P,Q) = \left| \ln \frac{|PS|/|PT|}{|QS|/|QT|} \right| \tag{6.2}$$

where $|PS|$ is the Euclidean distance between points P and S, (or the Euclidean length of PS), and ln is the natural logarithm.

Theorem 6.6. L is a metric on $\mathcal{P}$.

Proof. Notice first that when $P = Q$, $\frac{|PS|/|PT|}{|QS|/|QT|} = 1$, and $\ln 1 = 0$. Note further that by the definition, the L-distance between points is always nonnegative.

The triangle inequality and condition (2) of Definition 6.2 are easy to show using properties of logarithms and absolute values. Likewise, if P, Q determine a type-1 line, it is easy to show that $L(P,Q) = 0$ implies $P = Q$. We leave those verifications to the exercises. We must show that condition (1) holds if P, Q determine a type-2 line.

Suppose then that $P \neq Q$ lie on a type-2 line in $\mathcal{P}$. Notice that $\overrightarrow{SP}$ and

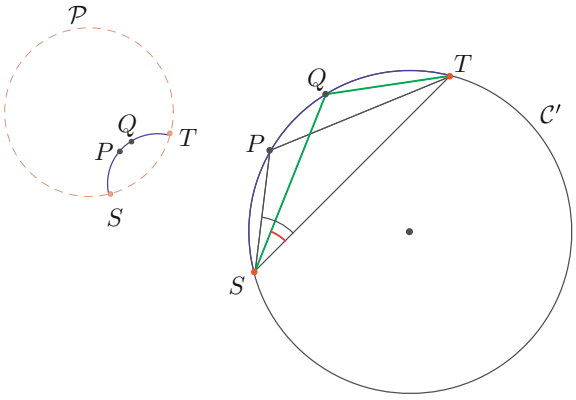


Figure 6.6: A type-2 line in $\mathcal{P}$ and the underlying set in $\mathcal{E}$

$\overrightarrow{SQ}$ must be different, otherwise S, P, Q would lie on both a line and a circle, in violation of Prop. III.10. Since $\overrightarrow{SQ}$ and $\overrightarrow{SP}$ lie on the same side of ST, $\angle TSP$ and $\angle TSQ$ must be different.

Assume now that $L(P,Q) = 0$. We then have

$$\frac{|PS|/|PT|}{|QS|/|QT|} = 1, \text{ that is, } \frac{|PS|}{|QS|} = \frac{|PT|}{|QT|}.$$

By Prop. III.21, $\angle SPT \cong \angle SQT$. We thus have SAS for similar Euclidean triangles (Prop. VI.6) $\Delta STP \sim \Delta STQ$. That implies $\angle TSP \cong \angle TSQ$, which is a contradiction. We conclude that if $L(P,Q) = 0$, then $P = Q$ as desired. □

As we proceed through our discussion, $|PQ|$ will always designate the Euclidean length of the line segment PQ, or equivalently, the Euclidean distance between P and Q.

The notion of a metric allows us to consider what it means for mappings to be continuous on spaces that are more general than $\mathbb{R}$ with the Euclidean metric.

Definition 6.7. Let f be a mapping from a metric space $(\mathcal{S}_1, d_1)$ to a metric space $(\mathcal{S}_2, d_2)$. Then f is **continuous at a point** $x \in \mathcal{S}_1$ provided for any real number $\epsilon > 0$ there is a real number $\delta > 0$ so that if for some $y \in \mathcal{S}_1$, $d_1(x,y) < \delta$, then $d_2(f(x), f(y)) < \epsilon$.

Continuous mappings on metric spaces have many of the properties we are used to from calculus. We have to be careful, though, to bear in mind that an arbitrary metric space may not come equipped with operations such as addition or multiplication, for example. In general, a sum of mappings between metric spaces may not make sense. Composition of mappings, though, makes sense as long as the domain of the second mapping contains the range of the first.

Lemma 6.8. Let $(\mathcal{S}_1, d_1)$, $(\mathcal{S}_2, d_2)$, and $(\mathcal{S}_3, d_3)$ be metric spaces. Let $f : \mathcal{S}_1 \to \mathcal{S}_2$ be a continuous mapping and let $g : \mathcal{R} \to \mathcal{S}_3$ be a continuous mapping, where $\mathcal{R} \subseteq f(\mathcal{S}_1)$. Then $g \circ f$ is a continuous mapping from $\mathcal{S}_1$ to $\mathcal{S}_3$.

Proof. Let x be in $\mathcal{S}_1$. Given $\epsilon > 0$, we must produce $\delta > 0$ so that if $d_1(x,y) < \delta$ for $y \in \mathcal{S}_1$, then $d_3(g(f(x)), g(f(y))) < \epsilon$.

Since g is continuous, we know there is $\xi > 0$ so that if $d_2(f(x), f(y)) < \xi$ for $f(y) \in f(\mathcal{S}_1)$, then $d_3(g(f(x)), g(f(y))) < \epsilon$. With that ξ fixed, we can in turn choose $\delta > 0$ so that if $d_1(x,y) < \delta$ for some $y \in \mathcal{S}_1$, then $d_2(f(x), f(y)) < \xi$, from which it follows that $d_3(g(f(x)), g(f(y))) < \epsilon$, as was to be shown. □

Now that we have a metric on $\mathcal{P}$ and a notion of continuity, we can approach the problem of coordinatizing L-lines using $\mathbb{R}$. We start by considering the same problem in the more familiar setting of $\mathbb{R}^2$.

Let ℓ be the line in $\mathbb{R}^2$ that passes through points (x_0, y_0) and (a, b). When we think about identifying the points of ℓ with the points of $\mathbb{R}$, we are usually thinking of something more than just a point-by-point correspondence. An "identification" should show some respect to distances between points of ℓ and distances between corresponding points of $\mathbb{R}$. Even so, there are many different ways to make this sort of identification using a parametrization. We can parametrize ℓ with a variable t and then think of a point X on ℓ as identified with the associated $t \in \mathbb{R}$. For example,

$$x = x_0 + t(a - x_0),\ y = y_0 + t(b - y_0),\ t \in \mathbb{R}. \tag{6.3}$$

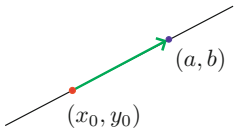


Figure 6.7: A line in $\mathbb{R}^2$ with a "starting point" and a direction

Here the point (x_0, y_0) is identified with 0, and the point (a, b) is identified with 1. The vector $\mathbf{v} =< a - x_0, b - y_0 >$ (shown in Fig. 6.7 in green) determines the positive direction on ℓ: As t increases, the associated points on ℓ can be found farther to the right of (x_0, y_0). Note, moreover, that points on ℓ that are close, are associated to values of t that are also close. Equation (6.3) thus gives us a way of assigning coordinates to the points on ℓ, so that distances between coordinates of points bear a certain fidelity to distances between points on ℓ.

Let us make this more precise.

Definition 6.9. A **coordinatization** of a curve $\mathcal{S}$ in a space with a metric d is a continuous bijective mapping $f : \mathcal{S} \to \mathbb{R}$.

The bijectivity condition guarantees a point-by-point correspondence between $\mathcal{S}$ and $\mathbb{R}$. The continuity condition guarantees that a coordinatization respects the notions of nearness on the curve $\mathcal{S}$ and in $\mathbb{R}$.

We get to our coordinatization through a sequence of lemmas.

Lemma 6.10. Let ℓ be an L-line in $\mathcal{P}$. Let T, S be the points at infinity of ℓ. Let $|ST| = \tau$. Define $R : \ell \to \mathbb{R}$ by $R(X) = |XT|$. If we view ℓ as a subset of $\mathbb{R}^2$ with the Euclidean metric, then R is a continuous bijective mapping of ℓ onto the interval $(0, \tau) \subset \mathbb{R}$.

Proof. Let everything be as hypothesized and suppose X is fixed on ℓ. Let $\epsilon > 0$ be given. If we take $\delta = \epsilon$, then if $|XY| < \delta$ for some $Y \in \ell$, Lemma 6.3 gives us

$$d(R(X), R(Y)) = ||XT| - |YT|| \leq |XY| < \epsilon,$$

which shows that R is continuous.

Next we show that R is bijective onto $(0, \tau)$. The case in which ℓ is a type-1 line is routine and we leave it to the reader. Suppose then that ℓ is a type-2 line.

To show that R is one-to-one we suppose it is not and seek a contradiction. Suppose then that for two different points X, Y on ℓ, $R(X) = R(Y)$. Since $|XT| = |YT|$, ΔXYT is isosceles. By Exercise 5.4.7, the median of ΔXYT from T (shown in red in Fig. 6.8) is the perpendicular bisector of XY. But XY is a chord of $\mathcal{C}'$, the Euclidean circle that determines ℓ. By Corollary 5.2, the perpendicular bisector of XY passes through the center of $\mathcal{C}'$. Since it also passes through T and a point on $\overset{\frown}{ST}$, that means the center of $\mathcal{C}'$ is interior to

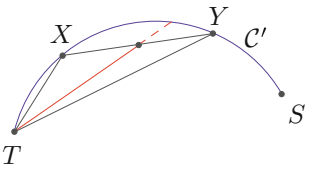


Figure 6.8: $|XT| = |YT|$ implies ΔXYT is isosceles

$\mathcal{C}$, the circle that determines $\mathcal{P}$. By Lemma 5.21 that is impossible because $\mathcal{C}'$ and $\mathcal{C}$ are orthogonal. We conclude that R is indeed one-to-one.

Finally, we show that R is onto the interval $(0, \tau)$. Notice that for $X \in \ell$, $|XT| \leq |XS| + |ST|$ so $|XT| < |ST|$. This shows that $R(X) \in (0, \tau)$. Let $\rho \in (0, \tau)$. Consider the circle in $\mathcal{E}$ centered at T with radius ρ. That circle must intersect $\overset{\frown}{ST}$ at some point X. It follows that $R(X) = \rho$, which establishes that R is onto $(0, \tau)$ and completes the proof. □

The proof of our next major result benefits from a theorem we learn in middle school or high school. This is close to Prop. II.11, 13, although Euclid does not mention cosines.

> **The Law of Cosines** Let a, b, c be the lengths of sides of a triangle and suppose the side with length a subtends the angle α. Then
>
> $$a^2 = b^2 + c^2 - 2bc\cos\alpha.$$

Lemma 6.11. Let ℓ be an L-line in $\mathcal{P}$ with points at infinity S and T. Let $\tau = |ST|$. If ℓ is a type-2 line, let $\varphi = \angle TXS$, for any X on ℓ. If ℓ is a type-1 line, let $\varphi = \pi$. If $x = |XT|$ and $y = |XS|$, then

$$y = x\cos\varphi + \sqrt{\tau^2 - x^2\sin^2\varphi}.$$

Proof. When ℓ is a type-1 line, $\cos\varphi = -1$ and $\sin^2\varphi = 0$. The result in this case follows nearly immediately.

Suppose ℓ is a type-2 line determined by a circle $\mathcal{C}'$. Since the base on the circumference of $\mathcal{C}'$ for $\overset{\frown}{ST}$ is the major arc associated to ST, φ is obtuse, so $\cos\varphi < 0$. (The verification that φ is obtuse is left as an exercise.)

The Law of Cosines gives us $\tau^2 = x^2 + y^2 - 2xy\cos\varphi$ so that

$$y^2 - (2x\cos\varphi)y + (x^2 - \tau^2) = 0.$$

By the quadratic formula,

$$y = \frac{2x\cos\varphi \pm \sqrt{4x^2\cos^2\varphi - 4(x^2 - \tau^2)}}{2} = x\cos\varphi \pm \sqrt{x^2(\cos^2\varphi - 1) + \tau^2}.$$

Since $\cos\varphi < 0$, the positive square root must be what gives us y. Note too that $\cos^2\varphi - 1 = -\sin^2\varphi$. This implies the result. □

Notice that by Prop. III.21, the angle φ in Lemma 6.11 is constant.

Now we dust off a few more tools from single variable calculus.

Recall that an *algebraic function* is a mapping from a subset of $\mathbb{R}$ to $\mathbb{R}$ that is defined using arithmetic (addition, subtraction, multiplication, and division), powers, and roots. Examples of algebraic functions are polynomials, rational functions, $f(x) = \sqrt{x}$, and $g(x) = \sqrt[3]{\frac{x^2+1}{3x^2+5}}$. Trigonometric functions, hyperbolic functions, exponentials, and logarithms are examples of non-algebraic functions. Algebraic functions are continuous on their domains.

Recall also the following important theorem from Calculus I.

Theorem 6.12 (Intermediate Value Theorem). Let f be a continuous function from an interval $I \subseteq \mathbb{R}$ to $\mathbb{R}$. If $f(a) < f(b)$ for some a, b in I, then for any $y \in (f(a), f(b))$ there is x between a and b so that $f(x) = y$.

Lemma 6.13. Let ℓ be an L-line in $\mathcal{P}$ with points at infinity S and T. Let $\tau = |ST|$. If ℓ is a type-2 line, let $\varphi = \angle TXS$, for any X on ℓ. If ℓ is a type-1 line, let $\varphi = \pi$. For $x \in (0, \tau)$, define

$$g(x) = \frac{x}{x\cos\varphi + \sqrt{\tau^2 - x^2\sin^2\varphi}}.$$

Then g is a continuous bijection from $(0, \tau)$ onto $\mathbb{R}_+$.

Proof. That g is continuous on its domain follows from the fact that g is algebraic in x. (Note that τ, and φ are constants so $\cos\varphi$ and $\sin\varphi$ are also constants.) In Lemma 6.11 we established that the denominator of g is $|XS|$, thus for $x \in (0, \tau)$, it also lies strictly between 0 and τ. This confirms that g is defined and continuous on all of $(0, \tau)$.

Note that g is differentiable on its domain with

$$g'(x) = \frac{\tau^2}{(x\cos\varphi + \sqrt{\tau^2 - x^2\sin^2\varphi})^2\sqrt{\tau^2 - x^2\sin^2\varphi}}.$$

As $g'(x) > 0$ for all $x \in (0, \tau)$, g is strictly increasing. This shows that g is one-to-one.

Next notice that

$$\lim_{x\to 0^+} g(x) = 0,$$

and that

$$\lim_{x\to\tau^-} g(x) = \lim_{x\to\tau^-} \frac{1}{\cos\varphi + \sqrt{\frac{\tau^2}{x^2} - \sin^2\varphi}}.$$

Since $\cos\varphi < 0$, and $\sqrt{1 - \sin^2\varphi} = |\cos\varphi|$, the denominator $\cos\varphi + \sqrt{\frac{\tau^2}{x^2} - \sin^2\varphi}$ approaches 0 from above as x approaches τ from below, thus $\lim_{x\to\tau^-} g(x) = \infty$. By the Intermediate Value Theorem, g must assume all values in $\mathbb{R}_+$, thus, g is onto $\mathbb{R}_+$. □

Theorem 6.14. Let ℓ be an L-line in $\mathcal{P}$. Let P be a fixed point on ℓ and suppose S and T are the points at infinity of ℓ. Let

$$f(X) = \ln \frac{|PS|/|PT|}{|XS|/|XT|}. \tag{6.4}$$

Then f is a coordinatization of ℓ by $\mathbb{R}$.

Proof. Let everything be as hypothesized.

We may view f as a composition, $f = f_1 \circ f_2$, where $f_2 : \ell \to \mathbb{R}_+$ by $f_2(X) = \frac{|PS|/|PT|}{|XS|/|XT|}$, and $f_1 : \mathbb{R}_+ \to \mathbb{R}$ is the natural logarithm function. Bijectivity and continuity of $f : \ell \to \mathbb{R}$ will follow if we show that f_1 and f_2 are both bijective and continuous.

By our calculus courses, we know that the natural logarithm is a bijective continuous mapping from $\mathbb{R}_+$ onto $\mathbb{R}$, so we need only address the qualifications of f_2. Notice that

$$f_2 = \frac{|PS|}{|PT|} g \circ R,$$

where $|PS|/|PT|$ is a constant, $R : \ell \to (0, \tau)$, $\tau = |ST|$, is the mapping we discussed in Lemma 6.10, and $g : (0, \tau) \to \mathbb{R}_+$ is the mapping from Lemma 6.13. As f_2 is the composition of continuous bijections, it must also be a continuous bijection. We leave the details underlying that statement to the reader. □

Suppose now that ℓ is an L-line with the line coordinatization f given (6.4). We can order the points on ℓ according to their coordinates: For points A, B, C on ℓ, say B is between A and C provided $f(A) < f(B) < f(C)$ or $f(C) < f(B) < f(A)$. That Hilbert's axioms of line order and continuity apply in $\mathcal{P}$ then follows from properties of $\mathbb{R}$, properties one studies, for instance, in a first course in real analysis. It is not too difficult, and we leave it as an exercise, to show that Pasch's Axiom holds in $\mathcal{P}$.

The L-metric on $\mathcal{P}$ allows us to define congruence of segments: L-segments AB and CD are congruent if and only if $L(A, B) = L(C, D)$. Since we are transferring the problem of comparing segments to the problem of comparing numbers, difficulties that Hilbert had to manage disappear. For instance, congruence of segments is an equivalence relation because equality of numbers is an equivalence relation. Angles in $\mathcal{P}$ are identical to angles in $\mathcal{E}$, so the axioms that apply to angle congruence in $\mathcal{E}$ carry over to $\mathcal{P}$. Note that once Hilbert's axioms are established, the associated theorems follow. We do not have to prove any of them.

Determining that SAS holds in $(\mathcal{P}, L)$ is more interesting. This is where congruence of segments and congruence of angles meet, as it were. Our strategy for proving SAS involves the use of L-line reflections.

L-Reflections on $\mathcal{P}$

Definition 6.15. The **type-1 reflections** on $\mathcal{P}$ are Euclidean reflections determined by type-1 lines. The **type-2 reflections** on $\mathcal{P}$ are circular inversions

determined by type-2 lines. The type-1 and type-2 reflections together are called **L-reflections** on $\mathcal{P}$.

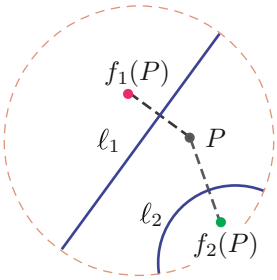


Figure 6.9: Two types of L-reflections

Just as Euclidean reflection leaves Euclidean distance unchanged, a type-2 reflection on $\mathcal{P}$ leaves L-distance unchanged. To understand this, we take a moment to consider the ratio of Euclidean segment lengths in (6.2). That ratio is the **planar cross ratio** of Euclidean points P, Q, S, T.

Lemma 6.16. The planar cross ratio of points in $\mathcal{E}\backslash O$ is invariant under circular inversion of $\mathcal{E} \setminus O$.

Proof. Consider points P, Q, S, T in $\mathcal{E} \setminus O$ and let P', Q', S', T' be their respective images under an inversion with radius a and center O. Assuming noncollinearity, we have the following four pairs of similar triangles, by Observation 5.16: $\Delta POS \sim \Delta S'OP'$, $\Delta POT \sim \Delta T'OP'$, $\Delta QOS \sim \Delta S'OQ'$, and $\Delta QOT \sim \Delta T'OQ'$. It follows that

$$\frac{|PS|}{|OP|} = \frac{|P'S'|}{|OS'|}, \text{ and } \frac{|PT|}{|OP|} = \frac{|P'T'|}{|OT'|}. \tag{6.5}$$

Dividing the second equation into the first we get

$$\frac{|PS|}{|PT|} = \frac{|P'S'|}{|P'T'|}\frac{|OT'|}{|OS'|}.$$

Similarly,

$$\frac{|QS|}{|QT|} = \frac{|Q'S'|}{|Q'T'|}\frac{|OT'|}{|OS'|}.$$

Dividing the latter two equations, we get

$$\frac{|PS|}{|PT|}\frac{|QT|}{|QS|} = \frac{|P'S'|}{|P'T'|}\frac{|Q'T'|}{|Q'S'|},$$

as desired.

Next, consider what happens if O, P, S are collinear. We may assume the points are labeled so that $|PS| = |OS| - |OP|$ and $|P'S'| = |OP'| - |OS'|$. Then $|PS||OS'| = a^2 - |OP||OS'|$, and $|P'S'||OP| = a^2 - |OS'||OP|$ so $|PS||OS'| = |P'S'||OP|$ implying that the first statement in (6.5) is still true. Thus, if any of the four "triangles" is not actually a triangle, we get the same equations and we can still conclude that the planar cross ratio is unchanged by inversion. □

Theorem 6.17. The L-metric is invariant under circular inversion.

Proof. The proof is immediate by Lemma 6.16. □

We study Euclidean reflections and rotations in some detail in later chapters after developing the requisite algebraic machinery. For now, we simply observe that under reflection across a diagonal, as well as rotation about its center, $\mathcal{P}$ is mapped to itself, L-lines are mapped to L-lines, and Euclidean length and angle are preserved. Our intuition tells us this must be so, and for now, we leave it at that.

Since Euclidean length is preserved, L-length is also preserved under type-1 reflections. (We leave the verification as an exercise.) Theorem 5.23 implies that a type-2 reflection on $\mathcal{P}$ leaves $\mathcal{P}$ fixed as a set. By Lemma 5.25, angles formed by L-lines are unchanged by inversion. L-reflections, then, map $\mathcal{P}$ to $\mathcal{P}$, and respect L-length and angle. The next result is less obvious.

Lemma 6.18. A type-2 reflection of $\mathcal{P}$ maps L-lines to L-lines.

Proof. We continue to use $\mathcal{C}$, with center O and radius 1, as the circle in $\mathcal{E}$ defining $\mathcal{P}$. Let ℓ be a type-2 line in $\mathcal{P}$ with underlying circle $\mathcal{C}'$ with center A and radius a. From Exercise 5.3.9, we know that $\mathcal{C}$ and $\mathcal{C}'$ are orthogonal if and only if

$$a^2 + 1 = b^2, \tag{6.6}$$

where $b = |OA|$.

Fix a coordinate system on $\mathcal{E}$ based on $\mathcal{C}'$, so that A has rectangular coordinates $(0, 0)$, O has coordinates $(b, 0)$, and $\mathcal{C}'$ is given by $x^2 + y^2 = a^2$. We use rectangular and polar coordinates to show that a type-1 line in $\mathcal{P}$ is mapped to a type-2 line under the type-2 reflection determined by $\mathcal{C}'$. Because circular inversion is an involution, this is enough to show that type-2 reflections map L-lines to L-lines in $\mathcal{P}$.

Note that a diameter of $\mathcal{C}$ belongs to a line that can be described in rectangular coordinates by

$$x - b = cy,$$

for some $c \in \mathbb{R}$. Switching to polar coordinates, the line is given by $r\cos\theta - b = cr\sin\theta$. Under inversion through radius a, the line is mapped to the circle in $\mathcal{E}$ given by

$$a^2 r\cos\theta - br^2 = a^2 cr\sin\theta + br^2.$$

Switching back to rectangular we have

$$x^2 - \frac{a^2}{b}x + y^2 + \frac{a^2 c}{b}y = 0,$$

which can be written

$$\left(x - \frac{a^2}{2b}\right)^2 + \left(y + \frac{a^2 c}{2b}\right)^2 = \frac{a^4(1+c^2)}{4b^2}.$$

The line through O is thus mapped to the circle $\mathcal{C}''$ in $\mathcal{E}$ with radius $a^2\sqrt{1+c^2}/(2b)$ and center Q (in rectangular coordinates) given by

$$Q = \left(\frac{a^2}{2b}, -\frac{a^2 c}{2b}\right).$$

Again invoking Exercise 5.3.9, we know that $\mathcal{C}''$ is orthogonal to $\mathcal{C}$ if and only if $|OQ|^2$ is the sum of the squares of the radii of $\mathcal{C}$ and $\mathcal{C}''$. We see then that ℓ, the L-line underlying the diameter of $\mathcal{C}$, is mapped to a type-2 line if and only if

$$|OQ|^2 = 1 + \frac{a^4(1+c^2)}{4b^2}.$$

With an eye toward Equation (6.6), we see that

$$|OQ|^2 = \left(\frac{a^2}{2b} - b\right)^2 + \left(-\frac{a^2 c}{2b}\right)^2 = \frac{a^4}{4b^2} - a^2 + b^2 + \frac{a^4 c^2}{4b^2}$$

$$= \frac{a^4(1+c^2)}{4b^2} + (b^2 - a^2) = \frac{a^4(1+c^2)}{4b^2} + 1,$$

as was to be shown. □

We can proceed now under the assumption that any L-reflection maps an L-line to an L-line.

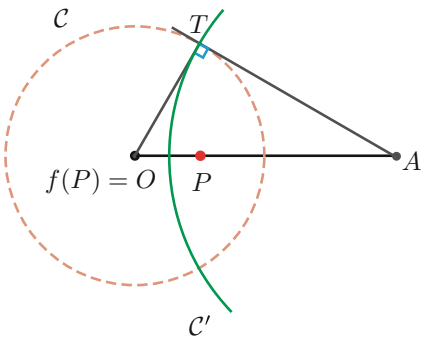


Figure 6.10: Every point can be L-reflected to O

Lemma 6.19. For every point P in $\mathcal{P}$ there is an L-reflection f with $f(P) = O$.

Proof. Given a point $P \in \mathcal{P}$ different from O, let $k = |OP|$. Since the radius of $\mathcal{C}$ is 1, $k < 1$. Let $A \in \overrightarrow{OP}$ satisfy $|OA| = 1/k$. (See Fig. 6.10.) Notice that since $0 < k < 1$, $1/k > 1$ so A is outside $\mathcal{C}$.

Form a tangent to $\mathcal{C}$ from A and say the point of tangency is T. Let $a = |AT|$. Notice that the circle $\mathcal{C}'$ with center A and radius a is indeed orthogonal to $\mathcal{C}$ so it does determine a type-2 line in $\mathcal{P}$, thus an L-reflection, f. We claim that $f(P) = O$. It suffices to show that $|Af(P)| = 1/k$.

We have $|AP||Af(P)| = a^2$. Note

$$|AP| = \frac{1}{k} - k = \frac{(1-k^2)}{k}.$$

By the Pythagorean Theorem, $|OA|^2 = 1 + a^2$ so

$$a^2 = \frac{1}{k^2} - 1 = \frac{1-k^2}{k^2} = \frac{(1-k^2)}{k}\frac{1}{k} = |AP|\frac{1}{k} = |AP||Af(P)|$$

in other words, $|Af(P)| = 1/k$, as was to be shown. □

Hilbert's Axiom III.5 says that if ΔABC and $\Delta A'B'C'$ satisfy $|AB| \cong |A'B'|$, $\angle BAC \cong \angle B'A'C'$, and $|AC| \cong |A'C'|$, then $\angle ABC \cong \angle A'B'C'$. In Hilbert's narrative, it is a short leap from Axiom III.5 to triangle congruence. With the tools in our hands, the route to the result that SAS implies L-triangle congruence is even more direct. This approach to the proof is from [18] and [23].

Theorem 6.20. SAS implies L-triangle congruence in $\mathcal{P}$.

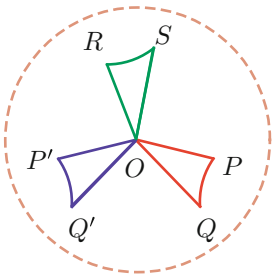


Figure 6.11: Euclidean reflection maps $\Delta OP'Q'$ to ΔOPQ. Rotation maps ΔORS to ΔOPQ

Sketch of proof. Let ΔABC and $\Delta A'B'C'$ be L-triangles in $\mathcal{P}$ with

$$L(A,B) = L(A',B'),\ L(A,C) = L(A',C'),\ \text{and}\ \angle BAC \cong \angle B'A'C'.$$

Let f be an L-reflection that maps A to O, the center of the Euclidean circle defining $\mathcal{P}$. Since L-reflections respect L-length, L-lines, and angle, $f(\Delta ABC) = \Delta OPQ$ is congruent to ΔABC.

Letting g be the L-reflection that maps A' to O, we get a fourth L-triangle

$$g(\Delta A'B'C') = \Delta OP'Q' \cong \Delta A'B'C'.$$

Note now that the sides OP, OQ, OP', OQ' all lie on diagonals of $\mathcal{C}$. Since $\angle POQ \cong \angle P'OQ'$, we can use either a Euclidean reflection or rotation to map $\Delta OP'Q'$ to a new L-triangle, $\Delta OP''Q''$, where P'' is on OP and Q'' is on OQ. Because

$$L(O,Q) = L(O,Q') = L(O,Q''), \text{ and } L(O,P) = L(O,P') = L(O,P''),$$

points P and P'' have to coincide, as do points Q and Q''. Since two points in $\mathcal{P}$ determine a unique L-line, the L-segments PQ and $P''Q''$ must coincide as well. Since the legs of $\angle QPO$ and $\angle Q''P''O$ coincide, the angles must be congruent as well. The same argument establishes that $\angle PQO \cong \angle P''Q''O$ so that

$$\Delta OPQ \cong \Delta OP''Q'' \cong \Delta OP'Q'.$$

Since congruence is an equivalence relation, we have $\Delta ABC \cong \Delta A'B'C'$, which proves the theorem. □

Authors who use this argument to prove SAS in $\mathcal{P}$ (see [11], besides [18] and [23], for example) usually note that it employs superposition, the method that Euclid used to argue Prop. I. 4. Remember that Euclid has been chastised through the ages for using superposition! Superposition is a valid technique if it is developed as part of the discussion of allowable transformations or *isometries* on the geometry in question. We have handled circular inversion fairly thoroughly, and appealed to intuition in our use of Euclidean reflections and rotations. In the next chapter, we meet transformations formally and in earnest.

Exercises 6.1. 1. Is the following statement true or false? Justify your answer with a proof, if it is true. If it is false, adjust the statement to make it true and supply details: "Any set of three noncollinear points in $\mathcal{P}$ is noncollinear in both the Euclidean and the hyperbolic sense, thus, by identifying a triangle with its vertices, we find that the collection of Euclidean triangles and the collection of L-triangles in $\mathcal{P}$ is the same."

2. There is a statement in the proof of Lemma 6.1 that says "P, Q and P' are noncollinear." Why is this true?

3. (a) Show that $d(x,y) = |x - y|$ is the Euclidean metric on $\mathbb{R}$.

   (b) Show that for x, y in $\mathbb{R}$, $|x + y| \leq |x| + |y|$.

4. Finish the verifications that were left out of the proof of Theorem 6.6.

5. Consider points P, Q in $\mathcal{P}$ with S, T the points at infinity of the associated L-line. What happens to $L(P, Q)$ if we keep Q fixed but move P closer (in the Euclidean sense) to S? What happens if we move P closer to T? Now think of orthogonal circles that are very small at the edge of $\mathcal{D}$ and their associated L-lines in $\mathcal{P}$. What does the L-distance tell us about the type-2 lines that those circles determine? In other words, would those type-2 lines somehow seem shrunken in a hyperbolic world, as the underlying Euclidean circles are so small?

6. Show that by parametrizing the line ℓ as we did in (6.3), we get a line coordinatization that satisfies Definition 6.9.

7. Parametrize the line in $\mathbb{R}^2$ that passes through the points $(-1, 1)$ and $(2, 3)$, in three different ways, using the formula as given in (6.3), as follows. In each case, identify the point on ℓ associated with the real number 2.

   (a) Identify $(-1, 1)$ with 0 and the positive direction with $\mathbf{v} =< 3, 2 >$;

   (b) Identify $(2, 3)$ with 0 and the positive direction with $\mathbf{u} =< -3, -2 >$;

   (c) Identify $(-1, 1)$ with 0 and the positive direction with $\mathbf{u}$ as given in part (b).

8. Show that R as defined in Lemma 6.10 is a bijection in the case in which ℓ is a type-1 line.

9. Finish the proof of Lemma 6.11 for the case of a type-1 line.

10. Verify this statement, as it appears in the proof of Lemma 6.11: "Since the base on the circumference of $\mathcal{C}'$ for $\overset{\frown}{ST}$ is the major arc associated to ST, φ is obtuse."

11. Show that g' is as given in the proof of Lemma 6.13.

12. Finish the proof of Theorem 6.14 by showing that $f_2 : \ell \to \mathbb{R}_+$ is bijective and continuous. (Hint: It may be easiest to argue that (1) multiplication by a constant is a continuous bijective mapping of $\mathbb{R}$ to $\mathbb{R}$, and (2) if $h_1 : A \to B$ and $h_2 : B \to C$ are bijections, then $h_2 \circ h_1 : A \to C$ is a bijection.)

13. Show that Pasch's Axiom holds in $\mathcal{P}$.

14. Show that any mapping that leaves Euclidean distance invariant also leaves L-distance invariant. Give an example of a mapping that leaves L-distance invariant but does not leave Euclidean distance invariant.

15. Figure 6.11 depicts two different ways congruent L-triangles with a vertex at O could be oriented relative to one another. (See the proof of Theorem 6.20.) What are the other possibilities? For each possibility, identify the reflection or rotation you would use to map one of the L-triangles to ΔOPQ. For instance, reflection across the angle bisector of $\angle Q'OQ$ would

map $\Delta OP'Q'$ to ΔOPQ. What mapping would take ΔORS to $\Delta OP'Q'$? What about other configurations, for example if the L-triangles share a side?

16. Fix a type-1 line and a point P not on it in $\mathcal{P}$. Sketch the limit parallels through P. Repeat the exercise for a type-2 line in $\mathcal{P}$.

17. Describe an L-circle in $\mathcal{P}$. Are L-circles Euclidean circles as well?

Chapter 7

Affine Geometry

7.1 Introduction

Two features of affine geometry make it an attractive topic in our studies at this point. First, affine geometry concerns the incidence relations in Euclidean geometry. This is a simple, accessible idea, and affine spaces are ubiquitous in mathematics. Second, affine geometry is based on linear algebra, which is fundamental in applications and in mathematics more generally. An introduction to affine geometry thus gives us an excuse to profer a quick refresher on the fundamentals of linear algebra.

Most students remember the calculations that arise in linear algebra: solving systems of equations, matrix multiplication, row reduction, etc. For our initial linear algebra review, we concentrate on the ideas that underlie the concept of a vector space. Few students come out of a first course in linear algebra with a good grip on the conceptual framework of the subject. We focus here on the ideas that are often mislaid and on techniques that prove useful in the long haul when we use linear algebra to study other types of mathematics.

Our refresher is meant to serve as a review. As such, it does not contain many of the proofs but does display some of the major theorems that arise in an introductory course.

7.2 Some Linear Algebra

Linear algebra is the study of vector spaces and the mappings on them. The formal definition of a so-called *real vector space* is as follows.

Definition 7.1. Let V be a set with two operations: **vector addition**, a mapping

$$+ : V \times V \to V,$$

and **scalar multiplication**, another mapping

$$\cdot : \mathbb{R} \times V \to V.$$

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M. I. Dillon, *Geometry Through History*,
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Write $+(\mathbf{v}, \mathbf{w}) = \mathbf{v}+\mathbf{w}$, and $\cdot(c, \mathbf{v}) = c\,\mathbf{v}$, for $\mathbf{v}, \mathbf{w} \in V$, and $c \in \mathbb{R}$. We say that V is a **vector space over** $\mathbb{R}$ or a **real vector space** provided the following axioms are satisfied.

(1) Vector addition is commutative and associative: For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$,

$$\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}, \text{ and } (\mathbf{v} + \mathbf{w}) + \mathbf{u} = \mathbf{v} + (\mathbf{w} + \mathbf{u}).$$

(2) V contains an additive identity called the **zero vector**: There is $\mathbf{0} \in V$ that satisfies $\mathbf{0} + \mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V.
(3) Every vector $\mathbf{v}$ has an additive inverse: Given $\mathbf{v} \in V$, there is $\mathbf{v}' \in V$ such that $\mathbf{v} + \mathbf{v}' = \mathbf{0}$.
(4) Scalar multiplication distributes over vector addition:

$$c(\mathbf{u} + \mathbf{v}) = c\,\mathbf{u} + c\,\mathbf{v}, \text{ for all } c \in \mathbb{R}, \mathbf{u}, \mathbf{v} \in V.$$

(5) For $c, c' \in \mathbb{R}$ and $\mathbf{v} \in V$, $(cc')\mathbf{v} = c(c'\mathbf{v})$, and $(c + c')\mathbf{v} = c\,\mathbf{v} + c'\mathbf{v}$.
(6) $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V.

Elements in $\mathbb{R}$ are **scalars**. Scalar multiplication is also called **scaling**. Elements in V are **vectors**. $\mathbb{R}$ is called the **base field**.

When we want to emphasize the vector addition and scaling associated to a vector space V, we write $(V, +, \cdot)$. Otherwise, we just refer to the space as V.

There are two sorts of objects of interest associated to a vector space, V: vectors, which are in V, and scalars, which are in the base field. Note that *scalar* is just another word for number in the present context.

Beware that there is a scalar 0 and a vector $\mathbf{0}$ associated to any vector space. In general, these are different objects: 0 is a number, $\mathbf{0}$ is a vector.

Until further notice, our vector spaces are over $\mathbb{R}$ and we refer to them simply as vector spaces.

Lemma 7.2. Let V be a vector space and let $\mathbf{v} \in V$.

(1) The additive identity in V is unique. Indeed, if there is $\mathbf{u} \in V$ such that $\mathbf{u} + \mathbf{v} = \mathbf{v}$, even for just one vector $\mathbf{v}$, then $\mathbf{u} = \mathbf{0}$.

(2) $0\,\mathbf{v} = \mathbf{0}$.

(3) The additive inverse of $\mathbf{v}$ is unique.

(4) The additive inverse of $\mathbf{v}$ is $-1\mathbf{v}$.

Proof. Suppose V is a vector space with $\mathbf{v} \in V$.

(1) Suppose there is $\mathbf{u} \in V$ with $\mathbf{u} + \mathbf{v} = \mathbf{v}$. If $\mathbf{v}'$ is the additive inverse of $\mathbf{v}$ we have

$$\mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u} + (\mathbf{v} + \mathbf{v}') = (\mathbf{u} + \mathbf{v}) + \mathbf{v}' = \mathbf{v} + \mathbf{v}' = \mathbf{0},$$

which proves (1).

(2) $\mathbf{v} = 1\,\mathbf{v} = (0+1)\mathbf{v} = 0\,\mathbf{v} + 1\,\mathbf{v} = 0\,\mathbf{v} + \mathbf{v}$ so by (1), $0\,\mathbf{v} = \mathbf{0}$.
(3) Suppose there are $\mathbf{u}, \mathbf{w}$ so that $\mathbf{u} + \mathbf{v} = \mathbf{0}$ and $\mathbf{w} + \mathbf{v} = \mathbf{v} + \mathbf{w} = \mathbf{0}$. We must show that $\mathbf{u} = \mathbf{w}$. We have

$$\mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{0} + \mathbf{w} = \mathbf{w},$$

which proves (3).
(4) First we claim that $\mathbf{0} = \mathbf{v} + -1\,\mathbf{v}$. We have

$$\mathbf{0} = 0\,\mathbf{v} = (1 + -1)\,\mathbf{v} = 1\,\mathbf{v} + -1\,\mathbf{v} = \mathbf{v} + -1\,\mathbf{v},$$

as claimed. This shows that $-1\,\mathbf{v}$ is an additive inverse for $\mathbf{v}$. By (3), $-1\,\mathbf{v} = \mathbf{v}'$, that is, $-1\,\mathbf{v}$ is the unique additive inverse of $\mathbf{v}$. This proves (4).

□

In light of Lemma 7.2.(4), we designate the additive inverse of $\mathbf{v}$ by $-\mathbf{v}$.

Example 7.3. For fixed $n \in \mathbb{Z}_+$, $\mathbb{R}^n$ is a vector space. If $n = 1$, the vector space and the base field are the same. Otherwise, the base field and the vector space are entirely different sets. Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ are often presented as columns of numbers. This makes matrix multiplication more intuitive, but algebraically, there is no difference if we use n-tuples, $(x_1, x_2, \ldots, x_n)$, $x_i \in \mathbb{R}$, to represent vectors in $\mathbb{R}^n$. In $\mathbb{R}^n$, $\mathbf{0} = (0, \ldots, 0)$ and for $\mathbf{v} = (x_1, \ldots, x_n)$, $-\mathbf{v} = (-x_1, \ldots, -x_n)$.

If $\mathbf{v} = (x_1, \ldots, x_n)$, then $x_i \in \mathbb{R}$ is the ith **component** or the ith **coordinate** of $\mathbf{v}$. We define addition and scaling in $\mathbb{R}^n$ **componentwise**, meaning that the arithmetic is done component by component. For example in $\mathbb{R}^2$, $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$ and $2(x, y) = (2x, 2y)$.

Definition 7.1 (2) guarantees that every vector space has at least one vector, namely, the zero vector, $\mathbf{0}$: The empty set cannot be made into a vector space. It is not difficult to show that $V = \{\mathbf{0}\}$ does satisfy the definition of a vector space. It is called the **trivial space**. We leave it as an exercise to show that the trivial space is the only vector space over $\mathbb{R}$ with finitely many vectors!

Definition 7.4. Let V be a vector space over $\mathbb{R}$ and let $\mathcal{S} \subseteq V$. A **linear combination of vectors in $\mathcal{S}$** is an expression of the form $c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, where $c_i \in \mathbb{R}$, and $\mathbf{v}_i \in \mathcal{S}$.

A problem that vexes many students beginning to study linear algebra is what it means for two linear combinations of vectors in $\mathcal{S}$ to be the same. The comparison of linear combinations of vectors in $\mathcal{S}$ is a comparison of the coefficients for the vectors in $\mathcal{S}$. Two **linear combinations of vectors in $\mathcal{S}$ are the same** provided a given vector from $\mathcal{S}$ has the same coefficient in each linear combination.

Note in particular that the order of the summands is immaterial: $\mathbf{u} + \mathbf{v}$ and $\mathbf{v} + \mathbf{u}$ are the same linear combination of $\mathbf{u}$ and $\mathbf{v}$. Also, any vector in $\mathcal{S}$ that does not appear in a linear combination may be considered part of the linear combination with a coefficient of zero.

Example 7.5. Let $\mathcal{S} = \mathbb{R}^2$. The following linear combinations of vectors in $\mathcal{S}$ are all considered the same.

(1) $0(1,2) + (-1,3) + 2(-3,0)$

(2) $2(-3,0) + (-1,3) + 0(2,-5)$

(3) $(-1,3) + 2(-3,0)$

(4) $(1/2)(-1,3) + 2(-3,0) + (1/2)(-1,3)$

Notice that we must combine terms to get a single coefficient of 1 for $(-1,3)$ in the fourth expression. When comparing linear combinations, we compare coefficients of the same vectors.

The following are all different linear combinations of elements in $\mathcal{S}$, though each adds up to $\mathbf{0}$.

(1) $1(0,0)$

(2) $2(0,0)$

(3) $0(1,1)$

(4) $-2(1,1) + (1,2) + -(1/3)(-3,0)$

If we want to compare two of these, say the second two, we would note that $2(0,0) + 0(1,1)$ and $0(0,0) + 0(1,1)$ have different coefficients.

The following is nearly immediate by Definition 7.1.

Lemma 7.6. Any linear combination of vectors in a vector space V is in V.

Linear combinations are really the defining operations on vector spaces. When you run across a nonempty collection of objects that seems to be closed under linear combinations, suspect that this collection comprises a vector space.

Example 7.7. Let $\mathcal{F}$ be the collection of functions that map $\mathbb{R}$ to $\mathbb{R}$. $\mathcal{F}$ is a vector space. Linear combinations of functions on $\mathbb{R}$ yield new functions on $\mathbb{R}$: Given $f, g \in \mathcal{F}$, and $c_1, c_2 \in \mathbb{R}$, the linear combination $c_1 f + c_2 g$ is defined by

$$(c_1 f + c_2 g)(x) = c_1(f(x)) + c_2(g(x)).$$

For instance, if $f(x) = \sin x$ and $g(x) = x^2$, $(2f + 3g)(x) = 2\sin x + 3x^2$ gives us another function on $\mathbb{R}$. The zero vector in $\mathcal{F}$ is the function $\mathbf{0}(x) = 0$ for all $x \in \mathbb{R}$. The additive inverse of $f \in \mathcal{F}$ is $-f$, where $(-f)(x) = -f(x)$. From here, it is easy to check that the axioms detailed in Definition 7.1 are all satisfied.

Example 7.8. Let $\mathcal{D}$ be the set of differentiable functions on $\mathbb{R}$. In calculus, you learn that if f and g are in $\mathcal{D}$, then so is any linear combination of f and g: If c_1 and c_2 are scalars, $(c_1 f + c_2 g)' = c_1 f' + c_2 g'$. The zero function is differentiable, and if $f \in \mathcal{D}$, $-f \in \mathcal{D}$ so $\mathcal{D}$ is a vector space sitting inside $\mathcal{F}$.

The last example suggests a definition. Before considering that, we clarify some terminology and notation.

Definition 7.9. Let W be a subset of a vector space $(V, +, \cdot)$. If $+$ maps $W \times W$ into W, we say that W is **closed under addition**. If $\cdot$ maps $\mathbb{R} \times W$ into W, we say that W is **closed under scaling**. If every linear combination of elements in W is again in W, we say that W is **closed under linear combinations**.

Notice that if $(V, +, \cdot)$ is a vector space and $W \subset V$ is closed under addition, then since $+$ maps $V \times V$ into V, we should recognize that restricting the domain of $+$ to $W \times W$ really gives us a different mapping. We might designate $+$ restricted to $W \times W$ by $+|_{W \times W}$, but we will not. Instead, we continue to use $+$ and $\cdot$ when referring to subsets of V that are closed respectively under addition and scaling, as addition and scaling are defined on V.

Definition 7.10. A **subspace** of a vector space $(V, +, \cdot)$ is a subset $W \subseteq V$ so that $(W, +, \cdot)$ is itself a vector space.

Every nontrivial vector space has two subspaces: itself, and the trivial subspace, $\{\mathbf{0}\}$. If V is a vector space and $W \subsetneq V$ is a subspace, then W is called a **proper subspace** of V. A proper, nontrivial subspace of V contains more than just $\mathbf{0}$, but does not contain every vector in V.

Lemma 7.11. If V is a vector space and $W \subseteq V$, then W is a subspace if and only if (1) $W \neq \emptyset$, and (2) W is closed under linear combinations.

Proof. If W is a subspace of V, then by Lemma 7.2.(1) it must contain the zero vector of V, so it is nonempty. By Definition 7.1, W must also be closed under linear combinations, which proves the lemma in one direction.

Suppose then that W is nonempty and closed under linear combinations. First, since there is $\mathbf{w}$ in W, Lemma 7.2.(2) implies $0\,\mathbf{w} = \mathbf{0} \in V$ is also in W, so Definition 7.1.(2) holds. Next, if $\mathbf{w}_1, \mathbf{w}_2$ are in W, then $\mathbf{w}_1 + \mathbf{w}_2 = c_1\mathbf{w}_1 + c_2\mathbf{w}_2$ where c_1, c_2 are both 1, so $\mathbf{w}_1 + \mathbf{w}_2$ is in W, which shows that W is closed under vector addition. For any $\mathbf{w}$ in W, $c\,\mathbf{w}$ is a linear combination of elements in W thus belongs to W, so W is closed under scaling. For $\mathbf{w} \in W$, $-1\,\mathbf{w} = -\mathbf{w}$ is a linear combination of elements in W thus is itself in W, establishing that Definition 7.1.(3) applies to W. Finally, notice that since Definition 7.1.(1), 4-6 hold throughout V they must hold in W, as well. This proves $(W, +, \cdot)$ is a vector space, thus a subspace of V. □

Notice that Lemma 7.2 precludes the possibility that a subspace of V could have a zero vector that is different from $\mathbf{0}$ in V.

Example 7.12. Consider the collection of all functions $x = x(t)$ that satisfy the differential equation $x'' + 4x' + 3x = 0$. That collection is called the *general solution* to the differential equation. Notice for example that $x(t) = e^{-t}$ is one solution because if $x = e^{-t}$, then $x' = -e^{-t}$, and $x'' = e^{-t}$ so

$$x'' + 4x' + 3x = e^{-t} - 4e^{-t} + 3e^{-t} = 0.$$

Note that $x(t) = 0$ for all t is also a solution. We leave it as an exercise for you to verify that the general solution for this differential equation is a subspace of $\mathcal{D}$.

The vector spaces $\mathcal{F}$ and $\mathcal{D}$ are examples of *function spaces*. It is natural to consider these because the study of linear algebra typically starts after one or two terms of calculus. As vector spaces, though, $\mathcal{F}$ and $\mathcal{D}$ are rather difficult to work with. The next examples are a bit more accessible algebraically.

Example 7.13. The set $\mathbb{R}[x]$ of all polynomials in x with coefficients in $\mathbb{R}$ is a vector space. It is actually a subspace of $\mathcal{D}$. The important things to notice are that polynomials contain the zero mapping and that polynomials are closed under linear combinations. Once we see that, we realize there is no reason to restrict to one variable: $\mathbb{R}[x, y]$, the collection of polynomials in two variables, also contains the zero mapping and is closed under linear combinations. Think of $x^2 + 2xy + y^2$, itself a linear combination of monomials x^2, xy, and y^2. Note that $\mathbb{R}[x, y]$, though, is not a subspace of $\mathcal{D}$. If we think of a polynomial in two variables as a mapping into $\mathbb{R}$, its domain is $\mathbb{R}^2$, not $\mathbb{R}$.

Every element in $\mathbb{R}[x]$ can be written uniquely as a linear combination of elements from the set $\mathcal{B} = \{1, x, x^2, x^3, \ldots\}$. For instance $p(x) = (1 - x)^2 = 1-2x+x^2$. Notice that by Definition 7.4, a linear combination is always a sum—that is, it always has finitely many summands.[1] Even though $\mathcal{B}$ is an infinite set, a linear combination of elements of $\mathcal{B}$ only sums finitely many vectors.

Example 7.14. If we fix n, we get a subspace of $\mathbb{R}[x]$, $\mathbb{R}_n[x]$, the set of polynomials of degree n or less. For instance, $\mathbb{R}_2[x] = \{a + bx + cx^2 \,|\, a, b, c \in \mathbb{R}\}$ is the space of polynomials of degree 2 or less.

Definition 7.15. The **span** of $\mathcal{S}$, when $\mathcal{S}$ is a subset of a vector space, V, is the set of all vectors in V that can be written as linear combinations of elements in $\mathcal{S}$. If $\mathcal{S} = \emptyset$ the span of $\mathcal{S}$ is defined to be $\mathbf{0}$. If W is the span of $\mathcal{S}$, we write $W = \operatorname{span} \mathcal{S}$. In this case, we say $\mathcal{S}$ **spans** W and that $\mathcal{S}$ is a **spanning set** for W.

Note that $\operatorname{span}\{\,\} = \mathbf{0} = \operatorname{span}\{\mathbf{0}\}$. The span of a single nonzero vector $\mathbf{v}$ in a vector space V is the set $\{t\mathbf{v} \mid t \in \mathbb{R}\}$. Its points are in one-to-one correspondence with elements in $\mathbb{R}$ so we can, and often do, view the span of $\mathbf{v}$ as a line through $\mathbf{0}$.

We have seen that $\mathbb{R}[x] = \operatorname{span}\{1, x, x^2, \ldots\}$ and $\mathbb{R}_2[x] = \operatorname{span}\{1, x, x^2\}$.

Lemma 7.16. The span of a subset of any vector space V is a subspace of V.

Proof. The proof follows immediately by Lemma 7.11. □

Typically, a vector space in its entirety is not something that we deal with visually. We understand it through algebra. A vector is an element in a set. In the spirit of Hilbert, we may feel it natural to think of a vector as a point.

[1] Sums are always finite. An "infinite sum" is a *series*.

$\mathbb{R}^2$ and $\mathbb{R}^3$, though, are different. They are the sets that underlie analytic geometry and that provide canvases for graphing in calculus. Rather than things we do not expect to visualize, $\mathbb{R}^2$ and $\mathbb{R}^3$ are themselves tools for visualization. In these settings, we may depict vectors either as points, or as arrows—that is, "vectors" as we think of them in physics. We often toggle between the two representations, depending on which is more convenient at a given moment.

When we view an element (a, b) in $\mathbb{R}^2$ as an arrow, we take the point (a, b) as the head of the arrow and $(0, 0)$ as its tail. We either write $\mathbf{v} = (a, b)$ or $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$. Similarly, $(a, b, c) \in \mathbb{R}^3$ can be sketched as a dot or as an arrow with its tail at $(0, 0, 0)$ and its head at the point (a, b, c).

When $\mathbf{v}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ is depicted as an arrow and positioned with its tail at the origin, it is in **standard position**.

Any nonzero vector in $\mathbb{R}^2$ or $\mathbb{R}^3$ can be depicted with its head or tail at any preferred point. For instance, $\mathbf{v} = (1, 2)$ can be depicted as an arrow emanating from $(5/2, -1)$ in $\mathbb{R}^2$. Its head is now at the point $(7/2, 1)$. Any choice of tail (x_0, y_0) and head (x_1, y_1) that satisfy $x_1 - x_0 = 1$ and $y_1 - y_0 = 2$ gives us a representation of $\mathbf{v} = (x_1 - x_0, y_1 - y_0) = (1, 2)$. Reverse the roles of the head and tail to get from $\mathbf{v}$ to $-\mathbf{v} = (-1, -2)$. Several different representations for $\mathbf{v} = (1, 2)$ and $-\mathbf{v}$ are shown in Fig. 7.1. The one in fuchsia is $\mathbf{v}$ in standard position.

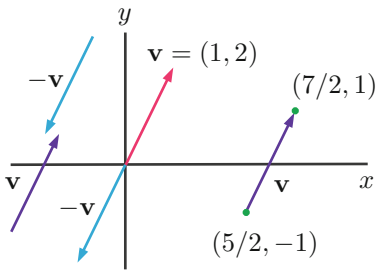


Figure 7.1: A vector in $\mathbb{R}^2$ can be positioned anywhere in the plane. In standard position, its tail is at the origin

Vectors in physics are often defined as entities determined by *direction* and *magnitude*. These are indeed vectors in the sense of Definition 7.1. In classical physics, vectors are typically elements in $\mathbb{R}^2$ or $\mathbb{R}^3$. Though we cannot visualize $\mathbb{R}^n$ for $n > 3$, we can visualize individual vectors in $\mathbb{R}^n$ as points or as arrows.

An element $(x_1, \ldots, x_n)$ in $\mathbb{R}^n$ has direction from the origin to the point $(x_1, \ldots, x_n)$. The magnitude of a vector in $\mathbb{R}^n$ is the Euclidean length of any arrow representing the vector. The direction and length of a vector in $\mathbb{R}^n$ are thus independent of its starting point and its endpoint.

Arbitrary vector spaces do not arrive on one's doorstep equipped with a device for measuring the length and/or direction of a vector but $\mathbb{R}^n$ does. The device is the *dot product*.

Definition 7.17. The **dot product** or **Euclidean inner product** on $\mathbb{R}^n$ is the mapping from $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ given by

$$(x_1, \ldots, x_n) \cdot (y_1, \ldots, y_n) = \sum_{i=1}^{n} x_i y_i.$$

Recall that for $\mathbf{v} = (a_1, \cdots, a_n)$ the **magnitude** or **length** of $\mathbf{v}$ is

$$|\mathbf{v}| = \sqrt{a_1^2 + \cdots + a_n^2}.$$

For $\mathbf{v} = (a_1, \ldots, a_n)$, and $\mathbf{w} = (b_1, \ldots, b_n)$, the **angle** between $\mathbf{v}$ and $\mathbf{w}$ is φ where

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}||\mathbf{w}| \cos \varphi. \tag{7.1}$$

Note that vectors $\mathbf{v}, \mathbf{w}$ are orthogonal when the angle between them is $\varphi = \pi/2$, that is, when $\cos \varphi = 0$. We can thus detect whether two nonzero vectors in $\mathbb{R}^n$ are orthogonal by checking to see whether their dot product is zero.

In precalculus courses, we verify that the description of the dot product in (7.1) conforms to that in Definition 7.17.

It is a short step to polar coordinates from thinking of vectors in $\mathbb{R}^2$ as determined by magnitude and direction. We first used polar coordinates in Chapter 5, starting with the discussion before Theorem 5.17. There and in Chapter 6, they served as a convenient device to help us understand circular inversion. Polar coordinates arise again in Chapter 10 when we discuss rotations. Here we interpret them using the dot product on $\mathbb{R}^2$.

We identify the **polar axis** in $\mathbb{R}^2$ with the positive x-axis. Each vector $\mathbf{v}$ in $\mathbb{R}^2$ is then determined by its magnitude or *modulus*, $r = |\mathbf{v}|$, and its *argument*, that is, the angle θ that it forms with the polar axis. Recall the conversions

$$x = r \cos \theta, \; y = r \sin \theta, \; r = \sqrt{x^2 + y^2}. \tag{7.2}$$

If c is any scalar, it is easy to verify that $|c\,\mathbf{v}| = |c||\mathbf{v}|$ so in $\mathbb{R}^n$, scaling a vector by 2 doubles its length, scaling by $1/2$ halves its length, and scaling a vector by a negative number c reverses its direction, and changes its length by a factor of $|c|$.

A **unit vector** in $\mathbb{R}^n$ has length 1. If $\mathbf{v} \neq \mathbf{0}$, then $\mathbf{u} = \frac{1}{|\mathbf{v}|}\,\mathbf{v}$ is a unit vector in the same direction as $\mathbf{v}$. The arrows representing $\mathbf{v}$ and $-\mathbf{v}$ in $\mathbb{R}^2$ have the same slope but as vectors, they have opposite directions.

Slope may seem like the best way to capture the parallel class of a line, but it is not. The concept of slope is endemic to the xy-coordinate plane. It is not even defined in $\mathbb{R}^n$ when $n \geq 3$, for example. Worse, the slope of a line in the xy-plane is not defined if the points of the line (x, y), do not satisfy $y = f(x)$, for some function f. Geometrically, a vertical line is like any other, but when we think of lines in terms of slope, vertical lines are treated as aberrations. To talk

about the parallel class of a line in $\mathbb{R}^n$, we use the direction of a vector—often a unit vector—that is parallel to the line.

Vectors are better than slope for describing the parallel class of a line but they are not perfect. If we identify the direction of a line ℓ with a unit vector parallel to ℓ, there is more than one vector to choose from. This is simply because nonzero vectors $\mathbf{v}$ and $-\mathbf{v}$ are themselves parallel. Looking at it from the other side, we realize that a line has two directions: A line that extends east also extends west. The correspondence between parallel classes of lines and unit vectors is not one-to-one; it is one-to-two.

We consider how best to determine an algebraic description of the points on a line in $\mathbb{R}^n$. While a single linear equation in x and y may seem the most natural way to describe the points (x, y) on a line in $\mathbb{R}^2$, the single equation model does not carry into higher dimensions. In other words, for $n > 2$, the collection of points $(x_1, \ldots, x_n) \in \mathbb{R}^n$ that satisfy an equation of the form

$$c_1 x_1 + \cdots c_n x_n = b \tag{7.3}$$

for $c_i, b \in \mathbb{R}$, does not describe a line in $\mathbb{R}^n$. Vector expressions, and parametric equations, can be used to describe lines in $\mathbb{R}^n$ for all $n \in \mathbb{Z}_+$. These are our preferred tools.

Example 7.18. We consider the line ℓ in $\mathbb{R}^2$ determined by points $(-1, -2)$ and $(3, 1)$. One direction vector for ℓ is in the direction of $\mathbf{v} = (3, 1) - (-1, -2) = (4, 3)$, shown in Fig. 7.2 in standard position and in a second position from point

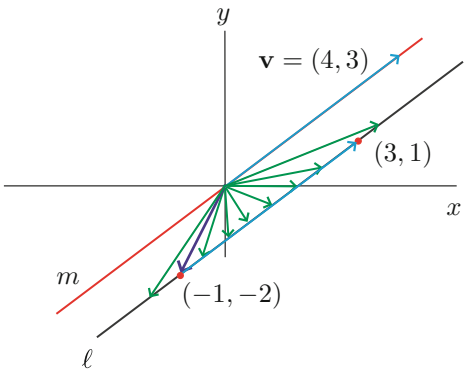


Figure 7.2: $\mathbf{v}$ is a direction vector for $m = \{t\mathbf{v} \,|\, t \in \mathbb{R}\}$ and all lines parallel to m

$(-1, -2)$ to $(3, 1)$. (Note that the slope of $\mathbf{v}$ is $3/4$, the slope of ℓ.) The span of $\mathbf{v}$ is $m = \{t\mathbf{v} | \, t \in \mathbb{R}\}$, a line through the origin parallel to ℓ. By adding a single vector to each vector in m, we translate m to ℓ. Any vector that, in standard position, has its head on ℓ may serve as a translation vector. We make

an arbitrary choice and take $\mathbf{w} = (-1, -2)$ (in dark blue) to write

$$\ell = \{t\mathbf{v} + \mathbf{w} | \, t \in \mathbb{R}\}. \tag{7.4}$$

Writing $\mathbf{v}$ and $\mathbf{w}$ in components, we have

$$t\mathbf{v} + \mathbf{w} = t(4, 3) + (-1, -2) = (4t - 1, 3t - 2),$$

so

$$\ell = \{(4t - 1, 3t - 2) | \, t \in \mathbb{R}\}.$$

From a slightly different point of view, we can describe ℓ as the collection of points (x, y) with

$$\begin{cases} x &= 4t - 1 \\ y &= 3t - 2, t \in \mathbb{R}. \end{cases} \tag{7.5}$$

We note that (7.4) gives us a **vector representation of the line** ℓ and (7.5) is a **parametric description** of ℓ.

A vector $\mathbf{v}$ lies on a line ℓ only if the head of $\mathbf{v}$ lies on ℓ when $\mathbf{v}$ is in standard position. Vectors that can be positioned to lie along the line ℓ are direction vectors for ℓ. Generally, a direction vector for a line does not lie on the line. If ℓ does not go through the origin—that is, if it is not a subspace—then no direction vector for ℓ lies "on" ℓ. In Fig. 7.2, the green vectors are on ℓ but $\mathbf{v}$ is not, although it appears to be when it is positioned from $(-1, -2)$ to $(3, 1)$. Every tool has its price and this is part of the price we pay to use vectors.

We can formulate this idea of direction for more general vector spaces where we would be less inclined to view vectors as arrows.

Definition 7.19. A **vector $\mathbf{v}$ is parallel to a line** $\ell \subseteq V$, V a vector space, provided $\ell = \{t\mathbf{v} + \mathbf{w} \, | \, t \in \mathbb{R}\}$, for some $\mathbf{w} \in V$. A **direction vector for a line** ℓ is a vector $\mathbf{v}$ parallel to ℓ. If $\mathbf{w} \neq \mathbf{0}$, $\mathbf{w}$ is a **translation vector** for ℓ. If ℓ and m are nonintersecting lines with the same direction vectors they are **parallel**.

We have seen that the span of a single nonzero vector in $\mathbb{R}^n$ is a line through the origin. Another way to say this is that a minimal spanning set for a line through the origin is a single nonzero vector on that line. This idea can be generalized.

Definition 7.20. A **basis** is a minimal spanning set for a vector space V.

What does it mean for a set $\mathcal{S}$ to be a *minimal* spanning set for V? It means that $\mathcal{S}$ spans V but no proper subset of $\mathcal{S}$ spans V.

It is not clear that every vector space should have a basis. A critical theorem in mathematics says that it does.

Theorem 7.21. Every vector space has a basis.

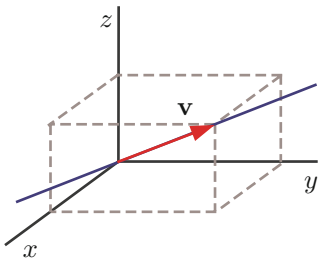


Figure 7.3: The span of a nonzero vector $\mathbf{v}$ in $\mathbb{R}^n$ is a line ℓ through the origin

This is one of the ideas that makes vector spaces so useful and so appealing. Proof of it is beyond the scope of our work here, and indeed, is rarely encountered in undergraduate courses in linear algebra.

Note that the empty set, $\{\,\}$, is a basis for the trivial vector space, $\{\mathbf{0}\}$.

Coordinates and bases go hand in hand. When we have an arbitrary vector in $\mathbb{R}^2$, we can —and often do—think of it as a linear combination of the elements in $\mathcal{B} = \{(1,0),(0,1)\}$. In other words, the coordinates of $\mathbf{v} = (a,b)$, give us the scalars we need to write $\mathbf{v} = a(1,0) + b(0,1)$, a linear combination of the elements in $\mathcal{B}$. Since we can write every element in $\mathbb{R}^2$ this way, and since no proper subset of $\mathcal{B}$ spans $\mathbb{R}^2$, then $\mathcal{B}$ must be a basis for $\mathbb{R}^2$. Since $a(1,0)+b(0,1)$ and $b(0,1)+a(1,0)$ are just different ways to write the same linear combination, we can say that there is one and only one linear combination of elements in $\mathcal{B}$ that yields $\mathbf{v}$.

We sometimes view a basis $\mathcal{B}$, not just as a set of vectors, but as a set of vectors with a particular ordering. In this case, we refer to $\mathcal{B}$ as an **ordered basis**. While $\{(1,0),(0,1)\}$ and $\{(0,1),(1,0)\}$ are the same set, they are different ordered bases for $\mathbb{R}^2$. This is convenient, for example, when we talk about coordinate vectors. In our example here, if we let $\mathcal{B} = \{(1,0),(0,1)\}$ be the ordered basis for $\mathbb{R}^2$, then the so-called $\mathcal{B}$-coordinates of $\mathbf{v}$, (a,b), determine $\mathbf{v}$ uniquely, and it is clear that a is the coefficient for the first element in $\mathcal{B}$, while b is the coefficient for the second element in $\mathcal{B}$. When the order of a basis is of some consequence, we use the locution *ordered basis*. When the order of a basis is immaterial, we leave off the word *ordered* and refer simply to a basis.

The ordered basis we have been discussing here, $\mathcal{B} = \{(1,0),(0,1)\}$, is called the **natural basis** or the **standard basis** for $\mathbb{R}^2$. In fact, the natural basis for any $\mathbb{R}^n$ is

$$\{(1,0,\ldots,0),(0,1,0,\ldots,0),\ldots,(0,\ldots,0,1)\},$$

with the indicated ordering. The following notation is in common usage as well:

$$\mathbf{e}_1 = (1,0,\ldots,0), \mathbf{e}_2 = (0,1,0,\ldots,0),\ldots,\mathbf{e}_n = (0,\ldots,0,1).$$

Example 7.22. Let $\mathcal{S} = \{(1,1),(1,0),(0,1)\}$. Note that $\mathcal{S} \subset \mathbb{R}^2$. We leave it as an exercise to show that because $\mathcal{S}$ contains a basis for $\mathbb{R}^2$, it spans $\mathbb{R}^2$.

Since the natural basis is a proper subset of $\mathcal{S}$, $\mathcal{S}$ is not a minimal spanning set for $\mathbb{R}^2$, thus cannot itself be a basis for $\mathbb{R}^2$. Notice that many different linear combinations of elements in $\mathcal{S}$ yield a given vector $\mathbf{v}$ in $\mathbb{R}^2$. For instance, while $(2,1) = 0(1,1) + 2(1,0) + 1(0,1)$, we also have

$$(2,1) = 2(1,1) + 0(1,0) + (-1)(0,1) = 1(1,1) + 1(1,0) + 0(0,1).$$

In fact, if there is more than one way to write any given vector as a linear combination of some set of vectors, there are infinitely many ways. We leave proof of that as an exercise.

The first linear algebra course focuses on consequences of the next theorem.

Theorem 7.23. If a vector space has a basis with n elements, then every basis for that vector space has exactly n elements.

This is one of the key results in elementary linear algebra, and allows us to make the following definition.

Definition 7.24. The **dimension** of a vector space V, $\dim V$, is the number of elements in a basis for V. If V is trivial, $\dim V = 0$. If V does not have a finite basis, V is **infinite-dimensional**.

Since $\mathcal{B} = \{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$, $\dim \mathbb{R}^n = n$.

Dimension is a measure of the "roominess" of a space, of how many degrees of freedom one has if one would like to change position in the space. Here, degrees of freedom are measured along lines. In other words, being trapped on a line, with the ability to move say, east-west, we have one degree of freedom. Being trapped in a plane and able to move east-west or north-south, we have two degrees of freedom. Notice that by combining the abilities to move both east-west and north-south, we can move along any point of the compass. In a three-dimensional space, we can move along any point of the compass, but we get a third degree of freedom that allows us, say, to change altitude when changing position.

An important theorem about subspaces and dimension follows. We omit the proof of this one as well but note that choosing a basis underlies the machinations in many of the proofs in elementary linear algebra.

Theorem 7.25. Let $\dim V = n$. The only subspace of V with dimension 0 is the trivial subspace. The only subspace of V with dimension n is V itself. Every proper subspace of V has dimension strictly less than n. For every nonnegative integer $k < n$, there are subspaces of V with dimension k.

We leave the proof of the following as an easy exercise.

Lemma 7.26. The intersection of subspaces of a vector space is itself a subspace.

Example 7.27. The set $\mathcal{B} = \{1, x, x^2, x^3, \ldots\}$ spans $\mathbb{R}[x]$ because every polynomial $p(x)$ can be written in the form $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ for some nonnegative integer n with coefficients $a_i \in \mathbb{R}$. We leave it as an exercise to show that no proper subset of $\mathcal{B}$ spans $\mathbb{R}[x]$, thus, $\mathcal{B}$ is a basis for $\mathbb{R}[x]$. By taking the spans of sets $\{1\}$, $\{1, x\}$, $\{1, x, x^2\}$, etc., we get a nested chain of proper subspaces of $\mathbb{R}[x]$ of dimensions 1, 2, 3, etc. Notice there are many subspaces of $\mathbb{R}[x]$ of a given dimension greater than 0. For instance, $\operatorname{span}\{1 + x\}$ is one-dimensional, but it neither contains $\operatorname{span}\{1\}$, nor is contained in $\operatorname{span}\{1\}$.

Since there is no finite subset of $\mathcal{B}$ that spans $\mathbb{R}[x]$, $\dim \mathbb{R}[x] = \infty$. We leave it as an exercise to show that $\mathcal{B}_n = \{1, x, x^2, x^3, \ldots, x^n\}$ is a basis for $\mathbb{R}_n[x]$. When it has the indicated ordering, $\mathcal{B}_n$ is called the **natural basis** or the **standard basis of** $\mathbb{R}_n[x]$. We then have $\dim \mathbb{R}_n[x] = n + 1$.

How can we tell whether a spanning set for a vector space is a basis? The minimality condition is actually tied to a uniqueness condition that we can check instead. This is captured in the idea of *linear independence*. Before considering what linear independence is, we make an observation about the consequences of having two different linear combinations of elements of $\mathcal{S}$ that yield the same vector in $\operatorname{span} \mathcal{S}$.

Observation 7.28. Suppose $\mathcal{S}$ is a spanning set for V and that we can write $\mathbf{v} \in V$ as two different linear combinations of elements in $\mathcal{S}$. Then

$$\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n, \tag{7.6}$$

for some $a_i, b_i \in \mathbb{R}$, and $\mathbf{v}_i \in \mathcal{S}$. If all the coefficients are identical, the linear combinations in (7.6) are the same so assume $a_1 \neq b_1$. Now we toss $\mathbf{v}$ aside entirely and notice that

$$\mathbf{v}_1 = \frac{b_2 - a_2}{a_1 - b_1}\mathbf{v}_2 + \frac{b_3 - a_3}{a_1 - b_1}\mathbf{v}_3 + \cdots + \frac{b_n - a_n}{a_1 - b_1}\mathbf{v}_n,$$

that is, $\mathbf{v}_1 \in \operatorname{span}(\mathcal{S} \setminus \mathbf{v}_1)$. We leave it as an exercise to show that this implies $\operatorname{span} \mathcal{S} = \operatorname{span}(\mathcal{S} \setminus \mathbf{v}_1)$. From there it follows that $\mathcal{S}$ cannot be a basis for V.

If every $\mathbf{v}$ in a vector space V can be written uniquely as a linear combination of elements in $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$, then the only way to write $\mathbf{0}$ as a linear combination of elements in $\mathcal{B}$ is to use the coefficient 0 for all $\mathbf{v}_i$. On the other hand, if $\mathcal{B}$ is a spanning set for V and the only way to write $\mathbf{0}$ as a linear combination of elements in $\mathcal{B}$ is $0\,\mathbf{v}_1 + \cdots + 0\,\mathbf{v}_n$, then if there is a vector $\mathbf{v}$ in V with

$$\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$$

for some choice of a_i and b_i in $\mathbb{R}$, we have

$$\mathbf{0} = (a_1 - b_1)\mathbf{v}_1 + \cdots + (a_n - b_n)\mathbf{v}_n$$

implying that $a_i - b_i = 0$ for all i, thus that $a_i = b_i$ for all i. We see then that if $\mathbf{0}$ is uniquely expressible as a linear combination of elements in $\mathcal{B}$, every vector

in span $\mathcal{B}$ is uniquely expressible as a linear combination of the elements of $\mathcal{B}$. In other words, if $V = \text{span}\,\mathcal{B}$, then to show $\mathcal{B}$ is a basis, we just have to show that there is only one way to write $\mathbf{0}$ as a linear combination of the elements in $\mathcal{B}$. This leads to our definition.

Definition 7.29. A set of vectors $\mathcal{S}$ is **linearly independent** provided the only way to express $\mathbf{0}$ as a linear combination of elements of $\mathcal{S}$ is to take all coefficients equal to 0. Any set that is not linearly independent is **linearly dependent**. If $\mathcal{S}$ is linearly dependent, and

$$\mathbf{0} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n, \tag{7.7}$$

where some c_i is nonzero, then (7.7) is called a **dependence relation**.

The following is a gathering of some of the basic results about linearly independent and dependent sets. The reader should supply the proofs.

Theorem 7.30. (1) A basis for a vector space must be linearly independent.

(2) Any set of vectors containing $\mathbf{0}$ is linearly dependent.

(3) A single vector forms a linearly independent set if and only if the vector is nonzero.

(4) A set of two nonzero vectors is linearly dependent if and only if one vector is a scalar multiple of the other.

(5) A set of nonzero vectors $\mathcal{S} = \{\mathbf{v}_1 \ldots, \mathbf{v}_n\}$ is linearly independent if and only if for each $\mathbf{v}_j \in \mathcal{S}$, $j = 2, \ldots, n$, $\mathbf{v}_j$ is not in $\text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}\}$.

(6) Every subset of a set of linearly independent vectors is linearly independent.

(7) Every set containing a linearly dependent set is dependent.

Showing linear dependence for a set with more than two nonzero vectors typically requires producing a dependence relation on the set.

Example 7.31. We saw that $\mathcal{S} = \{(1,1),(1,0),(0,1)\}$ in Example 7.22 was a non-minimal spanning set for $\mathbb{R}^2$. It follows that $\mathcal{S}$ is linearly dependent. Explicitly, we have the following dependence relation

$$\mathbf{0} = 1(1,0) + 1(0,1) + (-1)(1,1).$$

Example 7.32. Let $\mathcal{B} = \{(1,1),(1,-1)\}$. We claim that $\mathcal{B}$ is a basis for $\mathbb{R}^2$. Given $\mathbf{b} = (b_1, b_2)$ in $\mathbb{R}^2$, we must argue that there are unique real numbers x and y so that

$$x(1,1) + y(1,-1) = \mathbf{b}.$$

Working with components, we get two linear equations in two unknowns:

$$\begin{cases} x + y = b_1 \\ x - y = b_2 \end{cases}.$$

Adding the two equations, we get $2x = b_1 + b_2$, so $x = (b_1 + b_2)/2$. Subtracting the second equation from the first, we get $2y = b_1 - b_2$ so $y = (b_1 - b_2)/2$. In other words, we can always solve for x and y in terms of the coordinates of $\mathbf{b}$. (This shows that $\mathcal{B}$ spans $\mathbb{R}^2$.) Since the x- and y-coordinates are defined *uniquely* in terms of b_1 and b_2, $\mathcal{B}$ must be linearly independent. In particular, the only way to write $\mathbf{0}$ as a linear combination of the elements in $\mathcal{B}$ is by taking $x = y = 0$.

We cite another important theorem, and corollary, from an introductory linear algebra course.

Theorem 7.33. Let $\mathcal{S} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\} \subset V$, where $\dim V = n$.

(1) $\mathcal{S}$ is a basis for V if and only if $\mathcal{S}$ spans V.

(2) $\mathcal{S}$ is a basis for V if and only if $\mathcal{S}$ is linearly independent.

Corollary 7.34. If $\dim V = n$, then any set of $k > n$ vectors in V must be linearly dependent and no set of $k < n$ vectors can span V.

A set of vectors must satisfy two criteria if it is to be a basis for a finite-dimensional vector space. The theorem says that the criteria can be any two from among the following: (1) the set spans the space, (2) the set is linearly independent, (3) the number of vectors in the set is the same as the dimension of the space.

The following is another important theorem in applications. We omit the proof.

Theorem 7.35. Let V be finite-dimensional. (1) If $\mathcal{S}$ spans V, then $\mathcal{S}$ contains a basis for V. (2) If $\mathcal{S}$ is a linearly independent set of vectors in V, then $\mathcal{S}$ can be expanded to a basis for V.

Exercises 7.1.

1. Cite the axioms in Definition 7.1 that justify each step in the proof of Lemma 7.2.

2. Show that a nontrivial vector space over $\mathbb{R}$ contains infinitely many vectors.

3. Argue directly from Definition 7.1 that a linear combinations of vectors in a vector space is also in the vector space.

4. Sketch the vector $(-1, 2)$ in $\mathbb{R}^2$ two different ways: (1) emanating from the origin; and (2) emanating from the point $(1, 1)$.

5. Show that the formula given in (7.1) conforms to Definition 7.17.

6. Verify that if $\mathbf{v} = (a_1, \ldots, a_n)$ and $c \in \mathbb{R}$, then $|c\,\mathbf{v}| = |c||\mathbf{v}|$.

7. This problem refers to the line ℓ in Example 7.18.

   (a) Give an equation of the form $y = f(x)$ for ℓ.

(b) Find an equation for ℓ of the form $ax + by = c$, $a, b, c \in \mathbb{R}$. This is often called the *standard form* for a line ℓ in the xy-plane. What is the standard form for a line with undefined slope in the xy-plane?

(c) Find the vector description of ℓ when we translate m by $(0, -5/4)$. Use that description to get another parametric description of ℓ. Prove that the vector description and the parametric description you found here give you the same set of vectors/points as the descriptions given in Example 7.18.

(d) Find the vector description of ℓ using direction vector $(-1, -3/4)$ and translation vector $(0, -5/4)$. Use that vector description to get another parametric description of ℓ. Prove that the vector description and the parametric description you found here give you the same set of vectors/points as the descriptions given in Example 7.18.

8. Let ℓ be the line in $\mathbb{R}^3$ determined by the points $(-1, 2, 1)$ and $(2, 0, 3)$.

(a) Find two different direction vectors and two different translation vectors for ℓ. In each case, verify that ℓ has the form given in (7.4).

(b) Find two different ways to describe ℓ using parametric equations for x, y, z, coordinates of the points of ℓ.

9. The parallelogram rule for adding vectors in $\mathbb{R}^2$ is the same as adding components. The rule says that the sum of vectors $\mathbf{v}$ and $\mathbf{w}$ is the diagonal of the parallelogram that $\mathbf{v}$ and $\mathbf{w}$ determine. (See Fig. 7.4.) Use Euclidean geometry to show that the parallelogram rule for adding vectors $\mathbf{v} = (2, 0)$ and $\mathbf{w} = (1, 1)$ yields $(3, 1)$.

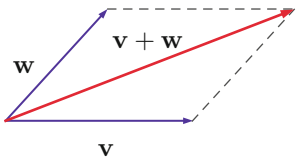


Figure 7.4: The parallelogram rule

10. Complete the verification that the solutions to $x'' + 4x' + 3x = 0$ is a subspace of $\mathcal{F}$.

11. Show that for a fixed positive integer n, $\mathbb{R}_n[x]$ is a subspace of $\mathbb{R}[x]$.

12. Show that there is no finite subset of $\mathcal{B} = \{1, x, x^2, x^3, \ldots\}$ that spans $\mathbb{R}[x]$. This is what we mean when we say that $\dim \mathbb{R}[x] = \infty$.

13. Show that if V is a vector space and $\mathcal{S} \subset V$ contains a basis for V, then $\mathcal{S}$ spans V.

14. Referring to Example 7.22, find two different linear combinations of the elements in $\mathcal{S}$, with all nonzero coefficients, that yield $(2, 1)$.

15. Show that if there is more than one way to write any given vector as a linear combination of some set of vectors, there are infinitely many ways.

16. Let $\mathcal{S}$ be a subset of a vector space V and suppose there is $\mathbf{v} \in \mathcal{S}$ so that $\mathbf{v} \in \operatorname{span}(\mathcal{S} \setminus \mathbf{v})$. Show that $\operatorname{span} \mathcal{S} = \operatorname{span}(\mathcal{S} \setminus \mathbf{v})$.

17. Argue that if a vector space has one basis with at least one element, then it has infinitely many different bases.

18. Prove Lemma 7.26.

19. Prove Theorem 7.30.

7.3 Fields

The scalars for an arbitrary vector space need not be real numbers, but they must form what is called a *field*. The notion of a binary operation is fundamental in the study of algebraic objects like vector spaces and fields so we linger over it a bit before taking up the definition of a field.

Definition 7.36. A **binary operation** $\circ$ on any set $\mathcal{S}$ is a mapping $\circ : \mathcal{S} \times \mathcal{S} \to \mathcal{S}$. In particular, for each pair of elements, a, b in $\mathcal{S}$, there must be a single element c in $\mathcal{S}$ so $a \circ b = c$. If $\circ$ is a binary operation on $\mathcal{S}$, we also say $\mathcal{S}$ is **closed** under $\circ$.

Addition and multiplication are binary operations on $\mathbb{R}$; the set of integers, $\mathbb{Z}$; the set of rational numbers, $\mathbb{Q}$; and the set of complex numbers, $\mathbb{C}$. Vector addition is a binary operation on any vector space, V. If V is a real vector space, then scalar multiplication is a mapping $\mathbb{R} \times V \to V$. *This is not a binary operation on* V! The domain of a binary operation on V is $V \times V$. Matrix addition and matrix multiplication are binary operations on $M_n(\mathbb{R})$, $n \times n$ matrices with entries in $\mathbb{R}$, or even $M_n(\mathbb{Z})$, $n \times n$ matrices with integer entries. Function addition, function multiplication, and function composition are all binary operations on $\mathcal{F}$.

Note that the composition of mappings is always associative, but rarely commutative.

Definition 7.37. A binary operation $\circ$ on $\mathcal{S}$ is **associative** provided $(a \circ b) \circ c = a \circ (b \circ c)$, for all $a, b, c \in \mathcal{S}$. The binary operation is **commutative** provided $a \circ b = b \circ a$ for all $a, b \in \mathcal{S}$. There is a $\circ$ **identity** e provided $a \circ e = e \circ a = a$ for all $a \in \mathcal{S}$.

Definition 7.38. A **field** $(\mathbb{F}, \oplus, \odot)$ is a set $\mathbb{F}$ with at least two distinct elements, 0 (zero) and 1, together with commutative and associative binary operations $\oplus$ (addition) and $\odot$ (multiplication) that satisfy the following axioms.

(1) 0 is the additive identity: $0 \oplus a = a$ for all $a \in \mathbb{F}$.

(2) For every $a \in \mathbb{F}$, there is an additive inverse $-a \in \mathbb{F}$: $a \oplus -a = 0$.

(3) 1 is the multiplicative identity: $1 \odot a = a$ for all $a \in \mathbb{F}$.

(4) For every nonzero element $a \in \mathbb{F}$, there is a multiplicative inverse $\hat{a} \in \mathbb{F}$: $a \odot \hat{a} = 1$.

(5) For all $a, b, c \in \mathbb{F}$, $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$.

The last axiom is called the **distributive law**.

When the binary operations $\oplus$ and $\odot$ are understood, we usually refer to a field $(\mathbb{F}, \oplus, \odot)$ simply as $\mathbb{F}$.

$\mathbb{R}$ is the most familiar field but $\mathbb{Q}$, the rational numbers, is a field as well. Since $\mathbb{Q} \subset \mathbb{R}$, and $\mathbb{Q}$ is a field under the same operations as in $\mathbb{R}$, we say $\mathbb{Q}$ is a **subfield** of $\mathbb{R}$.

We use the notation $\oplus$, $\odot$, and $\hat{a}$ when we talk about fields in the abstract. In $\mathbb{R}$ and $\mathbb{Q}$, $\oplus$ and $\odot$ are the usual addition and multiplication so we write them $+$ and $\cdot$. Multiplication is often designated by juxtaposition, so that $ab = a \cdot b$. In these fields, for $a \neq 0$, $\hat{a}$ is often written $1/a$ or a^{-1}.

$\mathbb{R}$ and $\mathbb{Q}$ are examples of **ordered fields**: Aside from satisfying the field axioms, they both have an ordering that respects the field operations of addition and multiplication. In particular, for $a, b, c \in \mathbb{R}$, $a < b$ implies $a + c < b + c$, and if $a > 0$, $b < c$ implies $ab < ac$. The ordering of the underlying field has implications for the geometries based on vector spaces over these fields. Beware, though: While any set can be ordered, most fields are not *ordered fields*, that is, they cannot be ordered in a way that meshes with the field operations.

The 0 in the next lemma is not to be confused with the real number 0. Here, 0 is the additive identity in an abstract field. We leave the proof of the lemma as an exercise.

Lemma 7.39. In any field $\mathbb{F}$, $0 \odot a = 0$ for all $a \in \mathbb{F}$.

Vector spaces over $\mathbb{Q}$ are not much different from vector spaces over $\mathbb{R}$. Pictures we might use to think about $\mathbb{R}^2$ work just as well if we are thinking about $\mathbb{Q}^2 = \{(a, b) | \, a, b \in \mathbb{Q}\}$, for instance. The differences between the two start to show up when we try to do Euclidean geometry, particularly when we consider the nature of a line.

Example 7.40. The complex numbers, $\mathbb{C}$, form a field. It has subfields $\mathbb{R}$ and $\mathbb{Q}$. We use $+$ and juxtaposition or $\cdot$ to denote addition and multiplication in $\mathbb{C}$. We use $1/z$ for the multiplicative inverse of a nonzero element z in $\mathbb{C}$. We often view $\mathbb{C}$ as the set $\{a + bi | \, a, b \in \mathbb{R}\}$, where i is a symbol that satisfies $i^2 = -1 \in \mathbb{R}$. If $z = a + bi \in \mathbb{C}$, where $a, b \in \mathbb{R}$, a is the *real part* and b is the *imaginary part* of z. In verifying that $\mathbb{C}$ is a field, one finds that the interesting thing to check is that a nonzero element in $\mathbb{C}$ has a multiplicative inverse in $\mathbb{C}$. We leave that verification as an exercise.

$\mathbb{C}$ often arises in its role as the **algebraic closure** of $\mathbb{R}$. This means that $\mathbb{R} \subset \mathbb{C}$ and that all polynomials with coefficients in $\mathbb{R}$ factor into degree one polynomials over $\mathbb{C}$, for example $x^2 + 1 = (x - i)(x + i)$. When we move from $\mathbb{R}$ to $\mathbb{C}$, we lose the ordered field structure. Like any set, $\mathbb{C}$ can be ordered, but it does not have an ordering that respects addition and multiplication.

Familiar sets that have two commutative associative binary operations that do not satisfy all the field axioms are $\mathbb{Z}$, $M_n(\mathbb{R})$, and $\mathcal{F}$. We can define multiplication on vectors in $\mathbb{R}^2$ to make it a field: $(a, b) \odot (a', b') = (aa' - bb', ab' + a'b)$ but when we do, it is because we are thinking of the points in $\mathbb{R}^2$ as complex numbers. As sets, and indeed as vector spaces over $\mathbb{R}$, $\mathbb{R}^2$ and $\mathbb{C}$ can be identified. When we make that identification, we call the set the **complex plane** instead of $\mathbb{R}^2$.

Example 7.41. If p is a prime number we let $\mathbb{F}_p = \{0, 1, 2, \ldots, p - 1\}$ with **addition and multiplication mod** p, that is, the sum or product of two elements in $\mathbb{F}_p$ is the remainder when we divide the usual sum or product by p. Then $\mathbb{F}_2 = \{0, 1\}$ with the following addition and multiplication tables.

$\oplus$	0	1
0	0	1
1	1	0

$\odot$	1
1	1

Since $0 \odot a = a \odot 0 = 0$ for all $a \in \mathbb{F}$, we leave 0 off the multiplication table for a field.

Similar to $\mathbb{F}_2$, we have $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$ with addition and multiplication mod 5 giving us the following tables.

$\oplus$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\odot$	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

We leave it as an exercise to verify that $\oplus$ and $\odot$ are well-defined on $\mathbb{F}_p$—that is, that they are binary operations on $\mathbb{F}_p$—and that they satisfy the axioms for a field.

All the definitions and results in Section 7.2 apply to vector spaces over arbitrary fields. The concepts of subspace and dimension apply with no change. We can apply the notion of coordinatizing a line in a vector space using $\mathbb{F}$, the underlying field. What we then find is that the cardinality of a line is the same as the cardinality of $\mathbb{F}$.

Figure 7.5 is a picture of $V = \mathbb{F}_2^3$. The only points in V are the eight red dots. We leave it as an exercise for you to write down all the one- and two-dimensional subspaces for this example.

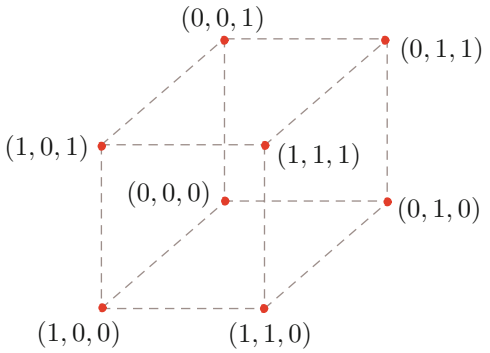


Figure 7.5: A three-dimensional vector space over the field with two elements

Associated to every finite field—and to certain infinite fields that we will not have occasion to meet—there is a positive integer k such that

$$\overbrace{1 \oplus \cdots \oplus 1}^{k \text{ times}} = 0. \tag{7.8}$$

For example, in $\mathbb{F}_2$, $1 \oplus 1 = 0$. If $\mathbb{F}$ is any field for which there is one such k, there must be a minimal value of k for which Equation (7.8) is true. That k is called the **characteristic of** $\mathbb{F}$. When there is no such k for a field $\mathbb{F}$, we say that $\mathbb{F}$ has **characteristic** 0. $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Q}$ all have characteristic zero. $\mathbb{F}_p$ has characteristic p. We leave it as an exercise to show that the characteristic of a field is either zero or a prime number.

Geometry associated to vector spaces over arbitrary fields is often subject to the assumption that the underlying field does not have characteristic 2 or 3. When we study affine geometries, we will see why.

Exercises 7.2. 1. Give an example of a binary operation on $\mathbb{R}$ that is not associative. Give an example that is not commutative.

2. Prove Lemma 7.39.

3. Find the multiplicative inverse of nonzero $z = a + bi \in \mathbb{C}$, that is, if $a, b \in \mathbb{R}$ are not both zero, find $c, d \in \mathbb{R}$ so that $1/(a+bi) = c+di$, where $i^2 = -1$.

4. Which of the field axioms does $\mathbb{Z}$ satisfy and which does it fail to satisfy? What about $M_n(\mathbb{R})$? What about $\mathbb{Z}_4$, with $\oplus$ and $\odot$ both defined mod 4? For the $\mathbb{Z}_4$ example, write out the addition and multiplication tables.

5. Complete the verification that $\mathbb{F}_p$ is a field when p is prime. Be sure to show that addition and multiplication mod p are binary operations.

6. Show that if $a \odot b = 0$, for $a, b \in \mathbb{F}$, $\mathbb{F}$ a field, then if a is nonzero, $b = 0$. (This shows that a field does not contain nonzero *zero divisors*.)

7. Let $\mathbb{F}$ be an arbitrary field. We can extend our definitions of $2 = 1 \oplus 1$ and $3 = 1 \oplus 2$, to any positive integer $n > 3$, defining $n = 1 \oplus (n-1)$, where 1 is the multiplicative identity in $\mathbb{F}$.

   (a) Using the field axioms, show that if a is any element in $\mathbb{F}$, and k is a positive integer, $k \odot a = \overbrace{a \oplus \cdots \oplus a}^{k \text{ times}}$.

   (b) For positive integers m, n, what does $m \odot n$ mean in a field? Frame your answer using $\oplus$.

   (c) If $k = mn$, where k, m, n are all positive integers and the multiplication is in $\mathbb{Z}_+$, show that for any $a \in \mathbb{F}$, $k \odot a = m \odot n \odot a$.

   (d) Argue that if k is a positive integer, and $k \odot a = 0$ for $a \in \mathbb{F}$, then $p \odot a = 0$ where p is some prime factor of k. This shows that the characteristic of a field is zero or prime.

8. Refer to Fig. 7.5 to see the points in $\mathbb{F}_2^3$.

   (a) What vectors are in $\text{span}\{(1,0,0)\}$?

   (b) What vectors are in $\text{span}\{(1,0,0),(0,1,0)\}$?

   (c) Find all one- and two-dimensional subspaces of $\mathbb{F}_2^3$.

7.4 Linear Transformations

The study of linear algebra centers on vector spaces and the mappings on vector spaces that respect vector addition and scaling. This section is devoted to a review of the latter.

Definition 7.42. Let V and W be vector spaces over a field, $\mathbb{F}$. A **linear transformation** $L : V \to W$, is a mapping that commutes with linear combinations. In other words,

$$L(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) = c_1L(\mathbf{v}_1) + c_2L(\mathbf{v}_2),$$

for all c_1, c_2 in $\mathbb{F}$ and $\mathbf{v}_1, \mathbf{v}_2$ in V.

Consider $V = \mathbb{R}_2[x]$ and $W = \mathbb{R}^3$. We can define $L : V \to W$ by

$$a_0 + a_1x + a_2x^2 \mapsto (a_0, a_1, a_2). \tag{7.9}$$

This must be a linear transformation because we form a linear combination of polynomials by taking the same linear combination of the polynomials' coefficients.

The mapping in (7.9) is an example of a *coordinate mapping*.

Definition 7.43. Let V be a vector space over $\mathbb{F}$ with ordered basis $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. The **coordinate mapping determined by** $\mathcal{B}$ is $L : V \to \mathbb{F}^n$ given by

$$L(\mathbf{v}) = (a_1, \ldots, a_n), \text{ where } \mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n, \text{ for } a_i \in \mathbb{F}.$$

In this case, $[\mathbf{v}]_{\mathcal{B}} = L(\mathbf{v})$ is called the **$\mathcal{B}$-coordinate vector** of $\mathbf{v}$.

Coordinate mappings belong to a larger class of linear transformations.

Definition 7.44. An **isomorphism** of vector spaces V and W over $\mathbb{F}$ is a bijective linear transformation $L : V \to W$. When there is an isomorphism $L : V \to W$, we say that V is **isomorphic** to W and we write $V \cong W$.

Note that isomorphisms are invertible linear transformations.

We leave it as an exercise to verify that coordinate mapping are isomorphisms.

The following theorem is a centerpiece of the elementary theory of vector spaces.

Theorem 7.45. Finite-dimensional vector spaces over a field $\mathbb{F}$ are isomorphic if and only if they have the same dimension.

The proof of Theorem 7.45 is immediate once we realize that the isomorphisms between two vector spaces are the linear mappings that send a basis to a basis. We leave the details to the reader.

A corollary is immediate.

Corollary 7.46. Isomorphism is an equivalence relation.

Example 7.47. $\mathbb{R}_n[x] \cong \mathbb{R}^{n+1}$ since $\dim \mathbb{R}_n[x] = n+1$ because $\{1, x, x^2, \ldots, x^n\}$, the standard basis for $\mathbb{R}_n[x]$, has $n+1$ elements. The coordinate mapping determined by the standard basis for $\mathbb{R}_n[x]$ is an example of one isomorphism from $\mathbb{R}_n[x]$ to $\mathbb{R}^{n+1}$.

We establish notation and terminology before getting matrices fully into the picture.

An $m \times n$ matrix over $\mathbb{F}$ has the form

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ & & \vdots & \\ a_{m1} & a_{n2} & \ldots & a_{mn} \end{bmatrix},$$

where $a_{ij} \in \mathbb{F}$. Rows go across and columns go down in a matrix so *row* i of A is

$$\begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}$$

and *column* j is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

We have been writing vectors in $\mathbb{F}^n$ as n-tuples which we identify with column vectors. (Distinguish a row vector $[x_1 \dots x_n]$ from an n-tuple $(x_1, \dots, x_n)$.)

If A is an $m \times n$ matrix and $\mathbf{v} \in \mathbb{F}^n$, the product $A\mathbf{v}$ can be defined in terms of the dot product, itself defined exactly as in $\mathbb{R}^n$.[2] The ith entry in $A\mathbf{v}$ is the dot product of the ith row of A with $\mathbf{v}$. For example, if $A = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$ and $\mathbf{v} = (-2, 6)$ in $\mathbb{R}^2$, then $A\mathbf{v} = (10, 30)$. When doing this sort of matrix arithmetic, we normally write vectors in $\mathbb{F}^n$ as columns instead of as n-tuples. In this example, then, we have

$$A\mathbf{v} = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 6 \end{bmatrix} = \begin{bmatrix} 10 \\ 30 \end{bmatrix}.$$

This arithmetic extends to give us the product of an $m \times n$ matrix A and an $n \times r$ matrix, B. The ij-entry of AB—that is, the entry in row i and column j—is the dot product of row i of A and column j of B. If we think of the columns of B as vectors, we have $B = [\mathbf{b}_1 \dots \mathbf{b}_r]$, so that

$$AB = [A\mathbf{b}_1 \; A\mathbf{b}_2 \dots \; A\mathbf{b}_r].$$

Note in particular that AB is an $m \times r$ matrix.

Suppose $L : \mathbb{F}^n \to \mathbb{F}^m$ is a linear transformation. Like $\mathbb{R}^n$, $\mathbb{F}^n$ has a natural (ordered) basis, $\mathcal{B} = \{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ where $\mathbf{e}_i = (0, \dots, 0, 1, 0, \dots, 0)$, 1 in the ith position. For each $\mathbf{e}_j \in \mathcal{B}$, we have

$$L(\mathbf{e}_j) = (a_{1j}, a_{2j}, \dots, a_{nj}), \text{ for some } a_{ij} \in \mathbb{F}.$$

Definition 7.48. The **standard matrix representation** of a linear transformation $L : \mathbb{F}^n \to \mathbb{F}^m$ is the $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ & & \vdots & \\ a_{m1} & a_{n2} & \dots & a_{mn} \end{bmatrix}$$

where the jth column of A is $L(\mathbf{e}_j) = (a_{1j}, a_{2j}, \dots, a_{mj})$.

[2] The dot product on $\mathbb{F}^n$ does not have certain key properties that it has on $\mathbb{R}^n$. For instance, *positive* and *negative* are not meaningful in arbitrary fields so we cannot say $|\mathbf{v}| \geq 0$ for all $\mathbf{v} \in \mathbb{F}^n$. This is one barrier to using the dot product on $\mathbb{F}^n$ for defining length, for instance.

Example 7.49. Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $L(a, b) = (a + b, a - 2b)$, for all $(a, b) \in \mathbb{R}^2$. L is linear because if $\mathbf{u} = (a, b)$ and $\mathbf{v} = (a', b')$, then for $c_1, c_2 \in \mathbb{R}$,

$$L(c_1\mathbf{u} + c_2\mathbf{v}) = L(c_1(a, b) + c_2(a', b')) = L((c_1a, c_1b) + (c_2a', c_2b')) =$$

$$L(c_1a + c_2a', c_1b + c_2b') = (c_1a + c_2a' + c_1b + c_2b', (c_1a + c_2a') - 2(c_1b + c_2b')) =$$

$$c_1(a + b, a - 2b) + c_2(a' + b', a' - 2b') = c_1L(\mathbf{u}) + c_2L(\mathbf{v}).$$

Note that $L(\mathbf{e}_1) = (1, 1)$ and $L(\mathbf{e}_2) = (1, -2)$. It follows that the standard matrix representation of L is

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}.$$

Linear transformations on finite-dimensional spaces are arithmetic. A matrix representation of a linear transformation contains the coding for the arithmetic so that instead of thinking of a mapping $L(\mathbf{v})$, we can think of doing the arithmetic required to calculate $A\mathbf{v}$.

Lemma 7.50. If A is the standard matrix representation of a linear transformation $L : \mathbb{F}^n \to \mathbb{F}^m$, then for all $\mathbf{v} \in \mathbb{F}^n$, $L(\mathbf{v}) = A\mathbf{v}$.

The proof is a verification that we leave to the reader.

The linear transformations of most interest to us are from $\mathbb{R}^n$ to $\mathbb{R}^n$. We continue the discussion referring to $\mathbb{F}^n$ because there is little difference in the algebraic mechanics if we allow the underlying field to be more general than $\mathbb{R}$. However, we focus our remarks now on the case in Definition 7.48 where $m = n$. In this case, we say that $L : V \to V$ is a *linear transformation on* V.

A matrix representation for a linear transformation L on $\mathbb{F}^n$ expresses the details that define L with respect to a specific basis for $\mathbb{F}^n$. The standard basis is not always ideal. We define a matrix representation for a linear transformation L on $\mathbb{F}^n$ in terms of an arbitrary ordered basis as follows.

Definition 7.51. Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be an ordered basis for $\mathbb{F}^n$. The **matrix representation** of $L : \mathbb{F}^n \to \mathbb{F}^n$ **with respect to** $\mathcal{B}$ is the $n \times n$ matrix A with jth column given by $[L(\mathbf{v}_j)]_{\mathcal{B}}$.

It is not hard to check the following. We leave the proof as an exercise.

Lemma 7.52. If A is the matrix representation of $L : \mathbb{F}^n \to \mathbb{F}^n$ with respect to an ordered basis $\mathcal{B}$, then for any $\mathbf{v} \in \mathbb{F}^n$, $[L(\mathbf{v})]_{\mathcal{B}} = A[\mathbf{v}]_{\mathcal{B}}$.

Example 7.53. Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be given by $L(a, b) = (a + b, 2a)$. Let $\mathcal{B} = \{(1, 1), (1, -1)\}$. In Example 7.32, we showed that $\mathcal{B}$ is a basis for $\mathbb{R}^2$. Here we find the matrix representation of L with respect to $\mathcal{B}$.

We have

$$L(1, 1) = (2, 2) = 2(1, 1) + 0(1, -1)$$

so $[L(1, 1)]_{\mathcal{B}} = (2, 0)$. Since

$$L(1, -1) = (0, 2) = (1, 1) + (-1)(1, -1),$$

$[L(1,-1)]_{\mathcal{B}} = (1,-1)$. The matrix representation of L with respect to $\mathcal{B}$ is then

$$A = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}.$$

Now let's check that A does what it is supposed to do, at least on one vector.

Consider $\mathbf{v} = (1,0) = (1/2)(1,1) + (1/2)(1,-1)$. Then

$$A[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}.$$

Since $L(\mathbf{v}) = (1,2)$, we just have to check that

$$(3/2)(1,1) + (-1/2)(1,-1) = (1,2),$$

which verifies that $[L(\mathbf{v})]_{\mathcal{B}} = A[\mathbf{v}]_{\mathcal{B}}$ in this case.

A linear transformation on $\mathbb{F}^n$ yields an $n \times n$ matrix but it works the other way as well. Any $n \times n$ matrix with entries in $\mathbb{F}$ defines a linear transformation $L : \mathbb{F}^n \to \mathbb{F}^n$ by $L(\mathbf{v}) = A\mathbf{v}$. In other words, matrix multiplication itself is a linear transformation on $\mathbb{F}^n$:

$$A(c_1\mathbf{u} + c_2\mathbf{v}) = c_1 A\mathbf{u} + c_2 A\mathbf{v}.$$

We leave it as an exercise for you to verify this in the simple case of $V = \mathbb{F}^2$.

Now we consider some of the important linear transformations in the geometry of $\mathbb{R}^2$.

Example 7.54. Fix an angle φ. Normally we would choose $\varphi \in (-\pi, \pi]$ but any angle will do. Let $A = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$. A defines a rotation on $\mathbb{R}^2$ that leaves the origin fixed. We can see that by comparing the polar coordinates of $\mathbf{v}$ to the polar coordinates of $A\mathbf{v}$. In a rotation that leaves the origin fixed, the length of $\mathbf{v}$ should remain fixed, but the angle associated to $\mathbf{v}$ should change by the addition of φ. We leave it as an exercise for you to check that this is the effect of the mapping $\mathbf{v} \mapsto A\mathbf{v}$.

Example 7.55. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Define $L : \mathbb{R}^2 \to \mathbb{R}^2$ by $L(\mathbf{v}) = A\mathbf{v}$. We have

$$L(\mathbf{e}_1) = \mathbf{e}_1 \text{ and } L(\mathbf{e}_2) = -\mathbf{e}_2.$$

In general $L(a,b) = (a,-b)$. Notice that this is the reflection on $\mathbb{R}^2$ through the x-axis.

Example 7.56. Let $\ell = \text{span}\{\mathbf{v}\}$ be an arbitrary fixed line through $\mathbf{0}$ in $\mathbb{R}^2$ where $\mathbf{v} = (a,b)$ is not $\mathbf{0}$. We may assume $\mathbf{v}$ is a unit vector so that $a^2 + b^2 = 1$. Let $\mathbf{w} = (-b,a)$. We leave it as an easy exercise to check that $\{\mathbf{v}, \mathbf{w}\}$ is a linearly independent set, thus, is an ordered basis for $\mathbb{R}^2$. Since $\mathbf{v} \cdot \mathbf{w} = -ab + ba = 0$, the vectors in $\mathcal{B} = \{\mathbf{v}, \mathbf{w}\}$ are orthogonal.

We can define a linear transformation $L : \mathbb{R}^2 \to \mathbb{R}^2$ by what it does to any basis so define $L(\mathbf{v}) = \mathbf{v}$ and $L(\mathbf{w}) = -\mathbf{w}$. We leave it as an exercise for you to verify that L reflects the plane through ℓ. Note that the matrix representation of L with respect to $\mathcal{B}$ is $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

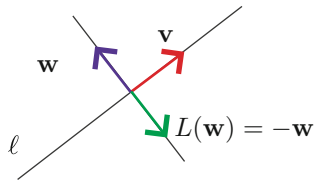


Figure 7.6: Reflection through a line ℓ in $\mathbb{R}^2$

To find the standard matrix representation for L, we can think about a three step process. First, map the basis $\mathcal{B} = \{\mathbf{v}, \mathbf{w}\}$ to the standard basis: $\mathbf{v} \mapsto \mathbf{e}_1$, and $\mathbf{w} \mapsto \mathbf{e}_2$. We leave it as an exercise to check that the standard matrix representation of this mapping is $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$.

The second step is to reflect across the x-axis, the line ℓ was mapped to under the first mapping. This is realized by the matrix from Example 7.55. The third step is to map the standard basis back to $\{\mathbf{v}, \mathbf{w}\}$: $\mathbf{e}_1 \mapsto \mathbf{v}$, $\mathbf{e}_2 \mapsto \mathbf{w}$. By performing the matrix multiplications, we can verify that the matrix representation of L with respect to the standard basis is

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a & b \\ -b & a \end{bmatrix} = \begin{bmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{bmatrix}. \tag{7.10}$$

Check that A has the desired effect: $A\mathbf{v} = \mathbf{v}$ and $A\mathbf{w} = -\mathbf{w}$.

The first and last matrices in the matrix product in Example 7.56 are *inverses* of one another. In general, if A and B are $n \times n$ matrices that satisfy

$$AB = BA = \begin{bmatrix} 1 & 0 & \dots & & 0 \\ 0 & 1 & 0 & \dots & 0 \\ & & \vdots & & \\ 0 & \dots & & 0 & 1 \end{bmatrix} = I_n$$

then A and B are matrix inverses. We write $A^{-1} = B$ and $B^{-1} = A$.

Matrix inverses represent inverse linear transformations. In Example 7.56, we saw the matrix $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ mapped $\mathcal{B}$ to the natural basis and its inverse, $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, mapped the natural basis to $\mathcal{B}$. If we understand the linear transformation effected by a given matrix, sometimes we can use that to find the

inverse of the matrix, if it is invertible. For example, knowing that a rotation through φ that leaves the origin fixed in $\mathbb{R}^2$ is given by $A = \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$ we can find

$$A^{-1} = \begin{bmatrix} \cos(-\varphi) & -\sin(-\varphi) \\ \sin(-\varphi) & \cos(-\varphi) \end{bmatrix} = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix}.$$

We check this by multiplying the two matrices together to get

$$\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix} = \begin{bmatrix} \cos^2\varphi + \sin^2\varphi & 0 \\ 0 & \cos^2\varphi + \sin^2\varphi \end{bmatrix}$$

$$= I_2.$$

Notice that if A and B are $n \times n$ invertible matrices, with respective inverses A^{-1} and B^{-1}, then

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I_n.$$

It follows that the product of invertible matrices is invertible.

There are many ways to characterize an invertible matrix. We note a few of them in the next theorem. This theorem and its proof, in the case where $\mathbb{F} = \mathbb{R}$, typically occupy an important place in an introductory linear algebra course.

Recall that the determinant of a square matrix is a number that we calculate using the entries in the matrix. The determinant of I_n is 1.

Theorem 7.57. Let A be an $n \times n$ matrix over a field, $\mathbb{F}$. Let $L : \mathbb{F}^n \to \mathbb{F}^n$ be given by $L(\mathbf{v}) = A\mathbf{v}$. The following statements are logically equivalent.

(1) A is invertible.

(2) $\det A \neq 0$.

(3) L is an invertible linear transformation.

(4) The columns of A form a basis for $\mathbb{F}^n$.

(5) $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis for $\mathbb{F}^n$ if and only if $\{L(\mathbf{v}_1), \ldots, L(\mathbf{v}_n)\}$ is a basis for $\mathbb{F}^n$.

(6) $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \subset \mathbb{F}^n$ is linearly independent if and only if $\{L(\mathbf{u}_1), \ldots, L(\mathbf{u}_k)\}$ is linearly independent.

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\det A = ad - bc$. Calculating determinants for larger $n \times n$ matrices usually involves cofactor expansion, with a few theorems thrown in to make the job a bit easier. Unless we are dealing with special matrices or small n, though, we usually find determinants using calculators or software. There is a trick for calculating 3×3 determinants which we work through in the exercises.

Like determinants, matrix inverses can present serious computational challenges that are often best left to machines. In the case of a 2×2 matrix, though, we can write down the inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, if the inverse is defined. It is

$$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Again, checking this is simply a matter of carrying out the calculation of the matrix product.

One of the most important theorems about determinants is that the determinant of the product of matrices is the product of the determinants:

$$\det(AB) = \det A \det B.$$

This implies, for instance, that the set of $n \times n$ matrices with determinant 1 is closed under multiplication. It also leads us directly to the observation that if A is an invertible $n \times n$ matrix, then

$$1 = \det I_n = \det AA^{-1} = (\det A)(\det A^{-1}),$$

from which it follows that $\det A^{-1} = 1/\det A$.

We have seen that there is a one-to-one correspondence between $n\times n$ matrices over $\mathbb{F}$ and linear transformations on $\mathbb{F}^n$. There is in fact a one-to-one correspondence between invertible $n \times n$ matrices over $\mathbb{F}$ and invertible linear transformations on $\mathbb{F}^n$.

Next we address the relationship between matrices that represent the same linear transformation with respect to different ordered bases. For this part of our discussion, fix ordered bases of $\mathbb{F}^n$, $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$, and $\mathcal{C} = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$.

Definition 7.58. The $\mathcal{B}$ to $\mathcal{C}$ **change of basis matrix** is $P = [[\mathbf{v}_1]_\mathcal{C} \cdots [\mathbf{v}_n]_\mathcal{C}]$.

We leave it as an exercise to check that if $\mathbf{x} \in \mathbb{F}^n$, then for P as in Definition 7.58, $P[\mathbf{x}]_\mathcal{B} = [\mathbf{x}]_\mathcal{C}$. We also leave it as an exercise to check that the columns of P form a basis for $\mathbb{F}^n$ so P is invertible. By arithmetic using P^{-1}, we have $[\mathbf{x}]_\mathcal{B} = P^{-1}[\mathbf{v}]_\mathcal{C}$, in other words, P^{-1} is the $\mathcal{C}$ to $\mathcal{B}$ change of basis matrix.

Now suppose L is a linear transformation on $\mathbb{F}^n$ and that A is the matrix representation of L with respect to $\mathcal{C}$. Given $\mathbf{x} \in \mathbb{F}^n$, and still using P as in Definition 7.58, we have

$$P^{-1}AP[\mathbf{x}]_\mathcal{B} = P^{-1}A[\mathbf{x}]_\mathcal{C} = P^{-1}[L(\mathbf{x})]_\mathcal{C} = [L(\mathbf{x})]_\mathcal{B}.$$

These are the elements that go into a proof of the following theorem.

Theorem 7.59. Let L be a linear transformation on $\mathbb{F}^n$. If A and B are matrix representations of L with respect to different ordered bases, then for some invertible matrix P, $B = P^{-1}AP$.

Corollary 7.60. Let A and B be matrix representations of a given linear transformation on $\mathbb{F}^n$ with respect to different ordered bases. Then $\det A = \det B$.

Proof. The result follows by applying the theorem and the multiplicative property of determinants. If $B = P^{-1}AP$, then

$$\det B = \det(P^{-1}AP) = (\det P^{-1})(\det A)(\det P)$$

$$= (\det P^{-1})(\det P)(\det A) = \det A.$$

□

Exercises 7.3. 1. Let A be a 2×2 matrix with entries in $\mathbb{F}$ and let $\mathbf{u}, \mathbf{v} \in \mathbb{F}^2$. Show that if $c \in \mathbb{F}$, then $A(c\,\mathbf{u} + \mathbf{v}) = cA\mathbf{u} + A\mathbf{v}$.

2. Verify that any coordinate mapping is an isomorphism.

3. Show that the composition of isomorphisms on a vector space V is itself an isomorphism on V.

4. Prove Theorem 7.45.

5. Show that the matrix A given in Example 7.49 maps $(a, b) \mapsto (a+b, a-2b)$.

6. Prove Lemma 7.50.

7. Prove Lemma 7.52.

8. Argue that A as given in Example 7.54 defines a rotation on $\mathbb{R}^2$ by determining that A leaves the length of any $\mathbf{v}$ in $\mathbb{R}^2$ unchanged and simply adds φ to the angle determined by the polar coordinates of $\mathbf{v}$. (See Fig. 7.7.)

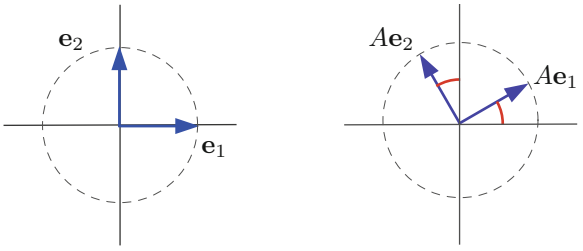


Figure 7.7: Rotation in $\mathbb{R}^2$ that leaves the origin fixed

9. Argue that rotations on $\mathbb{R}^2$ that fix the origin have determinant 1.

10. Let L, ℓ, $\mathcal{B} = \{\mathbf{v}, \mathbf{w}\}$, and A be as in Example 7.56.

    (a) Verify that L reflects vectors in $\mathbb{R}^2$ through the line ℓ.

(b) Show that $\mathcal{B}$ is a basis for $\mathbb{R}^2$.

(c) Verify that the matrix representation of L with respect to $\mathcal{B}$ is $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

(d) Find the $\mathcal{B}$ to $\mathcal{C}$ change of basis matrix if $\mathcal{C}$ is the standard basis.

(e) Find the inverse of the matrix you found above.

(f) Show that $\det A = -1$.

(g) Show that A is its own inverse. What does this say about L?

(h) Check that $A\mathbf{v} = \mathbf{v}$ and that $A\mathbf{w} = -\mathbf{w}$.

11. Let $\ell = \text{span}\{(1, 2)\}$. Follow Example 7.56 to find the standard matrix representation for the reflection on $\mathbb{R}^2$ through ℓ.

12. Calculate the inverse of $A = \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$ or determine that it is not invertible.

13. The determinant of an $n \times n$ matrix A is a sum of terms, each term a *signed* product of entries of A. Each product is formed by taking exactly one term from each row, and one term from each column of A. If

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ & & \vdots & \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix},$$

then a product of entries from each row and each column looks like

$$a_{1\sigma(1)}a_{2\sigma(2)} \dots a_{n\sigma(n)} \tag{7.11}$$

where σ is a bijection from $\{1, 2, \dots, n\}$ to itself. (A bijection from a finite set to itself is called a *permutation* on the set.) Since there are $n!$ permutations on $\{1, 2, \dots, n\}$, there are $n!$ terms in the determinant of an $n \times n$ matrix. The sign ($\pm$) associated to the term given in (7.11) is the so-called sign of σ.

(a) What are the permutations on $\{1, 2, 3\}$? We designate the identity permutation—that is, the one that does not rearrange the ordering of the elements in the set—as 123. Write down all the other permutations.

(b) A *transposition* is a permutation on a set that switches exactly two elements, leaving all other elements in the set unmoved. It is a fact that any permutation on a finite set is the result of applying a finite number of transpositions in sequence. We thus say that a permutation is a *product of transpositions*. For instance, the identity permutation is the result of 0 transpositions and the permutation

132 is the result of a single transposition—2 and 3 are switched, 1 is left alone. The permutation 231 is the product of the transposition that switches 1 and 2, to get 213, followed by the transposition that switches 1 and 3. The *sign* of a permutation is $(-1)^k$, where k is the number of transpositions that yield the permutation. What is not obvious, but is true, is that the sign of a permutation is well-defined. In other words, if you can use k_1 transpositions to effect a given permutation, and your friend finds k_2 transpositions that effect the same permutation, then $|k_1 - k_2|$ is an even number. Find the sign of each permutation that you found in part (a).

(c) Using permutations and their signs, write down the formula for $\det A$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

There should be $6 = 3!$ terms, each of the form

$$(-1)^k a_{1\sigma(1)} a_{2\sigma(2)} a_{3\sigma(3)},$$

σ a permutation on $\{1, 2, 3\}$, and $(-1)^k$ the sign of σ.

(d) Use the formula you just found to calculate $\det A$ if

$$A = \begin{bmatrix} 0 & 1 & -2 \\ 3 & 0 & 1 \\ -1 & 2 & 4 \end{bmatrix}.$$

(e) Figure 7.8 is a device to keep track of the terms, with the correct signs, in the determinant of a 3×3 matrix $A = [a_{ij}]$. To perform the calculation, add the products of entries connected with blue arrows and subtract the products of entries connected with red arrows. For instance, the product $a_{11}a_{22}a_{33}$ gets a positive sign. Note that the array in the figure is the entries in A with its first two columns repeated. Write down all the products, with signs, and check that you get the same formula as the one you found in part (c). (**Beware:** This trick does not work for larger matrices!)

(f) Use the trick now to find $|B| = \det B$ if $B = \begin{bmatrix} -1 & 2 & 3 \\ 3 & 0 & 4 \\ 2 & 5 & 1 \end{bmatrix}$.

14. Let P be as defined in Definition 7.58.

(a) Verify that for any $\mathbf{x} \in \mathbb{F}^n$, $P[\mathbf{x}]_{\mathcal{B}} = [\mathbf{x}]_{\mathcal{C}}$.

(b) Show that the columns of P form a basis for $\mathbb{F}^n$.

15. Show that if A and B are matrices such that $B = P^{-1}AP$ for some invertible matrix P, then $A = Q^{-1}BQ$ for some invertible matrix Q.

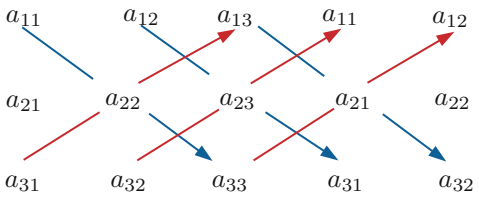


Figure 7.8: Products associated to the blue arrows have sign + and those associated to the red arrows have sign −

7.5 Affine Geometry

Affine geometry is the geometry associated to vector spaces. In the affine geometry determined by a vector space V—by which we mean a vector space over an arbitrary field, $\mathbb{F}$—vectors are points and the subspaces of V, and certain sets associated to them, are lines, planes, etc. Incidence is realized as nonempty set intersection. The question then is whether Hilbert's incidence axioms apply. Our concern for the rest of the chapter is addressing that question and exploring other devices for studying affine geometry.

The natural setting for us to work with Euclidean points, lines, and planes is $\mathbb{R}^n$, where a vector can model a geometric point and a line described as in (7.4) can be viewed as a copy of $\mathbb{R}$. We have seen that $\mathbb{R}^2$ is a model for the Euclidean plane. If we ask about the geometry of a two-dimensional vector space over a field $\mathbb{F}$ different from $\mathbb{R}$, we may no longer have notions of angle or distance, but we will always be able to use vectors to model points and we will always be able to say that a line is a set of points that can be described as in (7.4) with $\mathbb{F}$ replacing $\mathbb{R}$. Regardless of the base field, we will have incidence relations in any vector space.

Analytic geometry involves using coordinates in the xy-plane to understand Euclidean geometry. As sets, the xy-plane and $\mathbb{R}^2$ are identical, but when we call it the xy-plane or the coordinate plane, that often means we are ignoring the vector space structure and simply using the coordinate system as a grid. In affine geometry, we study the subspaces and related sets in $\mathbb{R}^2$—or more generally, in any finite-dimensional vector space over any field—to understand them as geometric objects.

All points are created equal, and all lines are created equal in the Euclidean plane. In a vector space, though, there is an important distinction between an arbitrary line or plane and a line or plane that passes through the origin: A line or plane that passes through $\mathbf{0}$ is a subspace, and subspaces have special status. If we are to move away from considering subspaces of vector spaces as algebraic objects of special interest, and toward considering lines and planes inside vector spaces purely as geometric objects, we must dethrone $\mathbf{0}$. Toward that end, we introduce *cosets*.

Definition 7.61. A **coset** of a nontrivial vector space V is a set of the form

$\mathbf{u}+W=\{\mathbf{u}+\mathbf{w}|\,\mathbf{w}\in W\}$, where W is a subspace of V, and $\mathbf{u}$ is a fixed vector in V. More particularly, $\mathbf{u}+W$ is the **W-coset determined by $\mathbf{u}$**. The vector $\mathbf{u}$ is called a **coset representative** of $\mathbf{u}+W$. The **dimension** of a coset $\mathbf{u}+W$ is the dimension of its underlying subspace, W.

Referring to Example 7.18, we can now say that the line in $\mathbb{R}^2$ through points $(-1,-2)$ and $(3,1)$ is the one-dimensional coset $(-1,-2)+W$ where W is the span of the vector $\mathbf{v}=(4,3)$. More generally, a one-dimensional subspace in V is a line through the origin in V, while a one-dimensional coset in V is an arbitrary line in V. This is the motivating idea behind Definition 7.61.

The next lemma is from [21].

Lemma 7.62. Let W be a subspace of a nontrivial vector space, V.

(1) Two W-cosets are identical if and only if the difference of their coset representatives is in W.

(2) Two W-cosets are either identical, or disjoint. That is, different W-cosets have empty intersection.

(3) Suppose W and W' are one-dimensional subspaces of V, where $\dim V=2$. If there are disjoint W- and W'-cosets, then $W=W'$.

Proof. Fix a nontrivial vector space V and let W be a subspace of V. Let $\mathbf{v}_1,\mathbf{v}_2\in V$.

(1) Suppose $\mathbf{v}_1+W=\mathbf{v}_2+W$. Then for any $\mathbf{w}_1\in W$, there is $\mathbf{w}_2\in W$ so that $\mathbf{v}_1+\mathbf{w}_1=\mathbf{v}_2+\mathbf{w}_2$. Since $\mathbf{v}_1-\mathbf{v}_2=\mathbf{w}_2-\mathbf{w}_1$, and W is a vector space, $\mathbf{v}_1-\mathbf{v}_2$ is in W. This proves that if two W-cosets are identical, then the difference of their representatives is in W.

Next suppose that $\mathbf{v}_1-\mathbf{v}_2=\mathbf{w}$ is in W. We must show that $\mathbf{v}_1+W=\mathbf{v}_2+W$. Since $\mathbf{v}_1=\mathbf{v}_2+\mathbf{w}$, then given any $\mathbf{w}_1\in W$, there is $\mathbf{w}_2=\mathbf{w}+\mathbf{w}_1$ in W with $\mathbf{v}_1+\mathbf{w}_1=\mathbf{v}_2+\mathbf{w}_2$. That says $\mathbf{v}_1+W\subseteq\mathbf{v}_2+W$. Reversing the roles of $\mathbf{v}_1$ and $\mathbf{v}_2$, we can argue that $\mathbf{v}_2+W\subseteq\mathbf{v}_1+W$, which proves $\mathbf{v}_1+W=\mathbf{v}_2+W$.

(2) Suppose $\mathbf{u}$ is in $\mathbf{v}_1+W\cap\mathbf{v}_2+W$ so that there are $\mathbf{w}_1,\mathbf{w}_2\in W$ with $\mathbf{u}=\mathbf{v}_1+\mathbf{w}_1=\mathbf{v}_2+\mathbf{w}_2$. The latter is true only if $\mathbf{v}_1-\mathbf{v}_2\in W$. By part (1), that is true if and only if $\mathbf{v}_1+W=\mathbf{v}_2+W$. This proves that if the intersection of W-cosets is nonempty, then the cosets are identical, which implies the result.

(3) Now suppose $\dim V=2$ and $\dim W=1$. Let W' also be a one-dimensional subspace of V. We prove the contrapositive of the statement, namely, that if $W\neq W'$, then $\mathbf{v}_1+W\cap\mathbf{v}_2+W'\neq\emptyset$. Assume then that $W\neq W'$.

Say $W=\text{span}\{\mathbf{w}\}$, and $W'=\text{span}\{\mathbf{w}'\}$. Note that $\mathcal{B}=\{\mathbf{w},\mathbf{w}'\}$ is a basis for V. Let $\mathbf{v}_1=c_1\mathbf{w}+c_2\mathbf{w}'$ and $\mathbf{v}_2=c_1'\mathbf{w}+c_2'\mathbf{w}'$. We have $\mathbf{v}_1+W=c_2\mathbf{w}'+W=\{c_2\mathbf{w}'+t\mathbf{w}|\,t\in\mathbb{F}\}$ and, similarly, $\mathbf{v}_2+W'=c_1'\mathbf{w}+W'=\{c_1'\mathbf{w}+t\mathbf{w}'|\,t\in\mathbb{F}\}$ so that $c_1'\mathbf{w}+c_2\mathbf{w}'$ is in $\mathbf{v}_2+W\cap\mathbf{v}_2+W'$, as claimed. □

While the subspace underlying a given coset must be unique, a coset has many different representatives. Lemma 7.62 tells us how different representatives for one coset must be related, and implies the following.

Corollary 7.63. An element is in a coset if and only if it is a representative for that coset.

The cosets for a fixed subspace of a vector space V actually determine an equivalence relation on V. The details are left to the exercises.

Lemma 7.64. Let V be a nontrivial vector space. Two vectors in V determine a unique one-dimensional coset in V.

Proof. Let $\mathbf{v}$ and $\mathbf{v}'$ be distinct vectors in V so that $\mathbf{v} - \mathbf{v}' \neq \mathbf{0}$. Let $W = \text{span}\{\mathbf{v}-\mathbf{v}'\}$. Notice that $\mathbf{v}'+(\mathbf{v}-\mathbf{v}') = \mathbf{v} \in \mathbf{v}'+W$. It follows by Corollary 7.63 that $\mathbf{v} + W = \mathbf{v}' + W$.

Suppose $\mathbf{u} + W'$ is another one-dimensional coset containing both $\mathbf{v}$ and $\mathbf{v}'$. Again by Corollary 7.63, $\mathbf{v} + W' = \mathbf{v}' + W' = \mathbf{u} + W'$, which implies by Lemma 7.62 that $\mathbf{v} - \mathbf{v}' \in W'$, thus that $W' = \text{span}\{\mathbf{v} - \mathbf{v}'\} = W$. □

Corollary 7.65. Let V be a vector space of dimension at least 2. Two distinct one-dimensional cosets in V intersect in at most one point.

Proof. The proof is immediate by the last lemma. □

Definition 7.66. Let V be a nontrivial vector space over a field $\mathbb{F}$. For any positive integer $k < \dim V$, an **affine k-plane** in V is a coset of a k-dimensional subspace of V. We say that an affine k-plane is **k-dimensional**. A **point** is an affine 0-plane, i.e., a vector in V. An **affine line** is an affine 1-plane. An **affine plane** is an affine 2-plane. Taken together, affine k-planes in V are **affine objects** for which V is the **ambient vector space**. We say two affine objects in V are **incident** provided the underlying cosets have a nonempty intersection. If $\dim V = n$, then $\mathbb{FA}^n$ is the collection of affine objects in V, together with their incidence relations. $\mathbb{FA}^n$ is **the affine geometry** associated to the ambient vector space V.

The notation suggests that $\mathbb{FA}^n$ depends only on $\mathbb{F}$ and n, not on the exact nature of V. This is correct. When $\dim V = n$, $V \cong \mathbb{F}^n$ by Theorem 7.45. Using a coordinate mapping, we can identify points in V with n-tuples over $\mathbb{F}$. For $k < n$, we can identify k-dimensional cosets in V with k-dimensional cosets in $\mathbb{F}^n$. Like all mappings, isomorphisms respect intersections of sets (see Exercise 5.3.1), thus an isomorphism from V to $\mathbb{F}^n$ respects the affine structure associated to V. We leave the details as an exercise.

We have occasion going forward to refer to *the real affine plane*, $\mathbb{RA}^2$, and *real affine 3-space*, $\mathbb{RA}^3$. $\mathbb{RA}^2$ could refer to the affine geometry associated to $\mathbb{R}^2$ or it could refer to the affine geometry of points and lines in a coset of any

two-dimensional subspace in a real vector space. The rest of this section will show that the affine geometries associated to the two settings are isomorphic.

When the context is $\mathbb{F}\mathbb{A}^n$, we often drop the adjective *affine* when referring to affine lines and affine planes.

Proceeding, we use vector notation for points and coset notation for affine k-planes.

We establish some terminology before approaching the rules of affine incidence.

Definition 7.67. The **direction of an affine line** ℓ in $\mathbb{F}\mathbb{A}^n$, is any vector $\mathbf{v}$ in the ambient vector space V, where, for some $\mathbf{x} \in V$, $\ell = \{\mathbf{x} + t\mathbf{v} | \, t \in \mathbb{F}\}$. A **direction in an affine** k*-plane* in $\mathbb{F}\mathbb{A}^n$, is a direction for an affine line in that k-plane. For $k \in \{1, \ldots, n-1\}$ fixed, **k-planes in $\mathbb{F}\mathbb{A}^n$ are parallel** provided they have the same directions and do not intersect. Two k-planes in $\mathbb{F}\mathbb{A}^n$ are **skew** if they have different directions and do not intersect.

Note that $\mathbf{v}$ and $\mathbf{w}$ are directions for a given line only if each vector is a nonzero scalar multiple of the other. Notice that k-planes share a direction if and only if the underlying subspaces have nontrivial intersection. We leave the verification of that statement as an easy exercise.

Theorem 7.68. (1) Two points determine a unique line in $\mathbb{F}\mathbb{A}^n$.

(2) If two distinct lines intersect in $\mathbb{F}\mathbb{A}^n$, their intersection is a single point.

(3) A point and a line not through the point lie in a unique plane in $\mathbb{F}\mathbb{A}^n$.

(4) Two parallel lines in $\mathbb{F}\mathbb{A}^n$ lie in a unique plane.

(5) If a plane in $\mathbb{F}\mathbb{A}^n$ contains two points, it contains the entire line the points determine.

(6) If a line and a plane in an affine 3-plane have empty intersection, then the direction of the line is a direction in the plane.

(7) If a line intersects a plane in $\mathbb{F}\mathbb{A}^n$, and the line does not lie in the plane, the intersection set is a point.

(8) Two intersecting lines in $\mathbb{F}\mathbb{A}^n$ lie in a unique plane.

(9) Three noncollinear points in $\mathbb{F}\mathbb{A}^n$ lie in a unique plane.

Proof. Let V be the ambient vector space for $\mathbb{F}\mathbb{A}^n$.

(1) Lemma 7.64 implies that two points determine a unique line in $\mathbb{F}\mathbb{A}^n$.

(2) Corollary 7.65 implies that the intersection of two lines in $\mathbb{F}\mathbb{A}^n$ contains at most one point.

(3) Let $\mathbf{u}$ be a point and $\ell = \mathbf{v} + \text{span}\{\mathbf{w}\}$ a line not through $\mathbf{u}$ in $\mathbb{FA}^n$. Then $\mathbf{u} - \mathbf{v} \notin \text{span}\{\mathbf{w}\}$ meaning $\{\mathbf{u} - \mathbf{v}, \mathbf{w}\}$ spans W, a two-dimensional subspace of V. It is clear that $\alpha = \mathbf{v} + W$ is a plane containing ℓ. Since $\mathbf{u} - \mathbf{v} \in W$, $\mathbf{v} + W = \mathbf{u} + W$, making it clear now that $\mathbf{u} \in \alpha$ as well.

If β is another plane containing $\mathbf{u}$ and ℓ, then since $\mathbf{v}$ and $\mathbf{v} + \mathbf{w} \in \ell$, we have

$$\beta = \mathbf{u} + W' = \mathbf{v} + W' = (\mathbf{v} + \mathbf{w}) + W',$$

where W' is some two-dimensional subspace of V. This means that both $\mathbf{u} - \mathbf{v}$ and $\mathbf{w}$ are in W' so $W \subseteq W'$, implying that $W = W'$. In other words, the plane containing $\mathbf{u}$ and ℓ is unique.

(4) Let $\ell = \mathbf{v} + \text{span}\{\mathbf{w}\}$ and $m = \mathbf{u} + \text{span}\{\mathbf{w}\}$ be parallel lines in $\mathbb{FA}^n$ so that $\mathbf{v} - \mathbf{u}$ is not in $\text{span}\{\mathbf{w}\}$. Consider the unique plane α determined by ℓ and $\mathbf{u}$, as guaranteed above. Since $\alpha = \mathbf{u} + \text{span}\{\mathbf{u} - \mathbf{v}, \mathbf{w}\}$, $m \subset \alpha$. Since any plane containing ℓ and m must contain $\mathbf{u}$ and ℓ, by (3), α is the only plane containing both ℓ and m.

(5) If α is a plane containing points $\mathbf{v}$ and $\mathbf{u}$, then for some two-dimensional subspace $W \subset V$,

$$\alpha = \mathbf{v} + W = \mathbf{u} + W,$$

so that $\mathbf{u} - \mathbf{v} \in W$. On the other hand, the line determined by $\mathbf{v}$ and $\mathbf{u}$ is $\ell = \mathbf{v} + \text{span}\{\mathbf{u} - \mathbf{v}\}$, so $\ell \subset \alpha$.

(6) Let $\mathcal{A} = \mathbf{x} + U$ be a 3-plane in $\mathbb{FA}^n$. Here U is a three-dimensional subspace of V and $\mathbf{x} \in V$. Let ℓ and α be a line and a plane, respectively, in $\mathcal{A}$. We leave it as an exercise to verify that we can write $\ell = \mathbf{y}_1 + \text{span}\{\mathbf{v}\}$, and $\alpha = \mathbf{y}_2 + W$ where $\text{span}\{\mathbf{v}\}$ and $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$ are one- and two-dimensional subspaces, respectively, of U.

Assume $\alpha \cap \ell = \emptyset$. If $\mathbf{v}$ is in W then the direction of ℓ is a direction in α. We assume, then, that $\mathbf{v}$ is not in W, seeking a contradiction. We start by claiming that $\mathcal{B} = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{y}_1 - \mathbf{v} - \mathbf{y}_2\}$ is a basis for U.

First we argue that $\mathcal{B} \subset U$. Since $\mathbf{y}_1, \mathbf{y}_2$ are in $\mathcal{A}$, Lemma 7.62.(1) implies $\mathbf{y}_1 - \mathbf{y}_2 \in U$. Since $\mathbf{v}$ is in U as well, $\mathbf{y}_1 - \mathbf{v} - \mathbf{y}_2 \in U$. This confirms that $\mathcal{B} \subset U$.

Next, we argue that $\mathcal{B}$ is linearly independent. If not, then $\mathbf{y}_1 - \mathbf{v} - \mathbf{y}_2 \in W$, but that implies that $\mathbf{y}_1 - \mathbf{v} \in \alpha$, making $\ell \cap \alpha$ nonempty, contrary to our assumption. This establishes that $\mathcal{B}$ must be linearly independent, thus a basis for U. There are then c_1, c_2, c in $\mathbb{F}$ so that

$$\mathbf{v} = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + c\,(\mathbf{y}_1 - \mathbf{v} - \mathbf{y}_2).$$

Since $\mathbf{v} \notin W$, $c \neq 0$. Then

$$-c\mathbf{y}_1 + (1 + c)\mathbf{v} = -c\mathbf{y}_2 + c_1\mathbf{w}_1 + c_2\mathbf{w}_2,$$

so that, dividing through by $-c$, we get

$$\mathbf{y}_1 + t\mathbf{v} = \mathbf{y}_2 + s_1\mathbf{w}_1 + s_2\mathbf{w}_2,$$

for some t, s_1, s_2 in $\mathbb{F}$. Since $\mathbf{y}_1 + t\mathbf{v} \in \ell$ and $\mathbf{y}_2 + s_1\mathbf{w}_1 + s_2\mathbf{w}_2 \in \alpha$, this contradicts $\alpha \cap \ell = \emptyset$, forcing us to conclude that $\mathbf{v} \in W$, that is, that the direction of ℓ is a direction in α.

(7) If a line intersects a plane in more than one point in $\mathbb{FA}^n$, then two of the points of intersection uniquely determine a line which, by (5) above, must lie in the plane.

(8) If ℓ and ℓ' are distinct intersecting lines in $\mathbb{FA}^n$, with point of intersection $\mathbf{v}$, then ℓ and ℓ' have different directions so we can write $\ell = \mathbf{v} + \text{span}\{\mathbf{u}\}$, and $\ell' = \mathbf{v} + \text{span}\{\mathbf{w}\}$, where $\{\mathbf{u}, \mathbf{w}\}$ is linearly independent. Then $W = \text{span}\{\mathbf{u}, \mathbf{w}\}$ is two-dimensional, and ℓ and ℓ' both belong to the affine plane $\alpha = \mathbf{v} + W$. Since ℓ and ℓ' are distinct, there must be a point $\mathbf{x}$ on one that is not on the other line. Let $\mathbf{x}$ be a point on ℓ that is not on ℓ'. Any plane containing ℓ and ℓ' must contain $\mathbf{x}$ and ℓ' thus by (3) is unique.

(9) Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be three noncollinear points in $\mathbb{FA}^n$. This means that $\mathbf{u}$ is not on $\ell = \mathbf{v} + \text{span}\{\mathbf{w} - \mathbf{v}\}$. By (3), there is a unique plane α determined by $\mathbf{u}$ and ℓ. We have to argue that $\mathbf{w} \in \alpha$.

The proof of (3) shows that $\alpha = \mathbf{u} + \text{span}\{\mathbf{w} - \mathbf{v}, \mathbf{u} - \mathbf{v}\}$. Since

$$\mathbf{w} = \mathbf{u} + (\mathbf{w} - \mathbf{v}) + (-1)(\mathbf{u} - \mathbf{v}),$$

$\mathbf{w}$ is in α. Any plane containing $\mathbf{u}, \mathbf{v}, \mathbf{w}$, must contain $\mathbf{u}$ and ℓ, so by (3) must be unique.

□

The proof of Theorem 7.68 gives us the following corollary.

Corollary 7.69. (1) The affine line determined by points $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{FA}^n$ is

$$\ell = \mathbf{v} + \text{span}\{\mathbf{u} - \mathbf{v}\} = \mathbf{u} + \text{span}\{\mathbf{v} - \mathbf{u}\}.$$

(2) The affine plane determined by noncollinear $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in $\mathbb{FA}^n$ is

$$\alpha = \mathbf{u} + \text{span}\{\mathbf{v} - \mathbf{u}, \mathbf{w} - \mathbf{u}\} = \mathbf{v} + \text{span}\{\mathbf{u} - \mathbf{v}, \mathbf{w} - \mathbf{v}\} = \mathbf{w} + \text{span}\{\mathbf{v} - \mathbf{w}, \mathbf{u} - \mathbf{w}\}.$$

There is an additional incidence theorem about affine 3-planes, Theorem 7.93, that we address in the next section, after we define and study affine groups. That result subsumes Hilbert's Axiom I.7, which says that if two planes have a point in common, they must have a second point in common, as well. (Note that Hilbert's context was Euclidean 3-space.)

The next theorem counts points in $\mathbb{FA}^n$. This result is guaranteed simply because a field has at least two elements: 0 and 1.

Theorem 7.70. (1) An affine line has at least two points.

(2) An affine plane has four points, no three of which are collinear.

(3) There exist four noncoplanar points in any affine 3-plane.

Proof. Let V be the ambient vector space for $\mathbb{FA}^n$.

(1) Any line in $\mathbb{FA}^n$ has the form $\ell = \{\mathbf{v} + t\mathbf{u} \,|\, t \in \mathbb{F}\}$, where $\mathbf{u} \in V$ is nonzero. Since $\mathbb{F}$ has at least two elements, there must be at least two points on ℓ: $\mathbf{v}$, and $\mathbf{v} + \mathbf{u}$.

(2) A plane in $\mathbb{FA}^n$ has the form $\alpha = \mathbf{v} + \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$ where $\{\mathbf{w}_1, \mathbf{w}_2\}$ is a linearly independent set in V. Notice that α contains points

$$\mathbf{v},\ \mathbf{v} + \mathbf{w}_1,\ \mathbf{v} + \mathbf{w}_2,\ \mathbf{v} + \mathbf{w}_1 + \mathbf{w}_2.$$

We claim that no three of these points are collinear.

Consider for instance $\mathbf{v}, \mathbf{v} + \mathbf{w}_1, \mathbf{v} + \mathbf{w}_2$. The line determined by $\mathbf{v}$ and $\mathbf{v} + \mathbf{w}_1$ is $\ell_1 = \mathbf{v} + \text{span}\{\mathbf{w}_1 - \mathbf{v}\}$ while the line determined by $\mathbf{v}$ and $\mathbf{v} + \mathbf{w}_2$ is $\ell_2 = \mathbf{v} + \text{span}\{\mathbf{w}_2 - \mathbf{v}\}$. If $\ell_1 = \ell_2$, then given $s \in \mathbb{F}$ there is $t \in \mathbb{F}$ such that $\mathbf{v} + s(\mathbf{w}_1 - \mathbf{v}) = \mathbf{v} + t(\mathbf{w}_2 - \mathbf{v})$, which implies $\mathbf{w}_1 - \mathbf{v} = t(\mathbf{w}_2 - \mathbf{v})$ for some $t \in \mathbb{F}$. Note then that $(t-1)\mathbf{v} = t\mathbf{w}_1 - \mathbf{w}_2$. Since $\mathbf{w}_1 \neq \mathbf{w}_2$, $t \neq 1$, so $\mathbf{v} \in \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$. This means α is a subspace so $\mathbf{v} = \mathbf{0}$. In that case, $\ell_1 = \text{span}\{\mathbf{w}_1\}$, and $\ell_2 = \{\mathbf{w}_2\}$ so $\ell_1 = \ell_2$ violates linear independence of $\{\mathbf{w}_1, \mathbf{w}_2\}$. The reader can check that the other cases lead to roughly the same conclusion so that α must indeed have four points, no three of which are collinear.

(3) Consider an affine 3-plane $\mathcal{A} = \mathbf{v} + \text{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$, for some linearly independent set $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} \subset V$. Let $\alpha = \mathbf{v} + \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$. We claim that $\mathbf{v} + \mathbf{w}_3$ is not in α. If it were, then there would be $s, t \in \mathbb{F}$ with

$$\mathbf{v} + s\mathbf{w}_1 + t\mathbf{w}_2 = \mathbf{v} + \mathbf{w}_3$$

implying that $\mathbf{w}_3 \in \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$, which is absurd. Since α contains $\mathbf{v}, \mathbf{v} + \mathbf{w}_1$, and $\mathbf{v} + \mathbf{w}_2$, it follows that $\mathbf{v}, \mathbf{v} + \mathbf{w}_1, \mathbf{v} + \mathbf{w}_2, \mathbf{v} + \mathbf{w}_3$, which are all in $\mathcal{A}$, are noncoplanar.

□

Hilbert's Axiom I.3 says that there are three noncollinear points. Theorem 7.70.(2) makes a stronger statement for any plane in $\mathbb{FA}^3$, thus implies that Hilbert's Axiom I.3 is true in $\mathbb{FA}^3$.

Corollary 7.71. $\mathbb{FA}^3$ satisfies Hilbert's Incidence Axioms I.1–6, and 8.

Proof. Hilbert's Incidence Axioms are outlined in Section 4.1. All except Axiom I.7 are proved or implied by Theorems 7.68 and 7.70. □

Let ℓ be a line in $\mathbb{FA}^n$. We have seen that if $\mathbf{u}$ and $\mathbf{v}$ are distinct points on ℓ, then

$$\ell = \mathbf{u} + \text{span}\{\mathbf{v} - \mathbf{u}\}. \tag{7.12}$$

If we take $\{\mathbf{u}, \mathbf{v}\}$ to be an ordered set, then taking arbitrary $\mathbf{x} \in \ell$, we have unique $\lambda \in \mathbb{F}$ with

$$\mathbf{x} = \mathbf{u} + \lambda(\mathbf{v} - \mathbf{u}) = (1 - \lambda)\mathbf{u} + \lambda\mathbf{v}.$$

Notice that $\lambda = 0$ when $\mathbf{x} = \mathbf{u}$ and $\lambda = 1$ when $\mathbf{x} = \mathbf{v}$.

Definition 7.72. The **affine line coordinates** for the point$\mathbf{x}$ on a line ℓ determined by the ordered pair of points $\{\mathbf{u}, \mathbf{v}\}$ in $\mathbb{FA}^n$ is the ordered pair $(1 - \lambda, \lambda)$ where $\mathbf{x} = (1 - \lambda)\mathbf{u} + \lambda\mathbf{v}$, for $\lambda \in \mathbb{F}$.

When we choose an ordered pair of vectors $\mathbf{u}$ and $\mathbf{v}$ on ℓ as in (7.12), $\mathbf{u}$ has coordinates $(1, 0)$ and $\mathbf{v}$ has coordinates $(0, 1)$. Notice that there is no point with affine line coordinates $(0, 0)$! In fact, we can identify ℓ with the line in $\mathbb{F}^2$ given by $\{(x, y) \,|\, x + y = 1\}$. This suggests the following definition.

Definition 7.73. If the characteristic of $\mathbb{F}$ is not 2, the **midpoint** of points $\mathbf{u}, \mathbf{v} \in \mathbb{FA}^n$ is $(1/2)(\mathbf{u} + \mathbf{v})$.

If the characteristic of $\mathbb{F}$ is 2, then $2 = 1 \oplus 1 = 0$ has no multiplicative inverse so the notion of $1/2$ is not meaningful. If the characteristic of $\mathbb{F}$ is not two, then $2 = 1 \oplus 1 \neq 0$ has a multiplicative inverse in $\mathbb{F}$. We use $1/2$ to designate the multiplicative inverse of $1 \oplus 1$ in this case.

Note that our definition of midpoint has nothing to do with the notion of a point being equidistant from two other points, although in a plane in which distance makes sense, we would like to know that our midpoint is the usual *midpoint* of a line segment. More generally, we would like some assurance that the midpoint of two points lies on the line determined by the two points in any affine space over a field of characteristic not 2.

Fix two points $\mathbf{u}, \mathbf{v}$ in $\mathbb{FA}^n$ and let ℓ be the line they determine. Since $\alpha\mathbf{u} + \beta\mathbf{v} \in \ell$ for all $\alpha, \beta \in \mathbb{F}$ with $\alpha \oplus \beta = 1$, the midpoint of $\mathbf{u}$ and $\mathbf{v}$ lies on ℓ provided $1/2 \oplus 1/2 = 1$ in $\mathbb{F}$. As long as the characteristic of $\mathbb{F}$ is not 2, we have, by the distributive law,

$$\frac{1}{2} \oplus \frac{1}{2} = \frac{1}{2} \odot (1 \oplus 1) = \frac{1}{2} \odot 2 = 1.$$

It follows that the midpoint of $\mathbf{u}$ and $\mathbf{v}$ is indeed on ℓ.

Next, we want to verify that in $\mathbb{R}^2$—the one setting where we have both intuition and a formula for distance—$(1/2)(\mathbf{u} + \mathbf{v})$ is the same distance from $\mathbf{u}$ as from $\mathbf{v}$. If $\mathbf{u} = (u_1, u_2)$, and $\mathbf{v} = (v_1, v_2)$ we have

$$\frac{1}{2}(\mathbf{u} + \mathbf{v}) = \left(\frac{u_1 + v_1}{2}, \frac{u_2 + v_2}{2}\right).$$

Using the distance formula to find the distance from $\mathbf{u}$ to $(1/2)(\mathbf{u}+\mathbf{v})$ in $\mathbb{R}^2$, we get

$$d\left(\mathbf{u}, \frac{1}{2}(\mathbf{u}+\mathbf{v})\right) = \sqrt{\left(u_1 - \frac{1}{2}(u_1+v_1)\right)^2 + \left(u_2 - \frac{1}{2}(u_2+v_2)\right)^2}$$

$$= \sqrt{\left(\frac{1}{2}(u_1-v_1)\right)^2 + \left(\frac{1}{2}(u_2-v_2)\right)^2} = \frac{1}{2}\, d(\mathbf{u},\mathbf{v}).$$

By symmetry, the distance is the same from $\mathbf{v}$ to $(1/2)(\mathbf{u}+\mathbf{v})$. This verifies that our definition of midpoint coincides with what we usually think of as the midpoint of two points in the one setting where we are confident that we know what it means.

Once we define what we mean by a triangle in $\mathbb{FA}^n$, we will be able to use midpoints to define medians.

Definition 7.74. A **triangle** in $\mathbb{FA}^n$ is a set of three noncollinear points and the three lines they determine. The points are the **vertices** of the triangle. The lines are the **sides** of the triangle.

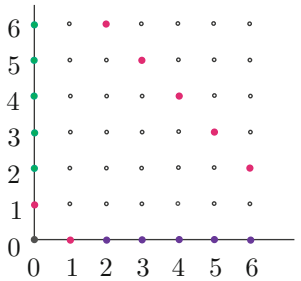


Figure 7.9: The triangle Δ determined by $\{(0,0),(1,0),(0,1)\}$ in $\mathbb{F}_7^2$ is comprised of the colored points in this figure. The points colored fuchsia comprise the side determined by $(1,0)$ and $(0,1)$. The other two sides of Δ are on the coordinate axes.

A corollary to Theorem 7.68 is immediate.

Corollary 7.75. A triangle determines a unique plane in $\mathbb{FA}^n$.

The corollary suggests that a study of triangles takes place most naturally in a planar setting. An interesting exception to that rule arises in the next chapter.

The notion of a line segment relies on the notion of betweenness but for vector spaces over arbitrary fields, the notion of betweenness does not necessarily make sense. Betweenness in $\mathbb{R}^n$ is realized using the fact that $\mathbb{R}$ is an ordered

field but as we have seen, most fields are not ordered fields. The sides of a triangle over an arbitrary field are thus lines, not line segments. In Fig. 7.9, for instance, we see a triangle Δ determined by $\{(0,0),(1,0),(0,1)\}$ in $\mathbb{F}_7^2$, a setting in which lines are sets of discrete points.

Lemma 7.76. Let V be the ambient vector space for $\mathbb{FA}^n$. A set of three vectors $\{\mathbf{u},\mathbf{v},\mathbf{w}\}$ in V determines a triangle in $\mathbb{FA}^n$ if and only if $\{\mathbf{v}-\mathbf{u},\mathbf{w}-\mathbf{u}\}$ spans a two-dimensional subspace of V.

Proof. We leave the proof as an exercise. □

Let α be the plane in $\mathbb{FA}^n$ determined by a triangle Δ with ordered vertices $\{\mathbf{u},\mathbf{v},\mathbf{w}\}$. For each $\mathbf{x}\in\alpha$, there are unique λ, $\mu\in\mathbb{F}$ with

$$\mathbf{x}-\mathbf{u}=\lambda(\mathbf{v}-\mathbf{u})+\mu(\mathbf{w}-\mathbf{u}).$$

This gives us

$$\mathbf{x}=(1-(\lambda+\mu))\mathbf{u}+\lambda\mathbf{v}+\mu\mathbf{w}.$$

Definition 7.77. The **affine plane coordinates** for the point$\mathbf{x}$ in the plane determined by the triangle $\Delta\subset\mathbb{FA}^n$ with ordered vertices $\{\mathbf{u},\mathbf{v},\mathbf{w}\}$ is the ordered triple $(1-(\lambda+\mu),\lambda,\mu)$ where $\mathbf{x}=(1-(\lambda+\mu))\mathbf{u}+\lambda\mathbf{v}+\mu\mathbf{w}$, for $\lambda,\mu\in\mathbb{F}$.

Notice that when $\mathbf{x}=\mathbf{u}$, $\lambda=\mu=0$ so that $\mathbf{u}$ has affine coordinates $(1,0,0)$ in the affine coordinate system based on this ordering of the vertices of Δ. There is no point with affine coordinates $(0,0,0)$! Indeed, when we fix a triangle, we are fixing an affine plane that we can identify with the plane in $\mathbb{F}^3$ given by $\{(x,y,z)\,|\,x+y+z=1\}$.

If Δ is the triangle with vertices $\{\mathbf{u},\mathbf{v},\mathbf{w}\}$, then $\mathbf{v}+\text{span}\{\mathbf{w}-\mathbf{v}\}$ is the side of Δ that is **opposite** the vertex $\mathbf{u}$. Now recall Definition 5.28, which applies in affine space as long as the characteristic of the base field is not 2: a *median of a triangle* is a line determined by a vertex and the midpoint of the opposite side.

We leave it as an exercise to show that when the characteristic of $\mathbb{F}$ is not two, the medians of a triangle Δ with vertices $\{\mathbf{u},\mathbf{v},\mathbf{w}\}\subset\mathbb{FA}^n$ are

$$\begin{aligned} m_1&=\mathbf{u}+\text{span}\left\{\frac{1}{2}(\mathbf{v}+\mathbf{w})-\mathbf{u}\right\}\\ m_2&=\mathbf{v}+\text{span}\left\{\frac{1}{2}(\mathbf{w}+\mathbf{u})-\mathbf{v}\right\}\\ m_3&=\mathbf{w}+\text{span}\left\{\frac{1}{2}(\mathbf{u}+\mathbf{v})-\mathbf{w}\right\}. \end{aligned} \tag{7.13}$$

We have had a glimpse of the peculiarities of affine spaces over fields with characteristic 2. Vector spaces over fields with characteristic 3 also suffer from certain limitations that give them strange geometries. To understand this, we

consider an example, $\mathbb{F}_3$, which has the following addition and multiplication tables.

$\oplus$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\odot$	1	2
1	1	2
2	2	1

Since $2 \oplus 1 = 0$, $-1 = 2$. Since $2 \odot 2 = 1$, $1/2 = 2$ in $\mathbb{F}_3$. If our affine space is $\mathbb{F}_3\mathbb{A}^n$, for instance, we have

$$\frac{1}{2}(\mathbf{v}+\mathbf{w}) - \mathbf{u} = 2(\mathbf{v}+\mathbf{w}) + 2\mathbf{u}$$

so that $\text{span}\{(1/2)(\mathbf{v}+\mathbf{w}) - \mathbf{u}\} = \text{span}\{\mathbf{u}+\mathbf{v}+\mathbf{w}\}$.

Now let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ determine a triangle in $\mathbb{F}_3\mathbb{A}^n$ with medians m_1, m_2, m_3. Take $\mathbf{x} = \mathbf{u}+\mathbf{v}+\mathbf{w}$. We can rewrite

$$m_1 = \mathbf{u} + \text{span}\{\mathbf{x}\},\ m_2 = \mathbf{v} + \text{span}\{\mathbf{x}\},\ m_3 = \mathbf{w} + \text{span}\{\mathbf{x}\}.$$

Recalling Lemma 7.62, we see then that the medians are either identical or parallel. We claim they are parallel, proving the claim by contradiction. Suppose then that $m_1 = m_2$.

Using arithmetic on coefficients from $\mathbb{F}_3$, we have

$$\mathbf{x} = \mathbf{u}+\mathbf{v}+\mathbf{w} = -2\mathbf{u}+\mathbf{v}+\mathbf{w} = (\mathbf{v}-\mathbf{u}) + (\mathbf{w}-\mathbf{u}).$$

Since $m_1 = m_2$, we have $\mathbf{u}-\mathbf{v} = c\mathbf{x}$, for some $c \in \mathbb{F}_3$ giving us

$$\mathbf{u}-\mathbf{v} = c\mathbf{x} = c(\mathbf{v}-\mathbf{u}) + c(\mathbf{w}-\mathbf{u}).$$

Since $\{\mathbf{v}-\mathbf{u}, \mathbf{w}-\mathbf{u}\}$ is linearly independent, c is unique. Note, though, that $\mathbf{u}-\mathbf{v} = 2(\mathbf{v}-\mathbf{u}) + 0(\mathbf{w}-\mathbf{u})$ forcing us to conclude both that $c = 2$ and $c = 0$, an impossibility in $\mathbb{F}_3$. It follows that the medians are distinct and parallel.

We see then that if the characteristic of $\mathbb{F}$ is 2, since $1/2$ does not exist in $\mathbb{F}$, the notion of midpoint, thus median, does not make sense and if the characteristic of $\mathbb{F}$ is 3, the medians of a triangle are parallel. Now we can reconsider Corollary 5.29, that the medians of a triangle in the Euclidean plane are concurrent, in $\mathbb{F}\mathbb{A}^n$.

Theorem 7.78. If the characteristic of $\mathbb{F}$ is not 2 or 3, the medians of a triangle in $\mathbb{F}\mathbb{A}^n$ are concurrent.

Proof. Let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ determine a triangle $\Delta \subset \mathbb{F}\mathbb{A}^n$. Say that α is the plane containing Δ. Denote by m_1, m_2, m_3 the medians of Δ. Any point$\mathbf{x}$ belonging to $m_1 = \mathbf{u} + \text{span}\{(1/2)(\mathbf{v}+\mathbf{w}) - \mathbf{u}\}$ has affine line coordinates $(1-\lambda, \lambda)$, unique to$\mathbf{x}$ and m_1. Making similar statements using the affine line coordinates with respect to m_2 and m_3, we can say that there is a point common to all three lines if and only if there are λ, μ, and ν in $\mathbb{F}$ with

$$\mathbf{x} = (1-\lambda)\mathbf{u} + (\lambda/2)\mathbf{v} + (\lambda/2)\mathbf{w},$$

$$\mathbf{x} = (\mu/2)\mathbf{u} + (1-\mu)\mathbf{v} + (\mu/2)\mathbf{w} \text{ and}$$
$$\mathbf{x} = (\nu/2)\mathbf{u} + (\nu/2)\mathbf{v} + (1-\nu)\mathbf{w}.$$

Equating coefficients of $\mathbf{u}$ we get $\mu/2 = \nu/2$, so that $\mu = \nu$. Equating coefficients of $\mathbf{w}$ we get $\lambda = \mu$. Going back to the coefficient of $\mathbf{u}$ in the first equation, we get $1 - \lambda = \lambda/2$ so that $\lambda = \mu = \nu = 2/3$. This gives us the point of concurrency

$$\mathbf{x} = \frac{1}{3}\mathbf{u} + \frac{1}{3}\mathbf{v} + \frac{1}{3}\mathbf{w}.$$

□

Recall that the point of concurrency of the medians of a triangle is the *centroid*.

Corollary 7.79. If the characteristic of $\mathbb{F}$ is not 2 or 3, the centroid of a triangle in $\mathbb{FA}^n$ has affine coordinates $(1/3, 1/3, 1/3)$ with respect to the triangle.

Proof. The affine coordinates of $\mathbf{x}$, the centroid as given in Theorem 7.78, are $(1-(\eta+\xi), \eta, \xi)$, where

$$\mathbf{x} = (1-(\eta+\xi))\mathbf{u} + \eta\mathbf{v} + \xi\mathbf{w} = \frac{1}{3}\mathbf{u} + \frac{1}{3}\mathbf{v} + \frac{1}{3}\mathbf{w},$$

which implies the result. □

Now that we understand something about affine geometry, we can revisit its relationship to Euclidean geometry. Affine geometry is based entirely on vector spaces and the incidences that must arise in vector spaces among the points, lines, planes, and k-planes therein. While vector spaces over $\mathbb{R}$ yield models for Euclidean geometry, arbitrary vector spaces do not, because Euclidean geometry relies on ideas of betweenness, length, and angle, notions that do not arise naturally in vector spaces over arbitrary fields. (Try identifying rays, for instance, in the vector space shown in Fig. 7.5.) When we view $\mathbb{R}^2$ as a model for Euclid's plane, we employ the distance and angle measures that we can define using the dot product. Properties of the dot product that give us Euclidean geometry are properties that arise as a consequence of the fact that $\mathbb{R}$ is an ordered field that satisfies the Completeness Axiom. We can view the Euclidean plane, then, as the real affine plane with the additional structure that arises from the dot product.

Exercises 7.4. 1. Let V be a vector space with subspace W.

(a) Show that the union of all W-cosets is V.

(b) Show that every W-coset is nonempty.

(c) Putting (a) and (b) together with Lemma 7.62.(2), we have that W determines a *partition* on V, that is, a collection of nonempty, nonintersecting subsets of V, the union of which is all of V. A partition on a set in turn defines an equivalence relation on the set: We say

that two elements in the set are equivalent provided they belong to the same subset in the partition. Show that $\mathbf{v} \sim \mathbf{x}$ if and only if $\mathbf{v} + W = \mathbf{x} + W$ does in fact define an equivalence relation on V.

2. Suppose that in $\mathbb{FA}^n$ we have a k_1-plane, $\mathcal{A} = \mathbf{v} + W$, and a k_2-plane $\mathcal{A}' = \mathbf{u} + W'$, where $k_1 + k_2 = n$. Suppose further that $\mathcal{A} \cap \mathcal{A}' = \emptyset$. Prove that $W \cap W'$ is a nontrivial subspace of V.

3. Let $L : V \to W$ be an isomorphism of vector spaces. Prove each of the following statements to show that L respects dimension, both of subspaces and of cosets, and the affine incidence structure in V and in W. This will verify that the model of affine geometry determined by an n-dimensional vector space V over $\mathbb{F}$ is isomorphic to the model of affine geometry determined by $\mathbb{F}^n$.

   (a) If $U \subseteq V$ is a subspace with $\dim U = k$, then $L(U) \subseteq W$ is a subspace with $\dim L(U) = k$.

   (b) If $\mathbf{v} + U$ is a k-dimensional coset in V, $L(\mathbf{v} + U)$ is a k-dimensional coset in W.

   (c) If A and B are objects in an affine space with ambient vector space V and $A \cap B = C$, then $L(A)$ and $L(B)$ are objects in an affine space with ambient vector space W and $L(A) \cap L(B) = L(C)$.

4. Verify that two k-planes, $\mathbf{v} + W$ and $\mathbf{u} + W'$, share a direction if and only if $W \cap W'$ is nontrivial.

5. Discuss the validity of Euclid's Prop. I.1 in $\mathbb{QA}^2$.

6. Let $\mathcal{A} = \mathbf{x} + U$ be an affine 3-plane in $\mathbb{FA}^n$. Here $\mathbf{x}$ is a vector and U a three-dimensional subspace in the ambient vector space , V. Suppose ℓ is a line and α is a plane in $\mathcal{A}$. Verify that we can write $\ell = \mathbf{y}_1 + \text{span}\{\mathbf{v}\}$, and $\alpha = \mathbf{y}_2 + W$ where W and $\text{span}\{\mathbf{v}\}$ are two- and one-dimensional subspaces, respectively, in U.

7. Show that the points $(1, 0, 0, \ldots, 0)$, $(0, 1, 0, \ldots, 0)$, $(0, 0, 1, 0, \ldots, 0)$ are noncollinear in $\mathbb{F}^n$, regardless of $\mathbb{F}$. Using the setting of $\mathbb{F}^3$, describe the lines these points determine, if you take two points at a time. How many lines are there?

8. Show that the points $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ are noncoplanar in $\mathbb{F}^3$, for any field $\mathbb{F}$. How many different planes do these four points determine? Argue that the planes you found must be distinct.

9. From what part of Theorem 7.68 does Corollary 7.75 follow and why?

10. Prove Lemma 7.76.

11. Finish the proof of Theorem 7.70.(2).

12. Let $\mathbf{v} = (a, b) \in \mathbb{F}^2$. Show that $(1-(a+b), a, b)$ are the affine coordinates of $\mathbf{v}$ with respect to the triangle with vertices $\{(0,0),(1,0),(0,1)\}$.

13. Show that if the characteristic of $\mathbb{F}$ is not 2, the medians of a triangle with vertices $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ in $\mathbb{FA}^2$ are given in (7.13).

14. Let Δ be the triangle in $\mathbb{F}_7^2$ with vertices $\{(0,0),(1,0),(0,1)\}$, as shown on the right in Fig. 7.10.

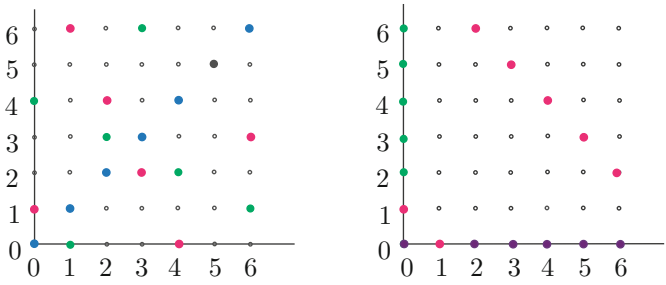


Figure 7.10: Δ determined by $\{(0,0),(1,0),(0,1)\}$ in $\mathbb{F}_7^2$ is on the right. Its medians are shown on the left, with the centroid indicated by a black dot.

(a) Verify that the line determined by $\{(1,0),(0,1)\}$ is comprised of the points $\{(1,0),(2,6),(3,5),(4,4),(5,3),(6,2),(0,1)\}$. (These are the fuchsia points in the picture on the right in Fig. 7.10.)

(b) Verify that the medians for Δ are as shown on the left in Fig. 7.10. Verify that the centroid is $(5,5)$. Verify that the centroid has affine coordinates $(5,5,5)$ with respect to Δ. Address the fact that affine plane coordinates must sum to 1.

15. Sketch the triangle with vertices $\{(1,0),(0,1),(1,1)\}$ in $\mathbb{F}_5^2$. Find the points on the lines determined by the sides. Find the medians and the centroid.

16. Find the points on the line in $\mathbb{F}_2^3$ given by

$$(1,0,1) + \text{span}\{(1,0,0)\}.$$

Make a sketch to show what a line looks like in $\mathbb{F}_2^3$.

17. Find the points on the plane in $\mathbb{F}_2^3$ given by

$$(1,0,1) + \text{span}\{(1,0,0),(0,1,0)\}.$$

Make a sketch to show what a plane looks like in $\mathbb{F}_2^3$.

7.6 The Affine Group

Groups were developed during the late nineteenth and early twentieth centuries and quickly became fundamental to modern geometry. Here we consider the groups associated to $\mathbb{FA}^n$. We start with a definition.

Definition 7.80. A **group** G is a set of mappings on a set $\mathcal{S}$ that satisfies the following conditions.

(1) G contains the identity mapping, that is e given by $e(s) = s$ for all $s \in \mathcal{S}$,

(2) G is closed under composition, and

(3) every element in G has an inverse in G.

If $g \circ h = h \circ g$ for all $g, h \in G$, then G is **abelian**.

Function composition, which is always associative, is the binary operation on a group: We say that G, as defined in Definition 7.80 is a group *under function composition*. When we want to emphasize the binary operation on G we write $(G, \circ)$. In a typical modern algebra course, a group is defined as a set G together with an associative binary operation for which G has an identity element, and so that each element in G has an inverse. **Cayley's Theorem** says that every group is a group of mappings on some set, hence, in our restricted context, we think of a group as a collection of mappings from the beginning.

Example 7.81. A vector space V is itself a group, in which composition is realized as vector addition. The challenge here is to see a vector space as a collection of mappings on itself.

Fixing $\mathbf{v} \in V$, define $t_{\mathbf{v}} : V \to V$ by $t_{\mathbf{v}}(\mathbf{u}) = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u} \in V$. These mappings $t_{\mathbf{v}}$ are *translations*. For $\mathbf{v}, \mathbf{u}, \mathbf{w} \in V$ we have

$$(t_{\mathbf{v}} \circ t_{\mathbf{w}})(\mathbf{u}) = t_{\mathbf{v}}(t_{\mathbf{w}}(\mathbf{u})) = t_{\mathbf{v}}(\mathbf{w} + \mathbf{u}) = \mathbf{v} + \mathbf{w} + \mathbf{u} = t_{\mathbf{v}+\mathbf{w}}(\mathbf{u})$$

so that $t_{\mathbf{v}} \circ t_{\mathbf{w}} = t_{\mathbf{v}+\mathbf{w}}$. This shows that the collection $G = \{t_{\mathbf{v}} \mid \mathbf{v} \in V\}$ is closed under composition and that composition of mappings corresponds to addition of the associated vectors. The identity mapping in G is $t_{\mathbf{0}}$ because $t_{\mathbf{0}} \circ t_{\mathbf{v}} = t_{\mathbf{v}}$, for all $t_{\mathbf{v}} \in G$. The inverse of $t_{\mathbf{v}}$ is $t_{-\mathbf{v}}$. G is abelian because vector addition is commutative: $t_{\mathbf{v}} \circ t_{\mathbf{u}} = t_{\mathbf{v}+\mathbf{u}} = t_{\mathbf{u}+\mathbf{v}} = t_{\mathbf{u}} \circ t_{\mathbf{v}}$.

Identifying $\mathbf{v} \in V$ with $t_{\mathbf{v}}$ allows us to view the elements in V as mappings on V. We must establish though that $\mathbf{v} \mapsto t_{\mathbf{v}}$ is a bijection. Define $f : V \to G$ by $f(\mathbf{v}) = t_{\mathbf{v}}$. It is clear that f is onto. For $\mathbf{u}, \mathbf{v} \in V$ suppose $f(\mathbf{u}) = f(\mathbf{v})$. Then for all $\mathbf{x} \in V$,

$$f(\mathbf{u})(\mathbf{x}) = t_{\mathbf{u}}(\mathbf{x}) = \mathbf{u} + \mathbf{x} = f(\mathbf{v})(\mathbf{x}) = t_{\mathbf{v}}(\mathbf{x}) = \mathbf{v} + \mathbf{x} \Longrightarrow \mathbf{u} = \mathbf{v},$$

so f is one-to-one, thus bijective.

This establishes that we can identify $(V, +)$ with $(G, \circ)$. We say that $(V, +)$ is an abelian group or that V is an abelian group under addition.

We can play the same trick to see that a field $(\mathbb{F}, \oplus, \odot)$ is an abelian group under $\oplus$ and that its nonzero elements form an abelian group under $\odot$. We leave the details to the exercises.

Next we turn to a group of mappings on an equilateral triangle, Δ. A mapping $\varphi : \Delta \to \Delta$ is a **symmetry** if it leaves Δ appearing to be positioned exactly as it starts, possibly with the vertices rearranged. We illustrate the symmetries on Δ in Fig. 7.11. Note that the labels on the vertices go along for the ride when the triangle is moved.

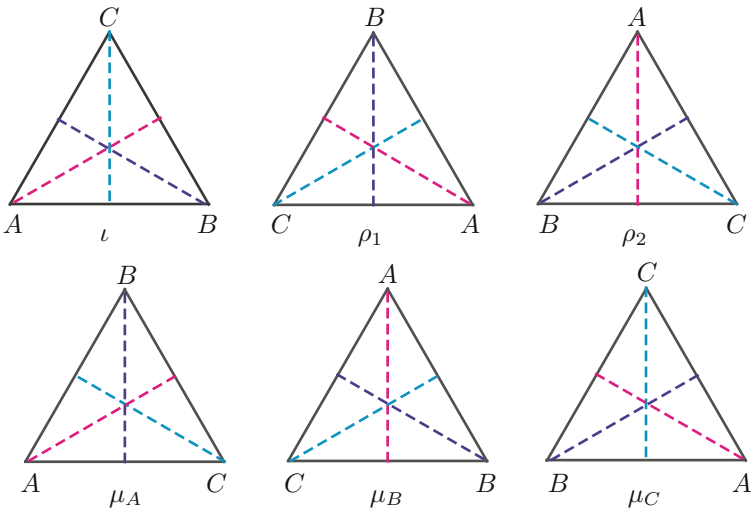


Figure 7.11: Symmetries on an equilateral triangle form a group called S_3

Example 7.82. Let Δ be an equilateral triangle in $\mathbb{RA}^2$, say with vertices A, B, C. The collection of symmetries on Δ forms a group called S_3, the **symmetric group on three letters**. Axes of symmetry for Δ are the rotation axis through the centroid of Δ, and the medians, each of which determines a reflection of Δ that leaves one vertex fixed and transposes the other two. Elements in S_3 are ι, the identity mapping that leaves Δ unchanged, ρ_1, the counterclockwise rotation through $2\pi/3$, centered at the centroid of Δ, and $\rho_2 = \rho_1^2$, the counterclockwise rotation through $4\pi/3$ or, equivalently, the clockwise rotation through $2\pi/3$. (The concern here is not with the actual motion, just its effect on Δ.) We designate the reflections μ_A, μ_B, and μ_C, each named for the vertex it fixes.

We leave it as an exercise to check that the following group table describes the compositions among symmetries in S_3.

$\circ$	ι	ρ_1	ρ_2	μ_A	μ_B	μ_C
ι	ι	ρ_1	ρ_2	μ_A	μ_B	μ_C
ρ_1	ρ_1	ρ_2	ι	μ_C	μ_A	μ_B
ρ_2	ρ_2	ι	ρ_1	μ_B	μ_C	μ_A
μ_A	μ_A	μ_B	μ_C	ι	ρ_1	ρ_2
μ_B	μ_B	μ_C	μ_A	ρ_2	ι	ρ_1
μ_C	μ_C	μ_A	μ_B	ρ_1	ρ_2	ι

(7.14)

Note that the following theorem does not depend on dimension. It is true for infinite-dimensional spaces as well as finite-dimensional spaces.

Theorem 7.83. The set of bijective linear transformations on any vector space is a group.

Proof. Note that the set of bijective linear transformations on a space is the set of isomorphisms from the space to itself. In Exercise 7.3.3 you showed that the set is closed under composition. Certainly the identity mapping is included in the set as is the inverse of any element in the set. This is enough to establish that the set is a group. □

Definition 7.84. The **general linear group on** V, $GL(V)$, is the set of bijective linear transformations on a vector space, V.

When $V = \mathbb{F}^n$, $GL(V)$ can be identified with the set of invertible $n \times n$ matrices with entries in $\mathbb{F}$. In this case we often write $GL(V) = GL_n(\mathbb{F})$.

Example 7.85. Since they have determinant 1, all the rotation matrices from Example 7.54 belong to $GL_2(\mathbb{R})$. Likewise, all reflection matrices from Examples 7.55 and 7.56 are in $GL_2(\mathbb{R})$. This follows from the fact that the determinant of a reflection matrix is always -1. What other sorts of mappings are in the general linear group? Contractions, dilations, and shears are all invertible linear transformations, so all of these mappings, which we explore in the exercises, are in $GL_2(\mathbb{R})$.

As Theorem 7.57 implies for $V = \mathbb{F}^n$, an element in $GL(V)$ maps any ordered basis for V to some other ordered basis for V. Conversely, two ordered bases of V determine a unique element in $GL(V)$ because a linear transformation is completely determined by where it maps a basis.

The result worked out in Exercise 7.4.3 above implies that elements in $GL(V)$ respect both the linear (vector space) structure of V, and the affine geometry determined by V. There are other mappings that respect affine structure, but not linear structure: translations.

We saw in Example 7.81 that the set V can be identified with translations by vectors in V. Here, we identify $\mathbf{v}$ with $t_{\mathbf{v}} : V \to V$ defined by $t_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \mathbf{v}$. To get a larger collection of affine transformations, we "add" translations to the group $GL(V)$.

Theorem 7.86. Let V be any vector space. Fixing $T \in GL(V)$ and $\mathbf{v} \in V$, let $A(T, \mathbf{v}) : V \to V$ be given by

$$A(T, \mathbf{v})(\mathbf{u}) = T(\mathbf{u}) + \mathbf{v} \text{ for all } \mathbf{u} \in V.$$

The collection of all such mappings, $\mathrm{Aff}(V)$, is a group.

Proof. We must verify that $\mathrm{Aff}(V)$ is closed under composition. Suppose T_1, T_2 belong to $GL(V)$, and that $\mathbf{v}_1, \mathbf{v}_2$ are in V. For arbitrary $\mathbf{x} \in V$, we then have

$$\begin{aligned}(A(T_2, \mathbf{v}_2) \circ A(T_1, \mathbf{v}_1))(\mathbf{x}) &= A(T_2, \mathbf{v}_2)(T_1(\mathbf{x}) + \mathbf{v}_1) \\ &= T_2(T_1(\mathbf{x}) + \mathbf{v}_1) + \mathbf{v}_2 = T_2(T_1(\mathbf{x})) + T_2(\mathbf{v}_1) + \mathbf{v}_2 \\ &= (T_2 \circ T_1)(\mathbf{x}) + (T_2(\mathbf{v}_1) + \mathbf{v}_2) = A(T_2 \circ T_1, T_2(\mathbf{v}_1) + \mathbf{v}_2)(\mathbf{x}).\end{aligned}$$

Then

$$A(T_2, \mathbf{v}_2) \circ A(T_1, \mathbf{v}_1) = A(T_2 \circ T_1, T_2(\mathbf{v}_1) + \mathbf{v}_2) \tag{7.15}$$

so $\mathrm{Aff}(V)$ is closed under function composition. The identity element in $\mathrm{Aff}(V)$ is $A(I, \mathbf{0})$, where I is the identity mapping on V. We can determine the inverse mapping of any $A(T, \mathbf{v})$ in $\mathrm{Aff}(V)$ by looking at Equation (7.15). We leave this as an exercise. This is enough to establish that $\mathrm{Aff}(V)$ is a group. □

Definition 7.87. The **affine group** for a vector space V, $\mathrm{Aff}(V)$, is the set of mappings $\{A(T, \mathbf{v}) \mid T \in GL(V), \mathbf{v} \in V\}$ where $A(T, \mathbf{v})(\mathbf{u}) = T(\mathbf{u}) + \mathbf{v}$.

When we say that $\mathrm{Aff}(V)$ is a group, we are implying that elements of $\mathrm{Aff}(V)$ are bijections. We leave the verification of that to the exercises.

As a set, $\mathrm{Aff}(V)$ is the Cartesian product of $GL(V)$ and V, in other words, an element of $\mathrm{Aff}(V)$ can be identified with an ordered pair $(T, \mathbf{v})$, $T \in GL(V)$, $\mathbf{v} \in V$. There are many different ways to define "products" of groups. The affine group is an example of what is called a *semi-direct product* of $GL(V)$ and V. This makes $\mathrm{Aff}(V)$ an interesting example from a group theoretic point of view but here we are more concerned with the effect of $\mathrm{Aff}(V)$ on the affine geometry of V.

Lemma 7.88. Elements of $\mathrm{Aff}(V)$ map points to points, lines to lines, and planes to planes in V.

Proof. The lemma is a consequence of the fact that elements in $\mathrm{Aff}(V)$ are bijections. For instance, let W be a subspace of V so that for $\mathbf{u}$ in V, $\mathbf{u} + W$ is a point, line, or plane in V. Our object is a point if and only if $\dim W = 0$, it is a line if and only if $\dim W = 1$, and it is a plane if and only if $\dim W = 2$. For $T \in GL(V)$ and $\mathbf{v} \in V$, we have

$$A(T, \mathbf{v})(\mathbf{u} + W) = T(\mathbf{u}) + \mathbf{v} + T(W),$$

where $\dim T(W) = \dim W$. □

The triangle, in some sense, forms the most basic organizational unit in an affine plane. That is one interpretation of Lemma 7.76 and one reason the next lemma is important.

Lemma 7.89. Elements in $\mathrm{Aff}(V)$ map triangles to triangles in V.

Proof. Let $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ determine a triangle in V so that $\{\mathbf{y} - \mathbf{x}, \mathbf{z} - \mathbf{x}\}$ is a basis for a two-dimensional subspace in V. Let $A(T, \mathbf{v}) \in \mathrm{Aff}(V)$. We have

$$A(T, \mathbf{v})\{\mathbf{x}, \mathbf{y}, \mathbf{z}\} = \{T(\mathbf{x}) + \mathbf{v}, T(\mathbf{y}) + \mathbf{v}, T(\mathbf{z}) + \mathbf{v}\}.$$

Note that $T(\mathbf{y}) + \mathbf{v} - (T(\mathbf{x}) + \mathbf{v}) = T(\mathbf{y}) - T(\mathbf{x}) = T(\mathbf{y} - \mathbf{x})$. Likewise,

$$T(\mathbf{z}) + \mathbf{v} - (T(\mathbf{x}) + \mathbf{v}) = T(\mathbf{z} - \mathbf{x}).$$

Since $T \in GL(V)$, Theorem 7.57 implies that $\{T(\mathbf{y} - \mathbf{x}), T(\mathbf{z} - \mathbf{x})\}$ is a linearly independent set, thus a basis for a two-dimensional subspace in V. Lemma 7.76 then implies that $A(T, \mathbf{v})\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a triangle in V, as claimed. □

We get a stronger result in the finite-dimensional setting.

Theorem 7.90. Let V be the ambient vector space for $\mathbb{FA}^n$. Given any two triangles in $\mathbb{FA}^n$, there is an element in $\mathrm{Aff}(V)$ that maps one to the other.

Proof. Let Δ_1 and Δ_2 be triangles in $\mathbb{FA}^n$. We must show that there is an affine transformation that maps the vertices of Δ_1 to the vertices of Δ_2 and the sides of Δ_1 to the sides of Δ_2. Suppose Δ_1 has vertices $\{\mathbf{x}_1, \mathbf{y}_1, \mathbf{z}_1\}$, and Δ_2 has vertices $\{\mathbf{x}_2, \mathbf{y}_2, \mathbf{z}_2\}$.

We know that $\mathcal{B}_1 = \{\mathbf{y}_1 - \mathbf{x}_1, \mathbf{z}_1 - \mathbf{x}_1\}$ and $\mathcal{B}_2 = \{\mathbf{y}_2 - \mathbf{x}_2, \mathbf{z}_2 - \mathbf{x}_2\}$ are linearly independent sets in V, thus can be expanded to ordered bases for V. Let $\mathcal{B} = \{\mathbf{y}_1 - \mathbf{x}_1, \mathbf{z}_1 - \mathbf{x}_1, \mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $\mathcal{B}' = \{\mathbf{y}_2 - \mathbf{x}_2, \mathbf{z}_2 - \mathbf{x}_2, \mathbf{u}'_1, \ldots, \mathbf{u}'_k\}$ be such ordered bases. There is then $T \in GL(V)$ mapping the ordered basis $\mathcal{B}$ to the ordered basis $\mathcal{B}'$.

We claim that $g = A(T, T(-\mathbf{x}_1) + \mathbf{x}_2)$ maps Δ_1 to Δ_2. We leave the details of the verification as an exercise. □

The next corollary is immediate.

Corollary 7.91. Let V be the ambient vector space for $\mathbb{FA}^n$. Given any two planes in $\mathbb{FA}^n$, there is an element in $\mathrm{Aff}(V)$ that maps one to the other.

Corollary 7.92. Let V be the ambient vector space for $\mathbb{FA}^n$. If ℓ is a line in $\mathbb{FA}^n$ containing a point $\mathbf{x}$, and ℓ' is a line in $\mathbb{FA}^n$ containing a point $\mathbf{x}'$, then there is an element in $\mathrm{Aff}(V)$ that maps ℓ to ℓ' and$\mathbf{x}$ to $\mathbf{x}'$.

Proof. The proof of Theorem 7.90 shows that if$\mathbf{x}$ is a vertex, and ℓ is a side going through that vertex in a triangle, and $\mathbf{x}'$ is a vertex and ℓ' is a side going through that vertex in a triangle, then there is an element in $\mathrm{Aff}(V)$ that maps $\mathbf{x}$ to $\mathbf{x}'$ and ℓ to ℓ'. This takes care of the result when $\dim V \geq 2$ since we can always form triangles starting with vertex-side pairs $\mathbf{x}$-ℓ and $\mathbf{x}'$-ℓ'. If $\dim V = 1$, the proof amounts to showing that we can map any vector in V to any other vector in V using an affine transformation. We leave the verification of that statement as an exercise. □

Groups can be handy for proving certain types of theorems in geometry. We apply them now to finish the proof that Hilbert's incidence axioms hold in $\mathbb{RA}^3$. Our next theorem implies that Hilbert's Axiom I.7 holds in $\mathbb{FA}^3$.

Theorem 7.93. Two distinct planes in $\mathbb{FA}^3$ intersect in a line or are parallel. If intersecting, the planes share exactly one direction.

Proof. Let α and β be distinct planes in $\mathbb{FA}^3$ and let V be the ambient vector space. Suppose $\alpha = \mathbf{u} + W$ and $\beta = \mathbf{v} + W'$, so that W and W' are two-dimensional subspaces of V.

Say that $W = \operatorname{span}\{\mathbf{w}_1, \mathbf{w}_2\}$ and $W' = \operatorname{span}\{\mathbf{w}'_1, \mathbf{w}'_2\}$. Notice then that the span of $\mathcal{C} = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}'_1, \mathbf{w}'_2\} \subset V$ is either two- or three-dimensional.

If $\dim \operatorname{span} \mathcal{C} = 2$, then Theorem 7.30.(5) implies $W = W'$. By Lemma 7.62, then $\alpha \cap \beta = \emptyset$. Since α and β have the same directions, they are parallel in this case.

If $\dim \operatorname{span} \mathcal{C} = 3$, then say that $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}'_1\} \subset \mathcal{C}$ is linearly independent. Let $c_1, c_2, c \in \mathbb{F}$ so that

$$\mathbf{u} - \mathbf{v} = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + c\mathbf{w}'_1.$$

Then $\mathbf{u} - (c_1\mathbf{w}_1 + c_2\mathbf{w}_2) = \mathbf{v} + c\mathbf{w}'_1$ is in $\alpha \cap \beta \neq \emptyset$. In this case, we can employ an element in $\operatorname{Aff}(V)$ to map the intersection set of α and β to the origin. In other words, invoking the affine group allows us to treat α and β as subspaces of V. Proceeding, we assume $\alpha = W$, $\beta = W'$. It remains for us to show that $W \cap W'$ is a line.

Since $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}'_1\} \subset \mathcal{C}$ is a basis for V, $\mathbf{w}'_2 = c_1\mathbf{w}_1 + c_2\mathbf{w}_2 + c_3\mathbf{w}'_1$, for some $c_i \in \mathbb{F}$, where the c_i are not all 0. This gives us a nonzero vector $\mathbf{y} = \mathbf{w}'_2 - c_3\mathbf{w}'_1$ common to W' and to W. As subspaces, W and W' must contain the line $\ell = \operatorname{span} \mathbf{y}$. Since α and β are distinct, Theorem 7.68.(3) implies that they have no other points in common. □

The corollary is immediate.

Corollary 7.94. Hilbert's Axiom I.7, holds in $\mathbb{FA}^3$: If two planes have one point in common, they must have a second point in common, as well.

The proof of the theorem highlights the importance of the dimension of the ambient space. Intersecting planes in $\mathbb{FA}^3$ must have a line in common because the union of bases for the two-dimensional subspaces underlying the planes must span the ambient three-dimensional vector space. Since that union of bases has four vectors, one of the four is a linear combination of the other three. That vector captures a direction common to the two planes. It is a direction vector for the line of intersection.

If we considered two planes in a higher-dimensional ambient space, the planes could intersect in a point. This will happen for instance if we take $\alpha = \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2\}$, and $\beta = \operatorname{span}\{\mathbf{u}_3, \mathbf{u}_4\}$ where $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ is a basis for the ambient space, V: Here, $\alpha \cap \beta = \mathbf{0}$. We can also construct skew planes

in a four-dimensional space, but they must have a common direction. For instance, let $\alpha = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$, and $\beta = \mathbf{x}+\text{span}\{\mathbf{u}_3, \mathbf{u}_4\}$, where $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ is a basis for the ambient space V and $\mathbf{x}$ is a nonzero vector in V. These have no common directions but do they intersect? There must be $c_i \in \mathbb{F}$ so that $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 + c_4\mathbf{u}_4$. Then $\beta = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \text{span}\{\mathbf{u}_3, \mathbf{u}_4\}$, so $\alpha \cap \beta$ contains $c_1\mathbf{u}_1 + c_2\mathbf{u}_2$. This is not a proof, but it suggests that an effort to construct skew planes with no common direction fails when the ambient space is four-dimensional, because the planes will wind up with a nonempty intersection. (We leave the details of a proof as an exercise.)

If we move to n-dimensional spaces for $n > 4$, it is easy to see that we can find skew planes with no directions in common.

The proof of Theorem 7.93 employs the affine group to map the planes in question to a location in the ambient space that makes things more convenient for us. This idea can be exploited further.

A group associated to a given geometry is a collection of mappings that can be used to define the objects and the relationships among them in the geometry. In the case of the affine geometry determined by a vector space $V \cong \mathbb{F}^n$, we could define the points of $\mathbb{FA}^n$ to be all translates of $\mathbf{0}$ under $\text{Aff}(V)$, the lines to be all translates of $\text{span}\{\mathbf{e}_1\}$ under $\text{Aff}(V)$, the planes to be all translates of $\text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ under $\text{Aff}(V)$, and the triangles to be all translates of a single triangle Δ with vertices $\{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2\}$. This is the idea behind Felix Klein's *Erlangen Program* of 1872, a philosophy of geometry described in a lecture in which Klein declared that groups and geometries are essentially equivalent.

It is easy to verify that if G is a group of mappings on a set $\mathcal{S}$, and if $A \subseteq \mathcal{S}$, then the set $H \subseteq G$ of elements in G that map $A \to A$ forms a group inside G with the same operations as in G and the same identity element. Thus, H is a **subgroup** of G. The elements of H are said to leave A fixed or *invariant.*

Let H be the subgroup of $G = \text{Aff}(V)$ that leaves one specific triangle invariant. Note that H then maps the triangle to itself possibly with a permutation of its vertices. A left H-coset in G is $gH = \{g \circ h | \, h \in H\}$, for some fixed $g \in \text{Aff}(V)$. Just as cosets of the span of a nonzero vector in V serve as lines in $\mathbb{FA}^n$, left cosets of H can serve as triangles in an affine geometry.

This approach is typical of the Erlangen Program, which Klein formulated during an era in which geometry was in disarray. The notion that groups could and should serve as a unifying force in geometry turned out to be profound. It remains at the heart of a great deal of mathematics being discovered today.

Exercises 7.5.

1. Let G be a group of mappings on a set $\mathcal{S}$. Show that every element in G is a bijection from $\mathcal{S}$ to $\mathcal{S}$.

2. Show that a linear transformation $L : V \to V$ always maps the zero vector to itself. (This shows that if a translation $t_{\mathbf{v}}$ is linear, then $\mathbf{v} = \mathbf{0}$.)

3. If $z = a + bi \in \mathbb{C}$, for $a, b \in \mathbb{R}$, then the *complex conjugate* of z is $\bar{z} = a - bi$. Consider $V = \mathbb{C}^2$ as a two-dimensional vector space over $\mathbb{C}$. Define $\varphi : V \to V$ by $\varphi(z_1, z_2) = (\bar{z_1}, \bar{z_2})$. Prove that φ sends lines to lines

and triangles to triangles in the affine plane determined by V but that it is not a linear transformation on V.

4. Show that a field forms an abelian group under $\oplus$ by identifying each element $a \in \mathbb{F}$ with an appropriately defined mapping $\lambda_a : \mathbb{F} \to \mathbb{F}$. Use the same type of trick to show that the nonzero elements in $\mathbb{F}$ form an abelian group under $\odot$.

5. Check the compositions of symmetries in (7.14).

6. Argue that elements in $GL(V)$ map points to point, lines to lines, planes to planes, and that they preserve incidence between objects in the affine space determined by V. In particular, prove that if two objects intersect in a line, then their translates under a mapping in $GL(V)$ also intersect in a line.

7. A *contraction* or *dilation* on $\mathbb{R}^2$ is a mapping that shrinks or stretches every vector in $\mathbb{R}^2$ by a factor of c, where c is some positive number. Contractions and dilations have no effect on the direction of any vector. Show that a contraction or dilation has standard matrix representation $\begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix}$. What values of c yield contractions and what values of c yield dilations? Show that contractions and dilations are in $GL_2(\mathbb{R})$.

8. Figure 7.12 shows the effect of a *shear*, L, on $\mathbb{R}^2$ in the x-direction. L is a linear transformation that leaves $\mathbf{e}_1$ unchanged, but adds a vector of the form $\mathbf{v} = c\,\mathbf{e}_1$ to $\mathbf{e}_2$.

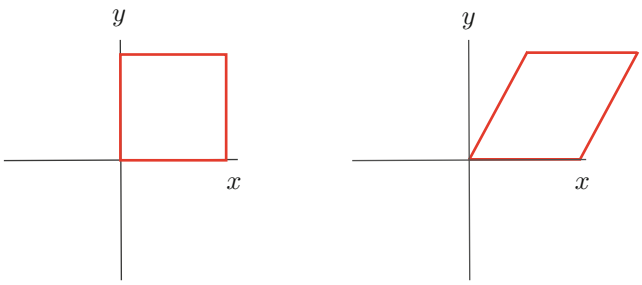


Figure 7.12: The effect of a shear in the x-direction on a rectangle at the origin in $\mathbb{R}^2$

   (a) Sketch the second image in Fig. 7.12 including the vector $\mathbf{v}$ in your picture.

   (b) Find the standard matrix representation of L.

   (c) Sketch a picture that shows a shear on $\mathbb{R}^2$, this one in the y-direction.

(d) What is the form of the standard matrix representation of the mapping you found in part (c)?

(e) Argue that shears, at least of these types, are in $GL_2(\mathbb{R})$.

9. Finish the proof that $\text{Aff}(V)$ is a group. Be sure to specify the inverse of an element $A(T, \mathbf{v})$ in $\text{Aff}(V)$. Check that the composition of your purported inverse and $A(T, \mathbf{v})$ is $A(I, \mathbf{0})$, using Equation (7.15).

10. Finish the proof of Theorem 7.90.

11. Prove that if $\mathbf{v}$ and $\mathbf{w}$ belong to a vector space V, there is an element in $\text{Aff}(V)$ that maps $\mathbf{v}$ to $\mathbf{w}$.

12. Prove that two skew planes in a four-dimensional affine space must have a common direction.

13. Give an example to show that there are skew planes with no directions in common in spaces of dimension 5 or higher.

14. Define a **complete quadrilateral** in $\mathbb{FA}^2$ to be a set of four points, no three of which are collinear, and the lines they determine. The four points are the *vertices* of the quadrilateral.

    A quadrilateral determined by vertices $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}\}$ is a *parallelogram* provided its vertices can be labeled so that $\mathbf{u} + \text{span}\{\mathbf{v} - \mathbf{u}\}$ is parallel to $\mathbf{x} + \text{span}\{\mathbf{w} - \mathbf{x}\}$ and $\mathbf{u} + \text{span}\{\mathbf{x} - \mathbf{u}\}$ is parallel to $\mathbf{v} + \text{span}\{\mathbf{w} - \mathbf{v}\}$. The lines $\mathbf{u} + \text{span}\{\mathbf{w} - \mathbf{u}\}$ and $\mathbf{v} + \text{span}\{\mathbf{x} - \mathbf{v}\}$ are the *diagonals* of the quadrilateral.

    (a) Prove that a quadrilateral determined by vertices $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}\}$ is a parallelogram if and only if its vertices can be labeled so that $\mathbf{x} = \mathbf{u} + \mathbf{v} - \mathbf{w}$.

    (b) Let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}\}$ determine a parallelogram in $\mathbb{FA}^2$, where the characteristic of $\mathbb{F}$ is not two. Show that the diagonals of the parallelogram intersect at their midpoints.

    (c) Prove that if the characteristic of $\mathbb{F}$ is not two, every parallelogram in $\mathbb{F}^2$ is the image under some affine transformation of the parallelogram determined by vertices $\{(0, 0), (2, 0), (2, 2), (0, 2)\}$.

15. In $\mathbb{R}^3$, let $\alpha = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ and let $\beta = (1, 1, 1) + \text{span}\{(1, -1, 1), (1, 0, 1)\}$.

    (a) Without doing any calculations, how can you tell that α and β intersect?

    (b) Give an analytic description of a plane $\alpha' = \mathbf{u} + \text{span}\{\mathbf{v}, \mathbf{w}\}$ that is parallel to α. Do the same for β. Call the plane you find here β'.

    (c) Find the line of intersection, $\ell = \alpha \cap \beta$.

    (d) Argue in this case that the only lines in α that are parallel to lines in β are those that are parallel to ℓ.

    (e) Find $\ell' = \alpha' \cap \beta'$. What do ℓ and ℓ' have in common?

Chapter 8

An Introduction to Projective Geometry

8.1 Introduction

Projective geometry has a long history going back to the late classical era with the work of Pappus. Early in the nineteenth century, Poncelet wrote the first treatise on the subject but it did not develop into one of the building blocks of modern mathematics until late in the nineteenth century. Nowadays, projective geometry is part of the everyday working tool set for mathematicians in geometry, topology, and algebra. It is also of interest in its own right.

Like affine geometry, projective geometry is a type of incidence geometry closely related to Euclidean geometry. It takes a mere three axioms to define a projective plane. Despite the seemingly spartan nature of such a configuration, a certain class of projective planes provides more than enough material to keep us occupied here.

The projective planes we study here are those that can be seen as sitting inside higher-dimensional projective spaces. Whether or not a given projective plane can be embedded naturally into projective 3-space turns on an algebraic property of the underlying coordinate system, which may be more general than a field. If the underlying coordinate system is a field, then we can model our projective plane using a vector space. Since finite-dimensional vector spaces always embed naturally into higher-dimensional vector spaces, in this setting, our projective plane embeds into projective 3-space.

This is not the first time we have seen the connection between the nature of the coordinatizing set for a line and features of the geometry itself. With a study of projective geometry, we come to a veritable precipice. Swaying above the turquoise depths of modern algebra roiling several stories below, we resist the impulse to jump, and content ourselves with a study of projective planes that arise from vector spaces over fields. (Jumpers may consult [8].)

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We start by defining projective space in terms of affine space, with the addition of so-called ideal points or points at infinity. We then show that we can shed the labels and treat all points in a projective space as one class of objects, all lines as another class of objects, all planes as another, and so on.

8.2 Projective Space

Because we need it for context, we start with a study of projective 3-space, rather than a projective plane. Before adding points to $\mathbb{FA}^3$, we reprise a definition we saw in Chapter 7.

Definition 7.67. The **direction of an affine line** ℓ in $\mathbb{FA}^n$, is any vector $\mathbf{v}$ in the ambient vector space V, where, for some $\mathbf{x} \in V$, $\ell = \{\mathbf{x} + t\mathbf{v} | \, t \in \mathbb{F}\}$. A **direction in an affine k-plane** in $\mathbb{FA}^n$, is a direction for an affine line in that k-plane. For $k \in \{1, \ldots, n-1\}$ fixed, **k-planes in $\mathbb{FA}^n$ are parallel** provided they have the same directions and do not intersect. Two k-planes in $\mathbb{FA}^n$ are **skew** if they have different directions and do not intersect.

A pair of distinct parallel lines in $\mathbb{FA}^3$ is contained in a unique plane. (This was part of Theorem 7.68.) There is no affine plane containing a pair of skew lines.

Partition the lines in $\mathbb{FA}^3$ by direction. For instance, lines in $\mathbb{R}^3$ with direction vector $(0, 0, 1)$ are the z-axis and all lines parallel to the z-axis. (See Fig. 8.1.) These comprise one equivalence class of lines in $\mathbb{R}^3$.

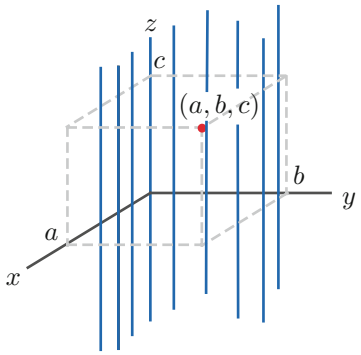


Figure 8.1: Lines parallel to the z-axis in $\mathbb{R}^3$

Add a single point, called an *ideal point* or *point at infinity*, to each line in a given equivalence class of parallel lines in $\mathbb{FA}^3$ and to each plane containing that line. The lines in each equivalence class now intersect in the unique ideal point associated to that class. Note that we add a single ideal point to $\mathbb{FA}^3$ for each different direction in $\mathbb{FA}^3$ and that this point is appended only to the lines and planes with that direction. We distinguish the points of the original affine space by calling them *ordinary points*.

Definition 8.1. The points of **projective 3-space over** $\mathbb{F}$, $\mathbb{FP}^3$, are the ordinary and ideal points associated to $\mathbb{FA}^3$. The **ideal plane** or **plane at infinity** is the collection of all ideal points in $\mathbb{FP}^3$. An **ordinary plane** in $\mathbb{FP}^3$ is the collection of all ordinary points associated to a plane in $\mathbb{FA}^3$ together with the ideal points associated to the directions in that affine plane. The collection of ideal points in an ordinary plane is the **ideal line** or **line at infinity** in that plane. A **projective plane** is an ordinary or ideal plane in $\mathbb{FP}^3$. An **ordinary line** is a line with ordinary points and a single ideal point. A **projective line** is an ordinary or ideal line in $\mathbb{FP}^3$. The projective points, lines, and planes in $\mathbb{FP}^3$ are its **projective objects**. Two projective objects in $\mathbb{FP}^3$ are **incident** provided the underlying sets have nonempty intersection.

An ordinary plane contains exactly one point at infinity for each direction in the underlying affine plane. It also contains exactly one line at infinity, the collection of all of its ideal points. The line at infinity for an ordinary plane α is the intersection of α with the plane at infinity.

Our first task is to establish the projective analogs to our main theorems on incidence in $\mathbb{FA}^3$, Theorem 7.68, and Theorem 7.93.

There is no concept of order in an arbitrary projective space so we will not have occasion to discuss segments in this chapter. As such, we use the notation AB for the line through the points A and B.

Theorem 8.2. (1) Two distinct points determine a unique line in $\mathbb{FP}^3$.

(2) If two lines intersect in $\mathbb{FP}^3$, their intersection is a single point.

(3) Two lines in a given plane in $\mathbb{FP}^3$ intersect.

(4) A point and a line not through the point lie in a unique plane in $\mathbb{FP}^3$.

(5) If two points are in a plane in $\mathbb{FP}^3$, the line they determine is in the plane.

(6) A line that does not lie in a plane in $\mathbb{FP}^3$ intersects the plane in one point.

(7) Two intersecting lines in $\mathbb{FP}^3$ lie in a unique plane.

(8) Three noncollinear points in $\mathbb{FP}^3$ lie in a unique plane.

(9) Two planes intersect in a unique line in $\mathbb{FP}^3$.

Proof. Let $V \cong \mathbb{F}^3$ be the ambient vector space for $\mathbb{FA}^3$.

(1) Two ordinary points in $\mathbb{FP}^3$ determine a unique affine line in $\mathbb{FA}^3$ by Theorem 7.68. The points of that affine line comprise the ordinary points on a unique ordinary line in $\mathbb{FP}^3$.

Suppose P is an ordinary point and Q_∞ an ideal point in $\mathbb{FP}^3$. P corresponds to a unique vector $\mathbf{v} \in V$, that is, a point in $\mathbb{FA}^3$, whereas Q_∞ corresponds to a direction in $\mathbb{FA}^3$ determined by a vector $\mathbf{w} \in V$. Then $\mathbf{v} + \text{span}\{\mathbf{w}\}$ is a line in $\mathbb{FA}^3$, comprising the ordinary points of an ordinary

line ℓ in $\mathbb{FP}^3$ that includes P and Q_∞. If there is any other line containing P and Q_∞ in $\mathbb{FP}^3$, it must be ordinary, as it contains P, and its ordinary points must coincide with $\mathbf{v} + \text{span}\{\mathbf{w}\}$ because this is the unique line in $\mathbb{FA}^3$ through $\mathbf{v}$ and $\mathbf{w} - \mathbf{v}$, which means it must be ℓ.

Suppose we are given P_∞ and Q_∞, two ideal points in $\mathbb{FP}^3$. There is no ordinary line containing them both as an ordinary line contains exactly one ideal point. P_∞ and Q_∞ correspond to directions in $\mathbb{FA}^3$, say given by vectors $\mathbf{v}$, $\mathbf{w}$ in V, where $\mathbf{w} \neq c\mathbf{v}$ for any $c \in \mathbb{F}$. Then $\{\mathbf{v}, \mathbf{w}\}$ is a basis for W, a two-dimensional subspace of V. The directions in W, and all planes in $\mathbb{FA}^3$ parallel to W, are identical, so every plane in the parallel class determined by W contains P_∞, Q_∞, and the unique ideal line determined by the set of directions common to all the planes in that parallel class. That unique line at infinity is the line determined by P_∞ and Q_∞.

(2) Suppose we have two intersecting lines in $\mathbb{FP}^3$. If both are ordinary and their ordinary points form parallel lines in $\mathbb{FA}^3$, then in $\mathbb{FP}^3$ the lines intersect in a unique ideal point that represents their common direction. If both are ordinary and they intersect in a unique ordinary point, then there can be no other ordinary point in the intersection, by Theorem 7.68. Since their ordinary points form lines with different directions, they cannot have an ideal point in common in $\mathbb{FP}^3$, so in this case, the intersection is a single ordinary point. If one line is ordinary and the other is ideal, then they can share at most one ideal point representing the direction of the ordinary line. If both lines are ideal, then one arises as the directions in one parallel class of affine planes, and the other as the directions in a second parallel class of affine planes. Nonparallel planes in $\mathbb{FA}^3$ intersect in a line and the direction of that line is represented as an ideal point in $\mathbb{FP}^3$, which must be the single point of intersection of our two ideal lines.

(3) Consider lines ℓ, m in a plane α in $\mathbb{FP}^3$. Suppose α is ideal so that ℓ and m are both ideal. In this case, ℓ and m represent parallel classes of affine planes in $\mathbb{FA}^3$. Let β', γ' be affine planes, one from each of these two parallel classes. By Theorem 7.93, $\beta' \cap \gamma' = \ell'$, a single line the direction of which is represented by a unique point in $\ell \cap m$.

If α is ordinary, then at least one of the lines must be ordinary. Say it is ℓ. Let ℓ' be the affine line comprised of the ordinary points of ℓ. If m is also ordinary, let m' be the affine line comprised of the ordinary points of m. Either $\ell' \cap m'$ is a single affine point or ℓ' and m' are parallel. If $\ell' \cap m'$ is a single affine point, then the two affine lines have different directions, thus, ℓ and m have no ideal points in common so $\ell \cap m$ is a single ordinary point. If ℓ' and m' are parallel, there is a single ideal point P in α representing the direction of the two lines. In this case, $\ell \cap m = P$. If m is the line at infinity in α, then it must contain a point P associated to the direction of ℓ'. In this case $\ell \cap m = P$.

(4) Suppose we are given a point and a line not through the point. If the point and line are both ideal, they lie in the unique ideal plane by construction.

If either the line, the point, or both are ordinary, let $\mathcal{S}$ be the union of the point and the points on the line. Since affine lines contain at least two points, and since there are at least three directions in $\mathbb{FA}^3$, $\mathcal{S}$ contains at least three points. Since at least one of our objects is ordinary, $\mathcal{S}$ must contain an ordinary point, P. From the remaining points in $\mathcal{S}$, pick two more, Q, R, so that P, Q, R are noncollinear. By part (1) above, P and Q determine a unique line ℓ in $\mathbb{FP}^3$. P and R determine a second unique line m in $\mathbb{FP}^3$. By assumption, ℓ and m are distinct lines, intersecting at P. Since they contain an ordinary point, P, ℓ and m are ordinary lines so their ordinary points determine a unique affine plane, α'. That affine plane underlies a projective plane with ideal points determined by the directions in α'. Since α' is unique in $\mathbb{FA}^3$, the associated projective plane is unique in $\mathbb{FP}^3$, which proves the result.

(5) Let P and Q belong to a plane α in $\mathbb{FP}^3$. Suppose both P and Q are ideal. If α is also ideal, it contains all ideal points, so contains the unique ideal line determined by P and Q. If α is ordinary, its ordinary points comprise an affine plane, α'. Let $\mathbf{v}$ and $\mathbf{w}$ be vectors in V with the directions represented by P and Q. Then $\alpha' = \mathbf{x} + \text{span}\{\mathbf{v}, \mathbf{w}\}$, for some $\mathbf{x}$ in V. The directions in $\text{span}\{\mathbf{v}, \mathbf{w}\}$, which determine the ideal line determined by P and Q, are the directions in α'. It follows that α contains the ideal line determined by P and Q.

If one point is ordinary and the other ideal, then α is ordinary and the ideal point represents a direction in α', the plane in $\mathbb{FA}^3$ comprised of the ordinary points of α. Say the ordinary point is associated to some $\mathbf{v}$ in V and the ideal point is associated to $\mathbf{w}$ in V. The affine line $\ell = \mathbf{v} + \text{span}\{\mathbf{w}\}$ is in α' so the associated projective line, which includes P and Q, must then be in α.

If both points are ordinary, they determine a unique affine line, ℓ', in $\mathbb{FA}^3$. The ordinary points of α comprise an affine plane that contains ℓ'. Then ℓ' is the set of ordinary points of a unique projective line determined by P and Q, and lying in α.

(6) Let α be a plane in $\mathbb{FP}^3$ and ℓ a line in $\mathbb{FP}^3$ not contained in α. The ideal plane contains all ideal lines so α and ℓ cannot both be ideal. Suppose both are ordinary. Let ℓ' be the affine line comprised of the ordinary points of ℓ. Let α' be the affine plane comprised of the ordinary points of α. We know that $\alpha' \cap \ell'$ is either empty or a single point, by Theorem 7.68. If empty, then ℓ' is parallel to lines in α' so $\alpha \cap \ell$ is the single ideal point associated to that class of parallel lines in $\mathbb{FA}^3$. If $\alpha' \cap \ell'$ contains a single point, then the direction of ℓ' is not in α'. It follows that $\alpha \cap \ell$ cannot contain an ideal point. Thus, in this case, there can be no more than the single ordinary point in $\alpha \cap \ell$.

If ℓ is ordinary and α ideal, then $\alpha \cap \ell$ is the unique ideal point associated to the parallel class of the affine line underlying ℓ. If ℓ is ideal and α ordinary, then ℓ is associated to a parallel class of planes in $\mathbb{FA}^3$ and α', the affine

plane comprised of the ordinary points of α, is not in that parallel class: If it were, ℓ would be contained in α. Let β' be an affine plane in the parallel class of affine planes determined by ℓ. Since α' is not parallel to β', $\alpha' \cap \beta' = \ell''$, where ℓ'' is an affine line. Let P be the ideal point associated to the direction of ℓ''. Notice that $P \in \alpha \cap \ell$. Moreover, if β'' is another affine plane in the parallel class determined by ℓ, then $\alpha' \cap \beta''$ is a line parallel to ℓ''. It follows that $P = \alpha \cap \ell$.

(7) Let ℓ and m be intersecting lines in $\mathbb{FP}^3$, with Q the point of intersection. Let P be a point on m that is not on ℓ. By (4), P and ℓ lie in a unique plane, $\alpha \subset \mathbb{FP}^3$. Since Q is on $\ell \subset \alpha$, (5) implies that $PQ = m$ is in α as well. Now any point on m, apart from Q, together with ℓ determine α and by (4), there is no other plane containing the point and ℓ. Since this is true for all points of m, except the one that also belongs to ℓ, it follows that α is the only plane containing both ℓ and m.

(8) This also follows by (1) and (7).

(9) Suppose α and β are ordinary planes in $\mathbb{FP}^3$. Either their ordinary points comprise planes in $\mathbb{FA}^3$ that are parallel or their ordinary points comprise planes in $\mathbb{FA}^3$ that intersect in a unique affine line, ℓ'. In the former case, every direction in α is a direction in β and vice versa so the two planes intersect in their common line at infinity. In the latter case, Theorem 7.93 guarantees that ℓ' comprises all the ordinary points in the intersection $\alpha \cap \beta$. By the same theorem, the direction of ℓ' is the only direction the ordinary points of the two planes have in common so the only ideal point the two can have in common is the one associated to ℓ'. In this case, then, $\alpha \cap \beta = \ell$, the projective line associated to ℓ'.

Now suppose α is ordinary and β is ideal. By definition, the collection of ideal points in α, the intersection of α and β, is the line at infinity for α.

□

Corollary 8.3. Three distinct planes in $\mathbb{FP}^3$ that do not have a common line intersect in a unique point.

Proof. The proof is left as an exercise. □

Next is a basic theorem about counting points in $\mathbb{FP}^3$. This is the projective version of Theorem 7.70.

Theorem 8.4. (1) A line in $\mathbb{FP}^3$ has at least three points.

(2) A plane in $\mathbb{FP}^3$ has four points, no three of which are collinear.

(3) There are four noncoplanar points.

Proof. Let α_∞ be the plane at infinity in $\mathbb{FP}^3$.

(1) An ordinary line contains two ordinary points by Theorem 7.70. In $\mathbb{FP}^3$, it has an additional ideal point so the total number of points on any ordinary line is at least three.

If ℓ_∞ is an ideal line in $\mathbb{FP}^3$, then it lies in some ordinary plane, α. Let $\alpha' \subset \mathbb{FA}^3$ be the underlying affine plane and let $V \cong \mathbb{F}^3$ be the ambient vector space for $\mathbb{FA}^3$. We can use an element from $\mathrm{Aff}(V)$, if necessary, to map α' to an isomorphic copy of $\mathrm{span}\{(1,0,0),(0,1,0)\}$, which we will call the xy-plane. A direction in the xy-plane can be identified with the nonzero scalar multiples of a nonzero vector in the plane. We identify three different directions in the xy-plane, then, corresponding to the vectors $(1,0,0),(0,1,0),(1,1,0)$. Since the correspondence between α' and the xy-plane is via an isomorphism of the affine geometry, there must be lines with three different directions in α'. Those three different directions in α' are represented by three different points on ℓ_∞.

(2) In Theorem 7.70.2. we established that in any affine plane, there are four points, no three of which are collinear. It follows that in any ordinary plane in $\mathbb{FP}^3$, there are four ordinary points, no three of which are collinear.

Next we consider α_∞. Let ℓ be a line in α_∞ and say that A, B, C are three points on ℓ. Note that ℓ represents directions in some parallel class of affine planes. Let α' be an affine plane in that parallel class. By Theorem 7.70, there are four noncoplanar points in $\mathbb{FA}^3$ so we can take an affine point not in α'. Take also any point in α'. The affine line determined by these two affine points has a direction that is not in α'. That direction is realized as a point, $Q \in \alpha_\infty$, where $Q \notin \ell$.

Let $m = AQ$. Notice that $m \subset \alpha_\infty$. Take D to be any third point on m. Now we have $\{A, Q, D\} \subset m$, $\{A, B, C\} \subset \ell$, $\ell \cap m = A$, and $\ell \cup m \subset \alpha_\infty$. Consider the set $\mathcal{S} = \{B, C, Q, D\}$. Any choice of three of these points gives us two points on ℓ and a point on m not on ℓ, or two points on m, and a point on ℓ not on m. This shows that $\mathcal{S}$ is a set of four points in α_∞, no three of which are collinear.

(3) Theorem 7.70 guarantees that $\mathbb{FP}^3$ has four ordinary points which do not lie in an ordinary plane.

□

Theorems 8.2 and 8.4 establish that there is no need to distinguish between ordinary and ideal objects in $\mathbb{FP}^3$. In terms of incidences, all points are alike, all lines are alike, and all planes are alike in $\mathbb{FP}^3$.

Our construction and the attendant theorems also show that any field can be the foundation for a projective space.

Appending ideal points to affine space is one way to construct projective space. Another approach is axiomatic. Here is a set of axioms we could use to define $\mathbb{FP}^3$.

Axioms for Projective Space

Axiom PS1. Two points determine a unique line.

Axiom PS2. Three noncollinear points lie in a unique plane.

Axiom PS3. If two points lie in a plane, the line they determine lies in that plane.

Axiom PS4. Two coplanar lines intersect in a unique point.

Axiom PS5. Two planes intersect in at least two points.

Axiom PS6. A plane and a line that does not lie in the plane intersect in exactly one point.

Axiom PS7. A line contains at least three points.

Axiom PS8. A plane contains four points, no three of which are collinear.

Axiom PS9. There exist four noncoplanar points.

The primary focus of our interest is $\mathbb{FP}^2$. We could have defined a projective plane using the following set of axioms. These apply to all projective planes, not just those constructed from affine space. Among projective planes defined with this set of axioms are those that are still the subject of active research.

Axioms for Projective Plane

Axiom PP1. Two points determine a unique line.

Axiom PP2. Two lines intersect in a unique point.

Axiom PP3. There exist four points, no three of which are collinear.

An advantage to using axioms is that they highlight the *principle of duality*, an intriguing feature of projective geometry.

Definition 8.5. The **dual** of a logical statement about a projective plane is the statement we get by switching the words "point" and "line" and the phrases "lie on" and "intersect in."

Note that in Axiom PP1, "determine" means "lie on." Axiom PP2 could be rephrased, *two lines determine a unique point.* Using the word "determine" in the two different senses, we highlight the duality of the two statements.

The following are dual to the axioms for a projective plane.

Dual Axioms for a Projective Plane

Axiom PP1*. Two distinct lines intersect in a unique point.

Axiom PP2*. Two distinct points determine a unique line.

Axiom PP3*. There are four lines, no three of which are concurrent.

The dual of *collinear*, a word that refers to the relative positions of points in a set, is *concurrent*, which refers to the relative positions of lines in a set. As we progress through our study of projective geometry, we will see that various locutions regarding points and lines have natural duals that we will not necessarily discuss. The dual of a statement captures a certain idea. The exact choice of words to express that idea is variable.

Principle of Duality. Any statement about a projective plane is true if and only if its dual statement is true.

Once we show that Axioms PP1–3 are equivalent to Axioms PP1*–3* in $\mathbb{FP}^2$, we will have established the principle of duality for projective planes constructed from affine planes.

Theorem 8.6. Axioms PP1–3 are equivalent to Axioms PP1*–3* in $\mathbb{FP}^2$.

Proof. It is clear that the pair Axioms PP1, 2 is equivalent to the pair Axioms PP1*, 2*. Suppose then that Axiom PP3 is true. Let $\{A, B, C, D\}$ be four points in $\mathbb{FP}^2$, no three of which are collinear. Consider the set of lines $\mathcal{S} = \{AB, BC, CD, AD\}$. (See Fig. 8.2.) By Axiom PP3, the four lines are distinct:

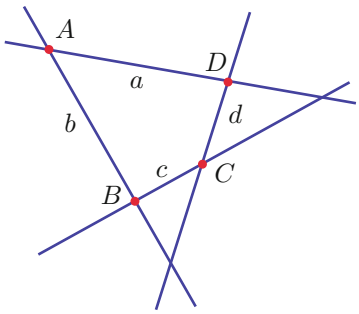


Figure 8.2: Four points, no three of which are collinear and four lines, no three of which are concurrent

for instance, because A, B, C are not collinear, AB and BC are distinct lines. To see that no three of the lines are concurrent, consider BC, CD, AD. We have $BC \cap CD = C$, but by assumption, C cannot belong to AD, so BC, CD, AD cannot be concurrent. We leave it as an exercise to show that lines in the other three three-line subsets of $\mathcal{S}$ are not concurrent. This shows that Axiom PP3 implies Axiom PP3*.

Now assume Axiom PP3*. Let $\mathcal{T} = \{a, b, c, d\}$ be four lines in $\mathbb{FP}^2$, no three of which are concurrent. Since any two lines have a common point, we have four points $A = a \cap b$, $B = b \cap c$, $C = c \cap d$, $D = d \cap a$. Since the point of intersection of two lines is unique, and since no three of these lines are concurrent, the four points must be distinct. We claim that no three of the points in the set $\mathcal{U} = \{A, B, C, D\}$ are collinear. Consider $\{B, C, D\}$. If these are collinear, then $D \in c$, but that would make a, c, d concurrent as $D = a \cap d$. The other three cases follow similarly. We leave them as an exercise. □

Duality manifests in higher-dimensional projective spaces as well. Given a statement about $\mathbb{FP}^3$, we obtain its dual by switching the words "point" and "plane" and the phrases "lie on" and "intersect in." Lines are self-dual in projective 3-space. We leave it as an exercise to dualize the axioms we stated above for projective 3-space.

Relatively simple assemblages of points and lines get a good amount of attention in projective geometry and turn out to be surprisingly subtle. The following definition applies in any kind of plane or space.

Definition 8.7. The **pencil of lines through a point** P is the collection of all lines through P.

A line ℓ in $\mathbb{FP}^2$ that does not go through P intersects each line in the pencil through P exactly once. We say ℓ is a *section* of the pencil. The dual of a pencil of lines through a point in a projective plane is the collection of points on a line. It is called the *range* of points on the line.

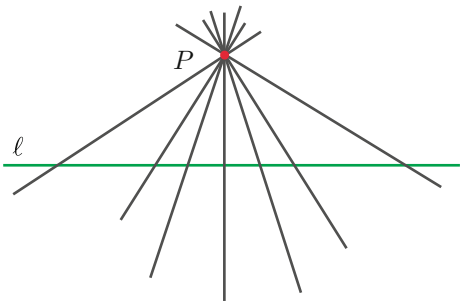


Figure 8.3: The pencil of lines through P is sectioned by ℓ

We noted in Chapter 7 that triangles form an important organizational unit in an affine plane. Triangles play a critical role in projective planes as well.

The definition of triangle does not vary much in different geometries but we repeat it here for the projective setting so that we can address the nature of its dual.

Definition 8.8. A **triangle** is a set of three noncollinear points (vertices) and the lines they determine (sides). A **trilateral** is a set of three nonconcurrent lines and the points they determine.

Triangles are **self-dual**, that is, the dual of a triangle—a trilateral—is the same collection of points and lines as the triangle itself.

Four-sided figures are not self-dual in projective geometry. We start with the four-sided analog of a triangle.

Definition 8.9. A **complete quadrangle** is a configuration in the projective

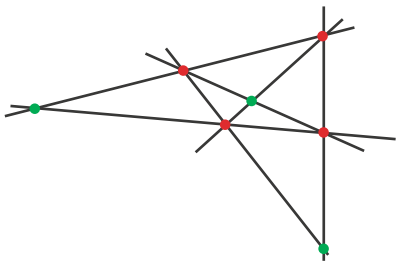


Figure 8.4: A complete quadrangle with diagonal points shown in green

plane composed of a set of four points (vertices), no three of which are collinear, together with the six lines (sides) they determine. **Opposite sides** of a complete quadrangle are two sides without a common vertex. The **diagonal points** of a complete quadrangle are the points where opposite sides intersect.

Note that a complete quadrangle is an assemblage of four points and six lines. If we say "quadrangle" in projective geometry we always mean "complete quadrangle," likewise for the dual of a complete quadrangle, a *complete quadrilateral*.

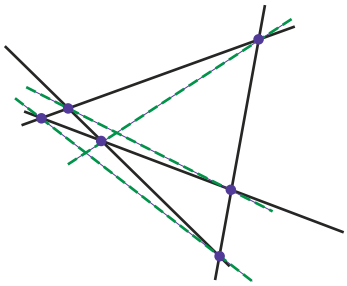


Figure 8.5: A complete quadrilateral with diagonals shown as broken green lines

Definition 8.10. A **complete quadrilateral** is a configuration in the projective plane composed of a set of four lines (sides), no three of which are

concurrent, together with the six points (vertices) they determine. **Opposite vertices** of a complete quadrilateral are two vertices without a common side. The **diagonal lines** of a complete quadrilateral are lines that pass through opposite vertices.

Exercises 8.1. 1. In the proof of part (4) of Theorem 8.2, it is suggested that after choosing an ordinary point in $\mathcal{S}$, P, we could choose two more points, Q, and R, so that P, Q, R are noncollinear. Why is this true?

2. Prove Corollary 8.3.

3. In the proof of Theorem 8.4, we refer to three different directions in $\mathbb{FA}^3$ given by vectors $(1,0,0), (0,1,0), (1,1,0)$. Write down parametric and vector versions of the line through $\mathbf{0}$ and each of the three given points in $\mathbb{R}^3$. Do the same, but also write a list of the points on each line in $\mathbb{F}_2^3$, and again in $\mathbb{F}_3^3$.

4. Consider the collection of four lines referred to in the proof of Theorem 8.6, $\mathcal{S} = \{AB, BC, CD, AD\}$. The proof addresses the fact that the lines in the subset $\{BC, CD, AD\}$ are not concurrent.

   (a) Write down every other subset of $\mathcal{S}$ that contains exactly three lines.

   (b) For each subset you found in part (a), argue that the lines are not concurrent.

5. Consider the collection of four points referred to in the proof of Theorem 8.6, $\mathcal{U} = \{A, B, C, D\}$. The proof addresses the fact that the points in the subset $\{B, C, D\}$ are not collinear.

   (a) Write down every other subset of $\mathcal{U}$ that contains exactly three points.

   (b) For each subset you found in part (a), argue that the points are not collinear.

6. Dualize each of Axioms PS1-9. Remember that in projective space, points and planes are dual to one another, and lines are self-dual.

7. Write the dual of each of the following objects and statements in a projective plane.

   (a) Four points, no three of which are collinear

   (b) Two lines, and a point on neither line

   (c) The line determined by a given point and the point of intersection of two given lines

   (d) All lines in the projective plane

   (e) Distinct concurrent lines have only one common point.

(f) Given a line and a point not on the line, distinct lines through the given point meet the given line in distinct points.

8. Write the dual for each of the following, as statements or objects in projective 3-space.

   (a) the set of points on a line
   (b) the set of points on a plane
   (c) the set of planes containing a given line
   (d) the set of lines passing through a common point
   (e) the set of lines lying in a given plane
   (f) the set of planes in space
   (g) the set of lines in space
   (h) the set of all points in space

9. Is a complete quadrangle self-dual? Explain.

10. How many sides go through a vertex of a complete quadrangle? How many sides go through a vertex of a complete quadrilateral?

11. Consider Fig. 8.6. This is a depiction of a planar geometry in which the

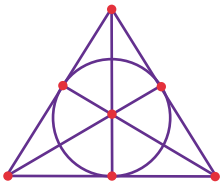


Figure 8.6: The Fano plane

points are represented as dots. The lines are the collections of points connected by the sides of the triangle, the cevians as shown, and what appears to be the incircle, as shown. The incidence geometry associated to the picture is called the *Fano plane*. Note that the triangle, segments, and circle are just schematic devices for capturing the incidences in the Fano plane.

   (a) Verify that the points and lines in the figure satisfy Axioms PP1–3. How many points and how many lines do we have in this plane?
   (b) Identify three different triangles in the Fano plane.
   (c) Identify a complete quadrangle in the Fano plane.
   (d) Identify a complete quadrilateral in the Fano plane.
   (e) Find the diagonal points on the complete quadrangle that you found above. Repeat the exercise with a different quadrangle. What do you notice about the diagonal points?

8.3 The Theorem of Desargues

Pappus's Theorem is often cited as the first theorem in projective geometry. Given that Pappus lived during the fourth century, and that projective geometry was not properly realized until the middle of the nineteenth century, it is clear that Pappus would not have been thinking about projective geometry per se. Pappus's Theorem has a distinctly projective flavor, though. The cleanest way to present it is in the projective plane, where intersection points of lines are guaranteed.

Desargues's Theorem is likewise often cited as the second theorem in projective geometry. Girard Desargues lived during the seventeenth century, so he is a bit closer to the development of modern projective geometry, which arose partly through the observations of Renaissance artists. Desargues was an architect, and belonged to a circle of thinkers that included the Pascals, Etienne (father) and Blaise (son), Fermat, Descartes, and Mersenne. Because his work was unorthodox—he invented his own terminology, for example—it was largely ignored except by the younger Pascal. Pascal's name is given to a result that is closely related to Pappus's Theorem. Desargues's Theorem, viewed in a certain light, is a generalization of Pappus's Theorem.

Pappus's proof of Pappus's Theorem involved heavy use of Euclidean geometry. Of course, in Pappus's time, there was no other geometry. Our approach is to use projective methods to prove Desargues's Theorem first. Deploying Desargues's Theorem after we develop projective transformations, we will find an easy proof of Pappus's Theorem.

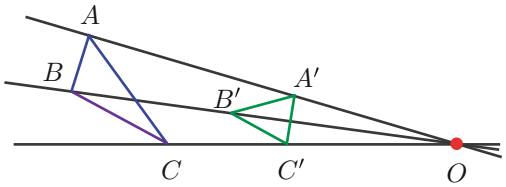


Figure 8.7: Triangles perspective from a point

Definition 8.11. Two triangles ΔABC, $\Delta A'B'C'$ are **perspective from a point** O provided AA', BB', CC' are concurrent at O. In this case, O is a **center of perspectivity**.

When triangles—or any other objects—are perspective from a point, they appear to overlap perfectly if one's eye is at the point of perspectivity. This is an essential idea in projective geometry and in art, especially Renaissance art, in which the effective illusion of space on a flat surface was an important goal.

Definition 8.12. Triangles ΔABC, $\Delta A'B'C'$ are **perspective from a line** ℓ

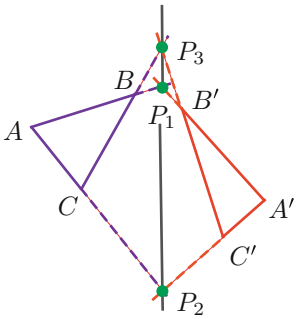


Figure 8.8: Triangles perspective from a line

provided the points $AB \cap A'B' = P_1$, $AC \cap A'C' = P_2$, and $BC \cap B'C' = P_3$ are collinear. In this case, ℓ is an **axis of perspectivity**.

Theorem 8.13 (Desargues). Two triangles are perspective from a point in $\mathbb{FP}^3$ if and only if they are perspective from a line.

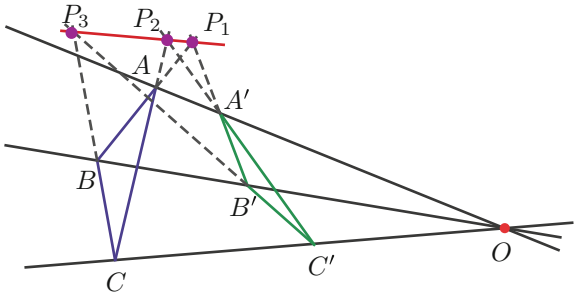


Figure 8.9: The theorem of Desargues

Proof. For the first part of the proof, we assume that the triangles lie in two different planes. Say ΔABC lies in the plane α and $\Delta A'B'C'$ lies in the plane α'.

If ΔABC and $\Delta A'B'C'$ are perspective from a point O, then since the lines AA' and BB' intersect at O, Theorem 8.2 guarantees that they determine a plane. That plane contains A, A', B, B' and O. Now the lines AB and $A'B'$ are in this plane so they intersect at a point P_1. Note that $P_1 \in \alpha \cap \alpha'$. The identical argument produces points $P_2 = AC \cap A'C'$ and $P_3 = BC \cap B'C'$, also in $\alpha \cap \alpha'$. This proves that $\alpha \cap \alpha'$ is the axis of perspectivity for the triangles.

Next suppose ΔABC and $\Delta A'B'C'$ are perspective from a line. As intersecting lines, AB and $A'B'$ lie in a plane β, again by Theorem 8.2. We then have A, B, A', B' all in β so AA' and BB' intersect in β. Similarly, AA' and CC' intersect in a plane β', and BB' and CC' intersect in a third plane β''. We must argue that the planes β, β', β'' intersect in a point, O, that is, that they do not share a line.

If the planes did share a line, that line would be $\beta \cap \beta'$, which contains AB, and at the same time, $\beta \cap \beta''$, which contains BC. As vertices of a triangle, A, B, C are noncollinear, so the line of intersection of β and β' must be different from the line of intersection of β and β''. It follows again by Theorem 8.2 that the intersection of the three planes is a single point, O. Now $AA' = \beta \cap \beta'$, $BB' = \beta \cap \beta''$, and $CC' = \beta' \cap \beta''$. The three lines AA', BB', CC' are thus concurrent at O.

Next, assume that ΔABC and $\Delta A'B'C'$ lie in a single plane, γ. The strategy for the proof in this case is to build another triangle in a different plane and then invoke the first part of the theorem.

Suppose the two triangles are perspective from a point $O \in \gamma$. Let ℓ be a line through O, not lying in γ. Choose P, P' distinct points on ℓ, both different from O. Let PAB be the plane determined by the point P and the line AB. Likewise define PAC, and PBC, each the plane containing P and one side of the triangle ΔABC.

Notice that PAB, PAC, and PBC must be distinct planes: If any two were equal, then P, A, B, and C would be coplanar, contradicting our choice of P. Moreover, $PAB \cap PAC = PA$, $PAB \cap PBC = PB$ and $PAC \cap PBC = PC$. Since the lines PA and $P'A'$ lie in the plane determined by intersecting lines AA', and PP', they must intersect in a point A''. We define points B'', C'' in a similar manner.

Next we argue that A'', B'', C'' cannot be collinear: If they were, then since $A''B''$ is in PAB, and $A''C''$ is in PAC, and $B''C''$ is in PBC, the three planes would have to intersect in a line containing P and lying in γ. This contradicts P lying outside γ. Thus, we have a new triangle, $\Delta A''B''C''$.

By construction, ΔABC and $\Delta A''B''C''$ are perspective from P, while $\Delta A'B'C'$ and $\Delta A''B''C''$ are perspective from P'. The line of perspectivity in both cases is the intersection of γ with the plane of $\Delta A''B''C$". This gives us that $A''B'' \cap AB$ and $A''B'' \cap A'B'$ fall on this line of intersection. Since $A''B''$ meets this line just once, AB and $A'B'$ intersect there as desired. The other sides of the two triangles meet on this line as well, for the same reasons. We conclude that ΔABC and $\Delta A'B'C'$ are perspective from the line $\gamma \cap \xi$, where ξ is the plane of $\Delta A''B''C''$.

We leave the last part of the proof to the reader. □

Notice that the proof of Desargues's Theorem is much easier when the triangles occupy different planes. Though it holds in Euclidean space, as long as all the intersection points exist, Desargues's Theorem is fundamentally about projective space. This business of embedding the plane in 3-space is not just a trick in the proof. The theorem is actually false in projective planes that cannot

be viewed as occupying a part of projective 3-space. Projective planes for which Desargues's Theorem holds are *Desarguesian* and those for which it does not hold are *non-Desarguesian*. Non-Desarguesian planes are not defined over fields. They are not based on vector spaces or, equivalently, affine spaces.

Figure 8.10 shows the arrangement in Fig. 8.9 without specifying a particular

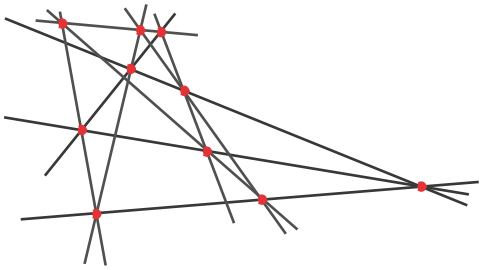


Figure 8.10: Any point can be the point of perspectivity in a Desarguesian configuration

role for any point or line. It turns out that any point in the configuration can play the role of O, the point of perspectivity. Likewise, any line can play the role of the axis of perspectivity. In the exercises, you will have a chance to explore these ideas a little bit.

Exercises 8.2. 1. In the proof of the second half of Desargues's Theorem, we have triangles in a single plane γ, perspective from a point O in γ. Then we let ℓ be a line not in γ, but through the point O. Why must such a line ℓ exist?

2. Make a sketch to illustrate the part of the proof of Desargues's Theorem that applies to triangles that lie in one plane.

3. Finish the proof of Desargues's Theorem.

4. Below we will learn that the Fano plane is Desarguesian. Identify two triangles perspective from a point in the Fano plane. Find the axis of perspectivity.

5. An assemblage of ten points and ten lines so that each line is incident to exactly three of the points and each point is incident to exactly three of the lines is a *Desarguesian configuration*. Verify that Fig. 8.9 depicts a Desarguesian configuration. Relabel the points in that figure so that the point of perspectivity is a point other than O. (See Fig. 8.10.) Similarly, any line can play the role of the axis of perspectivity. Relabel the points several different ways to illustrate this principle.

6. Referring to the previous exercise, does a given point of perspectivity uniquely determine the two triangles in perspective, if the choice is made in a Desarguesian configuration?

8.4 Harmonic Sequences

Desargues's Theorem is about triangles. In this section, we consider quadrangles, quadrilaterals, and associated arrangements of points and lines.

Let A, B, C, D be the vertices of a complete quadrangle, as prescribed by Definition 8.9. We designate this quadrangle simply as $ABCD$. The diagonal

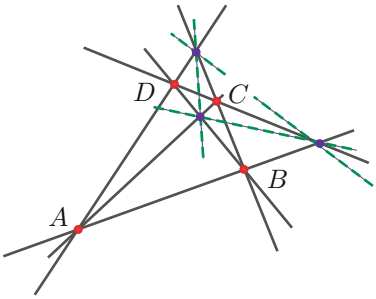


Figure 8.11: The quadrangle $ABCD$ with its diagonal points in blue and the sides of its diagonal triangle in green

points of $ABCD$ are $AB \cap CD$, $AC \cap BD$, and $AD \cap BC$. If our projective plane is defined over a field of characteristic different from two, these diagonal points are noncollinear. (This is not obvious and we omit the proof. See [9], for example, where it is proved using an underlying vector space.) The triangle they determine is called the *diagonal triangle* of the quadrangle. A complete quadrilateral also has an associated diagonal triangle, which you can define using duality. (See the exercises.)

Definition 8.14. A **harmonic sequence of points**, $H(P, Q; R, S)$, in a projective plane, is a set of four collinear points P, Q, R, S where P, Q are diagonal points of a quadrangle and R, S are on the sides of the quadrangle that pass through the third diagonal point. In this case, we say that S is the **harmonic conjugate** of R with respect to P and Q.

Dual to a harmonic sequence of points is a harmonic sequence of lines. When we dualize, we must remember that the dual of a quadrangle is a quadrilateral.

Definition 8.15. A **harmonic sequence of lines**, $H(p, q; r, s)$, in a projective plane, is a set of four concurrent lines p, q, r, s, where p, q are diagonal lines of a complete quadrilateral and r, s pass through the vertices of the third diagonal

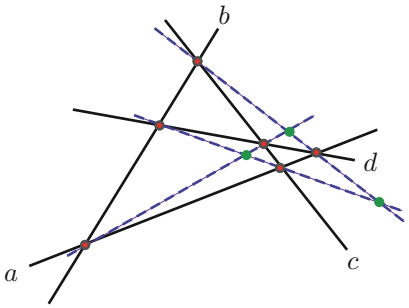


Figure 8.12: The quadrilateral *abcd* with its diagonal sides in blue, and vertices of the diagonal triangle in green

side of the quadrilateral. In this case, we say that s is the **harmonic conjugate** of r with respect to p and q.

The ordering of the points or lines in a harmonic sequence matters. While $H(P,Q;R,S)$ means $H(Q,P;S,R)$, it does not mean $H(P,R;Q,S)$, for instance.

The line determined by two of the diagonal points of a quadrangle must intersect the two sides of the third diagonal point in our projective planes. Consequently there is a harmonic sequence associated to any quadrangle. Indeed, a harmonic sequence of points is a *section* of the quadrangle: Every side of the figure intersects the line in one point.

Constructing a Harmonic Conjugate

We start our study by constructing a harmonic conjugate of a point. Our approach here closely follows [9].

Suppose P, Q, R are distinct points on a line ℓ in a projective plane. Choose C not on ℓ and $B \in QC$, B different from Q and C, as in Fig. 8.13. Form RB

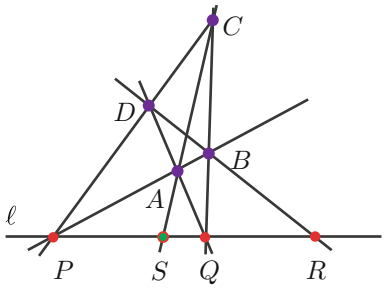


Figure 8.13: Construct the harmonic conjugate of R with respect to P and Q

and PC. Let D be their point of intersection. Next form PB and QD. Let A be their point of intersection. Form AC and let $S = AC \cap \ell$. We leave it as an exercise to verify that $ABCD$ forms a quadrangle and that S is a harmonic conjugate to R.

Our first theorem on the topic establishes that the harmonic conjugate of a point is unique.

Theorem 8.16. Given distinct points P, Q, R on a line ℓ in a projective plane, there is a unique point S on ℓ so that $H(P, Q; R, S)$.

Proof. Let P, Q, R be on ℓ in a projective plane. We have established that there is S so that $H(P, Q; R, S)$. Referring to the construction above, suppose $A'B'C'D'$ is a second quadrangle that produces a point S' so that $H(P, Q; R, S')$. We have $AC \cap \ell = S$ and $A'C' \cap \ell = S'$. Notice that $S' = S$ if and only if $AC \cap A'C'$ is on ℓ. (See Fig. 8.14.) We prove that $AC \cap A'C' \in \ell$ using Desargues's Theorem three times. First notice that ΔBCD and $\Delta B'C'D'$ are

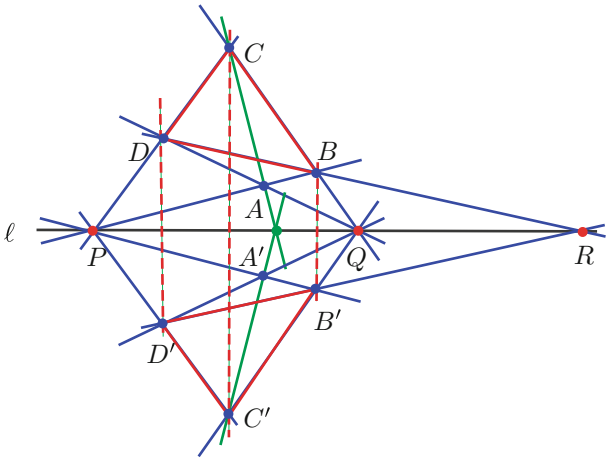


Figure 8.14: Since ΔBCD and $\Delta B'C'D'$ are perspective from ℓ, the lines BB', CC', DD' must be concurrent

perspective from ℓ, thus, BB', CC', and DD' are concurrent. (In Fig. 8.14 they appear to be parallel, so we can think of them as projectively concurrent.) Similarly, ΔABD and $\Delta A'B'D'$ are perspective from ℓ, so AA', BB', DD' are concurrent. From there we have AA', BB', CC' concurrent so that ΔABC and $\Delta A'B'C'$ are perspective from a point, thus, $AB \cap A'B' = P$, $BC \cap B'C' = Q$, and $AC \cap A'C' = S = S'$ are collinear, as desired. □

The principle of duality in a projective plane allows us to make the following statement without proof.

Theorem 8.17. If p, q, r are concurrent lines in a projective plane, then among the lines in the pencil at the point of concurrency, there is a unique line s so that $H(p, q; r, s)$.

Theorem 8.18. If a line ℓ meets concurrent lines p, q, r, s in the points P, Q, R, S in a projective plane, then $H(P, Q; R, S)$ if and only if $H(p, q; r, s)$.

Proof. Suppose p, q, r, s are lines concurrent at C. Let ℓ meet lines p, q, r, s in points P, Q, R, S, respectively so that $H(P, Q; R, S)$. Using C, we can form S from P, Q, R. In the process, we construct a quadrangle $ABCD$ so that pairs of opposite sides—AB, CD; and AD, BC—intersect in P and Q, and diagonals—AC and BD—intersect PQ at R and S, respectively. We have $P = AB \cap CD$, $Q = AD \cap BC$, $R = BD \cap PQ$, and $S = AC \cap PQ$.

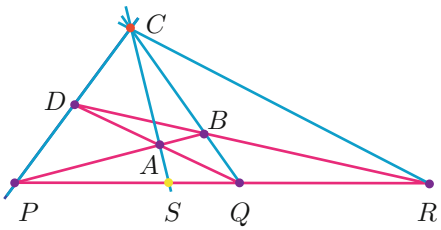


Figure 8.15: C is the point of concurrency of diagonal lines associated to the quadrilateral with pink sides

If we are to view p, q, r, s themselves so that $H(p, q; r, s)$, we need to see p, q as diagonals in a complete quadrilateral, and r, s as passing through the distinct vertices on the third diagonal side of the quadrilateral.

Consider the quadrilateral $\mathcal{Q}$ with sides $PB = PA$, $QD = QA$, $PQ = PR$, $BD = BR$, and vertices B, D, P, Q, A, R. Here $\{P, D\}$, $\{B, Q\}$, and $\{A, R\}$ are opposite vertex pairs. It follows that $PD = PC = p$ and $BQ = QC = q$ are diagonals for $\mathcal{Q}$ and that $r = RC$ and $s = AC$ satisfy $H(p, q; r, s)$.

The converse follows by duality. □

Exercises 8.3.

1. How many sides go through each vertex of a complete quadrangle in a projective plane? How many sides go through each diagonal point?

2. Use duality to define the diagonal triangle of a complete quadrilateral in a projective plane.

3. In Fig. 8.16 you can see Fig. 8.11 purely as a configuration. Likewise, Fig. 8.17 shows Fig. 8.12 purely as a configuration. Describe these configurations by specifying the number of points, the number of lines, and the incidences enjoyed by each point and each line. Are the two the same type of configuration? Are they dual? (See Exercise 8.2.5.)

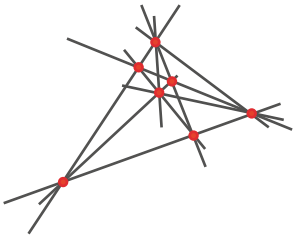


Figure 8.16: The configuration associated to a complete quadrangle and its diagonal triangle

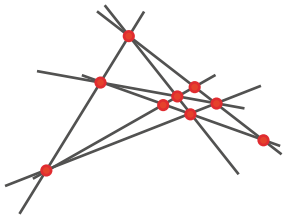


Figure 8.17: The configuration associated to a complete quadrilateral and its diagonal triangle

4. Verify that the line determined by the diagonal points of a quadrangle in a projective plane is a section of the quadrangle.

5. Verify that the construction we described as in Fig. 8.13 does in fact yield a complete quadrangle, $ABCD$, for which $H(P, Q; R, S)$.

8.5 Transformations and Pappus's Theorem

There are several different ways to approach projective geometry. We could have approached it strictly from the transformational point of view in which we identify the objects left invariant under certain transformations, declare those to be the objects of interest, and study the features of those objects that the transformations respect. This is Erlangen geometry. Our point of view is less algebraic, but from any point of view, the notions of perspectivity and projectivity are fundamental. As promised, these will lead us to Pappus's Theorem.

Our context here is a projective plane, $\mathbb{FP}^2$.

Definition 8.19. Let ℓ and ℓ' be lines in $\mathbb{FP}^2$. A **perspectivity** is a mapping from ℓ to ℓ' so that every line joining a point on ℓ to its image on ℓ' passes through a fixed point P, called the **center** of the perspectivity.

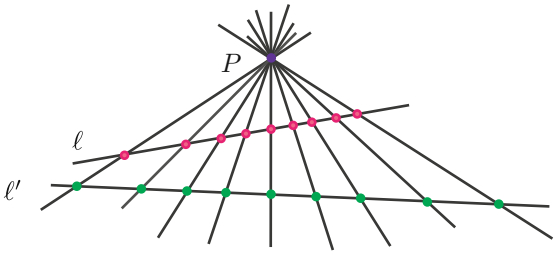


Figure 8.18: A perspectivity with center P

A perspectivity is its own inverse: In Fig. 8.18, we can view the mapping as going from ℓ to ℓ' or from ℓ' to ℓ. As long as ℓ and ℓ' are different lines, the center of a perspectivity is unique. We leave the verification of that statement as an easy exercise.

Definition 8.20. A **projectivity** or **projective transformation** is the composition of a finite number of perspectivities.

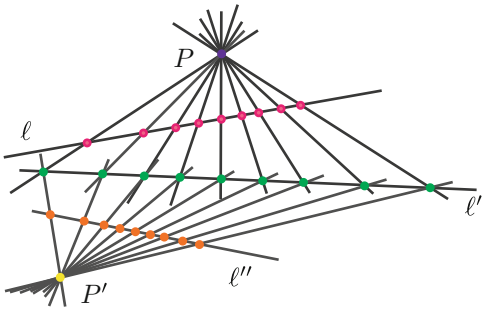


Figure 8.19: A projectivity from ℓ to ℓ'' (or from ℓ'' to ℓ) composed of two perspectivities, one through P, the second through P'

Since a perspectivity is its own inverse, a projectivity is also invertible: If $T = T_1 \cdots T_n$ is a product of perspectivities, then $T^{-1} = T_n \cdots T_1$.

If $A, B, C, \ldots$ are collinear points that are mapped to $A', B', C', \ldots$ respectively under a projectivity T, we write $T(A, B, C, \ldots) = (A', B', C', \ldots)$.

Theorem 8.21. Projectivities map harmonic sequences of points to harmonic sequences of points in a projective plane.

Proof. It is enough to argue that a single perspectivity sends a harmonic sequence of points in a projective plane to another harmonic sequence of points.

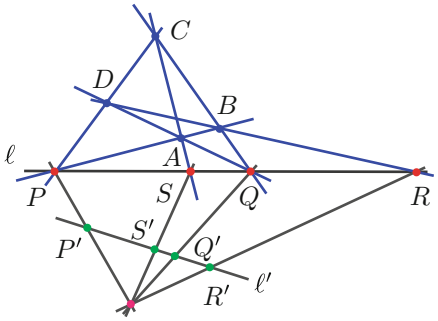


Figure 8.20: A perspectivity maps a harmonic sequence of points to a harmonic sequence of points

Suppose T is a perspectivity and $H(P,Q;R,S)$ where $\ell = PQ$. Referring to Fig. 8.20, we see this is the scenario addressed by Theorem 8.18: The line ℓ meets the concurrent lines PP', QQ', RR', SS' in the points P,Q,R,S with $H(P,Q;R,S)$ so $H(PP',QQ';RR',SS')$. By the same theorem, we also have $H(P',Q';R',S')$. □

As noted in [9], the content of the theorem suggests that there is no guarantee of the existence of a perspectivity that will map four arbitrary collinear points to another specified set of four collinear points. We can certainly map a given pair of points to any pair on another line via a perspectivity so pairs of points can be made to correspond using projective transformations, but not arbitrary quadruples. The next theorem answers the question about triplets.

Theorem 8.22 (Fundamental Theorem of Projective Geometry). Any set of three collinear points in a projective plane can be mapped to any other set of three collinear points in that plane by means of a unique projective transformation.

Sketch of proof. As in [9], we sketch the proof. First we argue for existence, then we address uniqueness. Let A,B,C and A',B',C' be arbitrary triples of collinear points in $\mathbb{FP}^2$.

Define $B'' = AB' \cap A'B$, $C'' = AC' \cap A'C$, and $A'' = AA' \cap B''C''$. If T_1 is the perspectivity through A', then $T_1(A,B,C) = (A'',B'',C'')$. Taking T_2 to be the perspectivity through A, we then have $T_2(A'',B'',C'') = (A',B',C')$. (See Fig. 8.21.) This gives us $T(A,B,C) = (A',B',C')$, where $T = T_2 \circ T_1$. The difficult part of the proof is arguing that T is unique, in other words, that knowing $T(A,B,C)$ is enough to know $X' = T(X)$ for any $X \in AB$. The approach is to argue that we can describe a given point on ℓ, or get arbitrarily close to a given point on ℓ, by means of harmonic sequences.

Theorem 8.16 implies there is unique D^* on ℓ so that $H(A,B;C,D^*)$. By Theorem 8.21, $H(A',B';C',D^{*\prime})$ where $T(D^*) = D^{*\prime}$. If $X = D^*$, we are done.

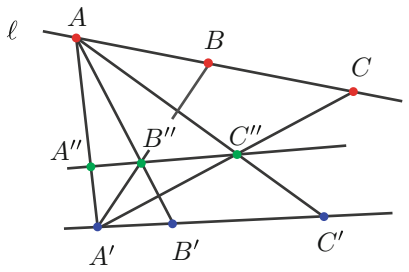


Figure 8.21: There is a projectivity relating any two sets of three collinear points

If not, repeat, noting there is unique E^* so that $H(B, C; D^*, E^*)$, and if $X \neq E^*$, there is unique F^* so that $H(C, D^*; E^*, F^*)$, etc. Depending on the nature of the underlying field—for example, whether it is $\mathbb{R}$ or $\mathbb{C}$ or finite—we either use up all the points on ℓ this way, eventually accounting for X as a member of a harmonic sequence, or we get arbitrarily close to X using harmonic sequences. Since each harmonic sequence of points is mapped to a well-defined harmonic sequence of points under T, this means $T(X)$ is determined by $T(A, B, C)$ as desired. □

Corollary 8.23. Let ℓ and ℓ' be distinct lines in $\mathbb{FP}^2$. A projectivity from ℓ to ℓ' that leaves a point on ℓ fixed is a perspectivity.

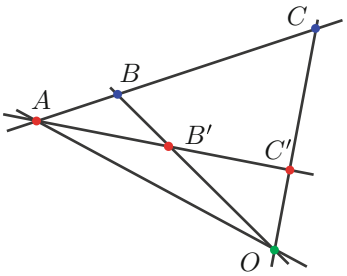


Figure 8.22: A nonidentity projectivity leaving one point fixed is a perspectivity

Proof. Let T be a projectivity with $T(A, B, C) = (A, B', C')$ for A, B, C collinear in some projective plane. Say $\ell = AB$ and $\ell' = AB'$ where $\ell \neq \ell'$. Let $O = BB' \cap CC'$. Then the perspectivity with center O is identical to T at A, B, C so by the theorem, that perspectivity must be T. □

Ironically, we conclude this section with the first theorem of projective geometry, Pappus's Theorem.

Theorem 8.24 (Pappus's Theorem). Suppose A, B, C, A', B', C' are points in a projective plane so that A, B, C are collinear, and A', B', C' are collinear. Then the points $AB' \cap A'B$, $AC' \cap A'C$, $BC' \cap B'C$ are collinear.

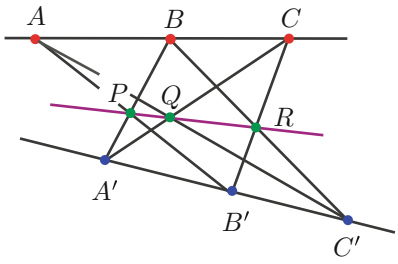


Figure 8.23: Pappus's Theorem says the green points are collinear

Proof. Let $P = AB' \cap A'B$, $Q = AC' \cap A'C$, $R = BC' \cap B'C$. We identify some additional points for the proof. Let $D = PQ \cap AC$, $E = AB' \cap A'C$, and $F = AC' \cap A'B$. Let T be the perspectivity from AB' to AC' with center A'. Then $T(A, P, E, B') = (A, F, Q, C')$. Let S be the perspectivity from AC' to PQ with center B, so that $S(A, F, Q, C') = (D, P, Q, R')$, where $R' = PQ \cap BC'$. We want to show that $R' = R$.

Composing the two perspectivities, we get the projectivity

$$(S \circ T)(A, P, E, B') = (D, P, Q, R').$$

By Corollary 8.23, $S \circ T$ is a perspectivity. Since AD and EQ meet at C, C is the center of the perspectivity and $R' = PQ \cap B'C$, implying $R' = R$, as desired. □

Exercises 8.4.

1. Show that the center of a perspectivity in a projective plane is unique.

2. Describe how to map a given pair of points in a projective plane to any pair of points on another line via a perspectivity.

3. If $T = T_1 T_2 \cdots T_n$ is a projectivity of a projective plane, with T_i a perspectivity, show that $T^{-1} = T_n \cdots T_2 T_1$.

4. Describe how to map a given pair of points in a projective plane to any other specified pair on the same line using a projectivity.

5. What is the dual of Pappus's Theorem?

6. Can you rearrange the labels A, B, C on one line and/or A', B', C' on the other line in Fig. 8.23 so that the line PQ changes? Keep the designation $P = AB' \cap A'B$, etc.

8.6 Homogeneous Coordinates

We return now to the idea that the projective planes we are studying here are those that arise from vector spaces. This time we take a slightly different point of view from the one we assumed when we first defined projective space as affine space with ideal points appended.

Let V be a three-dimensional vector space over $\mathbb{F}$ so $V \cong \mathbb{F}^3$. We use V itself to realize a projective plane over $\mathbb{F}$. In the process, we get an introduction to homogeneous coordinates. Again, remember that this cannot be done for every projective plane, just the ones based on fields. Note further that to realize a projective plane—a two-dimensional setting—we use a three-dimensional vector space.

A one-dimensional subspace of V is the span of any nonzero vector in V. In $\mathbb{R}^3$, a one-dimensional subspace looks like a line through the origin. In $\mathbb{F}_2^3$, it looks like a set of discrete points but we still refer to it as a line through the origin. **Projective points** in this model are lines through the origin in V. **Projective lines** in this model are two-dimensional subspaces of V, that is, the planes through the origin. **Incidence** is nonempty intersection. Before anything else, we establish that this model satisfies the axioms for a projective plane.

Theorem 8.25. The projective points and lines associated to $V \cong \mathbb{R}^3$ satisfy the axioms for a projective plane.

Proof. Axiom PP1 says two projective points determine a unique projective line. In our model, let $P = \text{span}\{\mathbf{v}\}$ and $Q = \text{span}\{\mathbf{w}\}$ be our two projective points. Here $\{\mathbf{v}, \mathbf{w}\}$ is a linearly independent set in V. P and Q determine the projective line $\ell = \text{span}\{\mathbf{v}, \mathbf{w}\}$. Since the plane in V that contains $\mathbf{v}$ and $\mathbf{w}$ is unique, so is ℓ. This proves Axiom PP1.

Next, consider two projective lines $\ell = \text{span}\{\mathbf{v}, \mathbf{w}\}$, and $m = \text{span}\{\mathbf{u}, \mathbf{x}\}$, each modeled on a two-dimensional subspace of V. Since $\dim V = 3$, $\{\mathbf{v}, \mathbf{w}, \mathbf{u}, \mathbf{x}\}$ is linearly dependent. We assume ℓ and m are distinct, implying that

$$\dim \text{span}\{\mathbf{v}, \mathbf{w}, \mathbf{u}, \mathbf{x}\} = 3.$$

There is thus a one-dimensional subspace common to $\text{span}\{\mathbf{v}, \mathbf{w}\}$ and $\text{span}\{\mathbf{u}, \mathbf{x}\}$. This one-dimensional subspace of V, a line through the origin, models a unique projective point, P, the point of intersection of ℓ and m. This shows that Axiom PP2 holds.

Next, let $P_1 = \text{span}\{(1,0,0)\}$, $P_2 = \text{span}\{(0,1,0)\}$, $P_3 = \text{span}\{(0,0,1)\}$, and $P_4 = \text{span}\{(1,1,1)\}$. Take any pair of these, for example P_1, P_2. These determine a unique projective line $\ell = \text{span}\{(1,0,0), (0,1,0)\}$. Notice that

neither P_3 nor P_4 lies on $\ell = \{(a, b, 0) | \, a, b \in \mathbb{R}\}$. The other two cases are easy to check, allowing us to conclude that our model contains four points, no three of which of are collinear. This verifies Axiom PP3. □

Next we coordinatize our model.

Define $\sim$ on nonzero $\mathbf{v}, \mathbf{w} \in \mathbb{F}^3$ by $\mathbf{v} \sim \mathbf{w}$ provided $\mathbf{v} = c\mathbf{w}$ for some $c \in \mathbb{F}$. We leave it as an exercise to check that $\sim$ is an equivalence relation on $\mathbb{F}^3 \setminus \mathbf{0}$.

Designate the equivalence class containing $\mathbf{v} = (a, b, c) \neq \mathbf{0}$ by $[\mathbf{v}] = [a, b, c]$. Each equivalence class corresponds to a unique projective point in our model. Each point in the model now has coordinates $[a, b, c]$, where not all a, b, c are zero.

Definition 8.26. The **homogeneous coordinates** for a point $P \in \mathbb{FP}^2$ determined by the span of a nonzero vector $\mathbf{v} = (a, b, c) \in \mathbb{F}^3$ is the equivalence class $[a, b, c] = \{t(a, b, c) \, | \, t \neq 0\}$.

If P is a projective point with $P = [a, b, c]$, at least one of the numbers a, b, c must be nonzero. There is thus always a representative of P that has one coordinate equal to 1.

It is instructive to reconcile this model of $\mathbb{FP}^2$ with the first model we had, where we appended ideal points to $\mathbb{FA}^2$. Because our experience is mostly with $\mathbb{R}^3$, we can use that to flesh out the connection between the two. This is purely for the convenience of familiarity. Nothing we say here depends on properties of $\mathbb{R}$ as the field of scalars.

Consider the pencil of lines through $\mathbf{0}$ in $\mathbb{R}^3$. This pencil corresponds to the set of projective points in our model. An arbitrary line in the pencil has the form $\ell = \text{span}\{(a, b, c)\}$, where $(a, b, c) \neq (0, 0, 0)$. If $c \neq 0$, $1/c$ is defined in $\mathbb{R}$ and we have the point $(a/c, b/c, 1)$ on ℓ. In other words, ℓ intersects the plane $z = 1$ at $(a/c, b/c, 1)$. In the usual coordinate system for $\mathbb{R}^3$, the plane $z = 1$ is one unit above, and parallel to, the xy-plane.

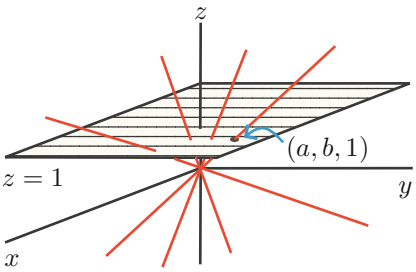


Figure 8.24: In the lines-through-the-origin model of $\mathbb{RP}^2$, the lines with homogeneous coordinates $[a, b, 1]$ correspond to affine points, and those with homogeneous coordinates $[a, b, 0]$ correspond to points at infinity

Any line ℓ in the pencil that does not intersect the plane $z = 1$ must be the span of a vector with the form $(a, b, 0)$. In other words, ℓ lies in the xy-plane. We thus have two kinds of lines in the pencil: those that intersect the plane $z = 1$, and those that lie in the xy-plane.

The projective points associated to the lines that intersect the plane $z = 1$ have homogeneous coordinates $[a, b, 1]$. The projective points associated to the lines in the xy-plane have homogeneous coordinates $[a, b, 0]$. Since $z = 1$ is a copy of $\mathbb{RA}^2$, we can identify the projective points having homogeneous coordinates $[a, b, 1]$ with the points of $\mathbb{RA}^2$, that is, the ordinary points of $\mathbb{RP}^2$. We then identify the projective points having homogeneous coordinates $[a, b, 0]$ with the ideal points appended to $\mathbb{RA}^2$.

There is yet another related model for $\mathbb{RP}^2$. Think about the pencil of lines through $\mathbf{0}$ in $\mathbb{R}^3$, this time in its relationship to the unit sphere in $\mathbb{R}^3$, also called the **unit 2-sphere**:

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 \,|\, x^2 + y^2 + z^2 = 1\}.$$

Each line intersects the surface of S^2 in exactly two Euclidean points. Those two points are called **antipodal points** on S^2. A pair of antipodal points on S^2 determines the single projective point that the line represents. In this S^2-based model of $\mathbb{RP}^2$, projective lines are based on the great circles on the sphere—that is, circles centered at $\mathbf{0}$. A projective line, then, is the set of antipodal point pairs on a great circle on S^2. Since any plane through $\mathbf{0}$ in $\mathbb{R}^3$ intersects S^2 in a great circle, and any great circle on S^2 determines a plane through $\mathbf{0}$, we have a neat correspondence between the elements in one model and the elements in the other model.

Antipodal point pairs on S^2 give us a good model for thinking about the physical nature of $\mathbb{RP}^2$, that is, what it would be like to live in $\mathbb{RP}^2$. Consider

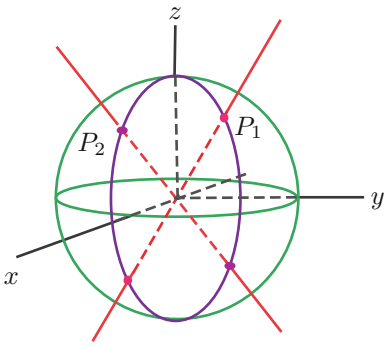


Figure 8.25: In the S^2-based model for $\mathbb{RP}^2$, a projective point is realized as an antipodal point pair on the sphere. A projective line is realized as the antipodal point pairs on a great circle.

a point P on S^2 and say P' is its antipodal point. If we move along any great circle through P, we must eventually arrive at P'. In $\mathbb{RP}^2$, then, if move along a line through a point, we meet back up again with that point. This suggests that if light travels in a line, then in a projective plane, observers who could see far enough would actually be looking at the backs of their own heads.

Finally, we employ homogeneous coordinates to do a calculation in $\mathbb{RP}^2$.

Example 8.27. Let $P = [1, 0, 0]$, $Q = [0, 1, 0]$, and $R = [1, 1, 0]$ in $\mathbb{RP}^2$. Notice that P, Q, R are collinear: This follows because the two-dimensional vector subspace of $\mathbb{R}^2$, $\alpha = \text{span}\{(1, 0, 0), (0, 1, 0)\}$ contains the one-dimensional subspace of $\mathbb{R}^2$, $\ell = \text{span}\{(1, 1, 0)\}$. We would like to find the harmonic conjugate of R with respect to P and Q.

Referring to the construction we carried out in Section 8.4, we choose a projective line through Q, different from PQ, and take two points on that line, different from Q. Let the projective line be modeled by the Euclidean plane

$$\beta = \text{span}\{(0, 1, 0), (0, 0, 1)\} = \{(0, a, b) |\ a, b, \in \mathbb{R}\}.$$

The projective points $B = [0, 1, 1]$ and $C = [0, 2, 1]$ are on that line and different from Q.

Let $D = PC \cap RB$. The projective line PC is determined by the Euclidean plane $\{(a, 2b, b) |\ a, b \in \mathbb{R}\}$. RB is determined by $\{(a, a + b, b) |\ a, b \in \mathbb{R}\}$. It follows that $D = [1, 2, 1]$. The projective line PB is represented by the Euclidean plane $\{(a, b, b) |\ a, b \in \mathbb{R}\}$, while QD is represented by $\{(a, b, a) |\ a, b \in \mathbb{R}\}$. The projective point $A = PB \cap QD$ is then $[1, 1, 1]$. We have AC represented by $\{(a, a + 2b, a + b) |\ a, b \in \mathbb{R}\}$ and PQ represented by $\{(a, b, 0) |\ a, b \in \mathbb{R}\}$. It follows that $S = AC \cap PQ$, the harmonic conjugate of R with respect to P and Q, is $[1, -1, 0]$.

Exercises 8.5.

1. Show that $\sim$ defined on $\mathbb{F}^3 \setminus \mathbf{0}$ by $\mathbf{v} \sim \mathbf{w}$ provided $\mathbf{v} = c\mathbf{w}$ for some $c \in \mathbb{F}$ is an equivalence relation.

2. If $V = \mathbb{R}^3$, the lines $\ell_1 = \text{span}\{(1, 0, 0)\}$ and $\ell_2 = \text{span}\{(0, 1, 0)\}$ determine what plane? What is the plane determined by lines ℓ_1 and $\ell_3 = \text{span}\{(0, 0, 1)\}$? What about the plane determined by lines ℓ_2 and ℓ_3? What is the plane determined by ℓ_1 and $\ell_4 = \text{span}\{(1, 1, 1)\}$? Describe it. How is it oriented in V?

3. Let $\alpha = \text{span}\{\mathbf{v}, \mathbf{w}\}$, and $\beta = \text{span}\{\mathbf{u}, \mathbf{x}\}$ be distinct planes in V where $\dim V = 3$. Detail the argument that $\{\mathbf{v}, \mathbf{w}, \mathbf{u}, \mathbf{x}\}$ is linearly dependent, thus that there is one-dimensional subspace in common to α and β. Use a dependence relation to describe the line of intersection of α and β.

4. Finish the proof of Theorem 8.25.

5. In Example 8.27, we note that if $C = [0, 2, 1]$ and $P = [1, 0, 0]$, then the projective line PC is determined by the Euclidean plane $\{(a, 2b, b) |\ a, b \in \mathbb{R}\}$. Verify that this is correct. Do the same for the rest of the Euclidean

planes in that example and verify that the intersection points, given in homogeneous coordinates, are correct. Note in particular the Euclidean plane underlying the projective line QD. Why can we say that plane contains any Euclidean point of the form (a, b, a)?

6. Rework Example 8.27 using different points B and C.

7. Rework Example 8.27 over $\mathbb{F}_2$ and again over $\mathbb{F}_3$.

8. Show that the Fano plane is a model for $\mathbb{F}_2\mathbb{P}^2$.

9. Use homogeneous coordinates to count the number of points in $\mathbb{F}_3\mathbb{P}^2$.

Chapter 9

Algebraic Curves

9.1 Introduction

Our studies to this point have been concerned almost entirely with points, lines, and circles. Here we turn to more general curves, but only those given by polynomials. These are *algebraic curves*. This is a brief introduction to a vast subject: We keep our scope modest, and our goal simple and clearly defined. We confine our attention to algebraic plane curves, that is, curves given by the zero sets associated to polynomials over $\mathbb{R}$ in two variables. Our goal is to appreciate Bézout's Theorem for plane curves, one of the cornerstones of the subject.

Bézout's Theorem is a generalization of the fact that two distinct lines in a projective plane intersect in one point. Getting to a correct statement of Bézout's Theorem for curves is enough to bring us face-to-face with the realization that the theory of curves in $\mathbb{RA}^2$ may be best understood from a point of view afforded by the real projective plane, or maybe even the complex projective plane, the algebraic and geometric completion of $\mathbb{RA}^2$.

When we discuss plane curves, the plane may be any affine or projective plane but for convenience, we confine our remarks almost entirely to $\mathbb{RA}^2$, which we identify with $\mathbb{R}^2$, and $\mathbb{RP}^2$, which we model using homogeneous coordinates in $\mathbb{R}^3$.

Curves are a source of much comment in calculus. Typically, in that context, we do not study curves in full. We study pieces of curves, pieces that can be described as graphs of functions. While conic sections may get some attention over the course of a typical calculus sequence, a curve given by an equation such as $y^2 - x^2y = 1$ is often sidelined for failing to arise as the graph of a function. To be sure, functions represent a tremendous advance in mathematics that happened over the course of the nineteenth century. They give the calculus impressive firepower, but functions cannot solve every problem. In this chapter, we explore other devices for studying curves.

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9.2 Polynomials, Rings, and Integral Domains

Polynomials may seem warmly familiar but they lay traps for the unwary. We approach them here from the point of view of modern algebra, but not with the ambition or rigor we would bring to a course in modern algebra. To begin with, we think of polynomials as having several variables and numerical coefficients. In short order, we will open up the pool of structures from which we may draw coefficients but largely, we are interested in polynomials in one or two variables with coefficients in $\mathbb{R}$.

We start with the notion of a **monomial in k variables** $\{x_1, \ldots, x_k\}$. This is an expression of the form $ax_1^{n_1} \cdots x_k^{n_k}$, where $n_1, \ldots, n_k$ are in $\mathbb{Z}_+$, and a is a constant, called the **coefficient** of the expression. If $a \in \mathbb{R}$, then our monomial is **over** $\mathbb{R}$.

Two monomials in $\{x_1, \ldots, x_k\}$, $ax_1^{n_1} \cdots x_k^{n_k}$ and $bx_1^{m_1} \cdots x_k^{m_k}$, are **distinct** provided $a \neq b$, or there is $n_i \neq m_i$, for some $i \in \{1, \ldots, k\}$: $2xy$ and $2yx$ are not distinct—that is, $2xy = 2yx$—but x^2y and xy^2 are distinct. Multiplication of the constants and variables in monomials is commutative and follows the rules of exponents: $(3x^2yz)(2y^2z^3) = 6x^2y^3z^4$, for example.

When $a \neq 0$, the **degree of the monomial** $ax_1^{n_1} \cdots x_k^{n_k}$ is $n_1 + \cdots + n_k$. A nonzero constant is considered a monomial of degree 0. While 0 is considered a monomial, its degree is not defined.

A **polynomial** is an expression that can be written as a sum of monomials. The **degree of a polynomial** is the highest degree from among its distinct monomial terms with nonzero coefficients. Note that $x^3 + x^2y^2 + 2xy^2$ has degree 4, the degree of the middle term. The **zero polynomial** has zero for every coefficient.

If it is necessary for clarity, we write $p(x_1, \ldots, x_k)$ for a polynomial in k variables and if it is not necessary for clarity, we may refer simply to p.

The highest degree term in a polynomial in one variable is the **leading term** and its coefficient is the **leading coefficient**. If the leading coefficient is 1, the polynomial is **monic**. When we refer to a single variable polynomial in the abstract, for instance, as

$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

we assume $a_n \neq 0$, unless we make an announcement to the contrary.

Arithmetic with polynomials involves addition and multiplication, both of which are commutative and associative. The distributive law and the laws of exponents apply when we perform arithmetic on polynomials: $(2x+y)(xy-1) = 2x^2y + xy^2 - 2x - y$.

Monomials that comprise summands in a polynomial are **like terms** if they can be written in the form $ax_1^{n_1} \cdots x_k^{n_k}$ and $bx_1^{n_1} \cdots x_k^{n_k}$ for (possibly different) coefficients a and b. In a polynomial, we combine like terms by adding their coefficients: $(x - y)(x + y) = x^2 - y^2$, for example because $xy, -yx$, are like terms and add up to 0.

Let $\mathbb{R}[x_1, \ldots, x_k]$ be the collection of all polynomials in variables $x_1, \ldots, x_k$ with coefficients in $\mathbb{R}$. If p is in $\mathbb{R}[x_1, \ldots, k_k]$, we can think of p as a mapping

from $\mathbb{R}^k$ to $\mathbb{R}$. This involves evaluating p at each point $\mathbf{a} = (a_1, \ldots, a_k) \in \mathbb{R}^k$. Here, a_i is in $\mathbb{R}$ so $p(\mathbf{a})$ is the number we get when we substitute a_i for x_i in p and do the attendant arithmetic: If $p = 3x^2 + y^2$, then $p(1,2) = 3 \cdot 1^2 + 2^2 = 7$. If p, q are in $\mathbb{R}[x_1, \ldots, x_k]$, and $\mathbf{a} \in \mathbb{R}^k$, then $(p+q)(\mathbf{a}) = p(\mathbf{a}) + q(\mathbf{a})$ and $(pq)(\mathbf{a}) = p(\mathbf{a})q(\mathbf{a})$.

The **zero set** of a polynomial $p \in \mathbb{R}[x_1, \ldots, x_k]$, is the set of all $\mathbf{x} \in \mathbb{R}^k$ such that $p(\mathbf{x}) = 0$. A point in the zero set of a polynomial p is called a **zero** of p. Context is important in understanding a zero set. For instance we can view the zero set for the expression $x - y$ as a subset of $\mathbb{R}^2$, $\mathbb{R}^3$, or $\mathbb{R}^n$, for $n > 3$. In $\mathbb{R}^2$, the zero set for $x - y$ is a line, $\ell = \{(x, x) \,|\, x \in \mathbb{R}\}$. In $\mathbb{R}^3$, though, the zero set of $x - y$ is the plane $\alpha = \{(x, x, z) |\, x, z \in \mathbb{R}\}$.

Another way to think about polynomials is as elements in a vector space. Recall that $\mathbb{R}[x]$ is a vector space but so is $\mathbb{R}[x_1, \ldots, x_k]$. The zero vector in a polynomial space is the zero polynomial.

Throughout calculus, we consider polynomials over $\mathbb{R}$ but we can extend the definitions to polynomials with coefficients in an arbitrary field, $\mathbb{F}$. We say that a polynomial in $\mathbb{F}[x]$ is a polynomial **over** $\mathbb{F}$, likewise for polynomials in several variables, $x_1, \ldots, x_k$. The arithmetic, terminology, and idea of zero set all carry over to the more general setting. We note that though we will not have occasion to study them, polynomials over finite fields can be a tricky business.

If a field $\mathbb{F}$ is a subfield of another field K, then an element in $\mathbb{F}[x]$ can also be viewed as an element in $K[x]$. For example, x^2+1 can be viewed as belonging to $\mathbb{Q}[x]$, $\mathbb{R}[x]$, or $\mathbb{C}[x]$, as its integer coefficients are elements in $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$. We can go a step further and make the underlying collection of coefficients more general than a field. For example, we often confine our attention to polynomials with integer coefficients. The integers are an example of a *ring*.

Definition 9.1. A **ring**, $(R, +, \cdot)$, is a set R with two binary operations, $+$ called addition and $\cdot$ called multiplication, satisfying the following properties.

(1) $(R, +)$ is an abelian group, that is

   (a) $+$ is commutative and associative;

   (b) there is an additive identity $0 \in R$ such that $a + 0 = a$ for all $a \in R$;

   (c) each $a \in R$ has an additive inverse $\hat{a}$, so that $a + \hat{a} = 0$.

(2) Multiplication is associative.

(3) Multiplication and addition together obey the distributive law:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \text{ and } (a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

If the multiplication is commutative, $(R, +, \cdot)$ is a **commutative ring.** If R has a multiplicative identity, $1 \neq 0$, $(R, +, \cdot)$ is a **ring with identity**.

When the binary operations on a ring are understood, we refer to $(R, +, \cdot)$ simply as R.

Lemma 9.2. In any ring R, the additive identity is unique, the additive inverse of any element is unique, and $0 \cdot a = 0$ for all $a \in R$.

Proof. Let R be a ring with additive identities 0 and a so that $a + b = b$ for all $b \in R$. Then $a = a + 0 = 0$ so the additive identity is unique.

Suppose there are $a, \hat{a}, b \in R$ so that $\hat{a}$ and b are both additive inverses of a. We have $a + \hat{a} = a + b = 0$. Adding $\hat{a}$ to the expression on the left, we get $\hat{a} + (a + \hat{a}) = \hat{a}$. Adding $\hat{a}$ to $a + b$, we get $\hat{a} + (a + b) = (\hat{a} + a) + b = b$. We conclude that $\hat{a} = b$, that is, that the additive inverse of a is unique.

Finally, $0 \cdot a = (0 + 0) \cdot a = (0 \cdot a) + (0 \cdot a)$ so if we add $\widehat{0 \cdot a}$ to the first and last expressions we get $0 = 0 \cdot a$. □

If R is a ring with $a, b \in R$, we usually use juxtaposition to indicate multiplication, that is, $ab = a \cdot b$.

Lemma 9.3. If R is a ring with elements $a, b \in R$, then $\widehat{ab} = \hat{a} \cdot b = a \cdot \hat{b}$.

Proof. Consider that $(\hat{a} \cdot b) + ab = (\hat{a} + a)b = 0 \cdot b = 0$. This shows that $\hat{a} \cdot b = \widehat{ab}$. Similarly, $(a \cdot \hat{b}) + ab = a(\hat{b} + b) = a \cdot 0 = 0$, proving that $a \cdot \hat{b} = \widehat{ab}$. □

Typically we designate the additive inverse of a by $-a$ instead of $\hat{a}$. Notice then that Lemma 9.3 says that $-(ab) = (-a)b = a(-b)$. This helps us exploit the idea of *subtraction* in rings. In particular, we think of subtraction of b as adding $-b$, but the question is whether the arithmetic has the properties we are used to if we use the notation that way. Certainly $a - b = -b + a$, but does $a(b - c) = ab - ac$ in an arbitrary ring? It is easy to verify that it does, and we leave that as an exercise. Note that we can also say that if $a - b = 0$ then $a + -b = 0$, so by adding b to both sides, we get $a = b$.

$\mathbb{Z}$ is a commutative ring with identity, as is any field. Our first lemma gives us many more examples.

Lemma 9.4. If R is a commutative ring with identity, then $R[x]$ is a commutative ring with identity. The additive identity in $R[x]$ is the additive identity in R. The multiplicative identity in $R[x]$ is the multiplicative identity in R.

Sketch of proof. Consider $p, q \in R[x]$, where $p = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n = \sum_{i=0}^{n} a_i x^i$ and $q = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m = \sum_{i=0}^{m} b_i x^i$. The degree k term of $p + q$ is defined to be $(a_k + b_k)x^k$, where $a_k = 0$ if $k > n$, and $b_k = 0$ if $k > m$. Associativity and commutativity of addition in $R[x]$ now follow associativity and commutativity of addition in R. The definition of addition makes it clear that the additive identity in $R[x]$ is 0, the additive identity in R, which we can identify with the zero polynomial. The additive inverse of p is then $-p = \sum_{i=0}^{n}(-a_i)x^i$.

We define the product pq using the distributive law on the monomial terms of p and q: In effect, multiplication in $R[x]$ is defined to be distributive. The coefficient of the degree k term of pq is $a_0b_k + a_1b_{k-1} + \cdots + a_kb_0$. Verification of associativity of multiplication follows a tedious but obvious path, while commutativity of multiplication follows rapidly on the heels of commutativity

of multiplication and addition in R. That the multiplicative identity in $R[x]$ is 1, the multiplicative identity in R, follows the definition of multiplication in $R[x]$. □

It follows from Lemma 9.4 that $\mathbb{R}[x]$ is a commutative ring with identity as is $R[y]$, where $R = \mathbb{R}[x]$. In fact, $\mathbb{R}[x, y] = R[y]$, where $R = \mathbb{R}[x]$. This device is central in a study of polynomials.

Example 9.5. We noted above that $p = x^3 + x^2y^2 + 2xy^2$ as an element in $\mathbb{R}[x, y]$ has degree four. We write $\deg p = 4$, and we think of the polynomial here as being over $\mathbb{R}$.

Writing the monomial terms of p in order of descending degree we have

$$p = x^2y^2 + 2xy^2 + x^3.$$

The first monomial has degree 4, the second two have degree 3 over $\mathbb{R}$. If we look at p as an element of $\mathbb{R}[x][y]$, we may write it

$$p = (x^2 + 2x)y^2 + x^3.$$

As a polynomial in y over $\mathbb{R}[x]$, its first term has degree 2, and its second (nonzero) term has degree zero.

If we view p as an element of $\mathbb{R}[y][x]$, we write it

$$p = x^3 + y^2x^2 + y^2x,$$

viewing the monomials, respectively, as degree 3, degree 2, and degree 1 in x over the ring $\mathbb{R}[y]$.

Employing this trick—viewing a polynomial in two variables over a field $\mathbb{F}$ as being a polynomial in one variable over a ring $R = \mathbb{F}[y]$—we must understand some of the algebraic properties of R. Indeed, the theory of commutative rings is fundamental to algebraic geometry, the larger mathematical area that includes algebraic curves.

Definition 9.6. A **subring** is a subset of a ring $(R, +, \cdot)$ that is itself a ring under $+$ and $\cdot$. A **proper subring** of R is a subring $S \subsetneq R$.

Notice that $\{0\}$ with $0+0 = 0$ and $0 \cdot 0 = 0$ is itself a ring, called the **trivial ring**. Every nontrivial ring then has at least two subrings: itself, and $\{0\}$. Note that the empty set cannot be made into a ring.

Notice that we can treat $\mathbb{Z}[x]$ as a subring of $\mathbb{R}[x]$.

Polynomial rings share certain critical properties with the integers and some other commutative rings.

Definition 9.7. A **zero divisor** in a ring R is a nonzero element $a \in R$ where $ab = 0$, for some nonzero $b \in R$. An **integral domain** is a commutative ring with identity that has no zero divisors.

The word "integral" in "integral domain" refers to the integers, $\mathbb{Z}$, the archetype for integral domains. Any field is also an integral domain. We usually refer to an integral domain simply as a *domain.*

There is a certain amount of arithmetic available to us in a domain that we cannot expect in a general ring.

Lemma 9.8 (Cancellation)**.** If a, b, c are elements in a domain D, and $a \neq 0$, then $ab = ac$ implies $b = c$. Similarly, if $ba = ca$, and $a \neq 0$, then $b = c$.

Proof. Suppose a, b, c are in a domain D and that $a \neq 0$. If $ab = ac$ then by Lemma 9.3, we have

$$ab + (-ac) = ab + a(-c) = a(b - c) = 0.$$

Since D contains no zero divisors, $a \neq 0$ implies $b - c = 0$, so $b = c$. This proves the first statement of the lemma. The second follows by commutativity of multiplication in D. □

An example of a commutative ring with identity that is not a domain is $\mathbb{Z}_4$, where we define both addition and multiplication mod 4. We have the following addition and multiplication tables for $\mathbb{Z}_4$.

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

·	1	2	3
1	1	2	3
2	2	0	2
3	3	2	1

Since $2 \cdot 2 = 0$, $2 \in \mathbb{Z}_4$ is a zero divisor.

Lemma 9.9. If R is a domain, $R[x]$ is also a domain.

Proof. It follows Lemma 9.4 that $R[x]$ is a commutative ring with identity if R is a domain. It remains to show that $R[x]$ has no zero divisors when R is a domain.

Recall that for $p = \sum_{k=0}^{n} a_k x^k$ and $q = \sum_{k=0}^{m} b_k x^k$, the coefficient of x^k in pq is $a_0 b_k + a_1 b_{k-1} + \ldots + a_k b_0$. Now suppose $pq = 0$ so that all the coefficients of pq are zero. Assume that q is nonzero. This means q has a nonzero coefficient. Let $k \in \{0, \ldots, m\}$ be least so that $b_k \neq 0$. Note then that the lowest degree term of pq that we need address is $(a_0 b_k)x^k$.

Since $pq = 0$, $a_0 b_k = 0$, with $b_k \neq 0$ implies $a_0 = 0$, as R is a domain. Consider next the degree $k + 1$ term of pq. Its coefficient is $a_0 b_{k+1} + a_1 b_k = 0$. Since $a_0 = 0$, the coefficient of the degree $k + 1$ term of pq is $a_1 b_k = 0$. Again, $b_k \neq 0$ implies $a_1 = 0$. So far, we have established that $p = 0 + 0x + a_2 x^2 + \cdots a_n x^n$. We leave it to the reader to complete the proof that $a_k = 0$, for $k \in \{2, \ldots, n\}$. This shows that when the product of polynomials is zero, one of the polynomials is zero, which means $R[x]$ has no zero divisors. □

The zero set of a polynomial in $\mathbb{R}[x, y]$ usually forms a curve in $\mathbb{R}^2$. There are exceptions: The zero set for x^2+y^2 is a single point $(0, 0)$ in $\mathbb{R}^2$, for example. The zero set of $x^2 - y^2$ in $\mathbb{R}^2$ is more interesting and we invite you to graph it in the exercises. We noted above that if the zero set of the polynomial can be written $y = p(x)$, then we can approach an understanding of the curve using the methods of classical algebra and calculus. In general, though, if we rummage around the standard-issue toolbox of undergraduate mathematics looking for resources we might bring to bear in a study of polynomials in two variables, we come up with precious little. We mentioned above one important device we can use, that is, to view $\mathbb{R}[x, y]$ as a polynomial ring in y over $\mathbb{R}[x]$ or, if it is more convenient, as a polynomial ring in x over $\mathbb{R}[y]$.

When we work with polynomials, we often address whether a given polynomial factors into polynomials of lower degree. If the polynomial in question has a single variable, x, then we know that over $\mathbb{C}$, for instance, the polynomial factors into linear factors. Indeed, this is how most of us became acquainted with $\mathbb{C}$ to begin with. Factoring over more general rings can be fraught, but if the coefficient ring is a domain, things run relatively smoothly.

Division and factoring are related but not identical processes. The next theorem addresses division of polynomials. A modified version of this arises in the *Euclidean Algorithm* in Book VII of *The Elements*, where Euclid seeks what we would call the greatest common divisor of two integers. In the next section, we apply the theorem to find the gcd of two polynomials.

Theorem 9.10 (Division Algorithm)**.** Let f and g belong to $D[x]$, D a domain. Suppose $g \neq 0$. There are a nonzero constant $a \in D$ and polynomials q and r in $D[x]$ with

$$af = gq + r, \text{ where } r = 0 \text{ or } \deg r < \deg g.$$

Proof. If $\deg f < \deg g$ or $f = 0$, take $a = 1$, $q = 0$, and $r = f$. If g is constant, we can take $a = g$, $q = f$, and $r = 0$.

Suppose that neither f nor g is a constant and that $\deg f \geq \deg g$. We prove the result by induction on $\deg f$, assuming it holds for any polynomial with degree less than that of f.

Say $f = a_0 + a_1x + \cdots + a_nx^n$, $g = b_0 + b_1x + \cdots b_mx^m$, and $n \geq m$. Consider that the leading term of b_mf is $a_nb_mx^n$ and the leading term of a_ng is $a_nb_mx^m$. If we divide the leading term of a_ng into the leading term of b_mf, we get x^{m-n}. Multiplying a_ng by that quotient and subtracting it from b_mf, we get $b_mf - a_nx^{n-m}g$, which is either 0 or a polynomial with degree strictly less than that of f.

If $b_mf - a_nx^{n-m}g = 0$, then we can take $a = b_m$, $q = a_nx^{n-m}$, and $r = 0$.

If $b_mf - a_nx^{n-m}g \neq 0$, then by assumption, there are a nonzero constant $a' \in D$ and polynomials $q', r \in D[x]$, with $r = 0$ or $\deg r < \deg g$, where

$$a'(b_mf - a_nx^{n-m}g) = gq' + r.$$

Then $a'b_mf = g(q' + a'a_nx^{n-m}) + r$, so that if we take $a = a'b_m \in D$, and $q = q' + a'a_nx^{n-m} \in D[x]$, we have the result. □

The Division Algorithm addresses division of f by g, so q is the *quotient* and r is the *remainder*.

The proof of the theorem reveals the algorithm, as we see in the next example.

Example 9.11. Let $f = x^3 + x^2 + 1$ and $g = 2x^2 - 5x + 3$, where $f, g \in \mathbb{Z}[x]$. We use the method outlined in the proof to find $a \in \mathbb{Z}$, $q, r \in \mathbb{Z}[x]$ so that $af = gq + r$, and either $r = 0$ or $\deg r < \deg g$. Since $m = \deg g = 2 < n = \deg f = 3$, we move to the step that says to replace f with $b_m f - a_n x^{n-m} g$.

As noted in the proof, x^{n-m} is the quotient when we divide the leading term of $a_n g$ into the leading term of $b_m f$. If we multiply $a_n g$ through by that quotient, x^{n-m}, and subtract it from $b_m f$, we get a remainder, $r_1 = c_0 + c_1 x + \cdots + c_\ell x^\ell$. If $\deg r_1 \geq \deg g$, we repeat, this time, dividing the leading term of $c_\ell g$ into the leading term of $b_m r_1$, multiplying that quotient, $x^{\ell-m}$, by $c_\ell g$, and subtracting from $b_m r_1$ to define the next remainder, r_2. We repeat until we arrive at a remainder $r = 0$ or r with $\deg r < \deg g$.

In the problem at hand, $n - m = 1$, $x^{n-m} = x$, $b_m = 2$, and $a_n = 1$ so

$$b_m f - a_n x^{n-m} g = 2f - xg = 2(x^3 + x^2 + 1) - x(2x^2 - 5x + 3) = 7x^2 - 3x + 2.$$

Our first remainder is then $r_1 = 7x^2 - 3x + 2$. Since $\deg r_1 = 2 = \deg g$, we repeat the division, this time dividing into the leading term of $b_m r_1 = 2r_1 = 14x^2 - 6x + 4$, by the leading term of $c_\ell g = 7g = 14x^2 - 35x + 21$. The quotient this time is 1. Multiplying $7g$ by 1 and subtracting from $2r_1$, we get our new remainder

$$2r_1 - 7g = 2(7x^2 - 3x + 2) - 7(2x^2 - 5x + 3) = 29x - 17.$$

Since $\deg(29x - 17) = 1 < \deg g = 2$, this is the last step and the final remainder is $r = 29x - 17$.

We would like to express all this in the form $af = qg + r$. We have $2f - xg = r_1$ and $2r_1 - 7g = r$ so $2(2f - xg) - 7g = 29x - 17$ which gives us $4f = (2x + 7)g + (29x - 17)$.

Notice that if we have to apply the algorithm k times, then $a = b_m^k$, where b_m is the leading coefficient of g. It follows that if g is monic, we can take $a = 1$.

The next two corollaries should be familiar from middle school. They follow the Division Algorithm nearly immediately.

Corollary 9.12 (Remainder Theorem)**.** Let D be a domain and let $c \in D$. If $f \in D[x]$ is divided by $x - c$, the remainder is $f(c)$.

Proof. Employing the Division Algorithm, we have $f = (x - c)q + r$ so that

$$f(c) = (c - c)q(c) + r(c) = r(c).$$

□

Example 9.13. Consider $f = x^3 - 2x^2 + 3x - 4 \in \mathbb{R}[x]$. If we divide f by $x - 1$, we get quotient $q = x^2 - x + 2$ and remainder $r = -2$ while $f(1) = 1 - 2 + 3 - 4 = -2$.

Corollary 9.14. If D is a domain then $x - c$ is a factor of $f \in D[x]$ if and only if $f(c) = 0$.

Proof. The result is immediate by the Remainder Theorem. □

Corollary 9.15. If D is a domain, a polynomial in $D[x]$ of degree n has at most n zeros.

Proof. The result is immediate by the previous corollary. □

Doing calculations with polynomials over $\mathbb{Z}$ may force us to deal with an inability to "divide through" by the coefficient of the highest degree term and the Division Algorithm suggests a way around that. Another approach is to broaden our coefficient ring to $\mathbb{Q}$, the smallest field where we can divide integers. For a more general domain, D, there is an analog to $\mathbb{Q}$ called the *field of quotients* or *quotient field* for D.

Constructing the Field of Quotients for a Domain

Fix an integral domain, D. Let $\mathcal{S} = \{(a, b) \mid a, b \in D, b \neq 0\}$. Define $\sim$ on $\mathcal{S}$ by

$$(a, b) \sim (a'b') \Leftrightarrow ab' = a'b. \tag{9.1}$$

We leave it as an easy exercise to show that (9.1) is an equivalence relation on $\mathcal{S}$. Let $[a, b]$ be the equivalence class containing (a, b). Let K denote the collection of equivalence classes in $\mathcal{S}$.

Notice right away that for $[a, b] \in K$, $[a, b] = [1, 1]$ only if $a = b$. Next, notice that $[a, b] = [0, 1]$ only if $a = 0$. These statements are easy verifications that we leave to the exercises.

Define addition on K by

$$[a, b] + [c, d] = [ad + cb, bd], \tag{9.2}$$

for any $[a, b], [c, d] \in K$. We must establish that (9.2) is well-defined.

Suppose, then, that $[a, b] = [a', b']$ and that $[c, d] = [c', d']$. We must show that the operation given in (9.2) yields the same sum in K whether we use summands $[a, b]$ and $[c, d]$ or summands $[a', b']$ and $[c', d']$. According to (9.2), we have

$$[a, b] + [c, d] = [ad + cb, bd] \text{ and } [a', b'] + [c', d'] = [a'd' + c'b', b'd'].$$

The two are the same in K provided

$$[a'd' + c'b', b'd'] = [ad + cb, bd],$$

that is, provided

$$(a'd' + c'b')(bd) = a'd'bd + c'b'bd = (ad + cb)(b'd') = adb'd' + cbb'd',$$

which is true since D is commutative.

It is easy to check that $[0,1]$ is an additive identity in K. Since

$$[a,b]+[-a,b]=[ab+(-a)b,b^2]=[0,b^2]=[0,1],$$

for any $[a,b]\in K$, it follows that every element in K has an additive inverse.

Define multiplication on K by

$$[a,b]\cdot[c,d]=[ac,bd]. \tag{9.3}$$

Again we must check that the operation is well-defined. Suppose $[a,b]=[a',b']$ and $[c,d]=[c',d']$ in K. By (9.3), we have

$$[a,b]\cdot[c,d]=[ac,bd] \text{ and } [a',b']\cdot[c',d']=[a'c',b'd'].$$

Multiplication is well-defined on K if and only if the two products yield the same element in K, that is, if and only if $acb'd'=a'c'bd$.

Since $[a,b]=[a',b']$ means $ab'=a'b$, and $[c,d]=[c',d']$ means $cd'=c'd$, we invoke commutativity and associativity in D to get

$$acb'd'=(ab')(cd')=(a'b)(c'd)=a'c'bd.$$

This verifies that multiplication on K is also well-defined. Since D has identity element $1\neq 0$, we have a multiplicative identity in K, $[1,1]$.

It is a routine matter to check that multiplication and addition are commutative and associative in K. (We leave the details as an exercise.)

Next, if $[a,b]\neq[0,1]$ in K, then $a\neq 0$ so $[b,a]\in K$ and

$$[a,b]\cdot[b,a]=[ab,ab]=[1,1].$$

In other words, every nonzero element in K has a multiplicative inverse.

We have yet to verify that the distributive law holds in K:

$$[a,b]\cdot([c,d]+[m,n])=[a,b]\cdot[cn+md,dn]=[acn+amd,bdn]$$

while

$$([a,b]\cdot[c,d])+([a,b]\cdot[m,n])=[ac,bd]+[am,bn]=[acbn+ambd,bdbn].$$

The distributive follows immediately on checking that

$$(acn+amd)(bdbn)=(acbn+ambd)(bdn),$$

which is a simple verification.

This establishes now that K is a field. We can identify D with a subring of K by

$$a\mapsto[a,1].$$

It is easy to check that this mapping respects both addition and multiplication in D, in other words,

$$a+b\mapsto[a+b,1]=[a,1]+[b,1],$$

and

$$ab \mapsto [ab, 1] = [a, 1][b, 1].$$

Finally, suppose Q is another field that contains a copy of D as a subring. In addition to containing an element that corresponds to each element of D, Q must contain a multiplicative inverse for each nonzero element in D. This means that Q must contain elements that correspond to $[a, 1]$, and to $[1, b] \in K$, $a, b \in D$, $b \neq 0$. Since Q must be closed under multiplication, it must have an element corresponding to each $[a, b] = [a, 1] \cdot [1, b] \in K$. In other words, if Q is another field containing D, it must contain K.

Definition 9.16. The **quotient field** or **field of quotients**, K, for a domain D is the minimal field containing D, in the sense that if $\mathbb{F}$ is an arbitrary field containing D, then $K \subseteq \mathbb{F}$.

Exercises 9.1.

1. Consider the polynomial $p = x^2 + x$ in $\mathbb{F}_2[x]$, where $\mathbb{F}_2$ is the field with 2 elements. Show that p is identically zero, that is, that for all $a \in \mathbb{F}_2$, $p(a) = 0$. Notice that $p = x(x + 1)$. Is either factor of p identically zero? Can you find a nonzero polynomial over $\mathbb{F}_3$ that is identically zero?

2. Sketch the zero set for $x^2 - y^2$ in $\mathbb{R}^2$. This is an example of a *reducible* algebraic curve. What do you think the word "reducible" means in this context? We also say that the curve $x^2 - y^2$ has two *irreducible components*. What do you think those components are?

3. Write the polynomials below as elements in $\mathbb{R}[x][y]$ and again as elements in $\mathbb{R}[y][x]$. In each case, identify the coefficients.

   (a) $x^2y - 2xy^2 + 3x + 2y - 7$

   (b) $3x^2 + 2xy + x - 3y + 1$

4. Show that $a(b - c) = ab - ac$ for a, b, c in any ring R.

5. Show that $a - (c - b) = (a - c) + b$ for a, b, c in any ring R.

6. Give an example to show that cancellation, as in Lemma 9.8, does not apply in a ring with zero divisors.

7. Show that a field is a domain.

8. Show that a subring of a domain is a domain.

9. Verify that when a, f, q, g, and r are as given in the Division Algorithm, for b_m the leading coefficient of g, we can take $a = b_m^k$, if we must apply the algorithm k times to find q and $r = 0$ or $r \in D[x]$ so that $\deg r < \deg g$, where $af = gq + r$.

10. Find the quotient and remainder in $\mathbb{Z}[x]$ if we divide $x^2 - 3x + 1$ into $2x^3 - 3x^2 + 4x - 5$.

11. Let D be a domain. This problem refers to Q, the set of so-called equivalence classes on $\mathcal{S} = D \times D \setminus 0$ defined by (9.1).

    (a) Show that $\sim$ is an equivalence relation on $\mathcal{S}$.

    (b) Verify that $[0, b] = [0, 1]$, for any nonzero $b \in D$.

    (c) Verify that $[a, b] = [0, 1]$ in Q only if $a = 0$ in D.

    (d) Show that addition and multiplication on Q are both associative and commutative.

12. What is the field of quotients for $\mathbb{R}[x]$?

9.3 Factoring and Division

Being able to run the Division Algorithm and not having to worry about zero divisors seem like a good a start in the direction of a theory of factoring. There are domains in which factoring is not as well behaved as we would like, though.

Example 9.17. Consider the set $D = \{a + bi\sqrt{3} \,|\, a, b \in \mathbb{R}\} \subset \mathbb{C}$. It is clear that D is closed under addition and a quick check reveals it is also closed under multiplication:

$$(a + bi\sqrt{3})(a' + b'i\sqrt{3}) = (aa' - 3bb') + (ab' + a'b)i\sqrt{3}.$$

It follows that D is a subring of $\mathbb{C}$, thus, a domain. (The proof of that statement was an exercise in the last section.) Notice now that $\mathbb{Z} \subset \mathbb{R} \subset D$ so in D, $2 \cdot 2 = 4$. But we also have $4 = (1 + i\sqrt{3})(1 - i\sqrt{3})$ in D so D does not enjoy unique factorization.

We can write any positive integer as a product of powers of distinct primes, in a way that is essentially unique. Where we are going, we need to know that we can do the same thing with polynomials. We start with vocabulary.

Definition 9.18. Let D be a domain. If a, b, c are nonzero elements in D and $c = ab$, then a **divides** c and we write $a|c$. We also say a is a **factor** or **divisor** of c. A **unit** in D is u such that $u|1$.

If $b \in D$ is nonzero, and $u \in D$ is a unit, then b and ub are **associates**. A nonzero element $a \in D$ is **irreducible** provided a is not a unit, and its only divisors are units and its own associates. A nonzero element in D is **reducible** if it is not irreducible.

The units in a domain are the elements that have multiplicative inverses. Indeed, a unit divides every element in a domain. (See the exercises.) The units in $\mathbb{Z}$ are 1 and -1 so the associates in $\mathbb{Z}$ are pairs of the form $\pm a$.

Since every nonzero element in $\mathbb{R}$ is a unit, any two nonzero real numbers are associates. The same is true in any field.

When we discuss factoring, we have to address the ambiguities that units and associates present: Every element in a domain can be factored in various

ways using units. When we talk about factoring integers, for instance, we often restrict our attention to $\mathbb{Z}_+$ where the only unit is 1 and no element has an associate other than itself. When working over arbitrary domains, we need the notions of units and associates to keep our remarks precise.

Definition 9.19. A **unique factorization domain**, UFD, is a domain in which every nonzero nonunit, a, has a factorization into irreducible nonunits, $a = a_1 \dots a_n$, which is unique up to order and associates. In other words, if $a = b_1 \dots b_s$, is another factorization in which each b_i is an irreducible nonunit, then $s = n$ and there is an ordering of the b_is so that a_i and b_i are associates.

$\mathbb{Z}$ is the model for UFDs. Since there are no nonzero nonunits in a field, every field is a UFD. The next theorem, which we cite without proof, suggests why it is important to understand that a field is a UFD.

Theorem 9.20. If D is a UFD, $D[x]$ is also a UFD.

It follows, for instance, that $\mathbb{Z}[x]$, $\mathbb{Q}[x]$, $\mathbb{R}[x]$, and $\mathbb{C}[x]$ are all UFDs. A corollary is immediate.

Corollary 9.21. If D is a UFD, $D[x_1, x_2, \dots, x_n]$ is also a UFD.

We say that factorization in a UFD is *unique up to associates*.

Example 9.22. Consider $x^2 - 1 \in \mathbb{R}[x]$. What are the units in $\mathbb{R}[x]$? They are the nonzero real numbers. The multiplicative inverse of a polynomial with positive degree is not a polynomial: $(x + 1)p(x) = 1$ is not true for any polynomial $p(x)$, for example. When we say that there is a unique factorization of $x^2 - 1$ over $\mathbb{R}$ into linear factors $x - 1$ and $x + 1$, we really do mean "up to associates" because $2(x - 1)(1/2)(x + 1)$ is a second factorization of $x^2 - 1$. Here, the associate pairs are $x - 1$ and $2(x - 1)$, and $x + 1$ and $(1/2)(x + 1)$.

The Euclidean Algorithm

We are ready now to construct and define the gcd for a pair of polynomials.

Repeated application of the Division Algorithm—first to the pair f, g, then to g, r, then to r, r', etc.—has to terminate eventually because of the reduction in the degree of the remainder at each step. That repeated application is called the *Euclidean Algorithm*.

Going forward, we use D to indicate a UFD, and K to designate the field of quotients for D.

Theorem 9.23 (Euclidean Algorithm)**.** Given polynomials f, g in $K[x]$, there is $h \in K[x]$ such that $h|f$, $h|g$, and if some other polynomial p also divides both f and g, then $p|h$. Further, there exist A, B in $K[x]$ so that $h = Af + Bg$.

Proof. We may assume $\deg f \geq \deg g$. Applying the Division Algorithm, we can produce polynomials q, r with $r = 0$ or $\deg r < \deg g$, so that for some nonzero constant a, $af = gq + r$. In K, a is a unit so we can divide through by a and

say that in $K[x]$, we have q_1, r_1, where $f = gq_1 + r_1$, again with $r_1 = 0$ or $\deg r_1 < \deg g$. If $r_1 = 0$, $f = gq_1$ so $g|f$, $g|g$ and we can write $g = 0f + g$. In that case, $h = g$.

If $r_1 \neq 0$, apply the Division Algorithm again, this time to g and r_1 to produce $q_2, r_2 \in K[x]$, $r_2 = 0$ or $\deg r_2 < \deg r_1$, with $g = r_1q_2 + r_2$. At the next step, we have $r_1 = r_2q_3 + r_3$, and so on. We must at some point get $r_{n+1} = 0$ because as i increases, the degree of r_i strictly decreases. Notice that our sequence of equations looks like this:

$$\begin{aligned} f &= gq_1 + r_1 \\ g &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ r_2 &= r_3q_4 + r_4 \\ &\vdots \\ r_{n-2} &= r_{n-1}q_n + r_n \\ r_{n-1} &= r_nq_{n+1} \end{aligned} \tag{9.4}$$

We claim that r_n has the requisite properties for h.

If we think of the algorithm as producing successive r_is, we can think of $f = r_{-1}$ and $g = r_0$. Then each equation in (9.4) can be written $r_i = r_{i+1}q_{i+2} + r_{i+2}$ or

$$r_{i+2} = r_i - r_{i+1}q_{i+2}, \tag{9.5}$$

where $i = -1, \ldots, n-1$, and $r_{n+1} = 0$.

The last equation in (9.4) says explicitly that $r_n|r_{n-1}$. Carrying this to the next equation going up the chain, we see it is also explicit in indicating that $r_n|r_{n-2}$. Iterating, we get to the top of the sequence in finitely many steps to conclude that $r_n|r_0$ and $r_n|r_{-1}$. This proves that $r_n|g$ and $r_n|f$.

Next suppose $p|f$ and $p|g$, that is, $p|r_{-1}$ and $p|r_0$. We want to argue that $p|r_n$. This time we work our way down from the top equation in (9.4), reading the equations as in (9.5). For $i = -1$, (9.5) says $r_1 = r_{-1} - r_0q_1$, so if $p|r_{-1}$ and $p|r_0$ as we assume, then $p|r_1$. For $i = 0$, we have $r_2 = r_0 - r_1q_2$, so since we now know that $p|r_0$ and $p|r_1$, we get $p|r_2$. Iterating, we get to the bottom of the sequence in (9.4) where we can say that if $p|r_{n-2}$ and $p|r_{n-1}$, then since $r_n = r_{n-2} - r_{n-1}q_n$, $p|r_n$, as desired. Since finitely many iterations are required to establish $p|r_{n-2}$ and $p|r_{n-1}$, we have proved that if $p|f$ and $p|g$, then $p|r_n$.

Finally we show that there are $A, B \in K[x]$ with $r_n = Af + Bg$. We substitute $r_{i+1} = r_{i-1} - r_iq_{i+1}$ into the right-hand side of (9.5). This gives us

$$r_{i+2} = r_i - (r_{i-1} - r_iq_{i+1})q_{i+2} = -q_{i+2}r_{i-1} + (1 + q_{i+1}q_{i+2})r_i. \tag{9.6}$$

When $i = n-2$, we get

$$r_n = -q_nr_{n-1} + (1 + q_{n-1}q_n)r_{n-2}, \tag{9.7}$$

but from (9.6), we can write r_{n-1} and r_{n-2} in the form $ar_{n-4} + br_{n-3}$, for $a, b \in K[x]$. Applying (9.6) finitely many times to the right-hand side of (9.7), we eventually get $r_n = Ar_{-1} + Br_0 = Af + Bg$, for some $A, B \in K[x]$.

This shows that r_n has the requisite properties so that for $h = r_n$, the theorem is proved. □

Definition 9.24. The **greatest common divisor or gcd** of polynomials f and $g \in K[x]$, is $h \in K[x]$ such that h divides both f and g, and if $p \in K[x]$ divides both f and g, p divides h.

The Euclidean Algorithm does more than guarantee the existence of a gcd for two polynomials. It guarantees that the gcd of f and g is a "linear" combination of f and g, where the coefficients come from $K[x]$. Since $K[x]$ is not a field, the term *linear combination* is not correct, but it helps capture this very important idea.

Example 9.25. We use the algorithm to find the gcd of $f = x^3 + 3x^2 + 3x + 2$, and $g = x^3 + 2x^2 + 2x + 1$.

Dividing g into f we get a quotient $q_1 = 1$ and remainder $r_1 = x^2 + x + 1$. Dividing r_1 into g, we get a quotient of $q_2 = x + 1$, with remainder $r_2 = 0$. By the Euclidean Algorithm, the gcd of f and g is thus $r_1 = x^2 + x + 1$. Checking to see that this is correct, we have $g = (x^2 + x + 1)(x + 1)$, and $f = g + (x^2 + x + 1) = (x^2 + x + 1)(x + 1) + (x^2 + x + 1) = (x^2 + x + 1)(x + 2)$.

How do we write $x^2 + x + 1$ in the form $Af + Bg$? We have $r_1 = f - gq_1$, so $x^2 + x + 1 = (x^3 + 3x^2 + 3x + 2) - (x^3 + 2x^2 + 2x + 1)$. In this case, $A = 1$ and $B = -1$.

Next we have a corollary to Theorem 9.23.

Corollary 9.26. Let $f, g, h \in K[x]$. If f is irreducible, and $f|gh$, then $f|g$ or $f|h$.

Proof. Let f, g, h be in $K[x]$. Assume that f is irreducible, and $f|gh$. Suppose f does not divide g.

Since f is irreducible, its only divisors are units, so the gcd of f and g is an element $a \in K$. By Theorem 9.23, there are A', B' in $K[x]$ with $a = A'f + B'g$. Divide through by a to get $1 = Af + Bg$ for some $A, B \in K[x]$. Then $h = Afh + Bgh$. Since $f|gh$, $h = Afh + BCf$, so $f|h$, which proves the result. □

Corollary 9.15 provides an upper limit on the number of zeros or *roots* of a polynomial in $K[x]$ but we know all too well that, depending on K, a polynomial in $K[x]$ may have no roots at all. For example, $x^2 - 2$ has no rational roots, meaning, as an element in $\mathbb{Q}[x]$, $x^2 - 2$ has no zeros. If we view $x^2 - 2$ as belonging to $\mathbb{R}[x]$, we get two zeros, $\sqrt{2}$ and $-\sqrt{2}$ but we have no zeros in $\mathbb{R}$ for $x^2 + 1$. In the *algebraic closure* of a field, we can always find the zeros of a polynomial in one variable.

Definition 9.27. A field K is **algebraically closed** provided every polynomial in $K[x]$, with degree greater than or equal to 1, has a root in K. If K is an arbitrary field, its **algebraic closure** is an algebraically closed field $\bar{K}$ that contains K as a subfield.

Like the field of quotients for a domain, the algebraic closure of a given field can always be constructed from the field. That story is much more delicate than the one that details the construction of a quotient field, but we can proceed on just two facts: (1) every field $\mathbb{F}$, has an algebraic closure, $\overline{\mathbb{F}}$, which is essentially unique, and (2) the algebraic closure of $\mathbb{R}$ is $\mathbb{C}$.

An interesting and fun fact is that the algebraic closure of any field is infinite.

Since every polynomial in one variable over an algebraically closed field has a zero in that field, we get the following theorem.

Theorem 9.28. If K is an algebraically closed field, and $f \in K[x]$ with $\deg f = n$, then there are $a_1, \ldots, a_n$ in K, not necessarily distinct, and $a \in K$, so that

$$f = a(x - a_1) \cdots (x - a_n).$$

Our last result in this section applies to polynomials in several variables. We note that the proof is by induction, but omit the details.

Theorem 9.29. Let $f \in D[x_1, \ldots, x_n]$. If $f(a_1, \ldots, a_n) = 0$ for all choices of a_i in some infinite subset of D, then f is identically zero.

The curves we care about are the zero sets of polynomials in two variables over a field. Remember that these polynomials then belong to $R[x]$, where $R = \mathbb{R}[y]$ is not a field.

We have already seen that there is a vast range of what we can expect when we look at the zero set of a polynomial in two variables over $\mathbb{R}$. This is germane to the problem of determining how many points of intersection we can expect to see between two curves, the subject of Bézout's Theorem.

Our first example about intersections conforms to expectations.

Example 9.30. Consider $f = x^2 - 2xy + y$ and $g = x^2 - 2x + y^2$ as elements in $\mathbb{R}[x, y]$. Graphing the zero sets of f and g in $\mathbb{R}^2$, we get the picture in Fig. 9.1. (The axes are not shown so that it is easier to see the points of intersection.) We

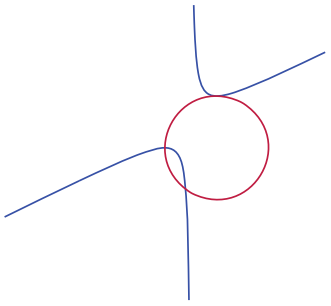


Figure 9.1: The curves given by $f = x^2 - 2xy + y$ and $g = x^2 - 2x + y^2$

leave it as an exercise to determine which curve is the zero set of f and which is the zero set of g. Both are conic sections: The blue curve is a hyperbola and the

red is a circle. The picture suggests that there are two transverse intersection points, that is, points where the curves pass through each other. There is a third, or maybe a third and a fourth intersection point, near the top of the circle, as well. With nothing but the picture, it is impossible to know whether this third place is actually two closely spaced transverse intersection points, whether the curves are tangent at that point, or whether the curves intersect at all at that point. As it happens, the curves are tangent at that point and the x-coordinate there is 1. That is easy to check using calculus and we leave it as an exercise, as well.

The next example may be less accessible intuitively.

Example 9.31. Consider the curve determined by $h = x^2 - xy - 2y^2$ in $\mathbb{R}[x, y]$, which is shown in Fig. 9.2. Why does this curve look like two lines? It is easy

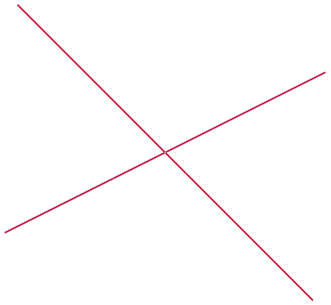


Figure 9.2: The curve given by $h = x^2 - xy - 2y^2$

to check that $h = (x - 2y)(x + y)$ so the zero set for h consists of the points in $\mathbb{R}^2$ on the line $x = 2y$ and the points on the line $-x = y$. The curve associated to h is then the union of curves associated to the irreducible factors of h. (The two lines are called *components* of the curve.)

The difference between the curve given by h in Example 9.31 and those given by f and g in Example 9.30 is that h is reducible while f and g are irreducible. Note that f and g are irreducible even if we look at them as polynomials over $\mathbb{C}$. When we go from coefficients in $\mathbb{R}$ to coefficients in $\mathbb{C}$, we are guaranteed that any single variable positive degree polynomial factors into degree one polynomials. There is no guarantee that a higher degree polynomial in $\mathbb{C}[x, y]$ factors into lower degree polynomials.

The advantage to viewing polynomials over $\mathbb{C}$, when their coefficients are in $\mathbb{R}$, is that the associated curves are *not degenerate*. A degenerate curve is a very small zero set. It could be empty or a single point. For instance, as an element of $\mathbb{R}[x, y]$, $p = x^2 + y^2$ has a degenerate curve: the single point $(0, 0)$ in $\mathbb{R}^2$. As an element of $\mathbb{C}[x, y]$, though, $x^2 + y^2 = (x + iy)(x - iy)$. In the complex plane, the curve determined by p is two lines.

We consider one more example of curves with a surprising intersection set.

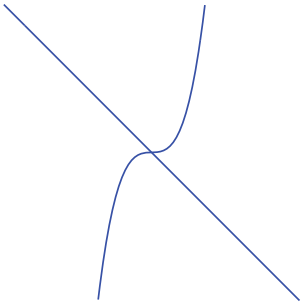


Figure 9.3: The curve given by $p = x^4 + x^3y - xy - x^2$

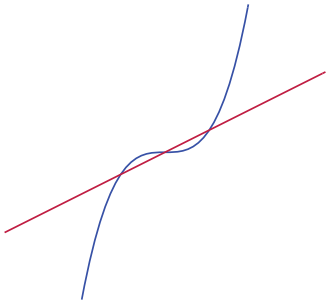


Figure 9.4: What remains when we ignore the common components of the curves given by $h = x^2 - xy - 2y^2$ and $p = x^4 + x^3y - xy - x^2$

Example 9.32. Let $p = x^4 + x^3y - xy - x^2 \in \mathbb{R}[x, y]$. The curve associated to p in $\mathbb{R}^2$ is shown in Fig. 9.3. We would like to consider any points of intersection of the curve given by p and the one given by $h = x^2 - xy - 2y^2$ in Example 9.31.

It is critical to notice that p and h have a common factor, $x + y$. All the points on the line $y = -x$ are then common to the two curves. Once we settle that, we can look at the remaining components of the curves to see if they have discrete points of intersection, more along the lines of what we saw in Example 9.30. As it turns out, there are three remaining points of intersection between the curves, as the picture in Fig. 9.4 suggests.

Exercises 9.2. 1. Is D from Example 9.17 a subfield of $\mathbb{C}$?

2. Let D be any domain. Prove the following statements for $a, b, c \in D$.

   (a) If $a|b$, and $b|c$, then $a|c$.

   (b) If $a|b$ and $a|c$ then $a|(k_1b + k_2c)$ for all $k_1, k_2 \in D$.

   (c) If u is a unit in D, then $u|a$ for all $a \in D$.

   (d) If b is an associate of a, then $b|a$.

   (e) If a and b are associates, then a is a unit if and only if b is a unit.

3. Show that any two nonzero elements in a field are associates.

4. Let D be a UFD and suppose $a, b, c \in D$. Show that if a is irreducible and $a|bc$, then $a|b$ or $a|c$.

5. Apply the Euclidean Algorithm to the integers 12 and 70 to produce their gcd. Verify using the prime factorization of 12 and 70.

6. Use the Euclidean Algorithm to find the gcd of f and g in $\mathbb{Q}[x]$, where $f = x^5 + 5x^4 + 8x^3 + 8x^2 - x - 21$ and $g = x^4 + 5x^3 + 8x^2 + x - 15$.

7. This problem is about the curves depicted in Fig. 9.1.

   (a) Use calculus and analytic geometry to sketch the curves in the xy-plane. Which is the zero set of f, and which is the zero set of g?

   (b) Verify that the curves have a common point where $x = 1$. Argue that the curves have a common tangent at that point.

   (c) We say that the *multiplicity of the intersection point* at $x = 1$ is two because the curves share not only a point at $x = 1$, but also a tangent. Can you detect an algebraic cue that suggests the number two should be associated to this intersection point?

8. Consider the curves given by $h = x^2 - xy - 2y^2$ and $p = x^4 + x^3y - xy - x^2$ as in Examples 9.31 and 9.32.

   (a) Verify that $x + y$ is a common factor of h and p.

   (b) Find the points of intersection of the components of the curves associated to h and p if we ignore the one associated to $x + y$.

9.4 The Resultant

We saw at the end of the last section that if two polynomials in $\mathbb{R}[x, y]$ have a common (nonunit) factor, the underlying curves have infinitely many points of intersection. One way to determine whether two polynomials have a common factor is to find their gcd, per Theorem 9.23. In this section, we discuss another tool that we can apply to this problem.

Recall that when we write $f = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, we assume $a_n \neq 0$.

We continue to use D to indicate a UFD.

Definition 9.33. Let $f = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, and $g = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m$ belong to $D[x]$. The **resultant of f and g**, $\mathcal{R}(f, g)$, is the

determinant

$$\begin{vmatrix} a_0 & a_1 & \dots & & a_n & & \\ 0 & a_0 & a_1 & \dots & a_{n-1} & a_n & \\ \vdots & & & & & & \\ b_0 & b_1 & \dots & b_{m-1} & b_m & & \\ \vdots & & & & & & \\ & & & 0 & b_0 & \dots & b_m \end{vmatrix}$$

where there are m rows of a_is and n rows of b_is.

The resultant of f and g, where $\deg f = n$ and $\deg g = m$, is thus the determinant of an $(m+n) \times (m+n)$ matrix.

Example 9.34. Let $f(x) = x^2 - 2x + 1$ and let $g(x) = x - 1$. Here $m = 1$, $n = 2$ so the resultant is the determinant of a 3×3 matrix. We start the matrix by filling in a row with the coefficients of f. We only need one row corresponding to f because the degree of g is 1. The next two rows record the coefficients of g. The second row corresponding to g is a copy of the first, but with the nonzero entries pushed over one entry to the right. This gives us

$$\mathcal{R}(f,g) = \begin{vmatrix} 1 & -2 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} = 0.$$

Note that in Example 9.34, $g|f$, in fact, $f = g^2$. As our next theorem will reveal, it is not an accident that $\mathcal{R}(f,g) = 0$. Before turning to that theorem, we have a quick review of the notion of a *homogeneous system of linear equations.*

Let $A = [a_{ij}]$ be an $m \times n$ matrix over $\mathbb{R}$, $\mathbf{b} = (b_1, b_2, \dots, b_m)$ an element in $\mathbb{R}^m$. Recall that the matrix equation $A\mathbf{x} = \mathbf{b}$ encodes the following system of linear equations.

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ & & & & \vdots & & & & \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

We view the components of $\mathbf{x} = (x_1, \dots, x_n)$ as unknowns. A solution to the system—or equivalently, the matrix equation—is $\mathbf{c} \in \mathbb{R}^n$ such that $A\mathbf{c} = \mathbf{b}$. If a system has a solution, it is *consistent.*

When $\mathbf{b} = \mathbf{0} \in \mathbb{R}^m$, both the matrix equation and the associated system of linear equations are *homogeneous.* Homogeneous systems of linear equations are always consistent because of the trivial solution $\mathbf{0} \in \mathbb{R}^n$. When $m < n$, the homogeneous system associated to $A\mathbf{x} = \mathbf{0}$ must also have nontrivial solutions. When $m = n$, the homogeneous system $A\mathbf{x} = \mathbf{0}$ has nontrivial solutions if and only if $\det A = 0$.

The proof of the next theorem casts the resultant as the determinant of a homogeneous system of equations. We sketch the proof and encourage the reader to try to understand its broad outline.

Theorem 9.35. Let f, g be nonzero polynomials in $D[x]$. $\mathcal{R}(f,g) = 0$ if and only if f and g have a nonconstant common factor.

Sketch of proof. Start by noting that f and g have a nonconstant common factor if and only if there are nonzero $\varphi(x)$ and $\psi(x)$ with

$$\deg \varphi(x) < \deg g(x),\ \deg \psi(x) < \deg f(x) \text{ and } f\varphi = g\psi.$$

Now think of trying to find $\varphi(x) = \alpha_0 + \alpha_1 x + \cdots + \alpha_k x^k$, and $\psi(x) = \beta_0 + \beta_1 x + \cdots + \beta_\ell x^\ell$, where α_i, β_i are unknowns. After doing the multiplications to write $f\varphi = g\psi$, we see that we have equations giving us $a_0\alpha_0 = b_0\beta_0$, $a_0\alpha_1 + a_1\alpha_0 = b_0\beta_1 + b_1\beta_0$, etc. Essentially, the resultant is the determinant of the coefficient matrix for this large homogeneous system of equations, with unknowns α_i, β_i. The system has a nontrivial solution—that is, there are nonzero φ, ψ so that $f\varphi = g\psi$—precisely when the determinant of this matrix, the resultant, is zero. □

We are most accustomed to curves given by polynomial functions $y = p(x)$ and $y = q(x)$. These are zero sets for polynomials of the form $f = p(x) - y$, and $g = q(x) - y$. To find intersection points for these sorts of curves, we look to solve two simultaneous equations, $f = 0$, $g = 0$. This amounts to solving $p(x) = q(x)$ or $p(x) - q(x) = 0$. While we start with simultaneous equations in two variables, we eliminate one variable and wind up with a polynomial equation in one variable. Often, this makes solutions accessible: First we solve $p(x) - q(x) = 0$, then we plug any solution x_0 into p or q to find the associated y-coordinate for the intersection point.

We are now thinking about curves given by more general polynomial expressions. These are zero sets for polynomials in two variables, $f = f(x,y)$ and $g = g(x,y)$ where f and g are not necessarily of the form $p(x) - y$. To find intersection points for these sorts of curves, we still look to solve two simultaneous equations, $f = 0$, $g = 0$. Eliminating a variable may be more difficult in these cases, though. This is where the resultant comes in.

If we treat $f(x,y)$, $g(x,y)$ as polynomials, say in x over $\mathbb{R}[y]$, then the resultant is a polynomial in y.[1] If the resultant has a zero y_0, then $f(x,y_0)$ and $g(x,y_0)$, which are polynomials in x, have a nonconstant common factor, thus a common zero, x_0. The point (x_0, y_0) then must be a point of intersection for the underlying curves.

Example 9.36. Let $f = x^2 - y$ and $g = y + 1$. We can view f and g as elements in $R[x]$ with $R = \mathbb{R}[y]$, or as elements in $R[y]$ with $R = \mathbb{R}[x]$. To use the resultant, we must be dealing with polynomials of positive degree. In $R[x]$,

[1] See Exercise 7.3.13 for a review of the determinant, if this statement seems at all mysterious.

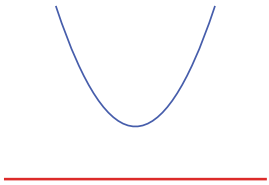


Figure 9.5: The curves given by $f = x^2 - y$ and $g = y + 1$

$\deg f = 2$ and $\deg g = 0$. In $R[y]$, $\deg f = 1$ and $\deg g = 1$ so we view f, g as polynomials in y, over $\mathbb{R}[x]$. The resultant is then a 2×2 determinant. To emphasize that we are viewing the entries in the determinant as coefficients of polynomials in y, we write $\mathcal{R}_y(f, g)$. Then

$$\mathcal{R}_y(f, g) = \begin{vmatrix} x^2 & -1 \\ 1 & 1 \end{vmatrix} = x^2 + 1.$$

Since $x^2 + 1$ is never 0 over $\mathbb{R}$, the resultant reveals that as polynomials in y over $\mathbb{R}[x]$, f and g have no nonconstant common factors, hence, no common zeros. We thus find what we already knew: The curves given by f and g have no intersection points in $\mathbb{R}^2$.

Example 9.37. Consider the curve given by $h = y^2 - 3y - x + 1$, and the one given by f in Example 9.36. We can look at $f = x^2 - y$ and h as polynomials

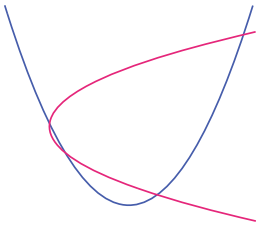


Figure 9.6: The curves given by $f = x^2 - y$ and $h = y^2 - 3y - x + 1$

in x, over $\mathbb{R}[y]$. Then

$$\mathcal{R}_x(f, h) = \begin{vmatrix} -y & 0 & 1 \\ y^2 - 3y + 1 & -1 & 0 \\ 0 & y^2 - 3y + 1 & -1 \end{vmatrix} = y^4 - 6y^3 + 11y^2 - 7y + 1.$$

Using a solver, we find that

$$y^4 - 6y^3 + 11y^2 - 7y + 1 = (y - 3.24698)(y - 1.55496)(y - 1)(y - 0.198062).$$

Viewing f and h then as polynomials in x with coefficients in $\mathbb{R}[y]$, we see that f and g have common nonconstant factors when $y = 3.24698, 1.55496, 1$, and

0.198062. This means that when y has any one of these four values, there are x-coordinates at which f and g are both zero. The resultant, then, allows us to identify four lines in $\mathbb{R}^2$—$y = 3.24698$, $y = 1.55496$, $y = 1$, and $y = 0.198062$—where we expect the curves associated to f and g to intersect each other.

We look for the x-coordinates of the intersection points by viewing f and h as polynomials in y, over $\mathbb{R}[x]$. We then calculate $\mathcal{R}_y(f, h)$, a function of x. Then

$$\mathcal{R}_y(f,h) = \begin{vmatrix} x^2 & -1 & 0 \\ 0 & x^2 & -1 \\ -x+1 & -3 & -1 \end{vmatrix} = x^4 - 3x^2 - x + 1.$$

Courtesy of a solver once again, we find

$$x^4 - 3x^2 - x + 1 = (x - 1.80194)(x - 0.445042)(x + 1)(x + 1.24698).$$

Checking which x-coordinates square to which y-coordinates (so that $f = 0$), we get the following points of intersection in $\mathbb{R}^2$ for the curves given by f and h:

$$(-1, 1), (1.80194, 3.24698), (0.445042, 0.198062), (-1.24698, 1.55496)$$

In Figure 9.6, we see four transverse points of intersection for the two curves.

The next theorem is something we see in calculus. Here, g' is the derivative of g with respect to x. Whenever we talk about calculus, the underlying field is $\mathbb{R}$.

Theorem 9.38. Let $f, g \in \mathbb{R}[x]$. If g is irreducible and not constant, then $g^2|f$ if and only if $g|f$ and $g|f'$.

Proof. If $g^2|f$, then say $f = g^2h$, $h \in \mathbb{R}[x]$. Certainly $g|f$. Using the product rule and chain rule, we have $f' = g^2h' + 2gg'h$, so $g|f'$ as well. Conversely, suppose $g|f$ and $g|f'$. We then have $f = gh$, for some $h \in \mathbb{R}[x]$, so $f' = gh' + g'h = gp$, for some $p \in \mathbb{R}[x]$. We leave it as an exercise for you to verify that this implies $h = gq$, for some $q \in \mathbb{R}[x]$. It follows that $f = g^2q$, so $g^2|f$ as desired. □

This gives us an immediate corollary about repeated roots or factors of a polynomial over $\mathbb{R}$.

Corollary 9.39. For $f \in \mathbb{R}[x]$, $(x-a)^2|f$ if and only if $(x-a)|f$ and $(x-a)|f'$.

We could use the resultant on f and f' to determine whether or not f has any repeated roots. That resultant is special: $\mathcal{R}(f, f')$ is the **discriminant** of f. Recall that the discriminant of a quadratic polynomial $f = ax^2 + bx + c$ is $b^2 - 4ac$. Let us compare this to $\mathcal{R}(f, f')$.

$$\mathcal{R}(f,f') = \begin{vmatrix} c & b & a \\ b & 2a & 0 \\ 0 & b & 2a \end{vmatrix} = 4a^2c - b(ab) = a(4ac - b^2).$$

The discriminant of a quadratic is zero if and only if the quadratic has a single repeated (real) root. Since $a \neq 0$, we see that the resultant tells the same story. The quadratic f has a repeated degree one factor—that is, a nonconstant factor in common with f'—if and only if $b^2 - 4ac = 0$.

Parametric equations are often a natural way to present a curve, especially in applications. In physics problems, the position of an object in the xy-plane at time t may be given by $(x(t), y(t))$. Recall for instance, that the unit circle may be described parametrically by $\{(\cos t, \sin t) | \, t \in [0, 2\pi)\}$. To determine whether such a curve is algebraic, we would like to eliminate t from the parametric equations describing x and y, and see if there is a polynomial in x and y that underlies the curve. This is the sort of job the resultant was made for.

The next example essentially establishes that if a curve is given parametrically as polynomials in t, then the curve must be algebraic. This example is taken from [19].

Example 9.40. Consider the curve $\mathcal{C}$ in $\mathbb{R}^2$ given parametrically by

$$x = 3t^2 + t + 1, \; y = t^4 - 4t^3 - 5.$$

A point (x, y) is on $\mathcal{C}$ if and only if

$$3t^2 + t - x + 1 = 0, \text{ and } t^4 - 4t^3 - y - 5 = 0.$$

Let $f = 3t^2 + t - x + 1$ and $g = t^4 - 4t^3 - y - 5$. We want to find a condition

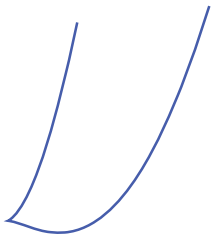


Figure 9.7: The curve given parametrically in Example 9.40

on x and y to make both $f = 0$ and $g = 0$. $\mathcal{R}_t(f, g)$, on the other hand, is zero precisely when the polynomials f and g have a common factor, $at + b$, that is, when the curves determined by f and g have an intersection point $(x(t), y(t))$ where $t = -b/a$.

Using software or an online calculator, we find

$$\mathcal{R}_t(f, g) = \begin{vmatrix} -x+1 & 1 & 3 & 0 & 0 & 0 \\ 0 & -x+1 & 1 & 3 & 0 & 0 \\ 0 & 0 & -x+1 & 1 & 3 & 0 \\ 0 & 0 & 0 & -x+1 & 1 & 3 \\ -y-5 & 0 & 0 & -4 & 1 & 0 \\ 0 & -y-5 & 0 & 0 & -4 & 1 \end{vmatrix} =$$

$$x^4 - 56x^3 - 18x^2y + 72x^2 - 84xy + 81y^2 - 580x + 899y + 2523.$$

This establishes that our curve is the zero set for

$$p(x, y) = x^4 - 56x^3 - 18x^2y + 72x^2 - 84xy + 81y^2 - 580x + 899y + 2523.$$

Example 9.40 generalizes to establish the following proposition.

Proposition 9.41. Let a plane curve $\mathcal{C}$ be given parametrically by $(x(t), y(t))$. If $x(t)$ and $y(t)$ are polynomials in t, let $f = x(t) - x$, and $g = y(t) - y$. Then $\mathcal{R}_t(f, g)$ is a polynomial and $\mathcal{C}$ is its zero set.

A second example from [19] calls for a bolder approach.

Example 9.42. Consider the curve given by $x = \cos(3t), y = \sin(2t)$. We would like to know whether the curve is algebraic. We leave it as an exercise to show that with trigonometric identities, we can rewrite these equations to describe x and y as polynomials in $\cos t$. This gives us

$$x = 4\cos^3 t - 3\cos t,\ y^2 = 4\cos^2 t - 4\cos^4 t. \tag{9.8}$$

Now we view this as a problem in eliminating $\cos t$ from the simultaneous equations in (9.8) to see whether we get a polynomial in x and y.

Let $T = \cos t$. We have $x = 4T^3 - 3T$, and $y^2 = 4T^2 - 4T^4$. Let $f = 4T^2 - 3T - x$, and let $g = -4T^4 + 4T^2 - y^2$. View f and g as belonging to $D[T]$, where $D = \mathbb{R}[x, y]$. Then $\mathcal{R}_T(f, g) = 16y^6 + 4x^4 - 24y^4 - 4x^2 + 9y^2$ so our curve is indeed algebraic and is the zero set for $16y^6 + 4x^4 - 24y^4 - 4x^2 + 9y^2$.

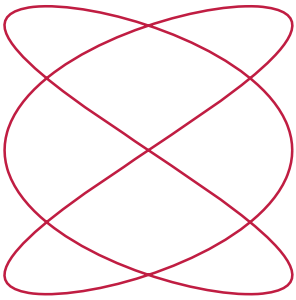


Figure 9.8: The curve described in Example 9.42

Exercises 9.3.

1. Check that if we eliminate x by hand from the equations $f = 0$, $h = 0$ in Example 9.37, we get $y^4 - 6y^3 + 11y^2 - 7y + 1 = 0$. Check that if we eliminate y by hand from the two equations, we get $x^4 - 3x^2 - x + 1 = 0$.

2. Sketch the curves determined by $f = x^2 - xy - 2y^2$ and $g = x^2 - 3xy + 2y^2$ in $\mathbb{R}^2$.

3. Let $f = 2x^3 + 5x^2 + 5x + 3$, $g = 2x^3 + x^2 - 5x - 3$.

   (a) What are the dimensions of the matrix you need to find $\mathcal{R}(f, g)$?

   (b) Write down the matrix.

   (c) Use a solver to find $\mathcal{R}(f, g)$ from the matrix.

   (d) What is the significance of $\mathcal{R}(f, g)$? What does it tell you about f and g?

   (e) What resultant would you use to find the intersection points of the two curves given by $y = 2x^3 + 5x^2 + 5x + 3$, and $y = 2x^3 + x^2 - 5x - 3$?

   (f) Calculate the resultant, either using the definition or eliminating a variable by hand. What does this tell you about intersection points of the two curves? How many intersection points are there?

   (g) Find the intersection points for the curves.

4. Without doing any calculations, find the resultant of $F = x^2 - 2xy + y^2$ and $G = x - y$. Now calculate the resultant from the definition.

5. In the proof of Theorem 9.38, we have $f' = gh' + g'h = gp$ where f, g, h all belong to $\mathbb{R}[x]$ and g is nonconstant and irreducible. We then claim that this implies $h = gq$, some $q \in \mathbb{R}[x]$. Justify that claim.

6. Using the definition of discriminant given in the text, find the discriminant of $x^3 + px + q$.

7. Use the resultant to find a polynomial equation in x and y for Newton's nodal cubic, which is given parametrically by $x = t^2 - 1, y = t - t^3$.

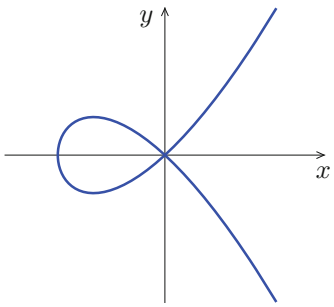


Figure 9.9: Newton's nodal cubic

8. Show that $\cos(3t) = 4\cos^3 t - 3\cos t$.

9. Show that if $y = \sin(2t)$, then $y^2 = 4\cos^2 t - 4\cos^4 t$.

10. Use the method of resultants to find a polynomial expression for the curve given by $x(t) = \cos(2t)$, $y(t) = \sin(4t)$.

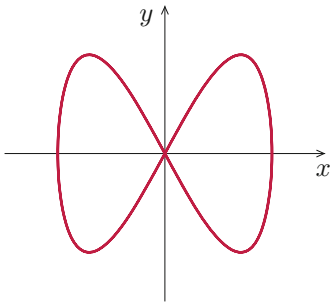


Figure 9.10: The curve given parametrically in Exercise 10

11. Consider the determinant

$$\begin{vmatrix} a_0 & a_1 & a_2 & \dots & a_{n-1} & a_n \\ -\lambda & 1 & 0 & \dots & 0 & 0 \\ 0 & -\lambda & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -\lambda & 1 \end{vmatrix}.$$

Use the definition of resultant to argue that this determinant must be $a_0 + a_1\lambda + a_2\lambda^2 + \cdots + a_n\lambda^n$.

9.5 Homogeneous Polynomials

When we appended ideal points to affine space to construct the associated projective space, it allowed us to view projective space as a geometric completion or closure of affine space. If we think of a pair of parallel lines in an affine plane as exceptional for missing a point of intersection, then we expunge their exceptionality when we append ideal points to form a projective plane. We need a method like that for placing an affine planar curve into a projective plane, so we can see it in this more complete geometric setting. The tool for lifting a curve from an affine plane to a projective plane is the topic of this section.

We have been using this definition, more or less, all along but we make it official and establish some more details.

Definition 9.43. A **plane affine algebraic curve** is the set of zeros in $\mathbb{FA}^2$ for a polynomial $f \in \mathbb{F}[x, y]$, where $\mathbb{F}$ is a field. If f is irreducible, the associated curve is also called **irreducible**. If f has nonunit factors $f_1, \dots, f_s$, then the curves determined by the f_is are **components** of the curve determined by f.

Recall that we can coordinatize $\mathbb{FP}^2$ using homogeneous coordinates in $\mathbb{F}^3$, $[a, b, c]$, where a, b, c are not all zero and for any nonzero $t \in \mathbb{F}$, $[ta, tb, tc] = [a, b, c]$. If $[a, b, c,]$ is going to belong to a curve in $\mathbb{FP}^2$, where the curve is the zero set of a polynomial, p, then (1) p should belong to $\mathbb{F}[x, y, z]$, and (2) p must

have the property that $p(a,b,c) = 0$ for some $(a,b,c) \neq (0,0,0)$, if and only if $p(ta,tb,tc) = 0$ for all nonzero $t \in \mathbb{F}$.

Definition 9.44. A polynomial $p(x_1,\ldots,x_n)$, that is not identically 0, is **homogeneous of degree** k provided $p(tx_1,\ldots,tx_n) = t^k p(x_1,\ldots,x_n)$. If p is not homogeneous we say it is **non-homogeneous**.

Example 9.45. Let $p = x^2 + 3x - y$. Then

$$p(tx,ty) = (tx)^2 + 3(tx) - (ty) = t^2x^2 + 3tx - ty$$

so p is non-homogeneous. Let $P = 2x^2 + 3xy - y^2$. Then

$$P(tx,ty) = 2(tx)^2 + 3(tx)(ty) - (ty)^2 = t^2(2x^2 + 3xy - y^2) = t^2P(x,y),$$

so P is homogeneous of degree 2.

We recognize a homogeneous polynomial of degree k in practice as a sum of monomials of degree k. All monomial summands of $P = x^2 + 3xy - y^2$ have degree 2 but $p = x^2 + 3x - y$ is a sum of monomials of degrees 1 and 2.

Definition 9.46. A **plane projective algebraic curve** is the set of zeros in $\mathbb{FP}^2$ for a homogeneous polynomial $p \in \mathbb{F}[x,y,z]$, where $\mathbb{F}$ is a field.

Recall that we can identify the affine plane $\mathbb{FA}^2$ with the plane $z = 1$ in $\mathbb{FA}^3$, which in turn corresponds to the set of points in $\mathbb{FP}^2$ with homogeneous coordinates $[a,b,1]$. This gives us a natural correspondence between the affine point (a,b) and the projective point $[a,b,1]$. We think of a projective point with the form $[a,b,0]$ as a point at infinity, relative to the affine plane.

There is an analogous way to associate homogeneous and non-homogeneous polynomials.

Let $f \in D[x_1,\ldots,x_n]$ be non-homogeneous. Suppose $\deg f = k$. Say $f = F_0 + F_1 + \cdots + F_k$ where F_i is homogeneous of degree i. Introduce a new variable x_{n+1} and let $F \in D[x_1,\ldots,x_n,x_{n+1}]$ be given by

$$F = x_{n+1}^k F_0 + x_{n+1}^{k-1} F_1 + x_{n+1}^{k-2} F_2 + \cdots + x_{n+1}F_{k-1} + F_k.$$

It is easy to check that F is homogeneous of degree k.

When we assign homogeneous $F \in D[x_1,\ldots,x_n,x_{n+1}]$ to non-homogeneous $f \in D[x_1,\ldots,x_n]$, we say that we **homogenize** f. In this case, we call f and F **polynomial associates**.

Let F be homogeneous of degree k in $D[x_1,\ldots,x_n,x_{n+1}]$. If there is an index i so that x_i does not divide F, we can assign a non-homogeneous polynomial associate to F, f, by substituting 1 for x_i in the expression for F. For instance if $i = n+1$, then f belongs to $D[x_1,\ldots,x_n]$.

While we have described exactly one way to assign a homogeneous associate to a non-homogeneous polynomial, there may be several ways to assign a non-homogeneous associate to a given homogeneous polynomial.

If we are presented with a curve $\mathcal{C}$ in $\mathbb{FA}^2$ determined by a polynomial $f \in \mathbb{F}[x, y]$, the homogeneous polynomial associate of f, $F \in \mathbb{F}[x, y, z]$, determines a projective curve. We say that F determines the **projectivization** of $\mathcal{C}$. We also say that we **lift** $\mathcal{C}$ to $\mathbb{FP}^2$. In this case, too, we say that f is an **affinization** of F or that the curve determined by f is an affinization of the projective curve determined by F.

Example 9.47. Let $f = x^2+3xy^2+x+y+1 \in \mathbb{R}[x, y]$. Note that f has degree 3. We homogenize f to get its polynomial associate $F \in \mathbb{R}[x, y, z]$,

$$F = x^2z + 3xy^2 + xz^2 + yz^2 + z^3.$$

For the homogeneous $P = x^2+3xy+y^2+xz+xy+z^2$, we get its polynomial associate $p = x^2 + 3xy + y^2 + x + xy + 1$ by taking $z = 1$. Since neither x nor y divides P, we get two other polynomial associates of P respectively, by taking $x = 1$, or $y = 1$. We leave it as an exercise to verify that associated polynomials always have the same degree.

Homogeneous polynomials have special properties. Our first theorem is a consequence of the fact that if you multiply non-homogeneous polynomials, you get a non-homogeneous polynomial.

Theorem 9.48. The factors of a homogeneous polynomial are themselves homogeneous.

Proof. Suppose we have $F = fg \in D[x_1, \ldots, x_n]$, not identically zero, where f is non-homogeneous. Say that $f = F_i + \cdots + F_r$, where the summands F_j are nonzero, homogeneous, and written in order of increasing degree. Assume for the moment that g is homogeneous. Notice then that fg has summands F_ig and F_rg. Since D is a domain, F_ig and F_rg are nonzero with $\deg F_ig < \deg F_rg$. It follows that F is non-homogeneous.

Now suppose that both f and g are non-homogeneous and not identically zero. Let $g = G_k+\cdots+G_s$, where the summands G_j are nonzero, homogeneous, and written in order of increasing degree. The product fg has lowest degree term F_iG_k and highest degree term F_rG_s, and these must have different degrees. It follows that F is non-homogeneous.

This proves that a nonzero product that includes non-homogeneous factors must itself be non-homogeneous, which implies the theorem. □

Example 9.49. $F = x^2 - y^2 + z^2 - 2xz = (x+y-z)(x-y-z)$. Since $\mathbb{R}[x, y, z]$ is a UFD, we know there is no other factorization of F that does not involve units.

Once we understand that the world of homogeneous polynomials is closed under multiplication, we can cite a corollary. We leave the proof to the reader.

Corollary 9.50. If F and f are polynomial associates, then the factors of F and f are polynomial associates. Moreover, F is irreducible if and only if f is irreducible.

The next result is also a corollary but because of its importance, we cite it as a theorem. We could state this in more generality for any algebraically closed field but our interest is in $\mathbb{R}$ and its algebraic closure, $\mathbb{C}$.

Theorem 9.51. Let F be homogeneous of degree k in $\mathbb{C}[x, y]$. Then F factors into k degree one polynomials with coefficients in $\mathbb{C}$.

Proof. Suppose F has a factor of x and/or y so that $F = x^r y^s G$, where G is homogeneous with no factor of x or of y. Let g be an associate of G. Then g is in either $\mathbb{C}[x]$ or $\mathbb{C}[y]$ so factors into degree one polynomials. It follows from Corollary 9.50 that G factors completely into degree one polynomials and from there we have the result. □

Think about these results now in terms of curves. An arbitrary polynomial in two variables, say over $\mathbb{C}$, need not factor into degree one polynomials. If it did, it would mean that every "curve" in $\mathbb{C}^2$ would be a union of lines.

Example 9.52. We claim that $f = xy + 1$ cannot be factored into positive degree polynomials over $\mathbb{C}$. Note that $\deg f = 2$. If f could be factored, then since it contains no x^2 or y^2 term, it must factor as $f = (ax + b)(cy + d)$, for some a, b, c, d in $\mathbb{C}$. Since f has no pure y term, $bc = 0$, implying $b = 0$ or $c = 0$. If $c = 0$, there is no xy term in the product so it must be the case that $b = 0$. But then $f = ax(cy + d)$ and the constant term in the product is zero, which contradicts $f = xy + 1$. It is thus clear that f does not factor, regardless of the underlying field.

Note that the theorem is not about a polynomial associate of $f = xy + 1$. It is about the homogeneous associate of a polynomial in one variable, for example $G = xy + x^2$ or $H = xy + y^2$. Both of these factor obviously, as the theorem guarantees.

If f and g are polynomials in x and y, and we care about the underlying affine curves, we may have occasion to consider resultants associated to the homogeneous polynomial associates of f and g, F and G. Note that F and G will be homogeneous polynomials in $\mathbb{R}[x, y, z]$. We may have up to three associated resultants. Viewing F and G as polynomials in x over $\mathbb{R}[y, z]$, we get $\mathcal{R}_x(F, G) \in \mathbb{R}[y, z]$. Viewing F and G as polynomials in y or z, respectively over $\mathbb{R}[x, z]$ or $\mathbb{R}[x, y]$, we get $\mathcal{R}_y(F, G) \in \mathbb{R}[x, z]$, or $\mathcal{R}_z(F, G) \in \mathbb{R}[x, y]$. The first part of Theorem 9.53 applies to any resultant. The proof, which we omit, uses properties of determinants.

Theorem 9.53. If F and G are homogeneous polynomials in $\mathbb{R}[x, y, z]$, with $\deg F = n$ and $\deg G = m$, the resultant $\mathcal{R}(F, G)$ is either 0 or a homogeneous polynomial in two variables with $\deg \mathcal{R}(F, G) \leq mn$. If we can write F with a term ax^n, and G with a term bx^m, for a, b nonzero in $\mathbb{R}$, then $\deg \mathcal{R}_x(F, G) = mn$.

The second statement in the theorem implies that if we can write F with a term au^n and G with a term bu^m, by choosing u judiciously from among x, y, z and taking a and b nonzero in $\mathbb{R}$, then $\deg \mathcal{R}_u(F, G) = mn$. We will run across some examples of the variability in $\deg \mathcal{R}(F, G)$ in the next section.

Exercises 9.4. 1. Let $f = 2xy - x^2 + x + y + 1$.

(a) Find F, the homogeneous polynomial associate for f in $\mathbb{R}[x, y, z]$.

(b) Find non-homogeneous polynomial associates of F that are different from f.

2. Show that if F and f are polynomial associates, then $\deg F = \deg f$.

3. Consider the equation $2x + 3y = 1$.

(a) What does this equation describe in $\mathbb{R}^2$?

(b) The object given by the equation is the zero set of what polynomial, f? Find a homogeneous associate, F, for f.

(c) What does $F = 0$ describe in $\mathbb{R}^3$?

(d) What is the projective curve determined by F?

4. Consider the set of points in $\mathbb{R}^2$ given by $(2t, t^2)$.

(a) What does this set of points describe?

(b) Write an equation that gives a relation satisfied by the x- and y-coordinates of the points on this curve.

(c) Homogenize the equation you found in part (b).

(d) Describe the set of points in $\mathbb{RP}^2$ that satisfy the homogeneous equation you found in part (c) using parameters s and t. (Hint: Think of embedding the points $(2t, t^2)$ into $\mathbb{R}^3$ by putting them in the plane $z = 1$. Now homogenize the resulting parametric description by introducing a second parameter, s.)

5. Supply the details of a proof for Corollary 9.50.

9.6 Bézout's Theorem

We have seen that if $f \in \mathbb{R}[x, y]$ is the product of n degree one polynomials $ax + by + c$, $a, b, c \in \mathbb{R}$, then the algebraic "curve" determined by f in $\mathbb{R}^2$ is the union of n lines. Suppose g is a second polynomial in $\mathbb{R}[x, y]$, also a product of m degree one polynomials. Suppose finally that neither f nor g has a repeated degree one factor, and that f and g do not share any degree one factors. If we projectivize the curves determined by f and g, we get a union of n projective lines for f and a union of m projective lines for g. Each of the n projective lines determined by f intersects each of the m lines determined by g exactly once: We expect the total number of intersections associated to the projective curves determined by f and g to be mn. This is the idea behind Bézout's Theorem.

Bézout's Theorem invokes the idea of the *multiplicity of an intersection point* of curves. In elementary examples, it is not difficult to understand what this means. Getting a precise definition, though, requires a more extensive apparatus

than what is at hand so we settle for an informal discussion and examples to develop some intuition about how multiplicity of intersection points manifests in algebraic curves.

Note first of all that a zero a for a polynomial $p(x)$ has *multiplicity* m provided $p(x) = (x-a)^m q(x)$, where a is not a zero for q. This is a starting point from which we can approach the more general idea of the multiplicity of an intersection point for two curves.

We start with some well-worn examples from analytic geometry and calculus.

Example 9.54. Consider the parabola $y = x^2$ and the line $y = x$, shown in Fig. 9.11. We find the points of intersection by solving $x^2 = x$ to get $x = 0, 1$.

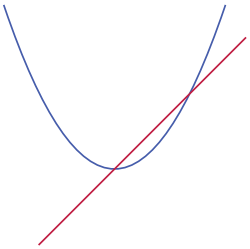


Figure 9.11: The curves $y = x^2$ and $y = x$

Since $x^2 - x = x(x-1)$, the intersection points are associated to roots of multiplicity one of $x^2 - x$. (It is easy to check that $x - x^2$ is also $\mathcal{R}_y(f, g)$ where $f = x - y$ and $g = x^2 - y$.)

Example 9.55. Next consider the curves given by $y = x^2$ and $y = 0$. In this

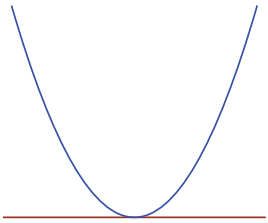


Figure 9.12: The curves given by $y = x^2$ and $y = 0$

case, there is one point of intersection, namely $(0, 0)$ but it has multiplicity two. We can see this manifested in the equation $x^2 = 0$ which has a multiplicity two zero at $x = 0$. We can also interpret a multiplicity two intersection point to be a point where the two curves are tangent. We press that story further in the next example.

Example 9.56. Consider the curves given by $y = x^3$ and $y = 0$. This time, we solve $x^3 = 0$ to find the intersection point. Since $x^3 = 0$ has a zero of multiplicity

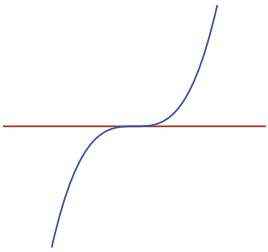


Figure 9.13: The curves given by $y = x^3$ and $y = 0$

three at 0, we say that the intersection point $(0, 0)$ has multiplicity three. As in the last example, the curves are tangent at the point of intersection. Notice the difference, though, in how the curves in this and the previous example bend away from each other near the point of intersection. (The scale is the same in the two figures.) While the curves in each picture share exactly one point of intersection at $(0, 0)$, the curves in Fig. 9.13 seem to have a more extended "hug." Why does that happen?

The functions $y = x^3$ and $y = 0$ do not only agree in value when $x = 0$, their first and second derivatives also agree at $x = 0$. When we get to the third derivative, $d^3(x^3)/dx^3 = 6$, while $d^3(0)/dx^3 = 0$. In Example 9.55, the functions underlying the parabola and the line agree in value at $x = 0$, as do their first derivatives, but their second derivatives are different: $d^2(x^2)/dx^2 = 2$, while $d^2(0)/dx^2 = 0$.

All of our curves so far arise as graphs of functions that are differentiable to all orders at the points of intersection. In this example and in Example 9.55, we can compare the Maclaurin series expansions for the two functions in question. Recall that the Taylor series for a function $y = f(x)$ at a point a where f has derivatives of every order is given by,

$$T(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2 + \cdots + \frac{f^{(n)}}{n!}(x - a)^n + \cdots .$$

The Maclaurin series for a function is the Taylor series at $a = 0$. If two curves intersect at a point $x = a$ where the underlying functions are differentiable to all orders, and where the Taylor series for the functions converge to the functions on an interval, then we can say that the multiplicity of the point of intersection is m, where the Taylor series of the two functions agree for the first m terms and disagree at the $m + 1$st term.

When we have $y = x^3$ and $y = 0$, we compare the Maclaurin series for x^3, which is $0 + 0 \cdot x + 0 \cdot x^2 + 1 \cdot x^3$, with coefficients of all higher degree terms equal to 0, and the Maclaurin series for 0, which is $0 + 0 \cdot x + 0 \cdot x^2 + 0 \cdot x^3$, with coefficients of all higher degree terms equal to 0. Since the Maclaurin series for the two functions are identical for the first three terms, and disagree at the fourth term, the intersection point at $(0, 0)$ has multiplicity three.

This business of comparing higher order derivatives to determine multiplicity

is natural. Recall from calculus that first and higher order derivatives of a function provide ways to measure how the underlying curve bends. If two curves intersect in a single point but seem to have a broad base of agreement in how much to bend at that point, then that intersection point should have higher multiplicity. This is the idea behind multiplicity. The problem is that it takes fairly elaborate machinery to make the idea precise and to measure it, in cases where the curves are not given by differentiable functions.

Example 9.57. Consider the curves given by $f = x^2 - y^3$ and $g = x$, shown in Fig. 9.14. The zero set of f can indeed be described by a function $h(x) = x^{2/3}$

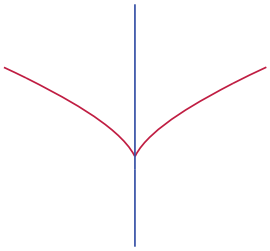


Figure 9.14: The curves given by $f = x^2 - y^3$ and $g = x$

but at 0, the graph does not have a defined tangent. This is a point where neither dy/dx nor dx/dy is defined if we take $x^2 - y^3 = 0$. Notice that though the curve given by f does not have a defined tangent at $(0, 0)$, both sides of the curve at that point are very steep, thus have some agreement with the vertical line about how much to bend. It is easy to check that in this example, $\mathcal{R}_x(f, g) = -y^3$. This actually allows us to measure the agreement between the two curves at the point of intersection, that is, to say that $(0, 0)$ is an intersection point for the two curves, with multiplicity three.

We explore more examples below and state Bézout's Theorem now with a sketched proof.

Theorem 9.58 (Bézout's Theorem)**.** Let F and G be homogeneous polynomials in $\mathbb{C}[x, y, z]$ where FG does not have a repeated factor with positive degree. If $\deg F = m$ and $\deg G = n$, then the curves determined by F and G have mn points of intersection where points are counted with multiplicity.

Sketch of proof. Let F and G be homogeneous in $\mathbb{C}[x, y, z]$ with $\deg F = m$ and $\deg G = n$. If FG has no repeated factors, then the projective curves associated to F and G have no common component. It follows that there are finitely many intersection points between the two curves. Though not necessarily obvious, it is true that via a change in variable, we can always choose a resultant $\mathcal{R}(F, G)$ so that $\deg \mathcal{R}(F, G) = mn$. (A change in variable effects a change in coordinates, but does not change the underlying curves.) By Theorem 9.51, that resultant factors completely into mn linear factors, counting with multiplicity. Setting each factor equal to zero, we determine a projective line in $\mathbb{CP}^2$ where F and G

have a common zero. If a factor of $\mathcal{R}(F,G)$ has multiplicity k, the associated line has k points, counting multiplicity, where the curves underlying F and G intersect. □

We would like to use Bézout's Theorem to understand all possible intersections between a pair of affine curves. By homogenizing polynomials, we lift our curves from $\mathbb{R}^2$ to $\mathbb{RP}^2$ when necessary to find intersections that do not appear in the affine plane. When faced with a polynomial that does not factor over $\mathbb{R}$, we expand the base field and go to $\mathbb{CP}^2$.

The next example is another from calculus but this time, we consider the projectivizations of the curves to see what materializes.

Example 9.59. Let $f = x^2 - y$ again and $g = 2x - 1 - y$. The associated curves are shown in Fig. 9.15. The curves are tangent at $(1,1)$ so have an intersection of

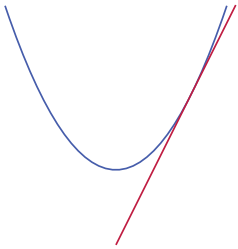


Figure 9.15: The curves $y = x^2$ and $y = 2x - 1$

multiplicity two at $(1,1)$. The multiplicity is apparent when we set $x^2 = 2x - 1$ and get $(x-1)^2 = 0$. We leave it as an exercise to verify that $\mathcal{R}_y(f,g) = 0$ yields $x^2 = 2x - 1$.

Since $(\deg f)(\deg g) = 2$, Bézout tells us that no new intersections arise in the projectivizations of the curves associated to f and g. To check to see how the multiplicity manifests there, though, we have projectivizations of f and g respectively given by $F = x^2 - yz$ and $G = 2x - z - y$. Then

$$\mathcal{R}_z(F,G) = \begin{vmatrix} x^2 & -y \\ 2x - y & -1 \end{vmatrix} = -x^2 + y(2x - y) = -x^2 + 2xy - y^2 = -(x-y)^2.$$

Taken by itself, this means that the curves will have 2 points of intersection—counting multiplicity—on the projective line $x = y$. Note that the projective line given by $x = y$ is the collection of points in $\mathbb{RP}^2$ with coordinates $[a,a,b]$, where a, b are not both zero.

To see where this leads, we substitute x for y in F and G. This gives us $F|_{y=x} = x^2 - xz = x(x-z)$ and $G|_{y=x} = x - z$. This means that the curve associated to F crosses the line $x = y$ at projective points $[0,0,1]$ and $[1,1,1]$, while the curve associated to G crosses the line $x = y$ at $[1,1,1]$. As the only projective point common to the two curves, the intersection point $[1,1,1]$ must have multiplicity two.

Our next example is one in which there is a shortage of intersection points in $\mathbb{R}^2$. The projectivized curves over $\mathbb{R}$ give us the complete picture.

Example 9.60. Let $f = x^2 - y$ and $g = x^2 + y$. These are shown in Fig. 9.16. Elementary methods reveal a multiplicity two intersection at $(0, 0)$. This time,

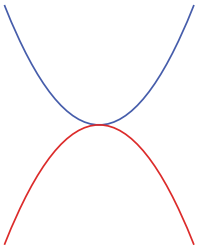


Figure 9.16: The curves $y = x^2$ and $y = -x^2$

we know there should be four points of intersection if we count correctly so we lift to the projectivization.

The projectivizations of f and g are respectively $F = x^2 - yz$ and $G = x^2 + yz$. With an eye toward Theorem 9.53, we see that $\mathcal{R}_x(F, G)$ will be degree four so we calculate $\mathcal{R}_x(F, G) = 4y^2z^2$. This tells us to expect two intersections on the projective line $y = 0$ and two on the projective line $z = 0$. When $y = 0$, we have $F|_{y=0} = x^2$ and $G|_{y=0} = x^2$, giving us the point in $\mathbb{RP}^2$ that corresponds to $(0, 0)$, $[0, 0, 1]$. As we already knew, this is a multiplicity two intersection point. If $z = 0$, $F|_{z=0} = x^2$ and $G|_{z=0} = x^2$ so x is zero in this case as well, and we get a second intersection point with multiplicity two, namely $[0, 1, 0]$.

Next we consider curves with no affine points of intersection at all.

Example 9.61. Let $f = x^2+1-y$ and let $g = x^2+1+y$. The curves determined by f and g are shown in Fig. 9.17. The two parabolas have no affine points in

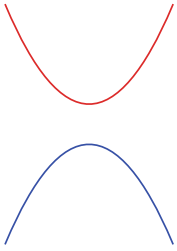


Figure 9.17: Nonintersecting parabolas in $\mathbb{R}^2$

common so if we count properly, we expect to find four projective points where their projectivizations intersect.

Projectivizations for the two curves are given by $F = x^2 + z^2 - yz$, and $G = x^2 + z^2 + yz$. We can say with certainty that F and G are irreducible

polynomials. (We leave the details as an exercise.) Note that the forms of F and G guarantee that both $\mathcal{R}_x(F,G)$ and $\mathcal{R}_z(F,G)$ have degree 4 so we can use either one to find lines in the projective plane that go through intersection points of the underlying curves.

$$\mathcal{R}_z(F,G) = \begin{vmatrix} x^2 & -y & 1 & 0 \\ 0 & x^2 & -y & 1 \\ x^2 & y & 1 & 0 \\ 0 & x^2 & y & 1 \end{vmatrix} = 4x^2y^2$$

From this we expect two intersection points on the projective line $x = 0$ and two on the projective line $y = 0$. Substituting $x = 0$ into F we get $z(z - y)$ and into G we get $z(z + y)$ so $x = 0$ implies $z = 0$. This gives us the projective point $[0, 1, 0]$. Since it is just one point, its multiplicity must be 2.

Substituting $y = 0$ into F we get $x^2 + z^2$, which we also get when we substitute $y = 0$ into G. So when $y = 0$, $x = iz$ or $x = -iz$. These conditions correspond to two points in $\mathbb{CP}^2$, $[i, 0, 1]$ and $[-i, 0, 1]$, each with multiplicity one.

The point $[0, 1, 0]$ with its intersection multiplicity of 2 is intriguing. Consider the affinization of the curves given by F and G by taking $y = 1$. We have $F|_{y=1} = x^2 + z^2 - z$ and $G|_{y=1} = x^2 + z^2 + z$. We leave it as an exercise to verify that the affine curves thus determined (in the xz-plane) are circles, one centered at $(0, 1/2)$ with radius $1/2$, the other centered at $(0, -1/2)$, with radius $1/2$. These are shown in Fig. 9.18. The circles are tangent at $(0, 0)$ so have a

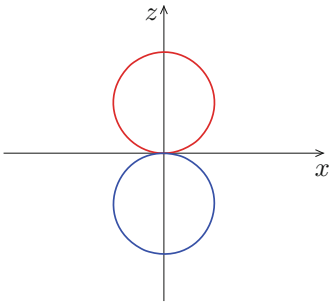


Figure 9.18: Circles with a multiplicity two intersection at the origin

multiplicity two intersection point there.

How does this show up if we address the problem of the intersecting circles from an elementary point of view? If we try setting $x^2 + z^2 - z = x^2 + z^2 + z$, we get $2z = 0$. Plugging $z = 0$ into $x^2 + z^2 - z = 0$ and $x^2 + z^2 + z = 0$, we get $x^2 = 0$, revealing the multiplicity of the intersection point $[0, 1, 0]$. Using another approach, we can describe the curves given by $F|_{y=1}$ and $G|_{y=1}$ locally as functions. The top circle is given by $z = 1 - \sqrt{1 - x^2}$ and the lower circle is $z = -1 + \sqrt{1 - x^2}$. Then we have $1 - \sqrt{1 - x^2} = -1 + \sqrt{1 - x^2}$ implying $1 = 1 - x^2$ implying $x = 0$, with multiplicity two.

Next is a relatively simple example in which the curves go up in degree.

Example 9.62. Consider the curves given by $f = x^3 - y$ and $g = x^4 - y$. We should be able to find 12 intersection points, if we count intersection multiplicities correctly. Since $f = g$ reduces to $x^3(x-1)$, the affine curves intersect at the origin with multiplicity three and at the point $(1,1)$ with multiplicity one. This gives us a total of 4 affine points of intersection. The other 8 must be in the projective plane.

Homogenize the polynomials to get $F = x^3 - yz^2$ and $G = x^4 - yz^3$. Since we are guaranteed that $\mathcal{R}_x(F,G)$ has degree 12, we calculate that to find

$$\mathcal{R}_x(F,G) = \begin{vmatrix} -yz^2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -yz^2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -yz^2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -yz^2 & 0 & 0 & 1 \\ -yz^3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -yz^3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -yz^3 & 0 & 0 & 0 & 1 \end{vmatrix} = y^3z^8(y-z).$$

The affine point $(1,1)$ shows up here via the $y - z$ term. When we substitute y for z in F we get $x^3 - y^3$ and in G we get $x^4 - y^4$, indicating that the projective point $x = y = z$ is a multiplicity one intersection point. That is, $[1,1,1]$, as we already knew. Now we consider the 11 points on the lines $y = 0$ and $z = 0$.

The factor of y^3 in the resultant is associated to an intersection point where $y = 0$. Looking at F and G, we see that when $y = 0$, $x = 0$. This is the origin, associated to the projective point $[0,0,1]$. We saw above that it has multiplicity three.

When $z = 0$, we also have $x = 0$. We conclude that $[0,1,0]$ must have intersection multiplicity 8 as it arises from the factor z^8.

Exercises 9.5.

1. Check that $\mathcal{R}_y(f,g) = 0$ is equivalent to $x^2 = 2x - 1$ in Example 9.59.

2. Affinize the polynomials F and G in Example 9.59 by taking $x = 1$, respectively $y = 1$. Analyze the resulting curves in the yz-plane, respectively, the xz-plane to get different arguments that $[1,1,1]$ has intersection multiplicity two.

3. Calculate $\mathcal{R}_x(f,g)$ in Example 9.54 and $\mathcal{R}_y(f,g)$, and $\mathcal{R}_x(F,G)$ in Example 9.59.

4. Referring to Example 9.60, calculate $\mathcal{R}_z(F,G)$ to see if you can use it to analyze intersections and their multiplicities for the curves associated to F and G.

5. Argue that $F = x^2 + z^2 - yz$, and $G = x^2 + z^2 + yz$ are irreducible by considering affinizations of F and G.

6. Calculate $\mathcal{R}_x(F,G)$ and $\mathcal{R}_y(F,G)$ from Example 9.61.

7. Referring to Example 9.61, if we let $F|_{y=1} = G|_{y=1}$, then near $(0,0)$ in the xz-plane, we claimed to have $1 - \sqrt{1-x^2} = -1 + \sqrt{1-x^2}$. Verify the claim. Now note that setting $\mathcal{R}_z(F|_{y=1}, G|_{y=1}) = 0$ is another way to eliminate z from the equations $F|_{y=1} = 0$, $G|_{y=1} = 0$. Redo that part of the example using the resultant to eliminate z. Which way is easier?

8. Find the intersection points of the projective curves given by $F = x^2 + y^2 - z^2$ and $G = x + y + z$.

9. Find the intersection points, including any that turn up in $\mathbb{RP}^2$ or $\mathbb{CP}^2$ if necessary, and determine the intersection multiplicities, for the curves given by $f = x^3 - y$ and $g = x^5 - y$.

10. Repeat the last exercise for the concentric circles in $\mathbb{R}^2$ centered at $(0,0)$ with radii 1 and 2 respectively.

Chapter 10

Rotations and Quaternions

10.1 Introduction

What is a rotation? A turn, an angle change, an event in which an object starts here and ends up there, where "here" and "there" are points that lie on a circular arc. It is hard to imagine a rotation without a journey, but for the most part, we care about where things start and where they end up, not how they actually found their way to their final positions.

Our rotations are mappings—either from $\mathbb{R}^2$ to itself or from $\mathbb{R}^3$ to itself—that keep the origin fixed. In $\mathbb{R}^2$, the origin is the only point fixed by a rotation that is not the identity mapping. In $\mathbb{R}^3$, there is a line through the origin that is fixed by a nonidentity rotation, the *axis* of the rotation. The idea of a rotation is applicable in higher-dimensional spaces, and we make some general remarks about the collection of rotations on $\mathbb{R}^n$, though our primary interest is rotations on $\mathbb{R}^2$ and $\mathbb{R}^3$.

Rotations in the plane are relatively straightforward but a close study of them lays the foundation for understanding what happens in 3-space, which gives us an excuse to study the quaternions. Until sometime relatively recently, quaternions were of interest mainly to mathematicians in certain specialties. Now they are of interest to computer graphics programmers, mechanical engineers, and other users who have to understand spatial rotations in applications. Our primary objective in this chapter is to understand how and why quaternions are used in applications to describe rotations. Along the way, we meet some other features of this curious world.

10.2 Background on $\mathbb{R}^n$

Rotations form a lovely nexus at the interface of algebra and geometry. We start with some reminders about the Euclidean geometry of $\mathbb{R}^n$, which is manifested in coordinates via the dot product. In Chapter 7, we touched briefly on the dot

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M. I. Dillon, *Geometry Through History*,
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product as it comes up in classical physics, which is often where people first see vectors.

We did not use the dot product in our discussion of affine geometry because affine geometry is the geometry of vector spaces and not every vector space has a dot product, or an analog that measures angle and distance. The dot product gives $\mathbb{R}^n$ its Euclidean structure.

If $\mathbf{v}$ and $\mathbf{w}$ are points in $\mathbb{R}^n$, the distance between them is

$$|\mathbf{v} - \mathbf{w}| = ((\mathbf{v} - \mathbf{w}) \cdot (\mathbf{v} - \mathbf{w}))^{1/2}.$$

The angle between the vectors $\mathbf{v}$ and $\mathbf{w}$ is φ where

$$\varphi = \cos^{-1}\left(\frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}||\mathbf{w}|}\right).$$

Definition 10.1. An **orthogonal mapping** is a linear mapping $R : \mathbb{R}^n \to \mathbb{R}^n$ that satisfies

$$R(\mathbf{u}) \cdot R(\mathbf{v}) = \mathbf{u} \cdot \mathbf{v}.$$

Orthogonal mappings are always invertible. This is part of an exercise that requires proving the following theorem.

Theorem 10.2. The orthogonal mappings on $\mathbb{R}^n$ form a group.

Fix $n \in \mathbb{Z}_+$. The **orthogonal group**, $O(n)$, is the set of all orthogonal mappings on $\mathbb{R}^n$.

Because they respect the dot product, orthogonal mappings on $\mathbb{R}^n$ respect distance and angle, the elements of Euclidean geometry that arise from Hilbert's axioms of order and congruence. To preserve the Euclidean geometry of $\mathbb{R}^n$, a mapping need not be linear, but it must preserve distance and angle.

Recall from Corollary 7.60 that matrices representing the same linear transformation on a finite-dimensional space have the same determinant. If $R \in O(n)$, it is not too difficult to argue that $\det R$ is ± 1. (See the exercises.)

Definition 10.3. A **rotation** is an orthogonal mapping $R : \mathbb{R}^n \to \mathbb{R}^n$ with determinant 1.

Thinking purely in terms of physics or geometry, we can convince ourselves that the rotations on $\mathbb{R}^2$ themselves comprise a group. If we perform two rotations in succession, we get the same effect using one rotation through the sum of the angles. That means that the composition of rotations is a rotation. Every rotation has an inverse: the rotation through the same angle but in the opposite sense. The identity mapping is considered a rotation through the angle 0. All this is true, but much less obvious, for rotations on $\mathbb{R}^n$. It is not difficult to show that rotations on $\mathbb{R}^n$ form a group, though, if we use the formal definition of rotation. We leave that as an exercise.

Definition 10.4. For a fixed $n \in \mathbb{Z}_+$, the **special orthogonal group** $SO(n)$ is the set of rotations on $\mathbb{R}^n$.

What distinguishes rotations from other mappings in $O(n)$? Reflections are in $O(n)$ but not $SO(n)$. How are they different from rotations? If we perform a reflection on a figure, how can we tell the figure was not the subject of a rotation? The answer to this is *handedness.*

If you are holding red balloons in your left hand, then after a reflection, the red balloons are in your right hand. (Look in the mirror!) *A reflection reverses handedness.* After a rotation, the red balloons are still in your left hand. *A rotation preserves handedness.*

How a linear transformation preserves, or does not preserve, handedness is captured mathematically by the sign of the determinant of a matrix representation for the transformation: Reflections have determinant -1 and rotations have determinant 1.

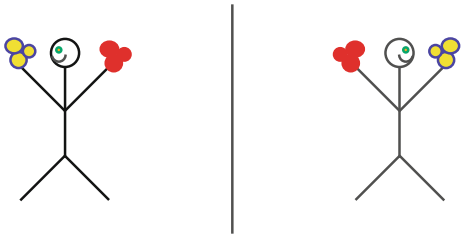


Figure 10.1: Reflection reverses handedness

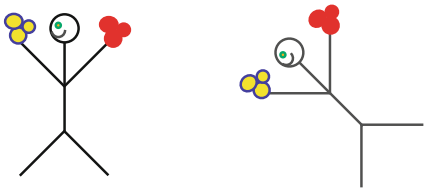


Figure 10.2: Rotation preserves handedness

We turn next to consider a second type of product on vectors in $\mathbb{R}^3$.

Definition 10.5. The **cross product** of $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ in $\mathbb{R}^3$ is $\mathbf{v} \times \mathbf{w} = (v_2w_3 - v_3w_2, v_3w_1 - v_1w_3, v_1w_2 - v_2w_1)$.

A mnemonic presents the cross product as a determinant:

$$\mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Here $\mathbf{i} = (1,0,0)$, $\mathbf{j} = (0,1,0)$, and $\mathbf{k} = (0,0,1)$ are just different labels for $\mathbf{e}_1, \mathbf{e}_2$, and $\mathbf{e}_3$, the elements in the standard basis for $\mathbb{R}^3$. (Physicists seem to prefer the $\mathbf{i}, \mathbf{j}, \mathbf{k}$ notation.)

The cross product of two vectors is perpendicular to both input vectors and the cross product of a vector with itself is $\mathbf{0} = (0,0,0)$. (Proofs are an exercise.)

The cross product is *anti-commutative*, that is, $\mathbf{v} \times \mathbf{w} = -\mathbf{w} \times \mathbf{v}$. (That is an easy exercise.)

The direction of $\mathbf{v} \times \mathbf{w}$ is given by the *right-hand rule*: Align the pinky side of your right hand in the direction of $\mathbf{v}$, then drag the same side of that hand toward $\mathbf{w}$, to make a fist. Now your thumb points in the direction of $\mathbf{v} \times \mathbf{w}$.

Note that the cross product is endemic to $\mathbb{R}^3$: It is not defined on other vector spaces, not even $\mathbb{R}^2$ or $\mathbb{R}^4$.

Definition 10.6. An **orthogonal basis** of $\mathbb{R}^n$ is a set $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ such that

$$\mathbf{v}_i \cdot \mathbf{v}_j = c_i \delta_{ij}$$

where $c_i \in \mathbb{R}_+$, and δ_{ij} is the **Kronecker delta**, that is,

$$\delta_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases} .$$

If for all $i = 1, \ldots, n$, $c_i = 1$, then $\mathcal{B}$ is an **orthonormal basis** for $\mathbb{R}^n$.

The natural basis $\mathcal{B} = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$, is an example of a *right-handed orthonormal basis* for $\mathbb{R}^3$. This is an orthonormal basis that forms a *right-handed* coordinate system. While *orthonormal* is a feature that comes from the dot product, *right-handedness* is a feature that comes from the cross product.

Note that the natural basis, $\mathcal{B}$, is an ordered basis. If we take the cross product of two elements in succession in $\mathcal{B}$—with some allowance about what "in succession" means—we get the third element: $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, and $\mathbf{k} \times \mathbf{i} = \mathbf{j}$. Figure 10.3 is another mnemonic for the cross products of elements in $\mathcal{B}$. Since the cross product is anti-commutative, we introduce a negative sign

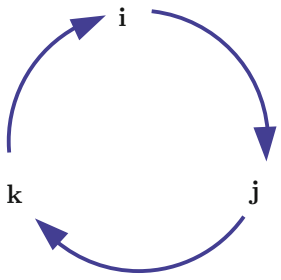


Figure 10.3: Rules for the cross product in $\mathbb{R}^3$

if we multiply successive elements in the counterclockwise direction.

Notice that the dot product of two vectors is a number, while the cross product of two vectors is another vector. Nevertheless, the dot and cross products

have algebraic properties in common. Both enjoy distributive laws with vector addition, for example

$$\mathbf{v} \times (\mathbf{u} + \mathbf{w}) = \mathbf{v} \times \mathbf{u} + \mathbf{v} \times \mathbf{u}.$$

Both the dot and the cross products respect scaling. If $a \in \mathbb{R}$,

$$a(\mathbf{v} \cdot \mathbf{w}) = a\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \cdot (a\mathbf{w}).$$

In working with cross products and dot products, bear in mind that the dot product of nonzero vectors is zero if and only if the vectors are orthogonal and that the cross product of two vectors is orthogonal to the input vectors.

Recall that an $n \times n$ matrix A is invertible provided there is an $n \times n$ matrix B with $AB = I_n$. In that case, we write $A^{-1} = B$.

The **transpose** of a matrix A, A^T, is the matrix we get by switching the rows with the columns of A. For example, if

$$A = \begin{bmatrix} 1 & 2 \\ -3 & 7 \end{bmatrix},$$

then

$$A^T = \begin{bmatrix} 1 & -3 \\ 2 & 7 \end{bmatrix}.$$

Notice that $I_n^T = I_n$ and that I_n is its own inverse.

An important and very useful fact about $SO(n)$ is that it can be viewed as $n \times n$ matrices A such that $A^{-1} = A^T$, and $\det A = 1$.

We always follow the *right-hand rule* for rotations, which is related to the right-hand rule for the cross product. If the thumb of the right hand points in the direction of the axis of rotation, then the fingers of the right hand curl in the direction of a positive rotation angle.

Exercises 10.1.

1. Show that the cross product of two vectors is perpendicular to both input vectors and that the cross product of a vector with itself is $\mathbf{0} = (0, 0, 0)$.

2. Give an example of a mapping on $\mathbb{R}^2$ that is nonlinear but that preserves angle and distance.

3. Though not entirely obvious, it is true that if A is an $n \times n$ matrix, then $\det A = \det A^T$. Prove this for $n = 2$ and $n = 3$.

4. Though not entirely obvious, it is true that if A and B are matrices for which the product AB is defined, then $(AB)^T = B^T A^T$. Prove this for the case where A is a 2×2 matrix and $B = \mathbf{v}$ is a 2×1 column vector. Then prove it when A and B are both 2×2 matrices.

5. The dot product of two vectors in $\mathbb{R}^n$ can be defined in terms of matrix arithmetic: $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$. Using this and the fact that for any $n \times n$ matrix, A, $\det A = \det A^T$, show that if $A\mathbf{v} \cdot A\mathbf{w} = \mathbf{v} \cdot \mathbf{w}$, for all $\mathbf{v}$, $\mathbf{w} \in \mathbb{R}^n$, then $\det A = \pm 1$.

6. Prove that an orthogonal mapping on $\mathbb{R}^n$ has determinant equal to ± 1.

7. Show that the collection of orthogonal mappings on $\mathbb{R}^n$ forms a group.

8. Show that $SO(n)$ is a group.

9. In $\mathbb{R}^2$, if we want to specify a rotation, it is enough to say where we would like to map a single nonzero vector.

   (a) Through what angle and in what direction is the rotation that maps $\mathbf{e}_1$ to $(-1/2, \sqrt{3}/2)$ Where does this rotation map $\mathbf{e}_2$?

   (b) Recall that the standard matrix representation of a linear transformation T on $\mathbb{R}^n$ is the $n \times n$ matrix for which the ith column is $T(\mathbf{e}_i)$. What is the matrix representation, A, for the rotation in part (a)?

   (c) Let $\mathbf{v} = (1, 1)$. Check that your answer in part (b) is correct by comparing $A\mathbf{v}$ to what you get by using trigonometry to determine where the rotation in part (a) sends $\mathbf{v}$.

10. Using the definitions of cross product and dot product, check that for $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$. This verifies that $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.

11. In the last problem, you interpreted the direction of $\mathbf{u} \times \mathbf{v}$ geometrically. In this problem, you will interpret its length. Verify that $|\mathbf{u} \times \mathbf{v}|$ is the area of the parallelogram with sides $\mathbf{u}$ and $\mathbf{v}$.

12. Find three different orthonormal bases for $\mathbb{R}^2$, then for $\mathbb{R}^3$.

13. Fix an ordering on each of the bases you found for the last problem and determine whether any of your examples forms a right-handed coordinate system for $\mathbb{R}^3$.

10.3 Rotations in $\mathbb{R}^2$

Rotations in $\mathbb{R}^2$ are much simpler than those in $\mathbb{R}^3$ because in $\mathbb{R}^2$, there is only one possible axis of rotation: the one going through the origin, perpendicular to the plane. Here we can work out all the details of rotations without too much trouble. That will help us set up the more delicate situation in $\mathbb{R}^3$, where any line through the origin can serve as an axis of rotation.

Let R be the rotation of $\mathbb{R}^2$ through the angle θ. From Chapter 7, we know that the standard matrix representation of R is

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{10.1}$$

It is easy to check, and we leave it as an exercise to verify, that a matrix of the form given in (10.1) satisfies the conditions of Definition 10.3. Indeed, it is

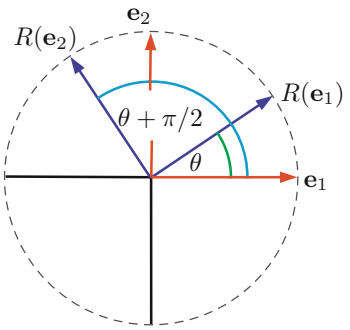


Figure 10.4: A rotation, R, in $\mathbb{R}^2$ is determined by $R(\mathbf{e}_1)$

natural to view $SO(2)$ as the collection of matrices A, as given in (10.1), with $-\pi < \theta \leq \pi$.

There are other ways to view $SO(2)$.

A rotation in $\mathbb{R}^2$ is determined by a single vector. Any nonzero vector will do, but say that we code a rotation according to where it sends $\mathbf{e}_1$. Standing at the origin and pointing your nose in the direction of $\mathbf{e}_1$, turn your head $\pi/2$ radians to your left and your nose will now point at $\mathbf{e}_2$. If R is the rotation through θ, then $R(\mathbf{e}_1) = (\cos\theta, \sin\theta)$ and $R(\mathbf{e}_2)$ is $\pi/2$ radians to the left of $R(\mathbf{e}_1)$. This allows us to identify the set of points on the unit circle,

$$S^1 = \{(x, y) \in \mathbb{R}^2 \,|\, x^2 + y^2 = 1\},$$

with the set of rotations, $SO(2)$: for $-\pi < \theta \leq \pi$,

$$(\cos\theta, \sin\theta) \leftrightarrow \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

The simplicity of identifying $SO(2)$ and S^1 is irresistible. But how do we view a point of S^1 as actually effecting a rotation on $\mathbb{R}^2$? How do we realize the transformation algebraically without going back to the matrix interpretation of elements of $SO(2)$? The answer is to use complex numbers.

Recall that identifying $\mathbb{C}$ with $\mathbb{R}^2$, we view (a, b) in $\mathbb{R}^2$ as representing the complex number $z = a + bi$. The points of S^1 then correspond to the unit complex numbers, $z = a + bi$ where $|z|^2 = a^2 + b^2 = 1$.

When we write $z = a + bi$, we say z is in *rectangular form*. Recall that every complex number can be written also in *polar form*

$$z = |z|e^{i\theta} = |z|(\cos\theta + i\sin\theta).$$

Here $|z|$ and θ are the polar coordinates of the point (a, b) in $\mathbb{R}^2$ and $e^{i\theta}$ is defined to be $\cos\theta + i\sin\theta$. (Here e is indeed the base of the natural exponential.) Just

as with polar coordinates for points in $\mathbb{R}^2$, we say $|z|$ is the *modulus* of z and θ is the *argument* of z.

We do not normally think of multiplying points of $\mathbb{R}^2$ but multiplication of complex numbers naturally follows use of the distributive law and the rule $i^2 = -1$. In terms of the real and imaginary parts of $z_1 = a_1 + b_1 i$, and $z_2 = a_2 + b_2 i$,

$$z_1 z_2 = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1) i. \tag{10.2}$$

In terms of moduli and arguments, if $z_1 = |z_1| e^{i\theta_1}$ and $z_2 = |z_2| e^{i\theta_2}$, then

$$z_1 z_2 = |z_1||z_2| e^{i(\theta_1 + \theta_2)}. \tag{10.3}$$

We leave it as an exercise to verify that these two views of multiplication in $\mathbb{C}$ are consistent.

Equation (10.3) implies that if z has modulus 1 and argument φ—that is, if z is a point on S^1—multiplication by z effects a rotation through φ. That is the key to understanding the unit complex numbers, S^1, as a model for $SO(2)$.

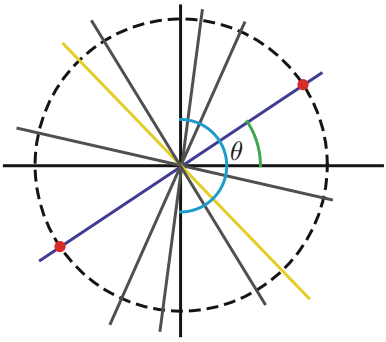


Figure 10.5: A rotation in $\mathbb{R}^2$ can be identified with a point on the real projective line. The red dots represent a single point in $\mathbb{RP}^1$

There is one more model for the group of rotations on $\mathbb{R}^2$. This one is based on the real projective line, $\mathbb{RP}^1$. A point in $\mathbb{RP}^1$ can be represented by a single line through the origin in $\mathbb{R}^2$. To the line at an angle of θ radians to the positive x-axis, associate the rotation through 2θ. Taking $\theta \in (-\pi/2, \pi/2]$, we get a 1-1 correspondence between projective points and rotations.

The interpretation of $SO(2)$ as the circle, S^1, or equivalently, as the real projective line, shows that $SO(2)$, a group, is also a *topological space*, that is, a set in which notions of open subsets and nearness make sense in a very particular way. This is not a statement about the effect of a rotation on $\mathbb{R}^2$. It is a statement about the rotation group itself. We will see this, as well, when we look at $SO(3)$. Topological structure becomes an important idea in deeper studies of $SO(n)$.

Exercises 10.2. 1. Verify that a matrix of the form given in (10.1) satisfies Definition 10.3.

2. Verify that the transpose of a matrix of the form given in (10.1) is its inverse.

3. Sometimes i, in $\mathbb{C}$, is defined as "the square root of -1." How would you then calculate the product $\sqrt{-1}\sqrt{-1}$? Could you add something to the rules for multiplying radicals or exponents that would correct the problem, so that you could use the radical notation in this context, as it often is used, $\sqrt{-1} = i$?

4. In this exercise, you will verify that multiplication by unit complex numbers corresponds to addition of angles, thus to rotations in $\mathbb{R}^2$.

   (a) Let $z = a + bi$ where $a, b \in \mathbb{R}$. We have $|z| = \sqrt{a^2 + b^2}$. What is the argument of z in terms of a and b?

   (b) If $z = |z|e^{i\theta}$ what are the real and imaginary parts of z in terms of $|z|$ and θ?

   (c) Convert $z_1 = |z_1|e^{i\theta_1}$ and $z_2 = |z_2|e^{i\theta_2}$ to rectangular form.

   (d) Verify that you get the same complex number $z_1 z_2$ whether you use (10.2) for z_1 and z_2 in rectangular form or (10.3) for z_1 and z_2 in polar form.

   (e) Verify that multiplication by a unit complex number has the effect of a rotation through θ, if we identify points in $\mathbb{R}^2$ with $\mathbb{C}$.

5. In Chapter 8, we discussed how we could identify an affine plane in the model for $\mathbb{RP}^2$ in which projective points are realized as lines through $\mathbf{0}$ in $\mathbb{R}^3$ and projective lines are realized as planes through $\mathbf{0}$ in $\mathbb{R}^3$. How can we identify an affine line in $\mathbb{RP}^1$ in the analogous fashion?

6. Again using lines through the origin in $\mathbb{R}^3$ as projective points, how did we define homogeneous coordinates in $\mathbb{RP}^2$? Mimic that process here to define homogeneous coordinates on $\mathbb{RP}^1$. What is the point at infinity?

7. If we use S^1 with antipodal points identified as a model for $\mathbb{RP}^1$, what corresponds to the point at infinity, as identified in the last problem? How do you identify the points of an affine line in this model? What model for $\mathbb{RP}^2$ corresponds to S^1 with antipodal points identified?

10.4 Rotations in $\mathbb{R}^3$

We have seen that among all $\mathbb{R}^n$, $\mathbb{R}^3$ has some special conventions. The region of $\mathbb{R}^3$ in which all coordinates are positive is called the *first octant*. (The other eight regions that are defined by the coordinate axes do not have standardized names.) The standard arrangement for coordinate axes in $\mathbb{R}^3$ is as in Fig. 10.6. The *coordinate planes* are the xy-plane, the yz-plane, and the xz-plane. With this choice of coordinate axes, the yz-plane is the plane of the page.

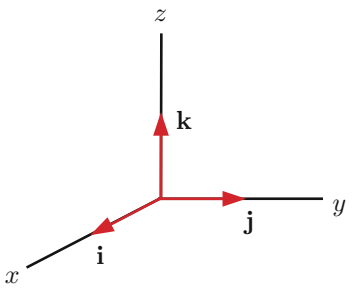


Figure 10.6: The standard axis arrangement for $\mathbb{R}^3$

If you think about a rotation in $\mathbb{R}^3$ in physical terms, there are several things you should notice. It might help if you think about what happens to the points on the unit sphere, S^2, the collection of points in $\mathbb{R}^3$ one unit from the origin. A rotation leaves S^2 invariant: Points on S^2 are mapped to other points on S^2. Note, moreover, that the relative positions of the points on S^2 do not change under rotation.

A rotation of $\mathbb{R}^3$ is always about some axis. (This is not the case in higher-dimensional $\mathbb{R}^n$!) That may seem obvious, but it becomes less so if you think of performing a sequence of rotations, and realize that the end result is a single rotation with a well-defined axis. (This is the content of a theorem of Euler.)

Perpendicular to the axis of rotation, there is a plane through the origin. That plane intersects S^2 in a great circle, the *equator* for the rotation. The equator for the rotation is the collection of points on S^2 that move the farthest under the rotation. Note that any point that moves under a rotation traces a circular arc centered at the axis of rotation.

Thinking of the plane of the equator as a copy of the xy-plane, we see that a rotation in $\mathbb{R}^3$ has an associated rotation in 2-space. We can think of this 2-space rotation as dragging everything in $\mathbb{R}^3$ along with it. If we know the axis of rotation, then we know what the plane of the equator is. If know the plane of the equator, and the angle of the rotation, then we should be able to write down a matrix that effects the rotation.

Recall that the standard matrix A for a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is given by $A = [T(\mathbf{e}_1)\ T(\mathbf{e}_2)\ T(\mathbf{e}_3)]$.

Example 10.7. Consider the rotation, R, through $\pi/6$ radians in which the axis of rotation is the z-axis. Note that $R(\mathbf{e}_3) = \mathbf{e}_3$. The x- and y-components for an arbitrary vector in $\mathbb{R}^3$, however, will be moved around in the xy-coordinate plane according to the matrix

$$\begin{bmatrix} \cos \pi/6 & -\sin \pi/6 \\ \sin \pi/6 & \cos \pi/6 \end{bmatrix} = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}.$$

The matrix that codes for the rotation in $\mathbb{R}^3$ is thus

$$A = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

In other words, for any $\mathbf{v} \in \mathbb{R}^3$, $R(\mathbf{v}) = A\mathbf{v}$.

It is a theorem in linear algebra that a general rotation in $\mathbb{R}^3$—one that is about an arbitrary axis through the origin—can be described using a sequence of three matrices, each of which describes a rotation about one of the coordinate axes. When rotations are handled this way in programming or mechanics, for example, users talk about employing *Euler angles* or *Euler rotations*. The advantage of using Euler angles to describe rotations is that the underlying idea is easy to understand: We code a single rotation by rotating first about the x-axis, then the y-axis, then the z-axis. Of course, any permutation of axes is available and one can get away with only using two axes, but generally, using Euler angles, there are three angles per rotation. Users find this appealing. Whatever problems they may encounter, at least they know what is going on mathematically.

The alternative, which seems more natural once you know some linear algebra, is to change coordinates. We did this in Chapter 7 when we discussed a reflection in $\mathbb{R}^2$ through an arbitrary line through the origin. Bringing this idea to bear here, we describe a rotation again using three matrices: The first and last matrices effect a change of basis, the middle matrix describes a single rotation about a coordinate axis.

Suppose, for example, that we want to describe the rotation R through $-2\pi/3$ radians about the line determined by the vector $\mathbf{n} = (1, 1, 1)$. We start by changing to a right-handed orthonormal basis that includes a unit vector in the direction of $\mathbf{n}$. That unit vector is $\mathbf{u} = (1/\sqrt{3})\mathbf{n}$. We need a unit vector orthogonal to $\mathbf{n}$. Call it $\mathbf{v} = (v_1, v_2, v_3)$ and notice that we must have

$$\mathbf{u} \cdot \mathbf{v} = 0, \text{ and } \mathbf{v} \cdot \mathbf{v} = 1.$$

We can take $\mathbf{v} = (1/\sqrt{2})(1, 0, -1)$. Taking $\mathbf{w} = \mathbf{u} \times \mathbf{v}$, we have

$$\mathbf{w} = (1/\sqrt{6})(-1, 2, -1).$$

Now $\mathcal{B} = \{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a right-handed orthonormal basis for $\mathbb{R}^3$.

Let $\mathcal{E}$ be the standard basis for $\mathbb{R}^3$. The $\mathcal{B}$ to $\mathcal{E}$ change of basis matrix is

$$P = [\mathbf{u}\ \mathbf{v}\ \mathbf{w}].$$

It is easy to check that this is also an element in $SO(3)$. Its inverse is thus its transpose.

We have $R(\mathbf{u}) = \mathbf{u}$. Let $R|_W$ be R restricted to the plane $W = \text{span}\{\mathbf{v}, \mathbf{w}\}$. The matrix representation of $R|_W$ with respect to the (ordered) basis $\{\mathbf{v}, \mathbf{w}\}$ for W is

$$\begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}.$$

The matrix representation for R with respect to $\mathcal{B}$ is then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/2 & \sqrt{3}/2 \\ 0 & -\sqrt{3}/2 & -1/2 \end{bmatrix}.$$

Using P and $P^{-1} = P^T$, we get the matrix representation of R with respect to the standard basis of $\mathbb{R}^3$:

$$\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ 0 & -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}. \tag{10.4}$$

You should work out the details to check that the product is correct and that this matrix effects a rotation of $\mathbb{R}^3$ through $-2\pi/3$ radians about $\mathbf{n} = (1, 1, 1)$, with respect to the standard basis.

Let's look at how we would get the same rotation using Euler angles. There are several ways to do this but we just need one. If we rotate first about the z-axis through π, then about the y-axis through $\pi/2$, then once again about the z-axis through $\pi/2$, we get the rotation we want. The three matrices that code for these mappings, in the correct order, are

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{10.5}$$

While the product of matrices is exactly the same as in (10.4), this approach gets us a prettier decomposition of the rotation, which we find appealing. There is a problem here, though, when users employ Euler angles for rotations. Typically, when studying transformations in geometry, we only care about the initial and the final positions of an object, not about its particular path. In mechanical and graphical systems, though, users do care about the particular path an object takes. This is where a difficulty may arise.

Suppose we rotate first about the x-axis, then the y-axis, then the z-axis. Let α, β, and γ be the angles of rotation. The matrix decomposition for Euler angles in this case looks like this.

$$\begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix} \tag{10.6}$$

At each step in the sequence, we specify α, β, γ, the angles of rotation about the x-, y-, and z-axes. If, at a particular step, $\beta = \pi/2$, then the middle matrix in (10.6) becomes

$$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \tag{10.7}$$

If we calculate the matrix product in (10.6), using the matrix in (10.7), then after applying trigonometric identities, we have

$$\begin{bmatrix} 0 & -\sin(\alpha+\gamma) & -\cos(\alpha+\gamma) \\ 0 & \cos(\alpha+\gamma) & -\sin(\alpha+\gamma) \\ 1 & 0 & 0 \end{bmatrix}. \tag{10.8}$$

At this stage, the difference between α, as an angle of rotation about the x-axis, and γ, an angle of a rotation about the z-axis, is lost. If we wanted to effect a change in α, it looks identical to effecting a change in γ. Two of our rotation axes have lined up, resulting in a loss of a degree of freedom called *gimbal lock*.

A gimbal is a ring that can spin on an axis upon which an object is mounted. Two and three gimbal rings can be connected to allow an instrument—for example a compass, or a gauge—to float freely when its environment tilts.

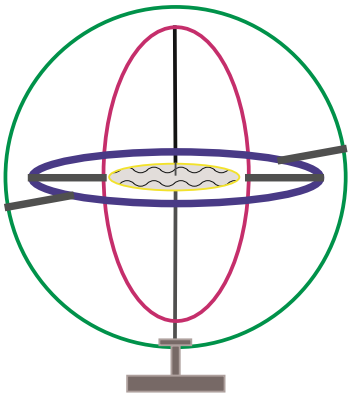


Figure 10.7: Three gimbal rings allow an object to float when the environment tilts

Gimbal rings securing objects on boats and planes are a common sight, for instance. A three gimbal ring system suffers gimbal lock when the axis of one gimbal aligns with the axis of one of the others resulting in a loss of a degree of freedom and awkward or labored motion of the mounted object.

The term "gimbal lock," as used in computer programming, refers to what happens when a program uses Euler angles to code a rotation and one of the angles is $\pi/2$. The effect of gimbal lock on an animation sequence is to cause an object undergoing rotation to take a strange looking path instead of one along a circular arc, as the viewer expects.

The problem of gimbal lock in mechanical engineering is typically solved using engineering methods. In computing, the solution to gimbal lock is to code rotations using *quaternions* instead of Euler angles. In terms of calculations, Euler angles are slightly more costly than quaternions. That slight difference becomes significant in long animation sequences, though. In addition to avoiding the problems associated to gimbal lock, programmers who use quaternions can enjoy enormous computational savings.

Definition 10.8. The **quaternions**, $\mathbb{H}$, is a four-dimensional vector space over $\mathbb{R}$. We designate its standard basis as $\{1, i, j, k\}$, so that a typical element in $\mathbb{H}$ can be written $q = a + bi + cj + dk$, where $a, b, c, d \in \mathbb{R}$. Here a is the *real part of* q. In addition to being a vector space, $\mathbb{H}$ is also a ring. Multiplication is determined by the distributive law, the fact that 1 is a multiplicative identity, and the following rules:

$$i^2 = j^2 = k^2 = -1$$

$$ij = k,\ ji = -k,\ jk = i,\ kj = -i,\ ki = j,\ ik = -j.$$

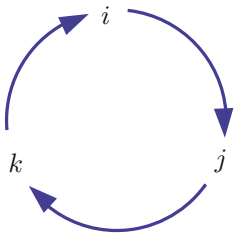


Figure 10.8: A mnemonic for multiplication in $\mathbb{H}$

Figure 10.8 shows a device for remembering the multiplications of i, j, k in $\mathbb{H}$. The similarity to the cross product of elements in the standard basis of $\mathbb{R}^3$ is not coincidental, as we will see below. As in $\mathbb{R}^3$, products performed in the clockwise direction are positive. Products performed in the counterclockwise direction are negative.

William Rowan Hamilton (1805–1865) discovered the quaternions after an arduous search inspired by the beautiful connection between the complex numbers and the geometry of $\mathbb{R}^2$. Seeking to extend this story to $\mathbb{R}^3$, Hamilton came up empty handed until he realized, at a famous moment while crossing a bridge in Dublin, that he could have what he sought if he viewed $\mathbb{R}^3$ as sitting inside $\mathbb{R}^4$. It was $\mathbb{R}^4$, with a multiplicative structure, that would be identified, not with a three-dimensional analog, but with a four-dimensional analog to $\mathbb{C}$. This four-dimensional space would be the foundation of Hamilton's quaternions.

While $\mathbb{C}$ and $\mathbb{H}$ are vector spaces, they are also rings containing $\mathbb{R}$ as a subring. Indeed, notice that $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$. In the case of both $\mathbb{C}$ and $\mathbb{H}$, we can view i as a basis element such that $i^2 = -1$.

Since $\mathbb{C}$ is a field, we might well ask whether $\mathbb{H}$ is a field as well. We see that the answer is "no" once we notice that $ij \neq ji$. (Recall that the nonzero elements in a field form an abelian group under multiplication.) The nonzero

quaternions do form a group under multiplication, but not an abelian group. Except for commutativity of multiplication, $\mathbb{H}$ satisfies all the other properties of a field. This makes $\mathbb{H}$ a *skew field*.

The multiplicative inverse of a quaternion is defined analogously to the multiplicative inverse of a complex number. Let $q = a + bi + cj + dk \in \mathbb{H}$ for $a, b, c, d \in \mathbb{R}$. The **quaternion conjugate** of q is $\bar{q} = a - bi - cj - dk$. Then

$$q\bar{q} = a^2 + b^2 + c^2 + d^2 = |q|^2,$$

where $|q|$ is the modulus or the *norm* of q, i.e., its length as a vector. If $w \in \mathbb{H}$ is nonzero, we can divide $q \in \mathbb{H}$ by w: $q/w = q\bar{w}/|w|^2$. The multiplicative inverse of nonzero $w \in \mathbb{H}$, $1/w = w^{-1}$, is then given by $1/w = \bar{w}/|w|^2$.

There are several distinguished subsets of $\mathbb{H}$ that we wish to identify. The set of **unit quaternions** is defined as $\mathcal{U} = \{q \in \mathbb{H} | \, |q| = 1\}$. $\mathcal{U}$ is not a subspace of $\mathbb{H}$, but is a multiplicative subgroup of the nonzero quaternions: If q_1, q_2 are in $\mathcal{U}$, then it is easy to check that $|q_1 q_2| = 1$ so that $q_1 q_2 \in \mathcal{U}$. The set of **pure quaternions** is $\mathcal{P} = \{bi + cj + dk | \, b, c, d \in \mathbb{R}\}$. $\mathcal{P}$ is a three-dimensional vector subspace of $\mathbb{H}$ but is not a subring of $\mathbb{H}$: for $b \in \mathbb{R}$, $bi \in \mathcal{P}$ but $(bi)(bi) = -b^2$ is not in $\mathcal{P}$.

If $a, b, c, d \in \mathbb{R}$, we may identify $q = a+bi+cj+dk \in \mathbb{H}$ with $(a, b, c, d) \in \mathbb{R}^4$. In this scenario, we can multiply $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^4$ according to the rule for quaternion multiplication. $\mathcal{U}$ is then identified with S^3, the 3-sphere in $\mathbb{R}^4$, that is, vectors in $\mathbb{R}^4$ with length 1. $\mathcal{P}$ is then identified with a three-dimensional subspace of $\mathbb{R}^4$, $\{(0, b, c, d) | \, b, c, d \in \mathbb{R}\}$.

We can use the idea of a *direct sum of vector spaces* to facilitate the connection between $\mathbb{H}$ and $\mathbb{R}^4$. Let $(U, +_U, \cdot_U)$ and $(V, +_V, \cdot_V)$ be vector spaces over $\mathbb{R}$. The **direct sum**, $U \oplus V = \{(a; b) | \, a \in U, b \in V\}$, is also a vector space over $\mathbb{R}$, where addition and scaling in $U \oplus V$ are defined componentwise, that is, if $a, a' \in U$, $b, b' \in V$, and $c \in \mathbb{R}$, we define $+$ and $\cdot$ on $U \oplus V$ by

$$(a; b) + (a'; b') = (a +_U a'; b +_V b') \text{ and } c(a; b) = (c \cdot_U a; c \cdot_V b).$$

Lemma 10.9. If U and V are vector spaces over $\mathbb{R}$, where $\dim U = m$, and $\dim V = n$, then $U \oplus V \cong \mathbb{R}^{m+n}$.

We leave the proof of the lemma as an exercise.

The connection between multiplication in $\mathbb{H}$ and the dot and cross product on $\mathbb{R}^3$ is revealed if we identify $\mathbb{H}$ with $\mathbb{R} \oplus \mathbb{R}^3$. Since $\dim \mathbb{H} = 4 = \dim(\mathbb{R} \oplus \mathbb{R}^3)$, we know that as vector spaces, $\mathbb{H} \cong \mathbb{R} \oplus \mathbb{R}^3$. The view of $q \in \mathbb{H}$ as a sum of a real number a and a pure quaternion $\mathbf{p} = bi + cj + dk$, for $b, c, d \in \mathbb{R}$ is realized here as the view of $q \in \mathbb{H}$ as an ordered pair, $(a; \mathbf{p}) \in \mathbb{R} \oplus \mathbb{R}^3$. Designating the dot product on $\mathbb{R}^3$ as; and the cross product as $\times$, we can use the distributive law to verify that, for $a, b \in \mathbb{R}$, and $\mathbf{p}, \mathbf{q} \in \mathcal{P}$,

$$(a; \mathbf{p})(b; \mathbf{q}) = (ab - \mathbf{p} \cdot \mathbf{q}; a\mathbf{q} + b\mathbf{p} + \mathbf{p} \times \mathbf{q}). \tag{10.9}$$

If $(a; \mathbf{p})$ is in $\mathcal{U}$, it is easy to show that its multiplicative inverse is $(a; -\mathbf{p})$. We leave that as an exercise.

Next we define a second kind of conjugation, different from quaternion conjugation. The dual-purpose terminology is unfortunate, but context should make it clear which type of conjugation is under discussion.

Fix nonzero $q \in \mathbb{H}$. We define a mapping $\mathbb{H} \to \mathbb{H}$ by

$$w \mapsto qwq^{-1},$$

for any $w \in \mathbb{H}$. This mapping is called **conjugation by a group element,** q, where in this case, q, belongs to the multiplicative group of nonzero elements in $\mathbb{H}$. Conjugation by an element comes up frequently in group theory, especially as it applies to geometry. We ran across conjugation with matrices when studying reflections, and again above, in (10.4), when we invoked a change of basis to find a rotation matrix. We leave it as an exercise to show that conjugation by a nonzero quaternion defines a linear transformation on the underlying vector space, $\mathbb{H}$.

Lemma 10.10. Conjugation by an element in $\mathcal{U}$ is a linear transformation on $\mathbb{H}$ that leaves the pure quaternions invariant.

Proof. We leave the first part of the proof as an exercise but show that conjugation by a unit quaternion maps $\mathcal{P}$ to $\mathcal{P}$. Note that we only need establish that on conjugation by a unit quaternion, an element in $\mathcal{P}$ is mapped to an element with real part 0.

Taking $q = (a; \mathbf{u}) \in \mathcal{U}$ and $v = (0; \mathbf{v}) \in \mathcal{P}$ we have

$$qvq^{-1} = (a; \mathbf{u})(0; \mathbf{v})(a; -\mathbf{u}) = (-\mathbf{u} \cdot \mathbf{v}; a\mathbf{v} + \mathbf{u} \times \mathbf{v})(a; -\mathbf{u})$$

$$= (-a(\mathbf{u} \cdot \mathbf{v}) + (a\mathbf{v} + \mathbf{u} \times \mathbf{v}) \cdot \mathbf{u}; \ldots) = (-a\mathbf{u} \cdot \mathbf{v} + a\mathbf{v} \cdot \mathbf{u} + \mathbf{u} \times \mathbf{v} \cdot \mathbf{u}; \ldots).$$

We leave it as an easy exercise to check that $-a\mathbf{u} \cdot \mathbf{v} + a\mathbf{v} \cdot \mathbf{u} + \mathbf{u} \times \mathbf{v} \cdot \mathbf{u} = 0$, which proves the result. □

We are ready now to view a rotation in $\mathbb{R}^3$ as arising from conjugation by a unit quaternion.

Theorem 10.11. Let R_θ be the rotation of $\mathbb{R}^3$ through an angle θ about an axis in the direction of the unit vector $\mathbf{u}$. Let $q = (\cos(\theta/2); \sin(\theta/2)\mathbf{u}) \in \mathbb{H}$. Then for $\mathbf{v} \in \mathbb{R}^3$, $R_\theta(\mathbf{v}) = \mathbf{w}$, where $\mathbf{w} \in \mathbb{R}^3$ is given by $(0; \mathbf{w}) = q(0; \mathbf{v})q^{-1}$.

Proof. If $\mathbf{u} \in \mathbb{R}^3$ is a unit vector, then $q = (\cos(\theta/2); \sin(\theta/2)\mathbf{u}) \in \mathcal{U}$, as

$$|q| = \sqrt{\cos^2(\theta/2) + \sin^2(\theta/2)|\mathbf{u}|^2} = 1.$$

Since $q \in \mathcal{U}$, its multiplicative inverse is $q^{-1} = (\cos(\theta/2); -\sin(\theta/2)\mathbf{u})$.

If conjugation by q does yield a rotation of $\mathbb{R}^3$ about the axis spanned by $\mathbf{u}$, then conjugation by q should fix $u = (0; \mathbf{u})$. This is easy to verify:

$$quq^{-1} = (\cos(\theta/2); \sin(\theta/2)\mathbf{u})(0; \mathbf{u})(\cos(\theta/2); -\sin(\theta/2)\mathbf{u}) =$$

$$(-\sin(\theta/2); \cos(\theta/2)\mathbf{u})(\cos(\theta/2); -\sin(\theta/2)\mathbf{u}) =$$

$$(-\sin(\theta/2)\cos(\theta/2)+\sin(\theta/2)\cos(\theta/2);\cos^2(\theta/2)\mathbf{u}+\sin^2(\theta/2)\mathbf{u})=(0;\mathbf{u}).$$

Next, we consider a matrix representation of R_θ. Let $\mathcal{B}=\{\mathbf{u},\mathbf{v},\mathbf{w}\}$ be a right-handed orthonormal basis for $\mathbb{R}^3$. Recall that we can form $\mathcal{B}$ starting with $\mathbf{u}$, by taking $\mathbf{v}$ orthogonal to $\mathbf{u}$, then scaling it to have length 1, and then taking $\mathbf{w}=\mathbf{u}\times\mathbf{v}$. The matrix representation of R_θ with respect to $\mathcal{B}$ is then

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}.$$

Notice then that R_θ maps

$$\mathbf{u}\mapsto\mathbf{u},\ \mathbf{v}\mapsto(\cos\theta)\mathbf{v}+(\sin\theta)\mathbf{w},\ \mathbf{w}\mapsto-(\sin\theta)\mathbf{v}+(\cos\theta)\mathbf{w}. \tag{10.10}$$

We just have to check that conjugation by q has the same effect on $\mathcal{B}$: As a linear transformation on $\mathbb{R}^3$, conjugation by q is completely determined by its effect on a basis.

Viewing $\mathbf{v}$ as belonging to $\mathcal{P}$, we have

$$q(0;\mathbf{v})q^{-1}=(\cos(\theta/2);\sin(\theta/2)\mathbf{u})(0;\mathbf{v})(\cos(\theta/2);-\sin(\theta/2)\mathbf{u})=$$

$$(0;\cos(\theta/2)\mathbf{v}+\sin(\theta/2)\mathbf{w})(\cos(\theta/2);-\sin(\theta/2)\mathbf{u})=$$

$$(0;\cos^2(\theta/2)\mathbf{v}+\cos(\theta/2)\sin(\theta/2)\mathbf{w}-\sin(\theta/2)\cos(\theta/2)\mathbf{v}\times\mathbf{u}-\sin^2(\theta/2)\mathbf{w}\times\mathbf{u})=$$

$$(0;(\cos^2(\theta/2)-\sin^2(\theta/2))\mathbf{v}+2\sin(\theta/2)\cos(\theta/2)\mathbf{w})=(0;(\cos\theta)\mathbf{v}+(\sin\theta)\mathbf{w})$$

We leave it as an exercise to show that conjugation by q maps

$$\mathbf{w}\mapsto-(\sin\theta)\mathbf{v}+(\cos\theta)\mathbf{w}.$$

That verifies the claim that conjugation by q effects R_θ. □

The calculations, and the evaporation of difficulties associated with coding a rotation using Euler angles, make it clear why $\mathbb{H}$ is the preferred device among programmers for coding rotations.

The Rotation Group on $\mathbb{R}^3$

The relationship between $SO(3)$, the group of rotations on $\mathbb{R}^3$, and the unit quaternions is analogous to that between $SO(2)$ and S^1, the unit complex numbers, except that the unit quaternions have a topological character—a shape, if you will—that is very different from that of $SO(3)$. (This is partly why the unit quaternions are better for coding rotations!) We jumped up two dimensions to get from $\mathbb{C}$ to $\mathbb{H}$. To approach an intuitive understanding of the topological nature of $SO(3)$—what it would feel like as a physical space if we were actually in it—we jump up two dimensions from S^1. One way to accomplish that is to use the **unit ball** in $\mathbb{R}^3$,

$$\mathbf{B}_0=\{(x,y,z)\in\mathbb{R}^3\,|\,x^2+y^2+z^2\leq 1\}.$$

If $P \in \mathbf{B}_0$ and $|P| < 1$, we say P is an *interior point* and if $|P| = 1$, we say P is a *boundary point* of $\mathbf{B}_0$.

Let O be the origin in $\mathbb{R}^3$. Associate a rotation on $\mathbb{R}^3$ to each point $P \in \mathbf{B}_0$ by taking $\overrightarrow{OP}$ as the axis of the rotation, and using the angle $\xi = |OP|\pi$. The boundary points, then, all determine rotations through π.

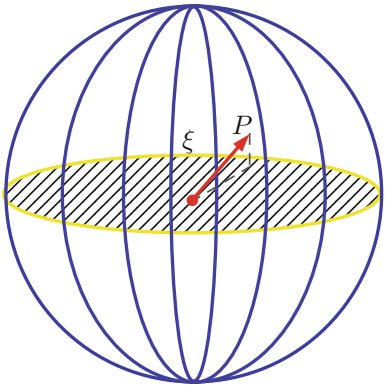


Figure 10.9: $\mathbf{B}_0 \subset \mathbb{R}^3$ can be identified with $SO(3)$, if we identify antipodal boundary points

Rotations are executed according to the right-hand rule. From a vantage point in the first octant, a viewer looking at the origin would see a counterclockwise rotation associated to any first octant point in $\mathbf{B}_0$. Points in the octant where x, y, z are all negative would then determine clockwise rotations.

Antipodal point pairs on $\mathbf{B}_0$ are boundary points on either end of a given diameter. Antipodal point pairs determine identical rotations through π radians. At one of the antipodal points, the rotation is clockwise. At the other, it is counterclockwise. As mappings, these rotations are the same. We get a bijection between rotations in $SO(3)$ and points in $\mathbf{B}_0$, if we identify each antipodal point pair as a single point.

A space is *simply connected* provided we can shrink any loop in the space—that is, any curve that starts and ends at the same point—down to a point without leaving the space and without breaking the loop. Examples of simply connected spaces are S^2 and $\mathbb{R}^n$. The circle S^1 is not simply connected and neither is its close friend, $SO(2)$. We would like to appreciate that $SO(3)$ is not simply connected, either.

Think of starting at a boundary point, P, on $\mathbf{B}_0$ and going straight through O along the course of the diameter determined by P right to the point P' that is antipodal to P. As points in $SO(3)$, $P = P'$ so the path we described is actually a loop in $SO(3)$. We cannot contract that loop to a point without breaking it: As soon as we start to shrink the loop, it is no longer a loop, because we lose the beginning/endpoint when it shrinks away from the surface of $\mathbf{B}_0$.

This story should seem familiar. If we think of antipodal point pairs on S^2 as a model for $\mathbb{RP}^2$, we uncover a similar plot. Start at a point on S^2, and

follow a path to its antipodal. This is a closed loop in $\mathbb{RP}^2$ that cannot be contracted to a point while remaining a loop. This convinces us that $\mathbb{RP}^2$ is not simply connected either. Indeed, $SO(3)$ can be identified with $\mathbb{RP}^3$. Just as we modeled $\mathbb{RP}^2$ using $\mathbb{R}^3$, we model $\mathbb{RP}^3$ using $\mathbb{R}^4$.

Using $\mathbb{R}^4$ to try to understand $\mathbb{RP}^3$, we have choices analogous to the ones we had when we were studying $\mathbb{RP}^2$. We can use homogeneous coordinates, or lines through the origin, or antipodal point pairs on S^3 to model projective points in $\mathbb{RP}^3$. We can make the connection between $SO(3)$ and $\mathbb{RP}^3$ using $\mathbf{B}_0$. In particular, we associate to $P \in \mathbf{B}_0$ an antipodal point pair on S^3.

Suppose the $\mathbb{R}^3$ coordinates for $P \in \mathbf{B}_0$ are (x, y, z). Since $x^2 + y^2 + z^2 \leq 1$, we can let

$$w = \sqrt{1 - (x^2 + y^2 + z^2)},$$

and we are guaranteed that $w \in \mathbb{R}$. If $w \neq 0$, P is an interior point of $\mathbf{B}_0$ and we assign to $P \in \mathbf{B}_0$ the antipodal point pair $(x, y, z, \pm w) \in S^3$. For instance, $\mathbf{0}$ is mapped to $(0, 0, 0, \pm 1)$. If $w = 0$, P is a boundary point on $\mathbf{B}_0$. In this case we assign to P and its antipodal the antipodals $(x, y, z, 0)$ and $(-x, -y, -z, 0)$ on S^3. This gives us a bijective correspondence between the points of $SO(3)$ and the points of $\mathbb{RP}^3$, a correspondence that actually gets at the topology. But that is another story for another course so a good place to end this journey.

Exercises 10.3.

1. Consider a rotation in $\mathbb{R}^3$ through π radians that maps $\mathbf{e}_1$ to $-\mathbf{e}_1$. How many different axes of rotation could there be for such a rotation? What is the criterion that determines such an axis?

2. Check that the matrix A given in Example 10.7 satisfies Definition 10.3. Verify that the angle between $(1, 1, 1)$ and $R(1, 1, 1)$ is actually $\pi/6$ radians.

3. Rework Example 10.7, this time using the y-axis as the axis of rotation with the rotation through the angle $3\pi/4$. Find the vectors $R(\mathbf{i})$ and $R(\mathbf{k})$. Find the rotation matrix, A and verify that it satisfies the definition of a rotation.

4. This problem is about the matrices in (10.4).

   (a) Verify that the outer matrices are themselves rotation matrices.

   (b) Verify that the outer matrices are inverses of one another.

   (c) Verify that the matrix on the left, maps $(1, 0, 0)$ to a vector on the line determined by $\mathbf{n} = (1, 1, 1)$.

   (d) Calculate the product of the three matrices by hand.

   (e) What is the net effect of the rotation on $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$?

5. Verify that the matrix product in (10.5) is the same as the product of the matrices given in (10.4).

6. Verify that the matrix in (10.8) does code for a rotation. In particular, if $-\sin(\alpha+\gamma) = \cos(\xi)$, and $\cos(\alpha+\gamma) = \sin(\xi)$, what is ξ? Verify that the axis for the rotation goes through the point $(0, 1, -\tan(\alpha+\gamma))$. (Discuss what is happening when $\alpha+\gamma = \pi/2$.)

7. Prove Lemma 10.9.

8. Suppose $q = (a; \mathbf{v}) \in \mathcal{U}$. Show that $q^{-1} = (a; -\mathbf{v})$.

9. Show that conjugation by $q \in \mathbb{H}$, $q \neq 0$, defines a linear transformation on $\mathbb{R}^4$.

10. Show that for any vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, and $a \in \mathbb{R}$,

$$-a\mathbf{u}\cdot\mathbf{v} + a\mathbf{v}\cdot\mathbf{u} + \mathbf{u}\times\mathbf{v}\cdot\mathbf{u} = 0.$$

11. Let $\mathbf{u}$ be a unit vector in $\mathbb{R}^3$. In the proof of Theorem 10.11, we mention some of the steps involved in building a right-handed orthonormal basis for $\mathbb{R}^3$ using $\mathbf{u}$. Fill in those steps. For example, if $\mathbf{u} = (a, b, c)$ is a unit vector, how could you choose $\mathbf{v}$? Then if we take $\mathbf{w} = \mathbf{u}\times\mathbf{v}$, is it automatic that $\mathbf{u}, \mathbf{v}, \mathbf{w}$ behave as $\mathbf{i}, \mathbf{j}, \mathbf{k}$, per the mnemonic given in Fig. 10.3? In other words, check that $\mathbf{v}\times\mathbf{w} = \mathbf{u}$, and $\mathbf{w}\times\mathbf{u} = \mathbf{v}$.

12. In the proof of Theorem 10.11, why did we not have to show that every vector in $\mathbb{R}^3$ underwent a rotation?

13. What conditions would we have to impose on q to get that conjugation by q on pure quaternions preserves norm? In other words, for what $q \in \mathbb{H}$ does $|qwq^{-1}| = |w|$ for $w \in \mathcal{P}$?

14. Find the explicit expression for q to effect the rotation in $\mathbb{R}^3$ through $\pi/6$, about an axis that is normal to the plane $x+y+z = 0$. Take the rotation that is counterclockwise when viewed along the axis from the first octant. Use q to determine where the point $(-1, 1, 2)$ is mapped to under the rotation. Give exact coordinates.

15. Check that the correspondences between $SO(3)$ and its various models are bijective.

16. Convince yourself that the correspondences that we describe between $SO(3)$ and its various models are actually continuous.

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